



The Marviken Full Scale Critical Flow Tests

Summary Report

Joint Reactor Safety Experiments in the Marviken Power Station Sweden

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ABSTRACT

The Marviken Full Scale CFT (Critical Flow Tests) are summarized. The program objectives are identified. The test facility hardware, instrumentation, procedures, matrix, data and error limits are discussed. The significant test results are summarized. Finally the test results are presented as a function of the universal stagnation quality to enable the scaling effects between the nine test nozzle geometries to be studied.

SUMMARY

The Marviken Full Scale CFT (Critical Flow Tests) were conducted between mid-1977 and December 1979 as a multi-national project at the Marviken Power Station. The 27 CFT experiments, together with the test procedures, equipment, and evaluation techniques are presented in a set of 35 reports. The content of the CFT program is summarized here. Data describing the nozzle inlet stagnation conditions and mass fluxes are presented. Of necessity, other data are only summarized if they are needed to enable data users to obtain a better understanding of the test behaviour.

The tests were conducted by discharging water and steam-water mixtures from a full sized reactor vessel through a large diameter discharge pipe that supplied the flow to the test nozzle. The nine test nozzle geometries all had rounded entrances followed by a nominally 200, 300 or 500 mm constant diameter cross-section. The nozzles ranged in length from 166 to 1809 mm.

Most tests were conducted with a nominal initial steam dome pressure of 5 MPa (4 MPa for one test) and with the water initially subcooled between 50°C and 1°C (with respect to the steam dome pressure). The water oxygen content was approximately the same for all tests, but it was decreased by a factor of 140 with respect to a reference test in order to determine the effect of the water air content on the critical mass flow rate.

The vessel, discharge pipe and nozzle were instrumented to determine the test behaviour and to provide a basis for evaluating the stagnation conditions and mass fluxes at the nozzle inlet. The instrumentation readings were recorded using a pulse code modulation system. Data error limits were evaluated for all measurements.

Using the recorded measurements, changes in the fluid state which occurred while the test fluid flowed from the

instrumentation station to the nozzle inlet were examined. The resulting nozzle inlet conditions are presented. The most accurate rates of mass flow through the nozzle were obtained using a pitot-static rake during the subcooled portion of the experiments and by tracing the liquid mass history during the saturated portion of the experiments. The peak mass flow rate measured during the CFT program was 14 Mg/s.

Nozzle pressure profiles showed a pressure recovery zone just downstream of the nozzle inlet during the subcooled test phase and usually a steadily decreasing pressure during the saturated test phase.

Examination of the subcooled data recorded for the nine nozzle geometries indicates that the different nozzle mass fluxes lie within the CFT program error limits at L/D (length to diameter) ratios greater than 1.5. However, the mass flux behaviour seems to be dependent on the diameter as well as the length at L/D's less than 1.5. A more clear dependence on length and diameter seems to apply for saturated flow with zero quality. The decrease in the initial water air content had no observable effect on the nozzle mass flux histories.

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AMERICAN MEDICAL ASSOCIATION

The American Medical Association is a national organization of physicians and surgeons, organized for the purpose of promoting the science and art of medicine, and of improving the medical education of the people. It was organized in 1847, and has since that time been engaged in a constant effort to advance the interests of the medical profession, and to secure the highest quality of medical education and practice. The Association has a long and distinguished history, and has been instrumental in the establishment of many of the most important medical institutions in the United States. It has also been successful in securing the recognition of the medical profession as a distinct and independent body, and in securing the highest standards of medical education and practice. The Association is now engaged in a new and important work, and is determined to continue its efforts to advance the interests of the medical profession, and to secure the highest quality of medical education and practice.

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1. INTRODUCTION

The Marviken Full Scale CFT (Critical Flow Tests) were initiated in mid-1977 to obtain critical mass flow rate data for short pipes of large diameter in the subcooled and low quality regions under steady or quasi-steady state conditions. The CFT Program was financed by the Forsøgsanlaeg Risø (Denmark), the Kernforschungszentrum Karlsruhe GmbH (Federal Republic of Germany), the Technical Research Centre of Finland (Finland), the Commissariat à l'Energie Atomique and the Electricité de France (France), N.V. tot Keuring van Electrotechnische Materialen (the Netherlands), the Institutt for Atomenergi (Norway), Studsvik Energiteknik AB (Sweden), the Electric Power Research Institute and the United States Nuclear Regulatory Commission (USA). The first test of the resulting program was conducted in January, 1978 and the test phase was completed in July, 1979.

Although extensive critical mass flow testing had been conducted with small scale nozzle geometries prior to the CFT program, no data were available to allow researchers to determine whether conclusions based on small scale results and phenomena observed there were applicable to full scale pipe geometries found in operating nuclear power stations. Thus the CFT Program sought to provide the data necessary to form a link between the available small scale data and existing operational hardware.

This report summarizes the data obtained during the CFT project. As such the report is of necessity limited to a general discussion of the data obtained. Details concerning individual tests or test equipment and procedures can be found by consulting one of the remaining thirty-four reports which document the CFT Program (see the Report List provided on the back cover).

The summary report is arranged to present a synopsis of the total aspect of the CFT program. The various sections of

the report summarize the objectives, the facility, the test program, the instrumentation/data acquisition systems, the data processing, the data error limits, the test results and finally the effect of parameter variations.

2. OBJECTIVES

The CFT program objectives were to obtain data on the critical mass flow rate as a function of upstream stagnation conditions and geometry:

$$\dot{m} = f(P, h, L, D, c)$$

where

$\dot{m}$ = mass flow rate

P = upstream stagnation pressure

h = upstream stagnation enthalpy

L = nozzle length

D = nozzle diameter

c = air concentration

Data were to be recorded for subcooled and low quality conditions upstream of the nozzle for short pipes (length-to-diameter ratios ≤ 4) of large diameter (200 to 500 mm). Water was to be used as the test fluid. The tests were to be conducted at steady or quasi-steady conditions at $P \leq 5$ MPa with the fluid subcooled $\leq 50^\circ\text{C}$. In addition, the effect of water air content on the nozzle critical mass flow rate was to be investigated.

As such, the objective of the Summary Report is to present the manner in which the CFT project fulfilled the program objectives. The relationships between the tests in the matrix and important results from individual tests are discussed. In general, only results which have been reported in the individual test reports are summarized. However, the defined stagnation conditions at the nozzle inlet and obvious scaling relationships between the test nozzles are also summarized.

3. TEST FACILITY DESCRIPTION

The relationships between the facility components are discussed. The test nozzles are described in some detail.

Finally, because a number of changes were made to the facility hardware in order to improve the test data as the CFT program was conducted, the modifications are listed and summarized.

3.1. Survey of the test facility prior to Test 1

Vertical cross-sectional views of the test facility and of the discharge pipe/test nozzle are shown in Figs 3:1 and 3:2 respectively. The major components of the facility are:

- a. The pressure vessel: net volume = 425 m^3 , maximum design pressure = 5.75 MPa, and maximum design temperature = 272°C .
- b. The discharge pipe: consisting of the ball valve and pipe spools which house the instrumentation upstream of the test nozzle.
- c. The test nozzles and rupture disc assemblies: a set of nozzles of specified lengths and diameters to which the rupture disc assemblies were attached.
- d. The containment and exhaust pipes: consisting of the drywell (net volume = 1934 m^3), the wetwell (net volume = 2144 m^3), the fuel element transport hall (net volume = 303 m^3), the ground level 3.2 m diameter exhaust pipe and the upper 0.4 m diameter exhaust pipe.

The interrelationships between the above components during the CFT were such that most of the test fluid was contained in the pressure vessel. A test was initiated by failing the discs contained in the rupture disc assembly. Measurements recorded in the vessel, the discharge pipe and the test nozzle provided the data to meet the test objectives, while the vessel fluid, in the form of a steam-water mixture, was exhausted through the test nozzle into the

containment. Most of the liquid discharge was retained in the containment while the containment pressure was relieved by discharging a steam-water mixture through the ground level and upper exhaust tubes to the atmosphere.

3:2 General description of the test nozzles and rupture disc assemblies

The CFT program was conducted using nozzles with nominal diameters of 200 to 500 mm and L/D's (length to diameter ratios) from 0.3 to 3.7. Nozzles in this range were chosen to provide full scale critical flow data for operational nuclear plants. It was acknowledged early in the CFT program that nozzle diameters larger than 500 mm could not be tested due to equipment constraints. Nozzle diameters less than 500 mm were chosen to allow extrapolation to larger diameters and to provide data in a range between existing data and the 500 mm CFT data. The range in nozzle L/D's was selected so as to provide data for short pipes only (i.e. L/D's approximately equal to 4 or less).

Each test nozzle consisted of a rounded inlet, having a radius of curvature equal to the nozzle radius, followed tangentially by a cylindrical section (see Table 3:1 and Fig 3:3). The inlet radius of curvature was defined as a compromise between all the possible alternatives which could be suggested as a representative inlet pipe section in a nuclear power plant.

Each nozzle was bolted to the lower pipe spool (see Fig 3:2). The nozzle outlet was equipped with an assembly containing two rupture discs installed in carrier rings, held in position by a ring fastener. The diameter of the discs was dependent on the test nozzle.

The first 12 critical flow tests were performed using test nozzles with flared outlets combined with rupture disc assemblies and flared ring fasteners. However, midway through the CFT program, concern was expressed because it was suspected that the fluid could not expand freely in the flared outlets downstream of the constant diameter test section. It was therefore decided to modify the nozzles and the rupture disc assemblies to obtain a constant flow area. Because constant diameter nozzle/rupture disc assemblies were not immediately available, Tests 13 and 14 were conducted with the constant diameter test nozzle with 200 mm diameter and $L/D = 3.0$ and a 400 mm diameter rupture disc assembly (see Fig 3:4). Finally, Tests 15 through 27 were conducted using a constant diameter test nozzle section, formed in part by the rupture disc assemblies and the straight ring fastener (see Fig 3:5).

The two rupture discs used in any test were identical. The carrier rings, which retained the rupture discs, were sandwiched between the test nozzle exit and the ring fastener. The upstream carrier ring was equipped with three vents to overpressurize the chamber between the rupture discs.

The hardware was manufactured with a surface finish equivalent to 25 μm . However, by the time some of the nozzles were tested, the metal surface was covered with a uniform coating of oxide which qualitatively had the roughness of galvanized metal. A short description of the surface condition of each nozzle is contained in Table 3:1.

Table 3:1 Test nozzles - basic dimensions and conditions

D mm	L mm	L/D	L1 mm	L2 mm	L3 mm	L4 mm	R mm	USED IN TESTS:	COMMENTS SEE FOOTNOTE	Figs
200	590	3.0	0	100	100	100	100	13, 14	a	3:4
300	290	1.0	55	150	150	150	150	6, 7	b	3:3
300	511	1.7	0	150	150	150	150	25, 26	b, c	3:5
300	895	3.0	55	150	150	150	150	1, 2, 12	a	3:3
300	1116	3.7	0	150	150	150	150	17, 18, 19	d, c	3:5
500	166	0.3	0	225	225	250	250	23, 24	e, f	3:5
500	730	1.5	0	225	225	250	250	20, 21, 22 27	g, f, h	3:5
500	1809	3.6	0	181	156	241	250	15, 16	i, f, h	3:5
509	1589	3.1	55	156	225	241	250	3, 4, 5, 8 9, 10, 11	i	3:3

NOTE: Tolerance = ± 4 mm

- a) Nozzle surface condition: homogeneous with only small undulations (less than 0.5 mm) evident in the surface, near the weld parallel to the nozzle centerline.
- b) Nozzle surface condition: step changes in diameter at planes at distances of 200, 270 and 293 mm from the nozzle inlet caused by welding. The step changes are irregular and range from 0.5 mm toward the centerline to 1 mm away from the nozzle centerline.
- c) Burrs on the rupture disc assembly rings protrude as much as 1.1 mm into the nozzle stream. The 004M281 tap was tested with a 0.5 mm peripheral burr.
- d) Nozzle surface condition: homogeneous excluding small region approximately 918 mm from the nozzle inlet with 0.5 to 1 mm undulation. One pressure tap (6 mm in diameter) at nozzle reference level, not connected until after Test 9, had a 1 mm burr.
- e) Nozzle surface condition: generally homogeneous but with slight irregularities and a prominent weld bead.
- f) Burrs on the rupture disc assembly protrude as much as 1 mm into nozzle stream.
- g) Nozzle surface condition: step changes in diameter at planes normal to the nozzle centerline and 276 and 524 mm from the nozzle inlet. Step changes at 276 mm from 0 to 1.4 mm away from the nozzle centerline. Step changes at 524 mm range from 0 to 1.3 mm toward the nozzle centerline. Between the 276 and 524 mm lengths a 1 mm high weld seam is parallel to the nozzle centerline. Weld splatter evident at four locations. Some surface irregularities.
- h) These nozzles have a decrease in diameter from 509 mm to 500 mm, beginning 55 mm upstream of the innermost rupture disc retaining ring (see Ref 1).
- i) Nozzle surface condition: homogeneous, but with a weld seam parallel to axis centreline approximately 1 mm high located between the 004M229 and 004M230 taps. The inlet radius of curvature of this nozzle is not standard (see Ref 1).

3.3 Summary of the test facility modifications following Test 1

During the course of the CFT project a number of modifications were made to the test facility. Such modifications are summarized in Table 3:2. It should be noted that comprehensive assembly drawings of all CFT hardware modifications, which occurred both prior to and during the program, are available to all participants if desired. The list of assembly prints is found in Ref 1.

Table 3:2 Summary of the modifications to the test facility which occurred after Test 1

FIRST TEST AFTER MODIFICATION	MODIFICATION	DESCRIPTION OF MODIFICATION - SEE REFERENCE
3	1. DP probe II installed	2
	2. Additional structural supports added to the discharge pipe/nozzle assembly	1
13	The test nozzle/rupture disc mounting geometry was modified	1
15	1. Hole in moderator tank bottom enlarged from 5.5 m ² to 9.0 m ²	1
	2. DP probe III installed	2
	3. Pitot-static rake installed in instrumentation ring II	2
	4. Larger exhaust pipe installed from top of containment to atmosphere	1
	5. The test nozzle/rupture disc assembly mounting geometry was modified	2

The test facility modifications with the most significant impact on the test nozzle inlet condition data are those which occurred just prior to Test 15. In particular, the enlargement of the hole in the moderator tank bottom (see Figs 3:2 and 3:6) was completed to improve the uniformity of the fluid conditions at the nozzle inlet. The test nozzle geometry was also modified as described in subsection 3.2.

Hole cuttings:

- 1 Floor beneath vessel
- 2 Drywell floor
- 3 Transport channel floor
- 4 Transport channel wall
- 5 Front wall
- 6 Cable passages

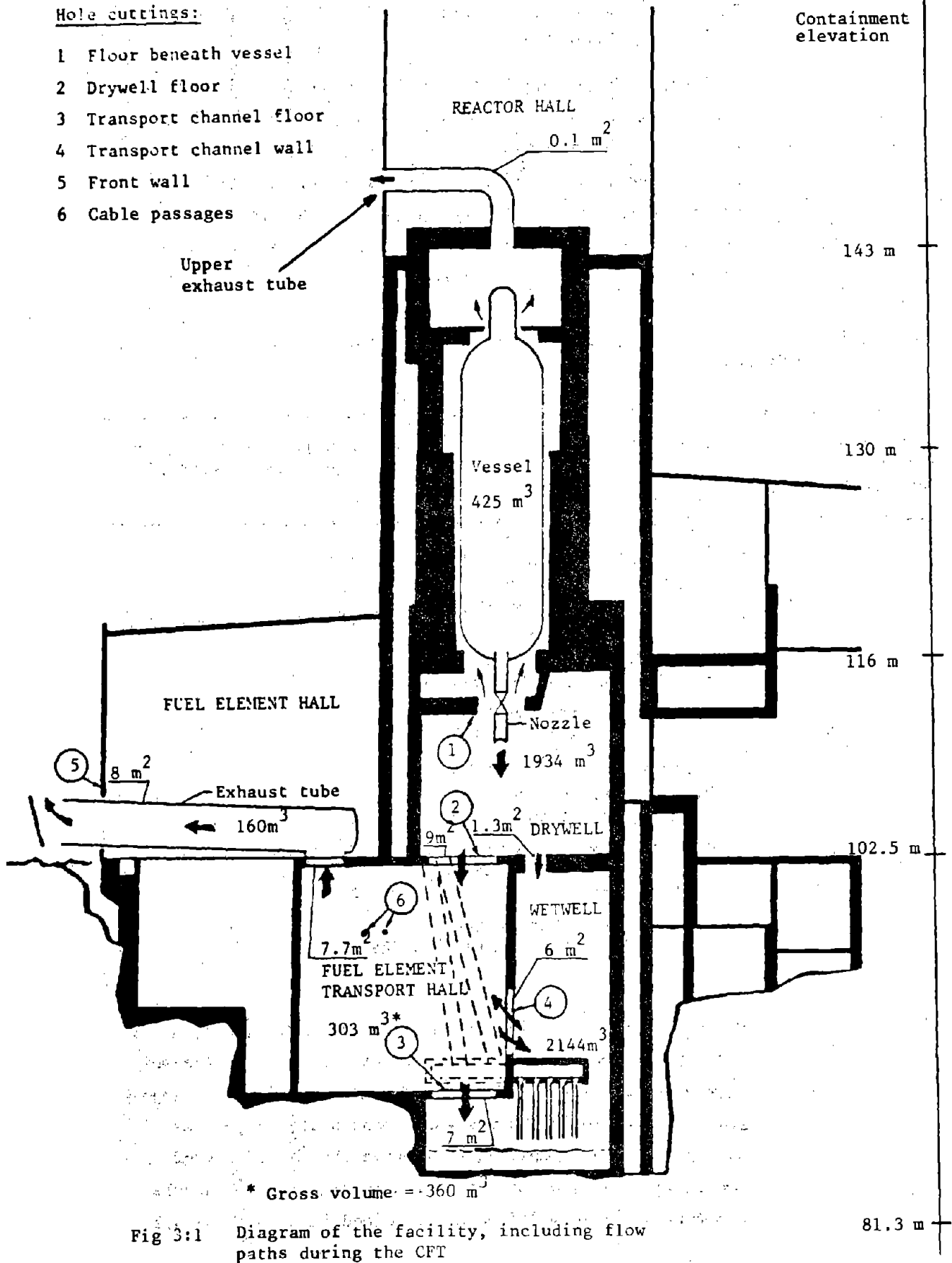


Fig 3:1 Diagram of the facility, including flow paths during the CFT

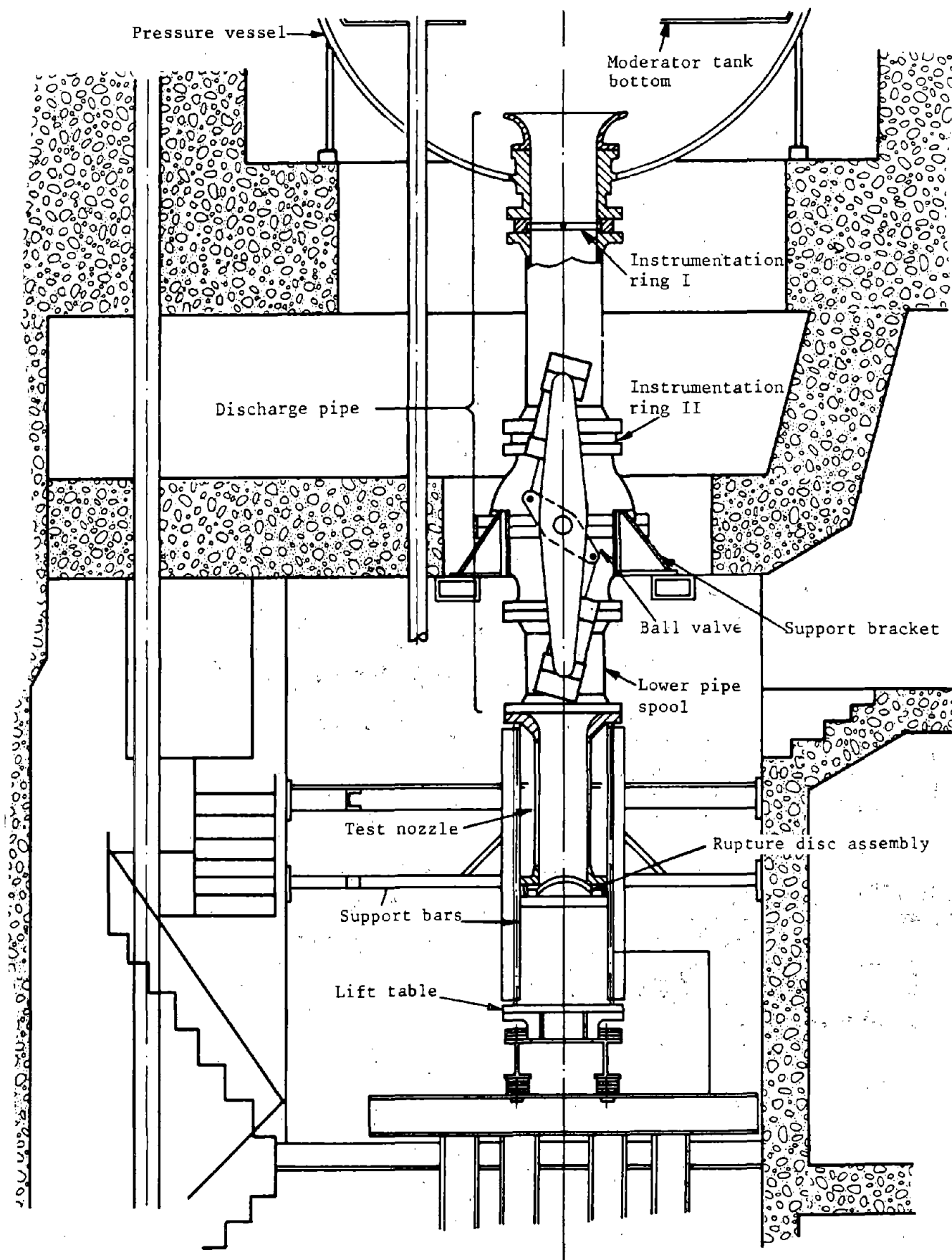


Fig 3:2 The discharge pipe including the test nozzle

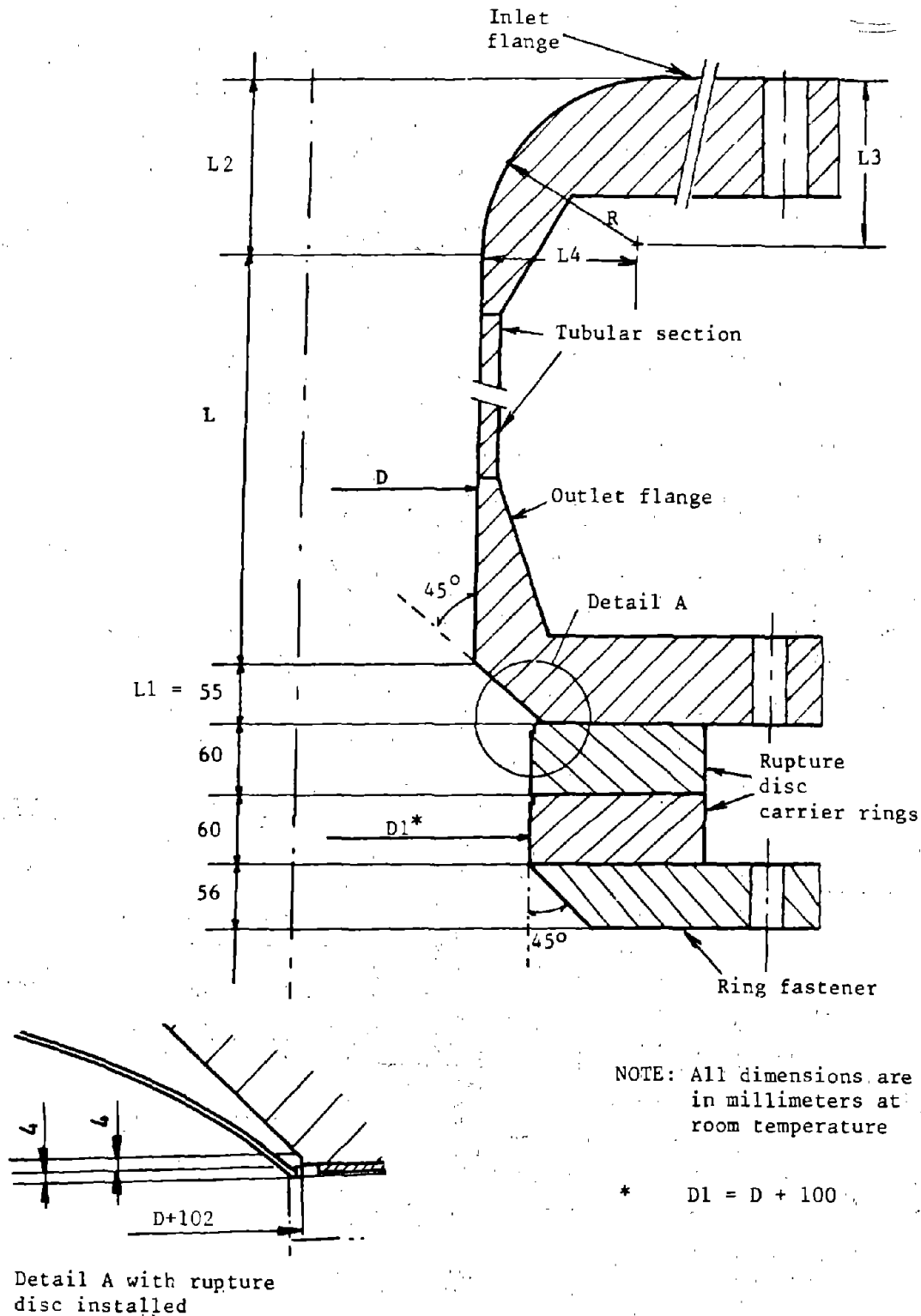


Fig 3:3 Dimensions of the nozzles used prior to Test 13

Note: All dimensions are in millimeters
at room temperature.

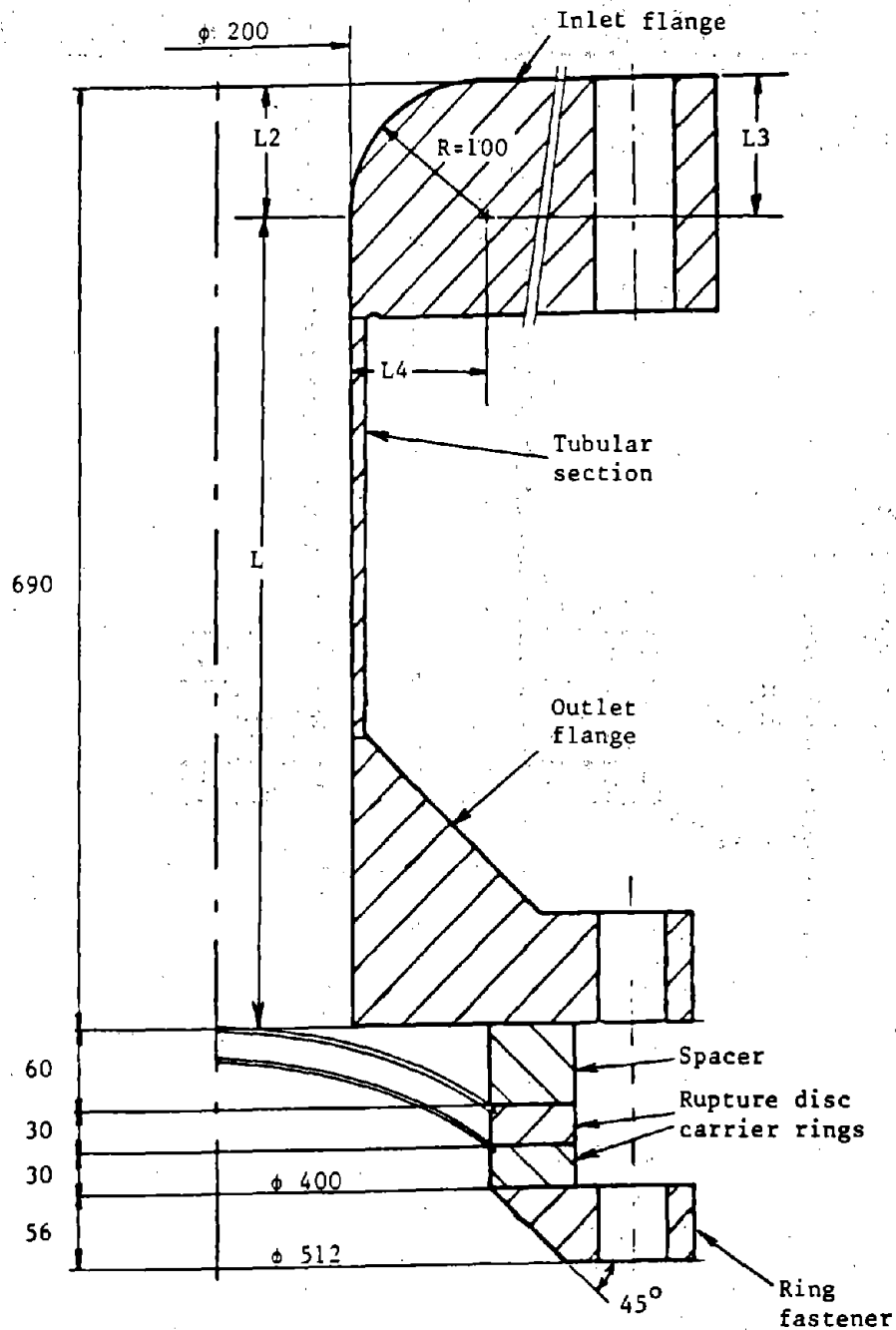


Fig 3:4 Dimensions of the test nozzle and rupture disc assembly for Tests 13 and 14

Note: All dimensions are in millimeters
at room temperature.

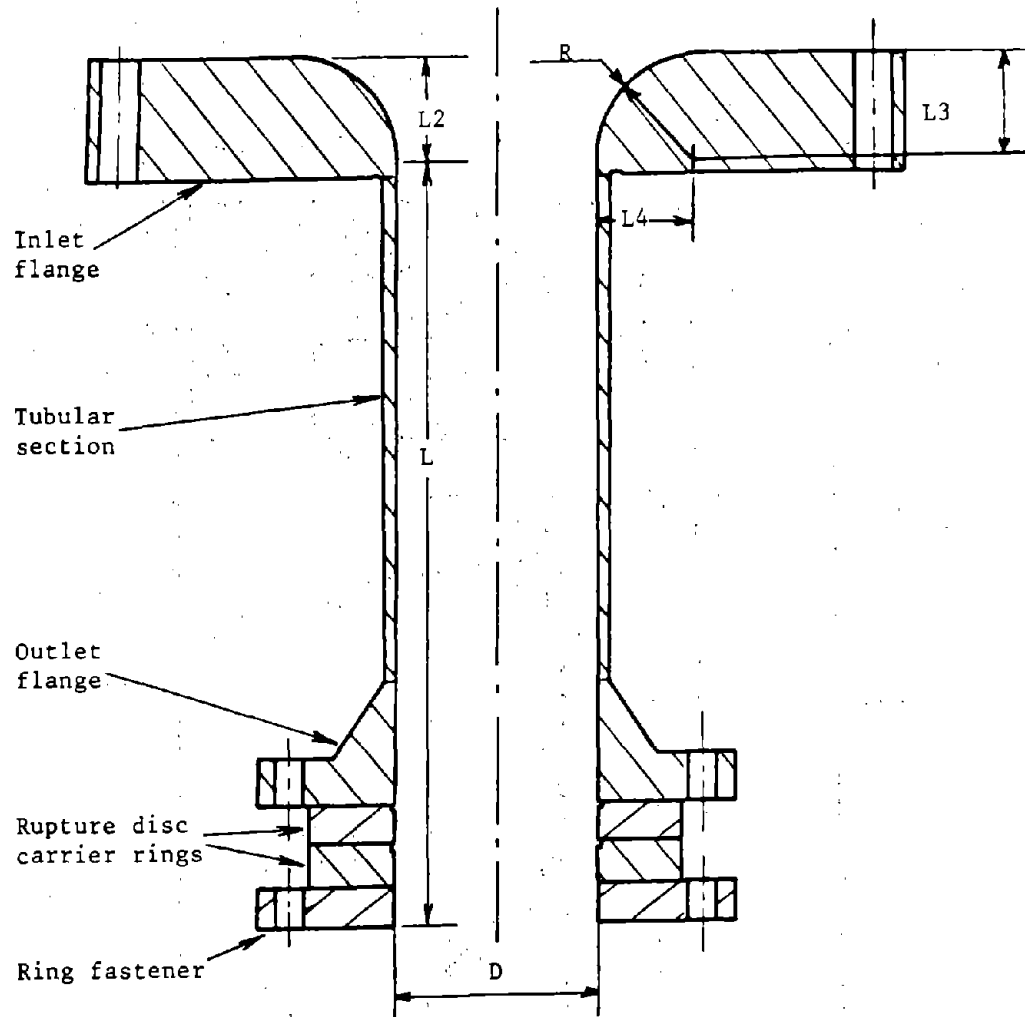


Fig 3:5 Dimensions of the test nozzles
used from Test 15 onwards

Note: Plan view looking downwards after transverse cut through the vessel at the 3 m elevation.

All dimensions are in millimeters at room temperature.

NORTH

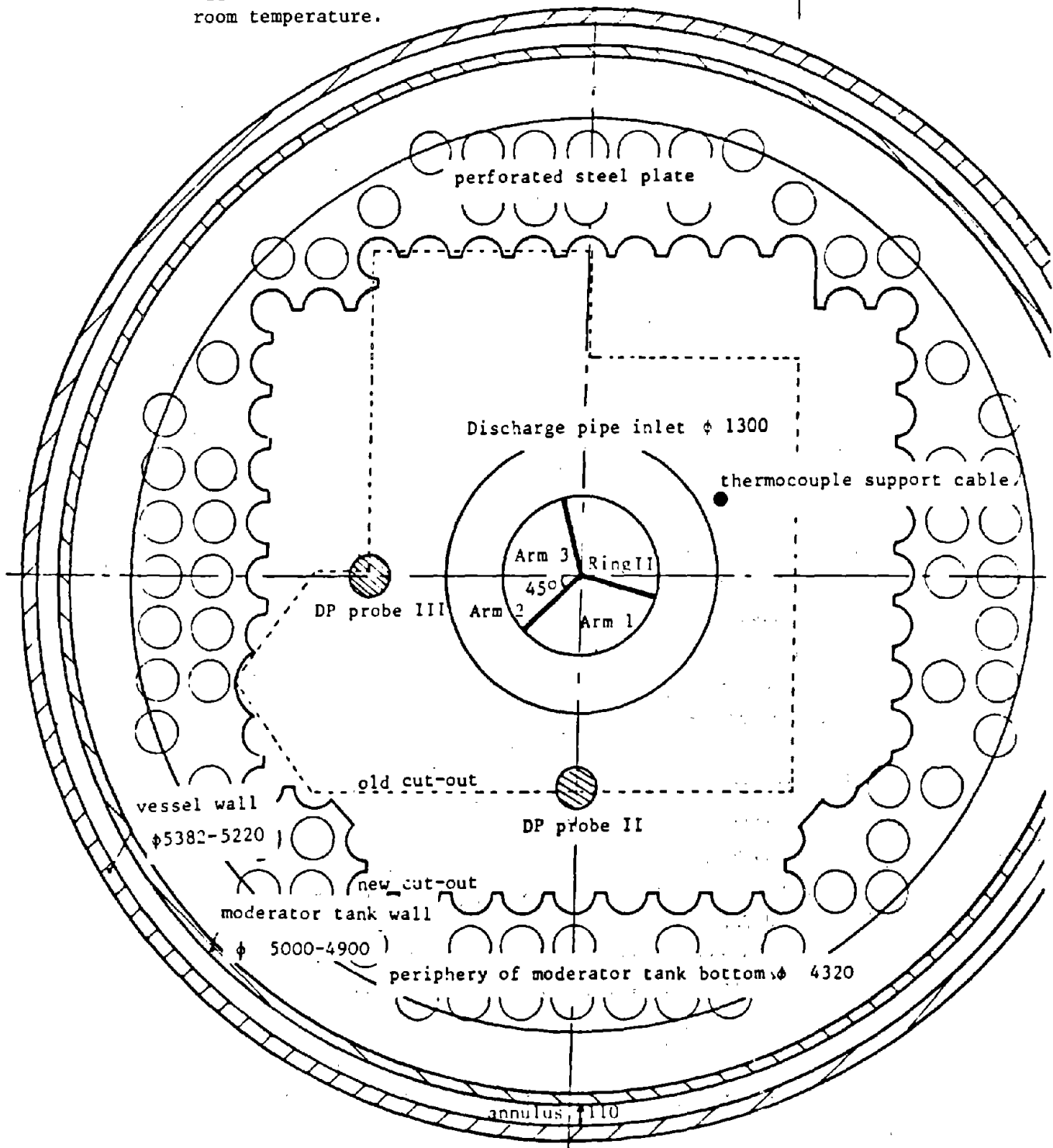


Fig 3:6 Comparison of the cut-outs in the moderator tank bottom before Test 15 and from Test 15 onwards

4. TEST PROGRAM DESCRIPTION

The test matrix and test procedure are summarized here. The purpose of conducting each test is listed and the relationships between the tests are discussed in subsection 4.1. The test procedure, the initial conditions and the changes in test procedure which occurred during the test program are discussed in subsection 4.2.

4.1 Test matrix description

The tests conducted during the CFT project are listed in Table 4:1 according to increasing nozzle size and increasing subcooling. The objective of each test is summarized.

In general the test matrix was constructed to define a test envelope with inlet conditions ranging from at least 30°C subcooled to low quality saturation for each nozzle. Tests conducted with water subcooled 15°C or more would give saturated data at pressures below 4.5 MPa, whereas tests with less than 5°C subcooling would give some saturated data at pressures greater than 4.5 MPa (usually as high as 4.8 MPa).

Midway through the test program, two changes were implemented which affected the test matrix. First, the nozzle exit geometry was modified (see Sections 3.2 and 3.3). Secondly, the initial water temperature profile in the vessel was changed for tests conducted with water subcooled 30°C or more. The tests conducted with a modified vessel temperature profile (see Fig 4:1 and Table 4:2 - item 13) are indicated in Table 4:1 as category II tests. Category I tests were conducted with water initially subcooled 15°C or more using the original temperature profile. Category III is the group of tests initially subcooled less than 5°C. Thus Tests 16 and 18 (cf Tests 15 and 17 respectively) were conducted to examine the effect of using category II

initial conditions as opposed to category I. In addition, most other tests after Test 14 were conducted to obtain a representative test envelope for the new test nozzle geometries.

Exceptions to the above test objectives were Tests 8, 10, 11, 12 and 27. Tests 8, 10, 11, and 12 were performed to examine repeatability. Test 27 was conducted to investigate the effect of the water content of dissolved air on critical mass flow rates.

4.2 Summary of test procedure

The test procedure was designed to obtain the specified initial steam dome pressure and fluid temperature profile (see Fig 4:1 and Table 4:2) The first step in test preparation was to fill the vessel with de-ionized water. With the vessel full, the integrity of the rupture discs was checked. Instrumentation sense lines were also bled to ensure that all air was removed. Water was then drained from the system until the specified elevation (at room temperature) was reached. The water was heated in three warm-up stages by circulating water from the bottom of the vessel through an external electric heater and re-introducing it into the vessel steam dome. The water from the heater was re-introduced along with other water also taken from the bottom of the vessel, but injected into the steam dome through a set of spray nozzles. Air was removed from the vessel by periodically venting the steam dome using the same procedure for each test with the exception of Test 27. Test 27 was degassed over a prolonged time period to reduce the air concentration below normal levels. Detailed information concerning the pre-test preparation procedures is given in Ref 1 and in each test report.

The warm-up process produced a temperature profile in the vessel with saturated conditions at 5 MPa in the steam dome (4 MPa in Test 5). The water level was between 16.5 and 20 m (see Table 4:2) in the vessel and had a saturated zone which existed to a specified elevation (see Fig 4:1), a transition zone (see Table 4:2) and a zone of relatively constant subcooling for category I and II tests. Initial water temperatures in category III tests were saturated throughout the vessel, excluding a small zone near the vessel bottom with fluid subcooled less than 5°C. The temperature of the discharge pipe water decreased with increasing distance from the vessel due to the heat losses. The initial temperatures at the test nozzle entrance are listed in Table 4:2.

About 4 hours before the initiation of a test, hot water was circulated through the discharge pipe to equalize the water temperatures in the subcooled zone in the lower part of the vessel. The circulation was terminated when the rupture discs were failed. However, the water circulation procedure in the discharge pipe produced a temperature zone in the lower part of the vessel which was not homogeneous and which affected the uniformity of the nozzle inlet stagnation conditions during the early portion of the subcooled discharge of each test. Consequently, the discharge pipe water circulation was terminated 1.5 h before each test after Test 13.

The measurement recording system was started approximately 180 s before the initiation of the test. The test was terminated when the ball valve began to close or when the steam entered the discharge pipe.

Table 4:1 Summary of the CFT Program test objectives

Nozzle Diameter ~ mm	L/D	Nominal* Subcooling ~ °C	Test Category	Test No	Test Objective
200	3.0	<5	III	14	Envelope definition
		30	I	13	Envelope definition
300	1.0	15	I	7	Envelope definition
		30	I	6	Envelope definition
300	1.7	<5	III	25	Envelope definition
		30	II	26	Envelope definition
300	3.0	15	I	1	Envelope definition
		30	I	2	Envelope definition
		30	I	12	Repeatability cf: Test 2
300	3.7	<5	III	19	Envelope definition
		30	I	18	Envelope definition
		30	II	17	Examine effect of new initial temp. profile cf: Test 18
500	0.3	<5	III	23	Envelope definition
		30	II	24	Envelope definition
500	1.5	<5	III	20	Envelope definition
		30	II	21	Envelope definition
		30	II	27	Examine effect of decreased water air content cf: Test 21
		50	II	22	Envelope definition
500	3.6	30	I	16	Examine effect of nozzle geometry change cf: Test 11
		30	II	15	Examine effect of new initial temp profile cf: Test 16
509	3.1	30	I	5 ⁺	Envelope definition
		<5	III	9	Envelope definition
		<5	III	10	Repeatability cf: Test 9
		15	I	3	Envelope definition
		30	I	4	Envelope definition
		30	I	8	Repeatability cf: Test 4
		30	I	11	Repeatability and also examine data at nozzle exit in rupture disc assembly cf: Test 8

* All tests performed at initial nominal steam dome pressures = 5 MPa except test 5.

+ Initial steam dome pressure = 4 MPa.

Table 4:2 Summary of initial and final conditions in the critical flow tests

1	Test No	1	2	3	4	5	6	7	8	9
2	Date of test performance	1978 01-19	1978 02-02	1978 02-24	1978 03-09	1978 03-21	1978 04-13	1978 04-27	1978 05-18	1978 06-01
3	Steam dome pressure ~ MPa	4.94	4.98	5.02	4.94	4.06	4.95	5.01	4.95	5.02
4	Saturation temperature ~ °C	263	264	264	264	251	263	264	263	264
5	Degree of nominal subcooling in the lower vessel (relative to steam dome saturation temperature) ~ °C	17-23	38	15-22	37	33	31	18	35	2
6	Min. fluid temperature in the vessel ~ °C	238	226	243	224	218	231	246	225	262
7	Initial temperature at nozzle inlet ~ °C	226	213	223	201	205	219	229	200	243
8	Mass of water and steam ~ Mg (incl the water in the discharge pipe)	287	284	274	286	286	289	286	285	286
9	Mass of steam ~ Mg	1.7	2.0	2.2	1.9	1.6	1.7	1.7	1.9	1.6
10	Mass of saturated water ~ Mg	114	104	100.5	109	110	113	125	108	256
11	Initial level in the vessel ~ m	17.84	17.41	17.06	17.59	17.44	17.81	17.86	17.51	18.15
12	Final level in the vessel ~ m	3.58	3.35	2.8	<0.74	<1.1	3.39	4.36	<0.74	<0.74
13	Nominal elevation of transition zone [†] ~ m ± 0.5 - see Fig 4:1	9-11	8-11	7-9.5	8-10.5	8-10.5	7.5-10.5	8-9.5	8-10	III
14	Oxygen content obtained after stabilization at 3 MPa* ~ mg O ₂ /kgH ₂ O	NA	NA	NA	NA	NA	NA	NA	NA	NA
15	Water electrical conductivity ~ μS/cm	0.7	1.2	NA	0.7	0.67	NA	NA	1.5	NA
16	Test period [†] ~ s	108	93	42	49	52	87	87	49	66

* Vessel water air content can be approximated assuming the nitrogen to oxygen percent volumetric composition ratio to be 79/21.

† Test period is the time from test initiation to when steam enters the discharge pipe (or the ball valve begins to close).

† = III indicates that little or no transition zone was present.

NA = not available.

Table 4:2 Summary of the initial and final conditions in the critical flow tests

1	Test No	10	11	12	13	14	15	16	17	18
2	Date of test performance	1978 06-20	1978 08-17	1978 08-29	1978 09-12	1978 09-21	1978 11-01	1978 11-16	1978 11-30	1978 12-12
3	Stream dome pressure ~ MPa	4.97	4.97	5.00	5.09	4.97	5.04	5.00	4.94	5.02
4	Saturation temperature ~°C	263.5	264	264	265	264	264	264	263	264
5	Degree of nominal subcooling in the lower vessel (relative to steam dome saturation temperature) ~°C	3	35	33	31	3	31	33	31	32
6	Min fluid temperature in the vessel ~°C	260	228	231	232	260	233	231	232	232
7	Initial temperature at nozzle inlet ~°C	242	202	215	170	170	177	180	174	174
8	Mass of water and steam ~ Mg (incl the water in the discharge pipe)	279	287	285	282	286	327	286	329	281
9	Mass of steam ~ Mg	1.8	1.8	1.9	1.9	1.6	0.6	1.9	0.7	2.0
10	Mass of saturated water ~ Mg	208	110	108	132	167	73.1	102	38.8	97.3
11	Initial level in the vessel ~ m	17.66	17.63	17.52	17.52	18.10	19.93	17.60	19.85	17.30
12	Final level in the vessel ~ m	<0.74	<0.74	<0.74	5.33	8.70	<0.74	<0.74	6.2	3.8
13	Nominal elevation of transition zone ~ m + 0.5 - see Fig 4:1	III	7.5-10	7.5-10.5	7-10	III	12.5-14	7.5-11.5	15.5-16.5	8-11
14	Oxygen content obtained after stabilization at 3 MPa* ~ mg O ₂ /kgH ₂ O	NA	NA	NA	NA	NA	0.8	1.3	0.9	0.5
15	Water electrical conductivity ~ mS/cm	0.8	NA	NA	NA	NA	NA	NA	2.0	NA
16	Test period [†] ~ s	64	48	126	148	146	55	49	90	87

* Vessel water-air content can be approximated assuming the nitrogen † = III indicates that little or no transition zone to oxygen percent volumetric composition ratio to be 79/21. was present.

+ Test period is the time from test initiation to when steam enters the discharge pipe (or the ball valve begins to close). NA = not available.

Table 4:2 Summary of the critical flow tests initial and final conditions

1	Test No	19	20	21	22	23	24	25	26	27
2	Date of test performance	1979 01-09	1979 01-25	1979 02-08	1979 02-27	1979 03-13	1979 03-29	1979 04-19	1979 05-03	1979 05-22
3	Steam dome pressure ~ MPa	5.06	4.99	4.94	4.93	4.96	4.96	4.92	4.91	4.91
4	Saturation temperature ~ °C	265	264	263	263	263	263	263	263	263
5	Degree of nominal subcooling in the lower vessel (relative to steam dome saturation temperature) ~ °C	4	7	33	52	3	33	6	34	33
6	Min fluid temperature in the vessel ~ °C	261	257	230	211	260	230	257	229	230
7	Initial temperature at nozzle inlet ~ °C	167	187	184	168	19	27	189	135	167
8	Mass of water and steam ~ Mg (incl the water in the discharge pipe)	267	262	330	334	314	330	313	320	329
9	Mass of steam ~ Mg	2.23	2.40	0.60	0.75	0.65	0.63	0.70	0.92	0.66
10	Mass of saturated water ~ Mg	262.6	257.3	48.9	35.7	310.7	39.4	309.8	38.8	46.9
11	Initial level in the vessel ~ m	16.99	16.65	19.95	19.64	19.85	19.88	19.73	19.31	19.82
12	Final level in the vessel ~ m	6.5	<0.74	<0.74	< .74	<0.74	<0.74	8.20	<0.74	<0.74
13	Nominal elevation of transition zone [†] ~ m ± 0.5 - see Fig 4:1	III	III	15.5-17	15.5-17	III	15.5-17	III	15.5-17	15.5-17
14	Oxygen content obtained after stabilization at 3 MPa* ~ mg O ₂ /kgH ₂ O	0.3	NA	0.7	0.2	0.7	0.5	0.9	0.3	0.005
15	Water electrical conductivity ~ μS/cm	NA	NA	NA	NA	NA	NA	NA	NA	0.52
16	Test period [†] ~ °	87	58	60	48	69	54	88	147	59

* Vessel water air content can be approximated assuming the nitrogen to oxygen percent volumetric composition ratio to be 79/21.

† Test period is the time from test initiation to when steam enters the discharge pipe (or the ball valve begins to close).

† = III indicates that little or no transition zone was present.

NA = not available.

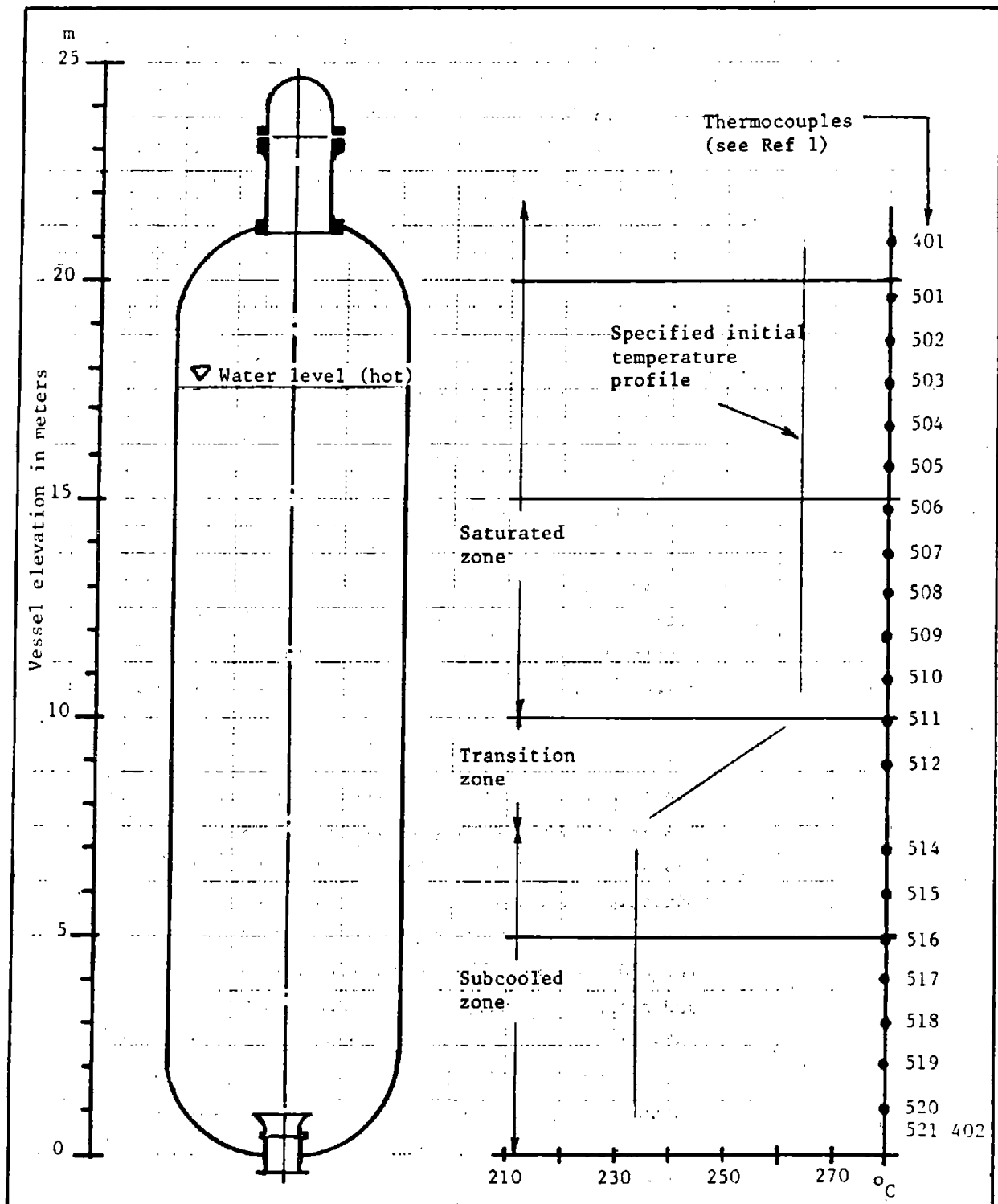


Fig 4:1 Typical initial temperature profile specification

5 INSTRUMENTATION AND DATA ACQUISITION SYSTEMS

The instrumentation and data acquisition systems used throughout the CFT program are summarized. The system components are described, the component specifications are summarized and the instrumentation locations are briefly discussed. Detailed instrumentation specifications and location descriptions can be found by consulting Ref 2 or the applicable test report.

5.1 Measurement system - a general summary

The measurement system consisted of transducers, signal conditioning units, data recording/reproducing parts and a process computer as shown in Fig 5:1.

The transducers for pressure and differential pressure measurements were of the strain gauge type. Chromel-alumel thermocouples were used for temperature measurements. The density was measured using a three beam gamma densitometer system.

The signals from the various instrumentation were processed through a signal conditioning unit. The channels of the signal conditioning unit were connected to a PCM (pulse code modulation) system. The PCM system consisted of modulators and demodulators together with a time code generator and receiver. The system worked with 10 bit data words and a scanning rate of 952 Hz per channel. The system also had the ability to reproduce the PCM-signals (recorded on the tape recorders in digital form) using a pen recorder, a light beam recorder or an oscilloscope.

The process computer was a system oriented, general purpose minicomputer. The core memory was 32 K (32768 words) and operated with a 16 bit word length. Available peripherals to the system were a fast matrix printer with key board, a magnetic tape transport and a disc memory.

During tests, data were recorded by the PCM-system on two analog tape recorders. After an experiment PCM-data were transferred digitally, via a PCM-demodulator, to the process computer. The digital tape was then sent to an external computer for evaluations.

5.2 Summary of the measuring devices

The instrumentation utilized to measure pressures, differential pressures, temperatures, density and other parameters during the tests is summarized. Manufacturers and measurement range data are listed in Table 5:1.

Pressure and differential pressure

Statham Instruments strain gauge pressure transducers with two different measuring ranges (see Table 5:1) were used to measure pressure. Differential pressures were measured using eight types of transducers. The pressure transducers were used in non-cooled and water-cooled mountings (see Ref 2).

Non-cooled transducers were placed outside the containment to avoid temperature effects and to avoid damage from missiles which could have been generated during an experiment. Even though transducers measuring containment pressure were located outside the containment, short sense lines were utilized to enhance their frequency response. Transducers located inside the containment, to reduce sense line length, were used with water-cooled mountings. Each pressure tap and transducer was connected by a sense line filled with oil or water.

Density

A model FM-800 three beam gamma densitometer system (manufactured by Measurements Incorporated, USA) was installed before Test 9 (see Table 5:2). The system consisted of a cesium-137 gamma source housed in a shielded cask, detector assemblies, shielded casks for the detectors, detector high voltage power supplies, an amplifier, control units, and an actuator box.

Temperature

Chromel-alumel thermocouples with insulated measuring junctions (Type "K") were used to measure temperatures in the vessel, the discharge pipe and the nozzle. The outer diameter of the junction was 0.5 mm for most thermocouples. For some of the measurements inside the containment, thermocouples with a 1.0 mm outer diameter were used.

Miscellaneous measurements

Data which were used to obtain initial values in tests were measured using a precision manometer, a barometer, a resistance thermometer and a rupture signal. The pressures and the temperatures were recorded prior to the test initiation while the rupture signal was recorded on the PCM-system (see Refs 1 and 2). The air content of the water was estimated by measuring the oxygen levels in water samples using either the Calometrie Indigo Karmin method or the Winkler method (see Ref 1).

5.3 Measuring point locations

The measuring point locations are described. The nomenclature used to designate instrumentation is described in the appendix of each test report. Table 5:2 lists the

major changes which were implemented on the instrumentation and data acquisition systems during the CFT program. It should be emphasized that not all the measurements shown in the following figures were recorded in each test.

Detailed listings of the measurement scheme for each test can only be obtained by consulting Ref 2 or the appropriate test report.

Pressure vessel

The absolute pressure was measured with three pressure transducers at the vessel top (001M101, 102, 103) and three transducers at the vessel bottom (001M104, 105, 106) as shown in Fig 5:2.

Seven transducers were used to measure the differential pressures between locations in the pressure vessel wall (001M201, 202, 203, 204 and 001M247, 248, 249). Three probes were located inside the vessel. DP (differential pressure) probe I was equipped with ten transducers, but was disconnected after Test 4 (due to equipment failures). DP probes II and III were equipped with six transducers (007M244 through 246 and 007M250 through 252) which together with transducers 001M204 and 007M253 measured the vessel differential pressure between 1.08 and 17.45 m in eight nearly equal increments. The measurements were used to calculate the vessel mass during each test. DP probe II was installed before Test 3 and DP probe III before Test 15.

Two temperature measuring points were located at the top of the vessel and one at the bottom of the vessel (001M401, 001R430, and 001R431). Between the top and bottom, 26 temperature measuring points were located (001M501 through 521, 402, 419 through 421, and 535) as shown in Fig 5:3. These thermocouples were used to define the initial temperature profile of the water in the vessel (see Fig 4:1).

Discharge pipe

The absolute pressure was measured in one position at instrumentation ring I (002M107) and in one position at instrumentation ring II (003M108). At the test nozzle inlet, the absolute pressure was measured at three locations (004M109 through 111) as shown in Fig 5:4.

The differential pressure between the inlet of the discharge pipe in the vessel bottom and instrumentation ring I was measured by one transducer (001M205). The differential pressure was also measured from the vessel bottom to instrumentation ring II (001M255).

The flow velocity was measured at ten positions at instrumentation ring I using differential pressures between the pitot tubes and the wall of the ring (see Fig 5:5). The pitot tubes were aligned parallel to the discharge pipe centerline. Between instrumentation ring I and the inlet of the discharge nozzle one transducer was used to measure the differential pressure (002M216). Between Tests 14 and 15 the pitot measurements were moved from ring I to ring II.

The differential pressure between the nozzle inlet and the entrance of the constant diameter part of the nozzle was measured by one transducer (004M217). Eleven temperature measuring points were located at ring I (see Fig 5:6), eleven at ring II and four at the nozzle inlet (004M405, 004M532, 004M534, and 004R432).

A three beam gamma densitometer system was used to evaluate the density below instrumentation ring I (003M601 through 603) after Test 8 (see Fig 5:4). The density was also measured at the discharge pipe inlet by utilizing the DP transducer 007M246 after Test 3. It should be noted that a change in the measurement characteristic of DP 246 was observed after the removal of a portion of the moderator

tank bottom (see Section 3) between Tests 14 and 15. Comparison between the densities obtained using DP 246 and the gamma densitometer showed better agreement after Test 15 than before.

Test nozzle

One absolute pressure was measured (004M118) as a reference, and five pressures were measured (004M121 through 125) at the nozzle outlet (see Figs 5:7 and 5:8). The constant diameter portion of the nozzle was equipped with a number of differential pressure transducers (004M218 and 004M222 through 230, and 004M281). The number of transducers was dependent on the length and diameter of the nozzle.

Containment and exhaust tube

In the containment and fuel channels, as well as in the exhaust tube, the absolute pressure was measured by seven pressure transducers (122M112, 110M113, 105M114 and 115, 102M116 and 006M119 and 120). Two transducers were used for differential pressure measurements between the drywell and wetwell (112M242, 243). Inside the containment and in the fuel channel seven temperature measurements were recorded (111M533, 112M406, 105M415 through 418, and 102M412). The location of the measuring points is given in Fig 5:9.

Table 5:1 Summary of the instrumentation specifications

Measurement	Manufacturer	Model/type	Range	Maximum	Location	Figure or Ref
Pressure	Statham Instruments, Inc, USA	PA856-100	0-0.689 MPa	1.378 MPa	Vessel and Containment	5:2, 5:9
		PA856-1M	0-6.89 kPa	10.342 MPa		
	Statham Instruments, Inc, USA	PM280TC±15-350	±105 kPa	±314 kPa	DP probes II/III containment	5:2, 5:9
		PM280TC±25-350	±175 kPa	±345 kPa		
Differential pressure	Statham Instruments, Inc, USA	PM280TC±50-350	±345 kPa	±689 kPa	DP probes II/III discharge pipe pitot-static nozzle, and containment	5:2, 5:4, 5:5 5:7, 5:8, 5:9
		PM8083±100	±689 kPa	±1387 kPa		
		PM8083±150	±1034 kPa	±2067 kPa		
		PM8083±500	±3450 kPa	±5168 kPa		
Density	Druck und Kraftmesstechnik, H.W. Keller, Switzerland	PM8052±15	±103 kPa	±1378 kPa	Pitot-static	5:5
		Keller PD-14	+400 kPa -0 kPa	+650 kPa -160 kPa	DP probe I	Ref 2
		FM-800	0.96-999.5 kg/m ³	500:1	Discharge pipe	5:4
		Measurements Inc. USA				
Temperature	Chromel/Alumel	Type K	Thermocouple	Time constants,* s		Vessel, containment, discharge pipe and pitot-static rake
				OD, mm	in water	
					in air	
				0.5	0.03	0.8 ~ 2.5
				1.0	0.08	1.8 ~ 5.0

* Velocity dependant - see Ref 2.

Table 5:2 Summary of the major changes to the instrumentation and data acquisition systems

First test after modification	Modification
3	DP probe II installed
5	DP probe I completely disconnected
9	Gamma densitometer installed
10	Pressure taps added to examine nozzle exit profile in measurements 004M121 to 123
15	1. DP probe III installed 2. Pitot-static rake installed at instrumentation ring II 3. Hole in the moderator tank enlarged. Effect of change noted in DP 246 data
23	The gamma densitometer cables and connectors were upgraded

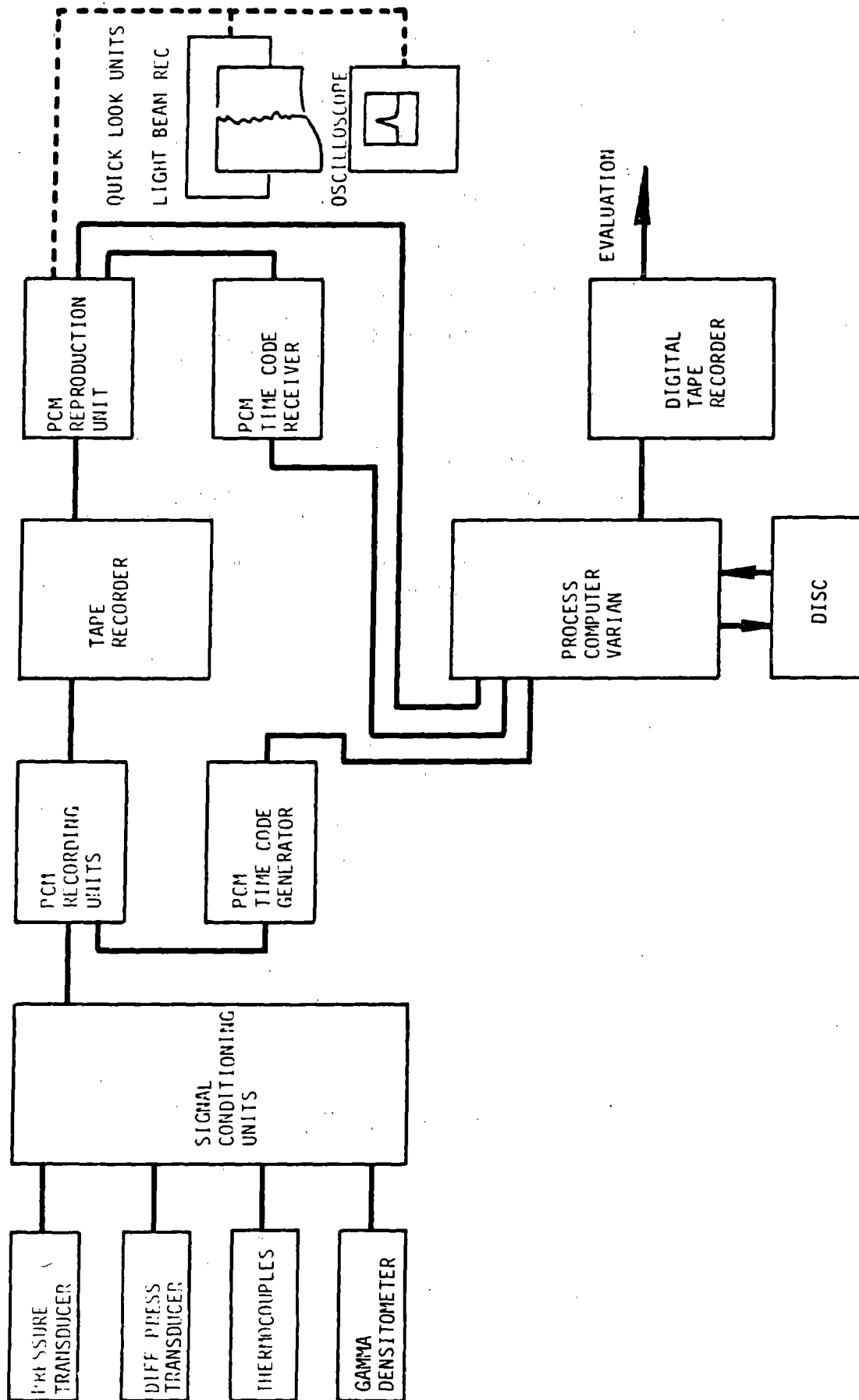


Fig 5:1 Measurement system

Note: All dimensions are in millimeters
at room temperature.

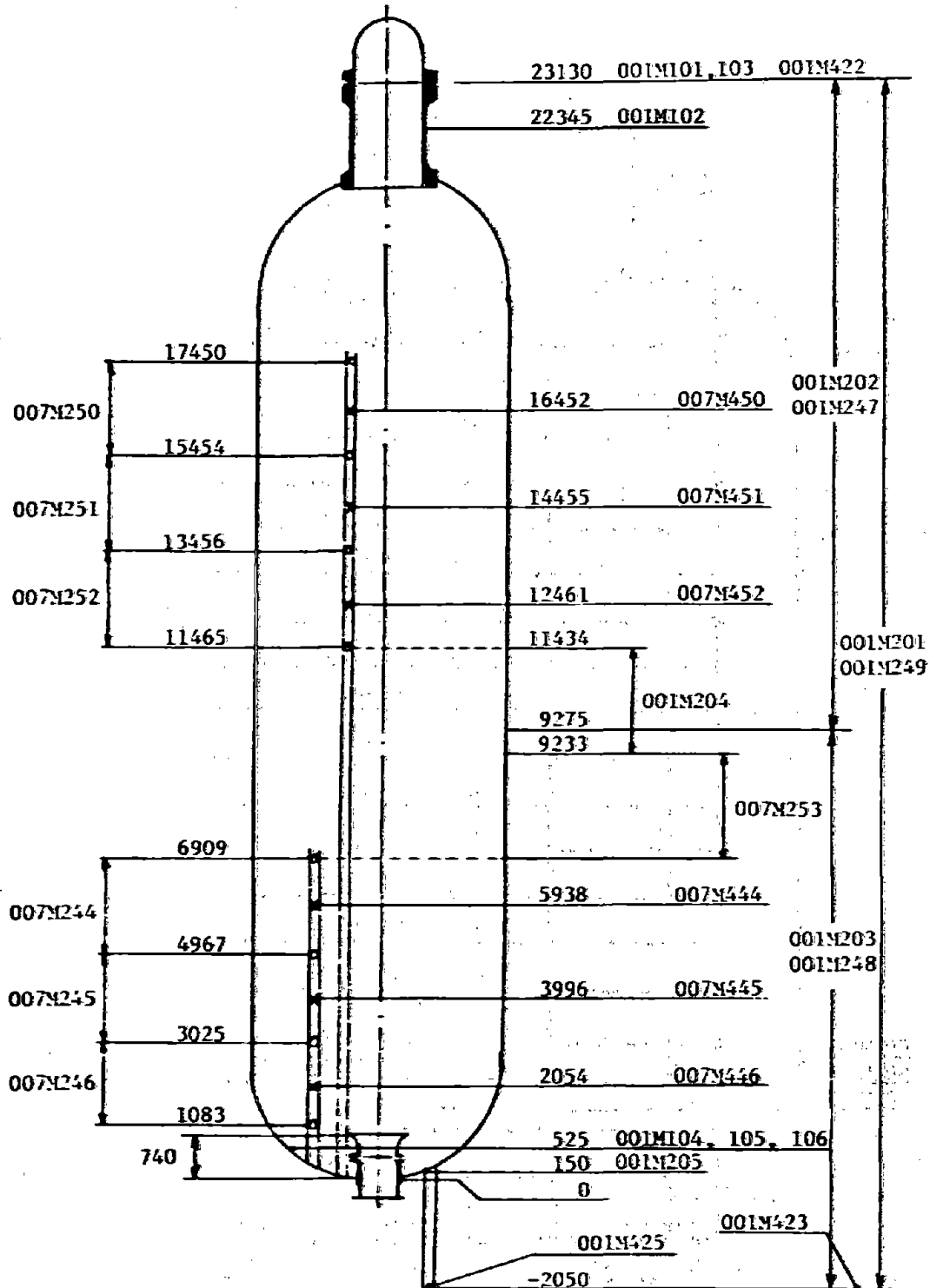


Fig 5:2 Locations of pressure and differential pressure measurements in the pressure vessel

Note: All dimensions are in millimeters
at room temperature.

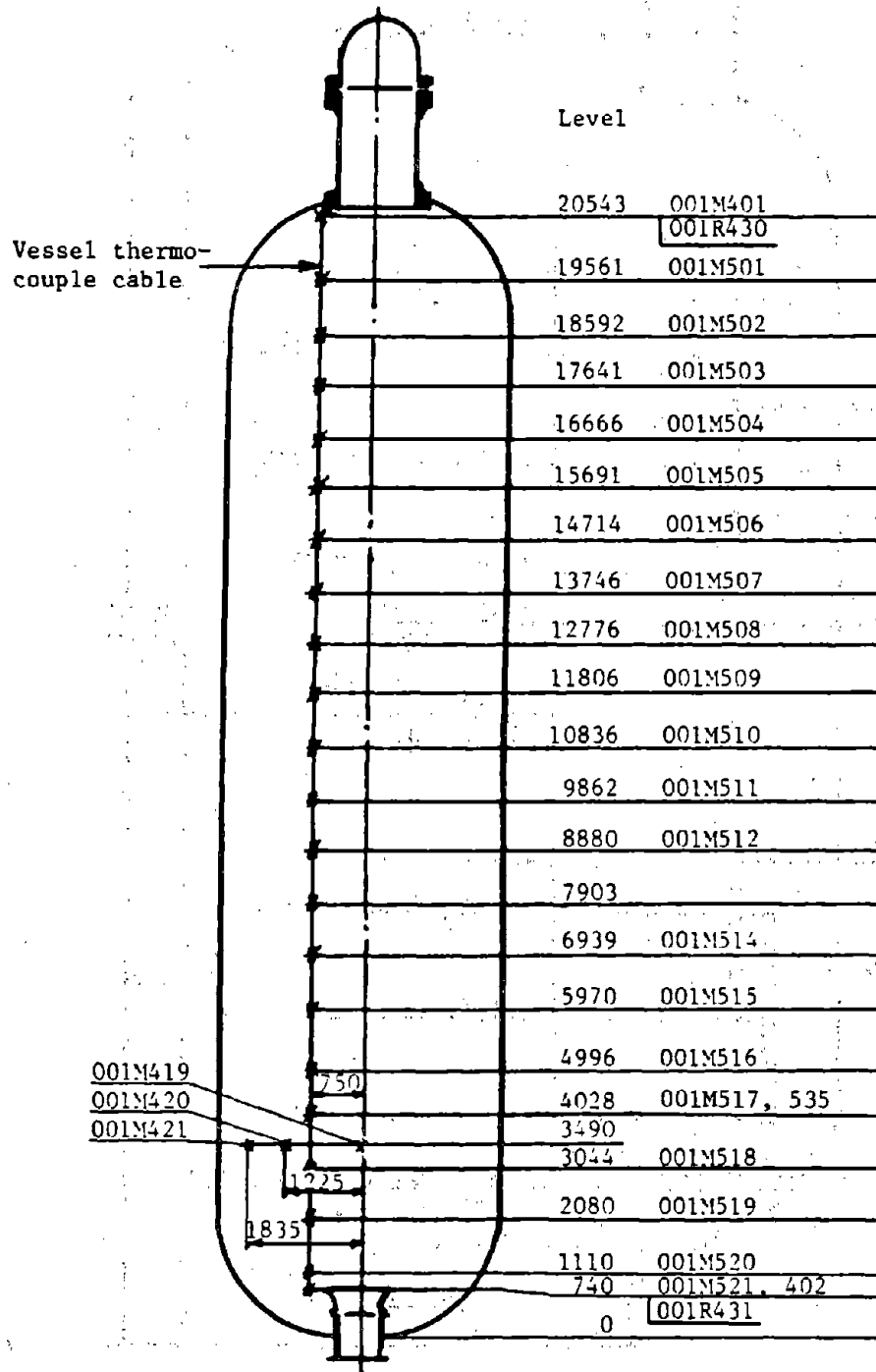


Fig 5:3 Locations of temperature measurements in the pressure vessel

Note: All dimensions are in millimeters
at room temperature.

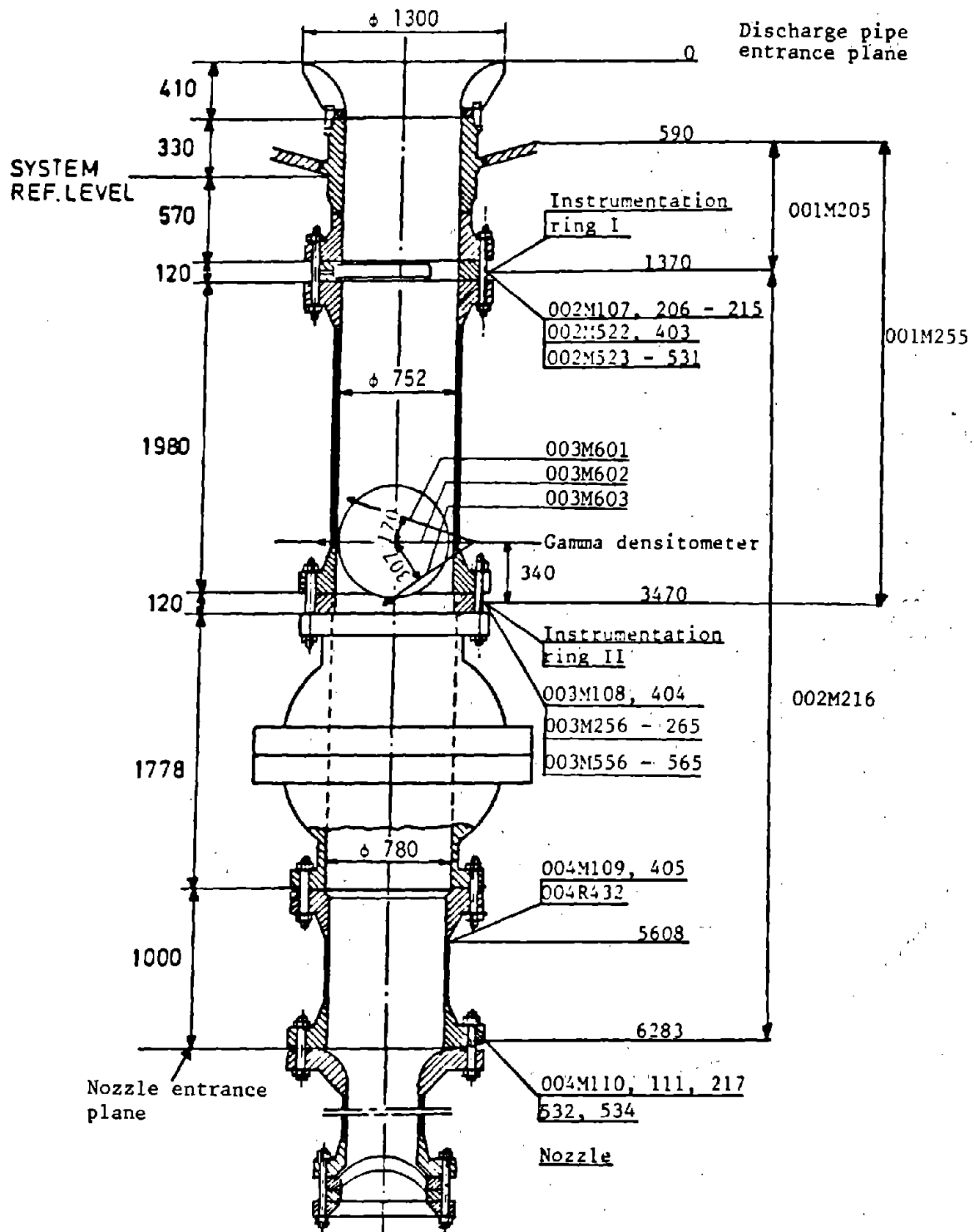


Fig 5:4 Locations of measurements in the discharge pipe

Note: All dimensions are in millimeters
at room temperature.

○ Arm number

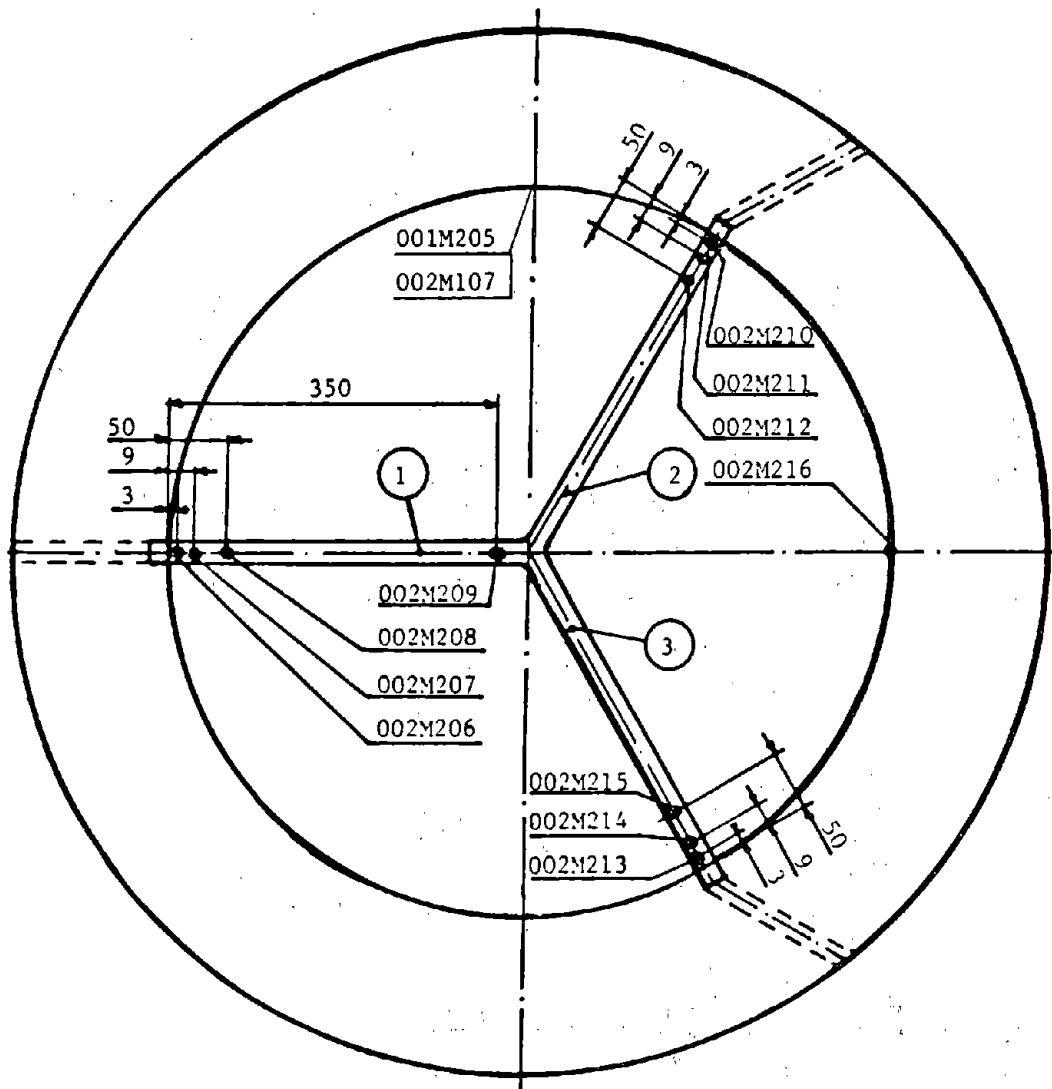


Fig 5:5 Locations of pressure and differential pressure measurements at instrumentation ring I

Note: All dimensions are in millimeters
at room temperature.

○ Arm number

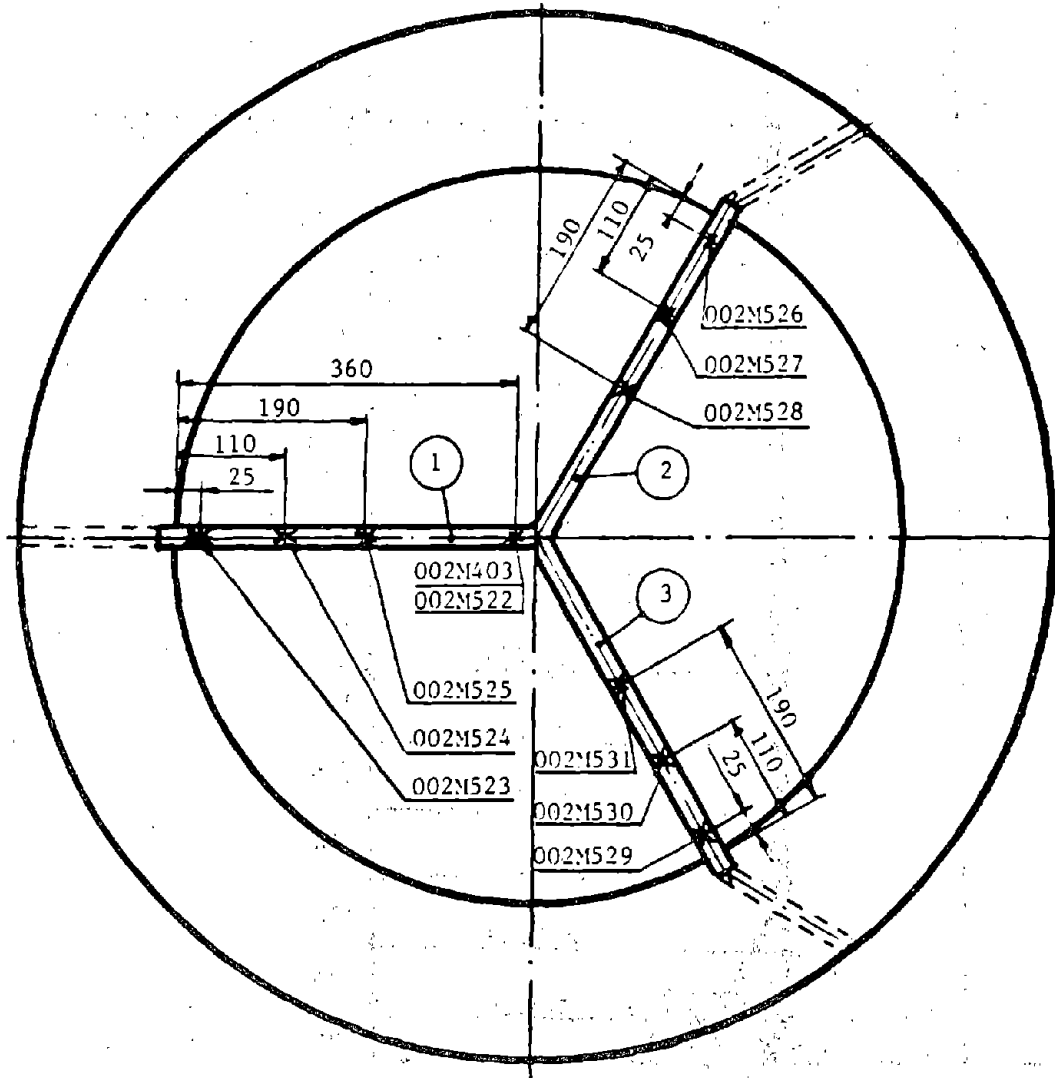


Fig 5:6 Locations of temperature measurements at instrumentation ring I

Note: All dimensions are in millimeters
at room temperature.

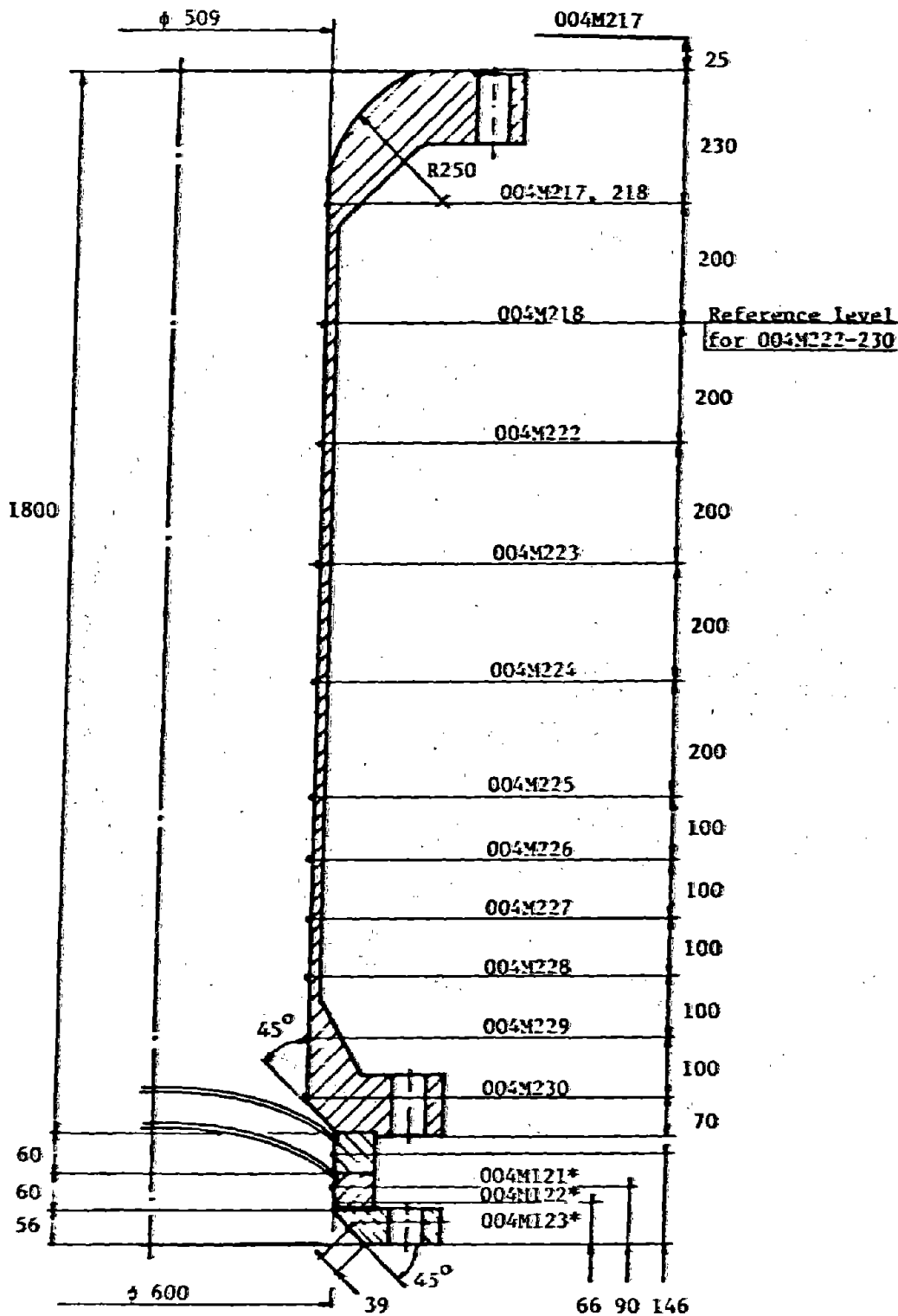


Fig 5:7 Locations of measurements in the nozzle with diameter 509 mm and L/D 3.1 (Tests 3, 4, 5, 8, 9, 10 and 11)

* 121, 122, and 123 only Tests 10 and 11.

Note: All dimensions are in millimeters
at room temperature.

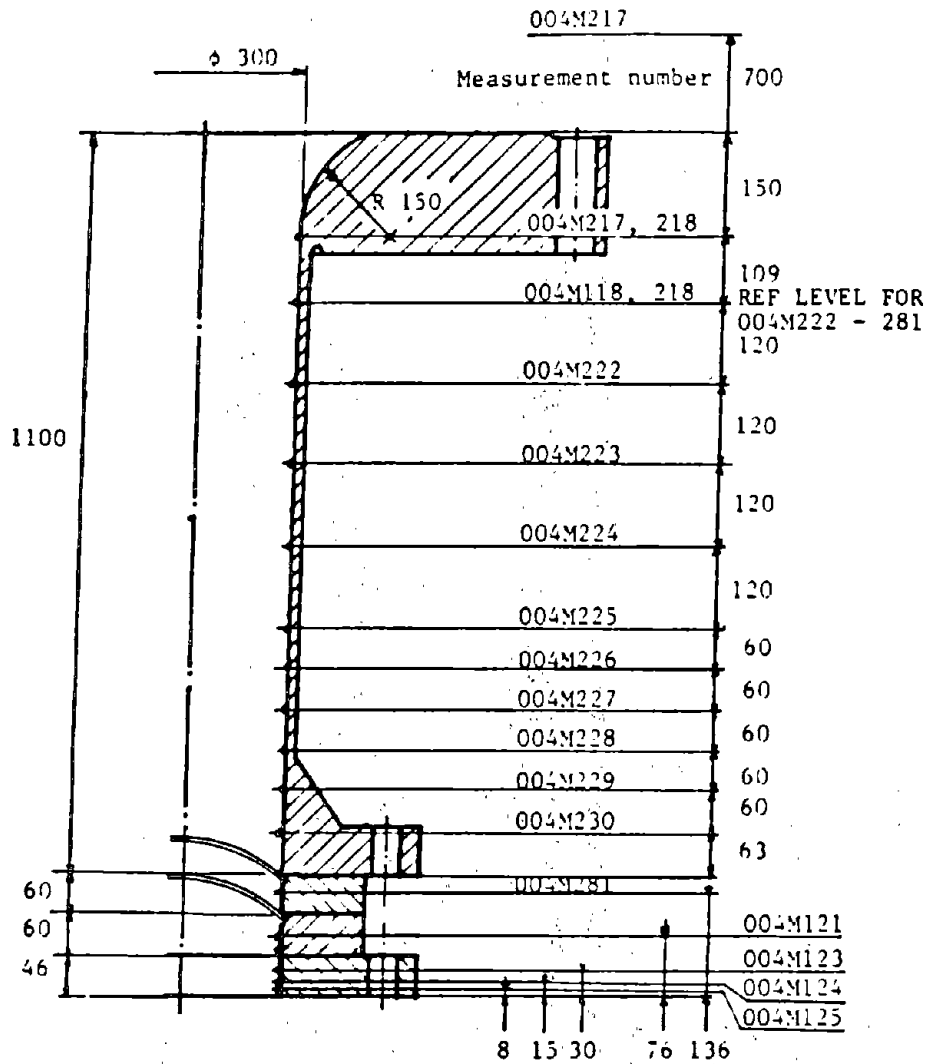


Fig 5:8 Locations of measurements in the nozzle with diameter 300 mm and L/D 3.7 (Tests 17, 18 and 19)

Note: All dimensions are in millimeters
at room temperature.

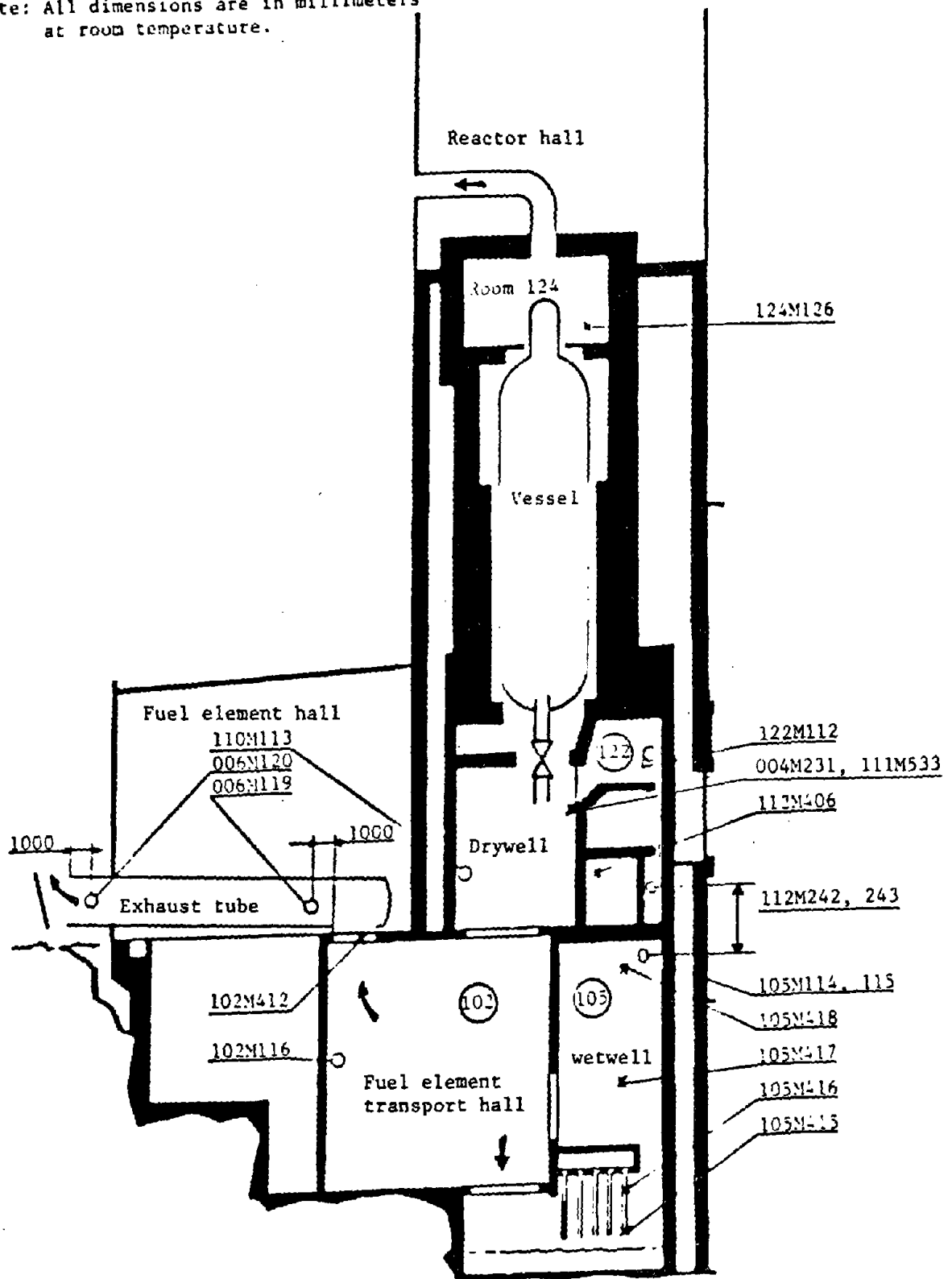


Fig 5:9 Locations of measurements in the containment

6. DATA PROCESSING AND EVALUATION

Data, measured using the instrumentation described in the previous section, were processed and further evaluated to obtain parameters such as the stagnation conditions at the nozzle inlet and mass flow rates. The procedures used to process the data, and to obtain fluid densities, states and mass flow rates are discussed. Detailed discussions of these topics are found in Ref 3.

6.1 Data processing and conditioning procedures

The measured data often exhibited oscillations of large amplitudes superimposed on the semi-static histories. In most of the applications only the semi-static histories were of interest. Thus the data had to be filtered before being used in the evaluations. The measured data were obtained at a rate of 952 Hz, and data decimation was therefore also necessary to be able to conduct the evaluations at reasonable computer costs.

Data averaging was the most common filtering/data decimation technique. Thus most data were averaged by computing unweighted mean values over time segments of the measured histories to generate a smoothed curve. The time histories were divided into small time intervals with an equal number of measured data (the sample size) in every interval. One average value was computed for each of the samples, and this was given a time equal to the mid-point of the corresponding time interval. The data decimation factor was thus equal to the sample size.

The data splining technique was used to smooth time histories with heavy oscillations so as to obtain values which could be used in the evaluations. The splining procedure was generally used to smooth differential pres-

tures in the DP-probes. The major advantage of the splining technique was that the smoothed histories also had smooth derivatives. This characteristic of the method was utilized to produce vessel pressure derivatives for the evaluation of mass flow rates using the steam dome expansion method.

6.2 Data evaluation procedures

The critical flow behaviour of the test nozzles was determined and the quality and consistency of the evaluated data were judged on the basis of a number of evaluated parameters derived from the basic measurements. The evaluated data included the fluid conditions at various locations in the system, the vessel mass histories, and the discharge flow rates.

System fluid conditions

Fluid conditions were evaluated at locations where fluid densities were obtained, e.g. at the local DP-measurements in the vessel and at the gamma densitometer. In addition, fluid conditions were evaluated upstream or downstream of specific density measurements assuming isentropic compression or expansion. Two-phase conditions were in general evaluated assuming the fluid to be in homogeneous equilibrium.

The fluid conditions in the vessel were evaluated using the DP-measurements from DP probes II and III. These measurements revealed that a distinct two-phase level existed in the vessel throughout the tests. The evaluations were in general carried out until the two-phase level reached the upper pressure tap of a measurement. The conditions evaluated using measurements 007M246 and 001M106, located in the vicinity of the discharge pipe entrance

(see Fig 5:1), were used as approximations of the stagnation conditions. The densities at the pitot-static rake were also obtained from these measurements assuming isentropic fluid expansion between the two elevations. These densities were used to evaluate mass flow rates when values from the gamma densitometer were not available.

Local fluid conditions in the discharge pipe were obtained using the three beam gamma densitometer. Densities were determined in a number of concentric radial zones corresponding to the number of gamma beams used. These densities were coupled with the pitot-static measurements to obtain mass flow rates. The average gamma densities were also used to obtain stagnation conditions at the discharge pipe entrance assuming isentropic compression.

The stagnation subcooled density at the nozzle entrance (see Fig 5:4) was obtained using an area-averaged temperature at the applicable instrumentation ring and pressure 001M106. Two-phase densities were evaluated from either 007M246 or the gamma densitometer assuming isentropic fluid expansion. The pressure variations along the discharge pipe (due to friction and gravity) and the fluid transit times were taken into account in these calculations.

Vessel mass histories

The vessel mass histories were evaluated using the external mass-DP measurements. The total mass in the vessel was determined by using the values of 001M201 and also by adding the masses obtained from 001M202 and 203 (see Figs 5:1). To account for variations in the vessel cross-sectional area with elevation, a complete separation of the phases was assumed, and the evaluated masses were based on a "collapsed liquid level". The steam was assumed to be saturated at the steam dome pressure, and the liquid

densities were based on a volume averaged temperature and the steam dome pressure. Corrections were made for sense line density variations. During the blowdowns with the largest nozzles the difference between the "collapsed liquid level" and the actual two-phase level could be more than 4 m.

Nozzle mass flow rates

The mass flow rates were determined using three methods. During the first tenths of a second of a test the mass flow rates were evaluated using the steam dome expansion method where the steam dome volume expansion was coupled with densities at the nozzle entrance to yield the mass flow rates. The volume expansion was determined from the depressurization rate assuming an isentropic process. Two sets of data were generated: one by assuming that thermodynamic equilibrium was maintained during the expansion, and one by assuming that no condensation occurred. The two sets of data thus obtained revealed a difference of 15 %, which is an expression for the uncertainty in the judgement of the thermodynamic process which occurred in the steam during the expansion. In fact, because the peak mass flow rate indicated by a pitot-static rake was often below both sets of data, the error limit for the steam dome expansion method was somewhat larger than 15 %. The method could be used until the liquid began to flash.

Mass flow rates were also obtained by the mass inventory method by differentiating the vessel mass histories.

The rate of change of the vessel mass with time was obtained as the slope of a linear function determined by the least-squares method (using 30 data points on either side of the given data point). In general, no reasonable mass flow rates could be calculated prior to 5 s due to oscillations of the differential pressures. The method performed best in time periods when the mass flow rates were fairly stable (e.g. during the saturated period) since the response to fast variations was slow.

Finally, mass flow rates were obtained using the pitot-static measurements located in the discharge pipe. The observed differential pressures were coupled with calculated or measured densities. Subcooled fluid densities were determined on the basis of the temperature and pressure of the fluid in the discharge pipe. Two-phase densities were evaluated either from the gamma densitometer or from O07M246 to obtain local mass fluxes. Second order polynomials were used to represent the mass flux profile between the radial locations of the measurements. The polynomials were determined using the local mass flux values and the condition that the derivatives should be continuous over the cross-section. The derivative at the center and the mass fluxes at the pipe wall were set to zero. The composite mass flux profile function was mathematically integrated over the cross-sectional area to obtain the mass flow rate. The pitot-static method was used throughout the tests. The method performed best in the subcooled portion of the tests since the density measurements had a large uncertainty during two-phase flow.

7 DATA ACCURACY

This section deals with the accuracy of the measured data and the most important non-measured quantities that are evaluated and presented in the test result reports. A more detailed account of the data accuracy is given in Ref 4.

Accuracy of the basic measurements

The accuracy of the measured data, specified in Table 7:1, is presented in terms of:

1. maximum errors calculated from:

$$(\Delta U)_{\max} = \pm \sum_{i=1}^m \left| \frac{\partial U}{\partial x_i} (\Delta x_i)_{\max} \right|$$

where

- U = $U(x_1, x_2, \dots, x_m)$ = the measured quantity
- m = total number of components used for the measurement of U
- x_i = signal value for component i
- $(\Delta x_i)_{\max}$ = maximum error for component i

and

2. probable errors defined as:

$$\sigma_U = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (U_i - \bar{U})^2}$$

where

- U_i = value obtained from measurement channel i
- $\bar{U}$ = mean value for the chosen group of channels
- n = total number of data channels in the group

The maximum error calculations are based on the manufacturer's specifications and give the error limits with high confidence. Nevertheless, it is occasionally seen in the diagnostic checks on the data that the data fall outside the maximum error limits. This is then noted and commented upon in each test report.

The probable error, here defined as one standard deviation, is a measure of the spread of data from identical or nearly identical channels measuring the same quantity. The probable error calculations for most channels could only be performed during time periods when the channel readings were not changing significantly with time. Thus calculations for the pressure and differential pressure channels were performed after the blowdown was terminated. Exceptions to the procedure were followed to evaluate the probable error of temperature and density channels since they could be analyzed using references during the transient. Obviously these calculations are not possible for unique data channels. They can, however, in most cases be assigned probable error limits on the grounds that they are similar to others with known probable errors.

The estimations of maximum errors and probable errors account only for static errors. However, the dynamic error which is related to the response time can normally be ignored in these quasi-steady tests. It should, thus, be noted that the error limits quoted in Table 7:1 apply only to the time averaged mean value and not to the instantaneous value at all time points. This should be remembered when considering the oscillations or the noise which are normally seen in the presented data. The amplitudes of the oscillatory data as shown are dependent on the signal filtering and the data averaging procedure.

Accuracy of evaluated data

The traditional methods used for evaluating the error limits for the basic measurements described above are not applicable to the evaluated data. Instead, the errors of these quantities, which are determined indirectly from measured data, are assessed from comparisons of data obtained from independent measurements and different methods and by considering the errors of the basic measurements along with an assessment of the evaluation methods applied. The error limits thus arrived at are considered to be conservative but naturally they cannot be given any confidence level. In this summary only the errors of mass flux and fluid density are considered.

Mass flux accuracy

The mass flux, constituting the key-parameter in these tests, was determined using three different methods:

1. The steam dome depressurization method
2. The vessel mass inventory method
3. The pitot-static method

The error limits of the mass flux as obtained by these different methods can be summarized as follows:

The steam dome depressurization method

This method is valid only during the initial phase of the test for a maximum period of 0.5 seconds and is used only to determine the peak mass flux. Irrespective of whether the mass flux is evaluated assuming thermodynamic equilibrium or no condensation in the dome, the maximum mass flux is considered to be accurate within $\pm 15\%$.

The vessel mass inventory method

This method becomes applicable after about 5 seconds. For most of the test, periods of slowly decreasing mass flux have an error limit of $\pm 6\%$. However, the rapid changes in mass flux which occur prior to 10 s can cause the error limit to reach $\pm 12\%$ in some tests in which 500 mm nozzles were used (see Ref 4).

The total integrated mass flow rate from the vessel, as measured with different mass inventory DP-channels, is usually observed to agree within $\pm 2\%$.

Rapid changes in the mass flux rate are not described accurately because of the low dynamic response of the evaluation method. However, comparisons with pitot-static mass fluxes indicate that the maximum error limits are within $\pm 10\%$, except when low water levels in the vessel (< 2.3 m) invalidate the mass inventory method.

The pitot-static method

This method is applicable to the whole test. The error limits of the subcooled pitot mass fluxes are between ± 3 to 10% depending on the magnitude of the mass flux (see Ref 3). The error limits fall within ± 8 to 15% as a function of the discharge pipe mass flux in the two-phase region. The error increases with decreasing mass flux in both cases.

Fluid density accuracy

The fluid density (which is used in combination with the pitot-static data) is known with an error of less than $\pm 1\%$ for the subcooled single-phase liquid flow when it is determined from the measured pressure and temperature. When the mean liquid density is evaluated using all three beams of the gamma densitometer, the weighted error is $\pm 12\%$.

When the fluid is saturated the density is determined either from the three beam gamma densitometer or from the vessel DP-probe channel measuring close to the discharge pipe entrance (007M246) plus the drop in flow pressure between the discharge pipe entrance and instrumentation ring I (001M205).

The error in the mean weighted density calculated from the three signals is of the order of $\pm 12\%$ (90 kg/m^3) provided that the densities do not deviate significantly from one beam to another. However, if only one or two beams of the densitometer are operational, the error increases. The

effect on the density error of using only one or two beams is treated in Ref 3. It should be noted that the gamma densitometer channels often produced unrealistic signals which are rejected.

When using DP channels 007M246 and 001M205 to evaluate the density, the error limits are $\pm 10\%$ ($\pm 75 \text{ kg/m}^3$), considering the DP errors only. However, these errors become considerably larger when taking into account the facts that 007M246 does not measure the density exactly at the entrance plane of the discharge pipe and that measurement disturbances are caused by the bottom of the moderator tank. Occasional deviations of up to 25% (200 kg/m^3) have been observed.

Table 7:1 Error limits of the basic measurements

PARAMETER	LOCATION	RANGE	MAXIMUM ERROR	PROBABLE ERROR
PRESSURE	VESSEL, DIS- CHARGE PIPE, NOZZLE	6890 kPa	± 90 kPa	± 9 kPa
PRESSURE	NOZZLE EXIT	6890 kPa	± 90 kPa	± 6 kPa ¹⁾
PRESSURE	CONTAINMENT	689 kPa	± 8 kPa	± 0.5 kPa
DIFF PRESSURE	VESSEL MASS, EXTERNAL	≤ 689 kPa	± 8 kPa	± 0.6 kPa
DIFF PRESSURE	VESSEL, DP- PROBE	± 103 kPa	± 2.5 kPa	± 0.5 kPa
DIFF PRESSURE	PITOT-STATIC RAKE	± 1034 kPa	± 15 kPa	± 0.7 to ± 1.6 kPa ²⁾
DIFF PRESSURE	DISCHARGE PIPE, NOZZLE	± 1034 kPa ± 3445 kPa ± 6890 kPa	± 15 kPa ± 42 kPa ± 90 kPa	± 6 kPa
TEMPERATURE	VESSEL, DISCHARGE PIPE, CONTAINMENT	275°C	± 2 °C	± 0.6 °C ³⁾
DENSITY	DISCHARGE PIPE (GAMMA DENSITOMETER)	< 1000 kg/m ³	± 40 kg/m ³ ⁴⁾	± 50 kg/m ³ ⁵⁾

1) Possibly too low because the error limit measured is at the end of the test when the reading is only a fraction of a typical test reading (see Ref 3).

2) Dependent on the nozzle diameter.

3) Careful editing of the temperature data will produce a probable error limit less than ± 0.4 °C.

4) Based on manufacturer's specification. The Marviken Project found this value to be unrealistic during the CFT.

5) This value is evaluated during the test periods with subcooled conditions after rejection of unrealistic data.

8 TEST RESULTS

The recorded test measurements and results have been discussed in the individual test reports in great detail. Of necessity, only selected measurements are presented here. Apart from data which are shown in order to illustrate specific features or characteristics of the tests, the data included mainly concern the conditions at the test nozzle inlet and the mass flux histories.

The section includes discussions on stagnation state conditions at the discharge pipe inlet and the nozzle inlet, the most representative nozzle mass flow rate history for each test, the program test envelope, test repeatability and the nozzle pressure profiles. Other phenomena, such as fluid superheating, initial pressure undershoot and recovery, as well as the pressure oscillations observed during the subcooled flow period, are grouped together as miscellaneous topics.

8.1 Stagnation state history at the nozzle inlet

The data presented are arranged so as to simplify their interpretation. Thus the data from the 27 tests are divided into three categories, depending on the inlet conditions. Since the instrumentation used to define the nozzle inlet conditions was located somewhat upstream of the nozzle inlet, changes which occurred between measurement time and arrival time are discussed. The nozzle inlet conditions are presented, taking these factors into account.

8.1.1 Relationship between the initial conditions and the stagnation state history

The data recorded in the CFT project fell within three broad categories, depending on the initial conditions in the vessel. Figs 8:1, 8:2 and 8:3 illustrate the relationship between the initial temperature profile in the vessel and the stagnation state history at the discharge pipe inlet.

Tests conducted with initial vessel water levels at or below 18 m and nominally subcooled by 15°C or more had stagnation state histories characterized by four distinct zones:

<u>Zone</u>	<u>Characteristic</u>
A	Initial rapid depressurization terminated by the incidence of vessel water flashing. An increased flashing rate causes a pressure increase until the flashing rate is stabilized. The greatly subcooled water located in the discharge pipe is also exhausted.
B	An extended time period of slowly increasing temperature i.e. $\frac{dT}{dt} \lesssim 1^\circ\text{C/s}$ and decreasing pressure, for less than half of the subcooled blowdown.
C	A period of $\frac{dT}{dt} \gtrsim 1^\circ\text{C/s}$ with decreasing pressure, for over half of the subcooled blowdown.
D	Saturated blowdown.

Zones A through D are identified on Fig 8:1a. Fig 8:1b shows the approximate times when water at the subcooled temperatures and elevations on Fig 8:1a reaches the discharge pipe inlet. The behaviour shown in Fig 8:1b is characteristic of the data obtained for Tests 1 through 8, 11 through 13, 16 and 18. These tests form category I.

Category II test data are identified by initial vessel water levels between 19.5 and 20 m with fluid nominally subcooled by between 30 and 50°C. A characteristic initial

temperature profile and stagnation state history are shown in Fig 8:2. Category II data are similar to category I, but they do not have a significant zone C. In general, the presence of zone C data causes the category I tests to have less useful data than category II. Zone B data are present for over half of the subcooled blowdown in category II. Tests 15, 17, 21, 24, 26 and 27 fall into the second category.

Finally, a number of tests were conducted with some water initially subcooled by less than 5°C . These tests are characterized by having relatively short or even non-existent zones B and C. The bulk of the data describes saturated conditions (zone D). The relationship between the initial temperature profile and the resulting stagnation state history is shown in Fig 8:3.

8.1.2 Evaluation of the stagnation state at the nozzle inlet

The stagnation state histories shown in Figs 8:1 through 8:3 were obtained using instrumentation at or near the discharge pipe inlet. However, because of the equipment constraints (which resulted in the location of the ball valve between the instrumentation rings and the test nozzles), the instrumentation used to evaluate the nozzle inlet conditions was situated between 2.8 and 7.6 m from the nozzle inlet. Consequently, a number of factors must be considered when defining the true nozzle inlet conditions.

Temperature profile

Temperatures measured at the instrumentation rings and along the wall of the discharge pipe often showed significant radial and azimuthal variations. The effects

of radial and azimuthal temperature variations were eliminated as much as possible by calculating an average temperature (see Appendix A) at the instrumentation ring for each test. The accuracy of the average temperature was checked by comparing it with the saturation temperature calculated from the nearest static pressure. Temperatures measured by 002M403 (located near the center of instrumentation ring I - see Fig 5:6) were examined for each test to determine the time when saturation first occurred. The times at which saturation had occurred at all the thermocouples on instrumentation ring I or II and at 004M532 (located near the nozzle entrance) were also determined. These values, listed in Table 8:1, provide insight into the behaviour of the fluid during the tests by indicating the approximate times when saturated flow first began to enter the nozzle (t_T), when only saturated flow entered the nozzle (t_g) and when the fluid adjacent to the wall at the nozzle entrance became saturated (t_{532}).

Although no attempt has been made to quantify the temperature variation for each test, three factors contributed to a reduction in temperature variations during the test program:

- 1) Termination of the water circulation in the discharge pipe, approximately 1.5 h before the test (after Test 13) was effective in reducing the changes in water temperature observed in approximately the first quarter of the subcooled transient.
- 2) The hole in the moderator tank bottom was enlarged prior to Test 15 to reduce the azimuthal temperature gradient. Figs 8:4 and 8:5 show the temperatures measured at instrumentation ring I during Test 12 and at instrumentation ring II during Test 18 when testing nozzles with a diameter of 300 mm ($L/D = 3.0$ and 3.7 respectively). The initial conditions in the vessel were the same for both tests. Clearly the azimuthal temperature variations were lower in Test 18.

- 3) Radial temperature variations were reduced during the subcooled portion of the blowdown by conducting tests with category II initial conditions. Fig 8:6 shows the temperatures measured at instrumentation ring II during Test 17. Fluid temperature variations in Test 17 are significantly less than those observed in Test 18, even though both tests were conducted with the same test nozzle and the same nominal subcooling.

Pressure at the nozzle inlet

The stagnation pressure was measured in the lower vessel near the discharge pipe inlet using 001M106. Because pressure 106 proved to be a consistently good measurement throughout the test program, with significantly lower amplitude oscillations and noise than the pressure measurements in the discharge pipe, it was used to define the pressure at the nozzle inlet. The fluid stagnation pressure at the nozzle inlet differed from pressure 106; it could be obtained from the latter pressure by the addition of the hydrostatic head and by the subtraction of the irreversible losses between the discharge pipe and the nozzle inlet. Fig 8:7 compares a calculated nozzle inlet pressure with an average trace of pressure 106 for Test 22. Because Test 22 had the largest measured mass flow rates in the discharge pipe, the nozzle inlet stagnation pressure is less than pressure 106 during the early portion of the test, due to irreversible losses; later in the test it is larger than pressure 106 because of the hydrostatic head. The test evaluation of the nozzle inlet pressure is shown in Appendix B. The calculational technique is presented in Ref 3.

Density at the inlet

The fluid density during the subcooled portion of the test was determined using the average temperature defined by the thermocouples at instrumentation rings I or II. However, when saturated fluid first began to appear in the discharge pipe (see Table 8:1), the density measurements were taken from either the gamma densitometer or DP246.

The measurement techniques which are considered to give the most representative average fluid density during saturated flow are listed in Table 8:1. The gamma densitometer was considered to be the instrumentation with the greatest ability to measure the true average two-phase density. However, test experience, demonstrated that the gamma densitometer was often in error by large amounts. Consequently DP246, the backup density measurement, was used as the test standard whenever the gamma densitometer error was found to be unacceptable.

Transit time

The distances between the instrumentation locations and the nozzle inlet, combined with the fluid velocity in the discharge pipe, define a transit time from each measurement to the nozzle inlet. The largest transit times, i.e. 3.5 s, were present during the tests conducted with the 200 mm diameter nozzles (see Fig 8:8) while measuring the two-phase density using DP246 (differential pressure 246). Representative transit times for the 500 mm nozzle, obtained while measuring the upstream conditions using instrumentation ring I and the gamma densitometer (see Fig 8:9) were between 0.1 to 0.3 s.

Thermodynamic state at the nozzle inlet

Using the average fluid entropy as calculated at the instrumentation rings or at the site of the density measurement, combined with the transit time and the nozzle inlet pressure (see Ref 3 for calculation details), the stagnation state at the nozzle inlet was completely defined. The fluid density of the stagnation state (see Appendix C) was defined in the subcooled regime (prior to t_T : see Table 8:1) using the average temperature calculated from the instrumentation ring. After t_T , the density was calculated on the basis of the listed two-phase density measurement.

8.2 Mass flow rates

The mass flow rates obtained during the CFT program have been determined using three different techniques (see Section 6). Peak mass flow rates, which occurred shortly after the start of each experiment, were determined using the steam dome depressurization method and the pitot-static method. Mass flow rates during the remainder of each experiment were calculated using the pitot-static technique and the mass inventory technique. These results are summarized below.

8.2.1 Peak mass flow rates

Table 8:2 lists the peak mass flow rates calculated using the steam dome depressurization (assuming "no condensation" and "equilibrium" - see Ref 4) and the pitot-static methods. In most cases the pitot-static method agreed more closely with the "no condensation" approach. The maximum mass flow rate obtained during the CFT program was 14 Mg/s.

8.2.2 Representative test mass flow histories

Although the mass flow rates were calculated for each test using both the pitot-static and the mass inventory methods, the most representative mass flow rate for an entire test was found to be a combination of the two techniques. This result occurred because the pitot-static method had a lower error band during the subcooled period when the vessel mass/water level was influenced by the water flashing rate and the mass flow rate was changing relatively quickly. However, when the fluid entering

the nozzle became saturated, the error level of the pitot-static method was greatly increased by the high error level for the gamma densitometer method. Consequently, the mass inventory method was the most accurate technique after saturation. Mass flow rate histories for each test are shown in Appendix D. The flow rates shown were determined using the pitot-static method until the time at which all thermocouples indicated saturated flow (t_g , see Table 8:1), after which the mass inventory method was used.

8.3 CFT program test envelope

Using Table 4:1 as a basis, the stagnation conditions at the discharge pipe inlet (see Figs 8:10 through 8:18) and the nozzle inlet pressure/nozzle mass flux characteristics (see Figs 8:19 through 8:27) have been plotted on individual figures for each test nozzle. Ideally, the stagnation conditions at the nozzle inlet should be used to define the test envelope. However, the conditions at the discharge pipe inlet provide a good approximation. The figures are arranged in order of increasing diameter and increasing length.

The nozzle envelope shown in the above figures illustrates the fact that data are available for all nozzles at mass fluxes ranging from $28 \text{ Mg/m}^2 \text{ s}$ to $50 \text{ Mg/m}^2 \text{ s}$ and pressures ranging from 4.1 MPa to 4.9 MPa with at least 30°C subcooled fluid and at low positive qualities. The largest group of data are available for the 500 mm diameter nozzles. Specifically, the nozzle with a diameter of 500 mm and $L/D = 1.5$, and with a diameter of 509 mm and $L/D = 3.1$ provided the test geometries for a total of 11 tests.

The data shown in Figs 8:10 through 8:27 have been further refined (see Appendices A through D). Using the times t_T

and t_s (listed in Table 8:1), the test transient can be divided into three parts. Thus during the test period before temperature 403 (at t_T) showed saturation, the flow entering the test nozzle was fully subcooled. The test period after all the instrumentation ring thermocouples showed saturation (at t_s) consisted of fully saturated flow. Finally, the test period between the fully subcooled and fully saturated periods was a transition period when the fluid entering the nozzle was a mixture of subcooled and saturated flow.

Table 8:3 lists the data ranges available for each nozzle geometry in terms of $x^{[a]}$ (the universal stagnation quality) and the pressure at the nozzle inlet and the nozzle mass flux. The table format is arranged to show the minimum and maximum values of the quality, pressure and mass flux for both the subcooled and saturated periods. The minimum saturated quality is zero in each case. The quality, pressure and mass flux ranges of the transition period are generally between the values of the universal quality for the maximum subcooled period and zero saturated quality. Data ranges for the subcooled period are listed only after the initial pressure spike (see Fig 8:7) has undergone recovery. The data listed in Table 8:3 are plotted for each test in Appendix E.

[a] The universal stagnation quality, x , is defined by the equation:

$$x = [v_{v_1}(P)] / [v_v(P) - v_{l_1}(P)]$$

where v is the specific volume of the mixture, p is the stagnation pressure, and the saturated liquid and vapour specific volumes (v_{l_1} and v_v , respectively) are defined by the stagnation pressure only.

8.4 Repeatability tests

Data obtained during repeat tests can be studied from several viewpoints. For example, the instrumentation behaviour during two similar tests can be examined for consistency, or the variation of the nozzle mass flux under equal nozzle inlet stagnation conditions can be compared using the data in Appendix E. A third method of comparing data obtained in repeatability tests is to examine the similarities between the stagnation state histories at given times in the repeat tests. This method, which is used to determine the ability to repeat tests from an operational standpoint, will be summarized here.

Several tests were conducted in order to observe the degree of repeatability when the same initial conditions and test nozzle were used. Specifically, Tests 4, 8 and 11 were conducted (category I tests) with the nozzle having a diameter of 509 mm and $L/D = 3.1$ with the water initially subcooled by 30°C , Tests 2 and 12 (category I tests) were conducted with the nozzle having a diameter of 300 mm and $L/D = 3.0$ nozzle with 30°C subcooled water, and Tests 9 and 10 (category III tests) were conducted with the nozzle having a diameter of 509 mm and $L/D = 3.1$ with the water subcooled by less than 5°C . In addition, Tests 21 and 27 can be used to demonstrate repeatability using category II inlet conditions even though the primary reason for conducting these tests was to evaluate the effect of water air content on critical mass flow rates through the nozzle. Tests 21 and 27 were conducted using the nozzle having a diameter of 500 mm and $L/D = 1.5$ with 30°C subcooled water. Detailed comparisons between these tests are reported in Refs 5 through 9. The results of the comparisons are summarized below using pressure 106, temperature 403, DP246 (the differential pressure 246) and mass flow rates calculated using the pitot-static technique, unless otherwise indicated.

Tests 4 and 8

Initial vessel temperatures in Test 4 were usually within 1°C of those in Test 8, excluding the transition zone. The test 4 transition zone (zone C - see Fig 8:1) was approximately 0.5 m above that of Test 8, so that a greater subcooled mass was present during Test 4. The difference in initial conditions caused the stagnation conditions at the discharge pipe inlet to differ by as much as 3°C and 80 kPa (see Fig 8:18) during subcooled discharge and by as much as 40 kPa with quality differences of 0.0015 during saturated discharge. The tests indicated that saturation occurred within 0.2 s of each other. During the subcooled portion of the test, the mass flow rates differed by a maximum of 400 kg/s or 5.2 per cent (prior to 14 s) while during the saturated blowdown the mass flow rates differed by a maximum of 200 kg/s or 5.6 per cent (after 22 s).

Tests 8 and 11

The initial vessel fluid temperatures were within 1.3°C of each other at all elevations. Comparison between the stagnation state histories shows that the pressures and temperatures usually differed by no more than 80 kPa and 1.8°C (see Figs 8:18) respectively at equal times during the subcooled blowdown. The tests show that saturation occurred within 0.5 s of each other. During saturated blowdown the pressures were usually within 50 kPa and the qualities within 0.001 of each other at equal times. The mass flow rates during the subcooled period, evaluated prior to 14 s, were within 2 per cent of each other.

Comparison of the flow rates evaluated on the basis of the mass inventory showed that the tests agreed to within 10 per cent between 20 and 32 s and to within 2 per cent from 32 to 45 s.

Tests 2 and 12

Although the vessel temperatures above the 11.8 m elevation agreed to within 0.8°C , the initial transition zone for Test 2 began approximately 0.7 m above that for Test 12.

Below 11.8 m, the vessel water in Test 2 was initially subcooled 2.5 to 10°C more than in Test 12 at equal elevations. The differences in initial subcooled temperature were reflected in a comparison between the stagnation state histories at the discharge pipe inlet. Differences in the subcooled temperature history, combined with the fact that no instrumentation was available in Test 2 to allow the saturated flow history to be monitored, make these tests poor operational repeats of each other (see Fig 8:13).

Tests 9 and 10

Initial vessel temperatures in Tests 9 and 10 agreed to within 1.5°C above the 4.9 m elevation. Below 4.9 m, the temperatures in the vessel differed by between 2 and 3°C. Saturation was indicated at 4.5 s in Test 10 and at 6.1 s in Test 9. Following saturation, pressures and qualities agreed to within 100 kPa and 0.001 (see Fig 8:18) at equal times for the two tests. Comparison of the saturated mass flow rates for these tests shows a maximum difference of 13 % during the transient using the vessel mass inventory method.

Tests 21 and 27

The vessel thermocouples indicated that the initial vessel water temperatures agreed to within 1.8°C above the 2 m elevation, excluding the 16.7 m elevation. Differences in the temperature at the 16.7 m elevation showed that Test 21 had a slightly higher transition zone. Comparison between the stagnation state histories shows that the two tests usually had pressures and temperatures that agreed to within 80 kPa and 1.5°C respectively at equal times (see Fig 8:16). Test 27 reached saturation first (at 25 s), but only 3 s prior to Test 21. Saturated stagnation state histories agreed to within pressures and qualities of 50 kPa and 0.0005 respectively. Comparison of the subcooled mass flow rates shows that the two tests agreed within the error limits for the pitot-static method. The saturated mass flow rates agreed within the error limits of the mass inventory method.

8.5 Nozzle pressure profiles

The pressure data recorded in the test nozzles are discussed. In particular, the profiles recorded during the subcooled and saturated periods, the effect of nozzle geometry irregularities and the profiles recorded with straight and flared exit geometries are summarized. While reading the following sections and reviewing the data presented, the reader should note that only wall static pressures are available for evaluation of the nozzle test conditions. Thus the discussions are limited by a lack of knowledge about the pressure profile over the nozzle cross-section.

8.5.1 Nozzle pressure profiles during saturated and subcooled flow

Nozzle pressure profiles exhibited during the saturated and subcooled portions of the transient are quite different from each other. The profiles during the subcooled flow regime show a large drop in pressure just downstream of the nozzle entrance (seen clearly in the 004M217 data) followed by pressure recovery further downstream in the nozzle. In addition, nozzle pressure profiles obtained during subcooled flow often include other locations which show lower static pressure upstream and short zones of pressure recovery further downstream. On the other hand, nozzle pressure profiles obtained during saturated flow are characterized by an absence of pressure recovery zones. Instead, the pressure decreases slowly with increasing length. However, data recorded during both the saturated and subcooled flow regimes indicate that the pressure drops rapidly between the last 60 to 100 mm of each straight section of the nozzle.

Examples of the differences between the nozzle pressure profiles obtained during subcooled and saturated flow

are shown in Fig 8:28 using Test 11 data. The pressure ratio was calculated by dividing the local pressure at each tap in the nozzle (see Fig 5:7) by the stagnation pressure at the vessel bottom (001M106). The profiles recorded at 1.5 and 10 s were during periods of subcooled flow, while the profile at 40 s was during the period of saturated flow. The characteristics discussed above are all evident in Fig 8:28.

Interpretations of the pressure profiles which exist during the subcooled and saturated flow regimes vary widely amongst the people who have studied the data.

Interpretations offered are:

1. The large pressure drop observed at the nozzle entrance is evidence of a vena contracta caused by the nozzle entrance configuration.
2. The large pressure drop observed at the nozzle entrance during subcooled flow is sufficient to cause the local pressure at the nozzle entrance to fall below saturation. Thus the data at the inlet provide evidence of the beginning of cavitation in the nozzle test section. In fact, the pressure recovery zone could be due to local high pressure zones, resulting from the rapid formation and collapse of vapor bubbles as the nozzle fluid cavitates. Thus, when the nozzle entrance conditions become saturated, fluid cavitation within the nozzle test section would cease for the most part and the pressure profile would become more characteristic of those seen in a pipe, indicating a pressure loss due to friction.

These observations are indicative of the fact that the data evidence to date is not sufficient to allow a complete description of the fluid conditions within the nozzle test section. A basic difficulty is the lack of information about the relationship between the wall and free-stream static pressures.

Indications of the relationship between the wall static pressure and the saturation pressure calculated at the nozzle entrance (using temperature 004M405) are shown in Fig 8:28. These data reveal the same behaviour as the pressure ratio based on the stagnation pressure at the discharge pipe inlet (see Fig 8:28).

8.5.2 Effect of nozzle surface irregularities on the nozzle pressure profile

Table 3:1 contains a comprehensive listing of the basic nozzle dimensions together with a limited qualitative description of the nozzle surface that was exposed to the test fluid. Of great interest is the effect of local surface irregularities on the nozzle pressure profile data. Although surface irregularities were found on all the test nozzles to some degree correlations between the surface irregularities and the available pressure profiles cannot always be made. However, several correlations between the data and the nozzle geometry are apparent:

All nozzles tested with a straight section, i.e. without a flared exit, had the rupture disc carrier rings as a part of the nozzle test section (see Fig 3:5 and Ref 1).

Because the rupture discs were brazed into position, a prominent burr (usually 1 mm in height) was left protruding into the nozzle free stream on the inner lip of each carrier ring following rupture disc failure. Consequently, the pressure measured by the 004M121 and 004M123 taps (see Fig 5:7), as well as the others downstream of the rupture discs, could be influenced by these irregularities (see Fig 8:30).

Another example of correlation between nozzle irregularities and pressure profile data is found by examination of the Tests 17, 18 and 19 data. Fig 8:30 (pressure profile recorded during Test 18) clearly shows an unexpected pressure decrease and rise between 0.9 and 1 m from the nozzle inlet in exactly the region that is indicated as being irregular in footnote d of Table 3:1.

8.5.3 Comparison between the straight and flared exit geometries

In general, the effect on the nozzle exit pressure of changing the nozzle exit from a flared to a straight section cannot be discussed in detail. Conclusive data cannot be shown because of the presence of the burrs on the rupture disc carrier rings in the nozzles with straight exit sections, and the lack of detailed pressure data from the straight section of the nozzles with flared exits. However, the alteration in the nozzle exit from a flared design (see Fig 3:3) to a straight design (see Fig 3:5) had no measurable effect on the nozzle mass flow rates.

8.6 Miscellaneous test results

During the course of the CFT program, several phenomena which could not be fully explained were found to occur in every test to some degree. Since the understanding of these phenomena contributes to the interpretation of the test data, they are discussed as miscellaneous test results. It is hoped that the work and the observations presented here may contribute to further progress by other investigators. Effects such as initial pressure undershoot and

recovery, local fluid superheating and the change in pressure oscillatory behaviour between the subcooled and saturated flow regimes are discussed. In addition, the containment data are briefly summarized.

8.6.1 Initial pressure undershoot and pressure recovery

In every test, the maximum flow rates occurred within the first 0.5 s after rupture disc failure. The flow rates themselves were a function of the fluid temperature and pressure in the nozzle and in turn resulted in a rapid vessel depressurization (see Fig 8:31). The extent to which the vessel depressurized before significant flashing occurred and reversed the process, causing an increase in the vessel pressure (see point B Fig 8:31), varied from test to test. Further, the length of time required before the vessel pressure peaked (see point C Fig 8:31) also varied from test to test.

Figure 8:32 shows the data in the form of the pressure drop (ΔP_{AB}) between points A and B, as a percentage of the initial pressure at point A (P_A), plotted as a function of the ratio of the quantity of steam in the dome (m_g) to the peak mass flow rate ($\dot{m}$) assuming equilibrium (calculated using the steam dome depressurization method). The relationships between these variables are such that the tests with decreased quantities of steam in the dome, with water levels above 19 m and with the greatest peak mass flow rates have the greatest pressure drop.

The variables seem to be related by the empirical equation:

$$\frac{\Delta P_{AB}}{P_A} \left(\frac{m_g}{\dot{m}} \right)^{1/2} = 3.5$$

However, the data scatter apparent in Fig 8:32 makes it obvious that other lines can be drawn to establish a means of predicting the total initial pressure drop which will occur in any given test. The above equation does suggest that the process is polytropic. Indeed, if the change in vessel pressure is examined in relation to the increase in the initial volume occupied by the steam, polytropic exponents can be determined as a function of the nozzle mass flow rate prior to t_B (see Fig 8:31).

Figure 8:32 also shows an apparent relationship between the recovery time, i.e. t_C , and the ratio $m_g/\dot{m}$. Those tests with large initial pressure drops have short recovery times. Exceptions only occur for tests conducted with the fluid subcooled by less than 5°C (category III tests). Category III tests (numbers 9, 10, 14, 19, 20, 23 and 25) have recovery times in excess of 3.4 s independent of the ratio $m_g/\dot{m}$. Further investigations should be able to establish a relationship between t_C and the quantity of fluid which flashes during t_C seconds, as obtained from accepted flashing rate calculations.

8.6.2 Local fluid superheating

Data on the fluid temperature in the vessel showed the presence of superheated fluid in the vicinity of the thermocouples located on the vessel cable (see Fig 5:3) during specific time periods for Category I and II tests. Examination of the data indicating local superheating (see Fig 8:33) shows that the amount of superheating was usually 2° to 4°C , although 5° to 9°C have been observed. Superheating was shown by the thermocouples in advance of the arrival of the collapsed liquid level. Thus the super-

heated fluid temperatures are believed to be indications of the arrival time of the water boiling boundary, which moves downward in the vessel with time (see Fig 8:33). Thus the upper thermocouples show superheating first, followed closely by those at lower elevations. The magnitude of the superheating may be related to the reaction time required by fluid which is subjected to a depressurization which causes the local pressure to reach and quickly fall below the saturation state, so that the liquid water is left in a nonequilibrium condition. In addition, the degree of fluid superheating seems to be related to the initial content of dissolved air in the water, since Test 27 (cf Test 21 - see Tables 4:1 and 4:2) exhibited the largest degree of superheating in any of the tests.

8.6.3 Oscillatory behaviour of subcooled and saturated data

Throughout the CFT program, the data showed the presence of oscillations. However, the character of the data oscillatory behaviour is marked by a distinct change which occurs when the fluid entering the test nozzle changes from the subcooled to the saturated regime (especially 004M109 data, see Fig 5:4). The nature of the mechanism which causes the change in oscillatory behaviour has been studied. Data analyses have shown that the observed frequency pattern cannot be explained by hydrodynamic noise (not including cavitation), the natural frequency characteristics of the vessel and discharge pipe, the nozzle geometry, measurement system vibrations or cavitation in the discharge pipe. However, there remains the possibility that the change in data oscillatory behaviour is caused by a change in the flow behaviour through the nozzle. Assuming that subcooled fluid entering the test nozzles would cavitate and that saturated fluid would not, the subcooled data would be marked by more significant noise (induced by

cavitation) than the saturated data. A study on cavitation noise induced in small plexiglass and stainless steel venturies (see Ref 10) revealed that significant cavitation noise was generated from 200 to 6000 Hz, but with maximum signal to noise ratios in the 4 to 6 kHz range with either mercury or water as the test fluids. If the change in data oscillatory behaviour observed in the CFT program was caused by cavitation in the test nozzle, a maximum signal to noise ratio in the 4 to 6 kHz range might also be expected in the CFT data. However, the measurement system was not mounted in a configuration which would allow data confirmation of this theory in any of the tests.

8.6.4 Containment data

Measurements were made in the containment, both to ensure that the building was not structurally overloaded and to monitor the fluid behaviour. The instrumentation (see Fig 5:9) was installed so that it measured differential pressures across several compartments in the containment and the exhaust pipe inlet, as well as pressures and temperatures in selected locations. Table 8:4 lists the maximum pressures measured in the containment and the times at which they occurred. Although the largest mass flow rates through the test nozzle were measured during Test 22 (see Table 8:2), the largest containment pressures, i.e. 360 kPa, were recorded during Tests 4, 8 and 11. The inconsistency is explained by the change (see Table 3:2) from a smaller to a larger exhaust pipe in the upper portion of the containment building (see Fig 3:1), which occurred between Tests 14 and 15.

Fluid behaviour in the containment was characterized by two other important observations:

1. Temperatures in the wetwell (thermocouples 105M415 through 418) were consistently lower than the saturation temperatures corresponding to the pressures measured at adjacent locations. This result might be due to air being trapped in the wetwell.
2. The containment data during some of the tests showed a pressure ratio across the large exhaust pipe inlet (see Fig 3:1) corresponding to choked flow. The choked condition would only occur during the early portion of a test.

Table 8:1 Nozzle inlet condition specifications: saturation times and primary instrumentation

Test Number	Test instrumentation ring used to determine the subcooled density	Two-phase ¹⁾ density measurement	$t_T^{2)} \sim s$	$t_s^{3)} \sim s$	$t_{532}^{4)} \sim s$
1	I	No data	49.2	60.0	~ 53.5
2		No data	46.1	55.9	~ 46.1
3		DP 246	12.6	19.5	15.0
4			15.4	22.2	20.3
5			17.5	22.0	21.6
6			38.5	52.2	49.4
7			35.9	47.0	48.1
8			15.6	20.0	16.0
9			6.1	11.0	7.1
10		G D	4.5	10.0	7.4
11			15.2	26.6	18.5
12			43.7	56.1	56.8
13			126.5	142.0	136.0
14		DP 246	40.0	73.7	~ 75.0
15	II		21.5	30.0	29.0
16			16.1	25.0	18.5
17			∞	∞	∞
18			46.0	59.5	59.4
19			13.5	34.0	20.0
20			1.6	9.6	10.9
21			28.0	28.5	28.0
22			26.2	33.8	31.6
23			4.9	6.8	12.0
24			21.7	28.5	24.3
25		G D	13.0	35.0	15.0
26			80.0	84.0	85
27			25.0	32.2	26

1) GD = gamma densitometer, DP 246 = 007M246

2) t_T = Time of first saturation of 002M4033) t_s = Time of final saturation of all thermocouples on test instrumentation ring.4) t_{532} = Time of saturation of 004M532.

Table 8:2 Summary of the peak nozzle mass flow rates

Test	$\dot{m}_{Eq}^{1)} \sim \text{kg/s}$	$\dot{m}_{Nc}^{2)} \sim \text{kg/s}$	$t^{3)} \sim \text{s}$	$\dot{m}_{Pitot-static}^{4)} \sim \text{kg/s}$	$t^{5)} \sim \text{s}$
1	4550	3900	0.20	3750	0.12
2	5340	4590	0.32	4440	0.21
3	11990	10280	0.26	11250	0.21
4	13840	11890	0.27	12500	0.21
5	12150	10480	0.29	10950	0.22
6	5300	4580	0.05	4560	0.10
7	5150	4450	0.06	4500	0.10
8	13190	11290	0.26	12300	0.22
9	9020	7730	0.10	8930	0.12
10	9250	8000	0.05 to 0.10	8800	0.12
11	13390	11500	0.25	12300	0.16
12	5130	4400	0.05 to 0.10	4380	0.12
13	2830	2430	0.04	2390	0.06
14	2690	2320	0.05	2180	0.06
15	12550	10750	0.17	11900	0.19
16	15900	13700	0.18	12400	0.18
17	5550	4800	0.05	4470	0.19
18	5670	4890	0.05 to 0.09	4650	0.16 to 0.19
19	6000	5170	0.06 to 0.07	5180	0.12
20	14550	12500	0.13	10050	0.12 to 0.16
21	13050	11200	0.14	12180	0.16
22	16000	13860	0.14	13700	0.15
23	12500	10720	0.09	11700	0.12
24	16900	13200	0.11	13000	0.12
25	5920	5120	0.03	5390	0.06
26	6190	5330	0.08	4530	0.09
27	14700	12600	0.14	12900	0.15

1) Obtained from the steam dome depressurization method assuming thermodynamic equilibrium

2) Obtained from the steam dome depressurization method assuming no condensation

3) Time of peak mass flow rate calculated by the steam dome depressurization methods

4) Peak mass flow rate - pitot-static method

5) Time of pitot-static peak mass flow rate

Table 8:3 Summary of inlet stagnation conditions

Nozzle D/L ~ mm	Test	Subcooled						Saturated				
		$x(10^4)$		$P \sim \text{MPa}$		$\dot{m}/A \sim \text{Mg/m}^2 \text{ s}$		$X(10^4)$	$P \sim \text{MPa}$		$\dot{m}/A \sim \text{Mg/m}^2 \text{ s}$	at x_{max}
		Min	Max	at x_{min}	at x_{max}	at x_{min}	at x_{max}		at $x=0$	at x_{max}		at x_{max}
200/ 590	14	-3.5	-1.5	5.09	4.91	36.94	32.48	0	4.78	4.49	29.60	26.40*
	13	-31.0	-1.7 ⁺	5.20	4.16	57.30	31.85	NO	DATA			
300/ 290	7	-13.0	-2.0	5.05	4.45	54.46	41.73	8.0	4.32	3.91	26.87*	22.35*
	6	-20.0	-4.0	5.02	4.27	60.11	43.85	6.0	4.11	3.75	26.88	22.21
300/ 511	25	-3.0	-1.0	5.02	4.91	35.36	34.09	0	4.76	4.37	26.87*	22.63*
	26	-19.4	-0.5	4.94	3.06	55.87	26.73	0	3.04	2.71	22.35	15.56
300/ 895	1	-14.0	-1.5	5.00	4.28	49.36	29.99	IF	4.18	3.87	23.76	18.39
	2	-20.5	-2.6	5.01	4.16	58.27	31.47	IF	4.05	3.70	28.29	20.08
300/ 1116	12	-20.5	-4.0	5.04	4.28	55.45	33.24	15.0	4.15	3.48	26.17*	17.26*
	19	-2.5	-0.5	5.12	5.00	33.24	30.27	0	4.86	4.43	26.87*	20.93*
500/ 166	18	-19.0	-2.7	5.03	4.22	53.75	32.25	0	4.10	3.89	22.63	20.23
	17	-18.5	-0.5	4.98	3.13	54.31	24.05	NO	DATA			
500/ 166	23	-1.5	-1.5	4.94	4.94	34.13	32.20	60.0 ⁺	4.90	3.71	26.59	18.75
	24	-17.0	-1.3	4.78	3.15	59.35	30.57	29.4	3.05	2.80	19.00	16.71

+ Large uncertainty

* Oscillatory saturated flow. Maximum and/or minimum values may exceed listed values

IF = instrument failure

Table 8:3 Summary of inlet stagnation conditions (Cont'd)

Nozzle D/L ~ mm	Test	Subcooled						Saturated					
		x(x10 ⁴)		P ~ MPa		ṁ/A ~ Mg/m ² s		x(x10 ⁴)		P ~ MPa		ṁ/A ~ Mg/m ² s	
		Min	Max	at xmin	at xmax	at xmin	at xmax	Max	at x=0	at xmax	at x=0	at xmax	
500/ 730	20	NO	DATA					35.5	4.86	3.89	23.43	18.34	
	21	-16.5	0.0	4.74	3.14	57.31	20.38	0.0	3.14	2.82	20.38	15.79	
	27	-17.0	-1.0	4.75	3.17	55.53	26.74	22.0	3.08	2.77	18.08	15.28	
	22	-24.0	-2.0	4.80	2.76	63.93	28.27	0.0	2.66	2.60	16.81	15.54	
	16	-17.5	-1.0	4.84	4.21	54.51	28.53	20.0	4.00	3.60	18.75	17.58	
500/ 1809	15	-16.0	-3.5	4.90	3.76	55.27	29.04	9.0	3.65	3.31	18.34	16.56	
	5	-14.0	-0.6	4.01	3.50	52.33	28.01	19.0	3.43	3.03	17.44*	15.72*	
	9	-2.0	-1.0	5.00	4.99	31.45	26.54	45.0	4.90	3.85	21.13*	17.69*	
	10	NO	DATA					17.0	4.87	3.80	22.06	16.81	
	3	-13.0	-1.5	4.87	4.51	47.42	32.92	18.0	4.34	3.82	20.15	17.20	
	4	-19.0	-1.0	4.80	4.19	57.74	31.45	34.5	4.05	3.61	18.43	16.46	
	8	-20.0	-1.5	4.99	4.19	57.25	24.32	40.0	4.13	3.62	19.07*	17.20*	
	11	-20.0	-2.5	4.85	4.22	55.77	29.48	30.0	3.98	3.59	18.13	16.71	

* Oscillatory saturated flow. Maximum and/or minimum values may exceed listed values.

Table 8:4 Measured peak pressures in the containment

Test	Maximum pressure ~ kPa	Measurement time of maximum pressure ~ s
1	160	11
2	180	12
3	330	12
4	360	12
5	320	15
6	190	13
7	140	12
8	360	12
9	246	6
10	253	7
11	360	13
12	173	12
13	122	4
14*	117	1
15	320	14
16	320	12
17	150	5
18	150	20
19	131	4
20	234	6
21	300	10
22	308	13
23	236	6
24	328	13
25	140	7
26	152	6
27	300	12

* The upper exhaust pipe was enlarged following Test 14

TEST 18 Fig 8:1a Vessel initial temperature profile -
Category I test

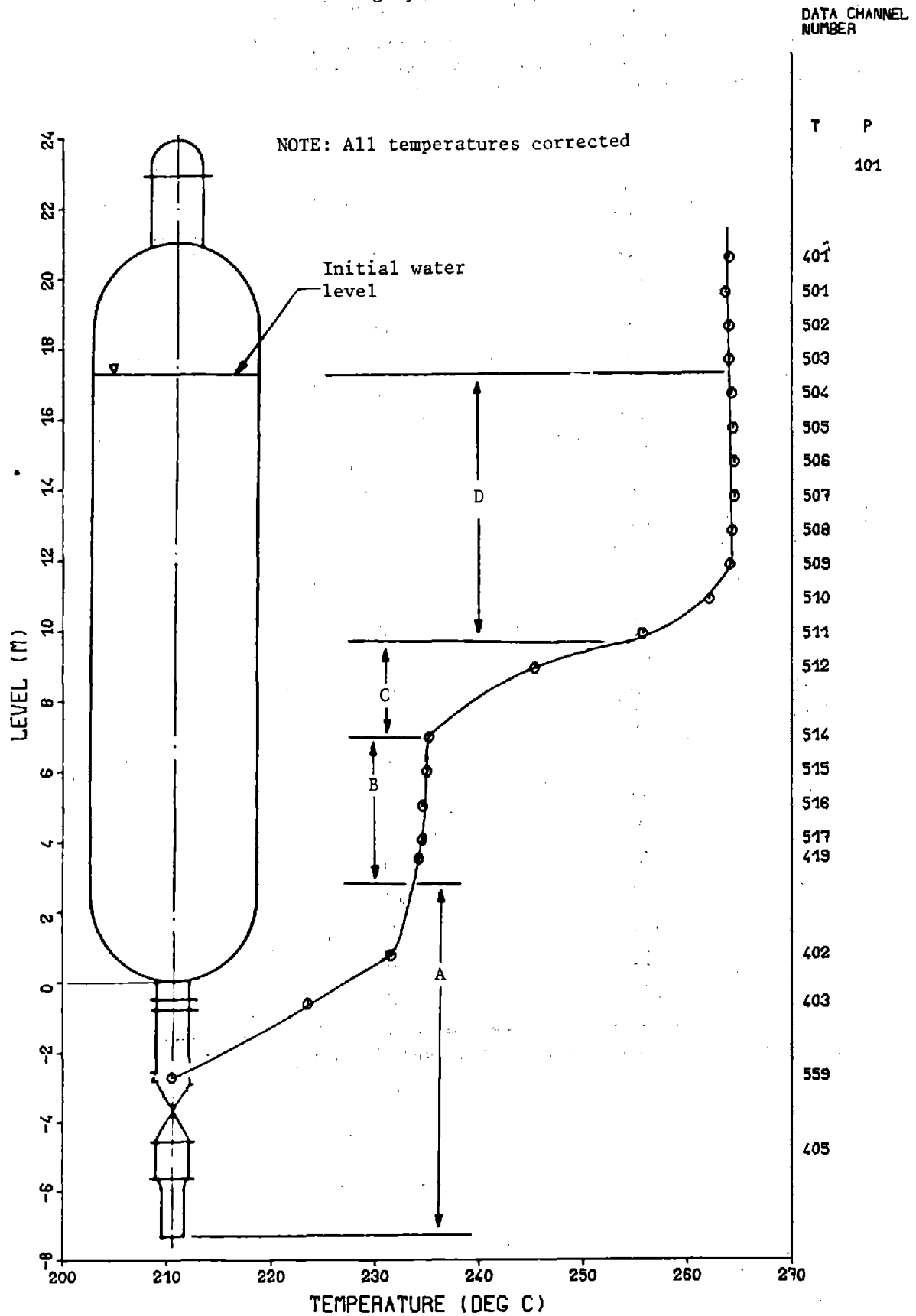
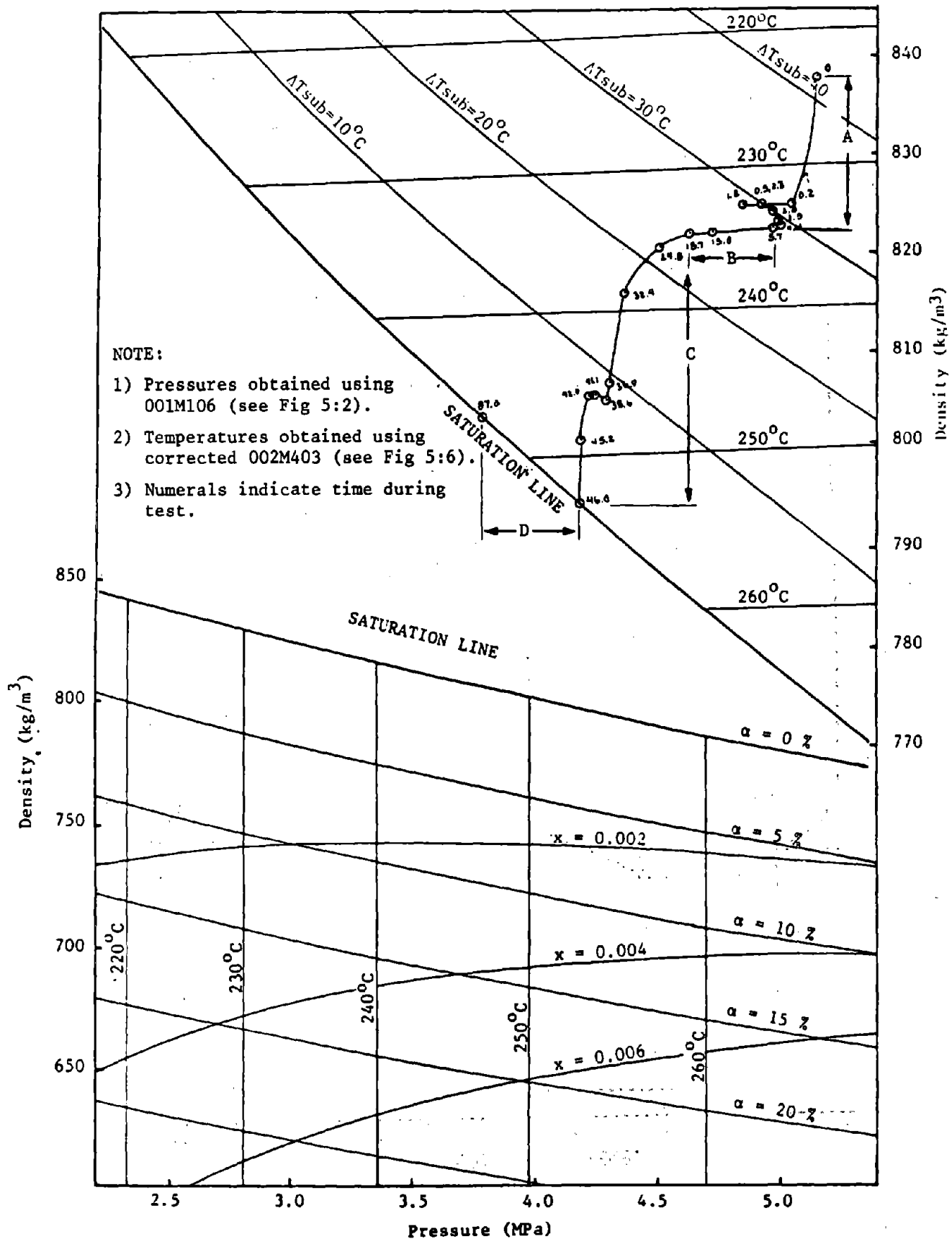


Fig 8:1b Category I test stagnation state history at the discharge pipe inlet



TEST 17 Fig 8:2a Vessel initial temperature profile -
Category II test

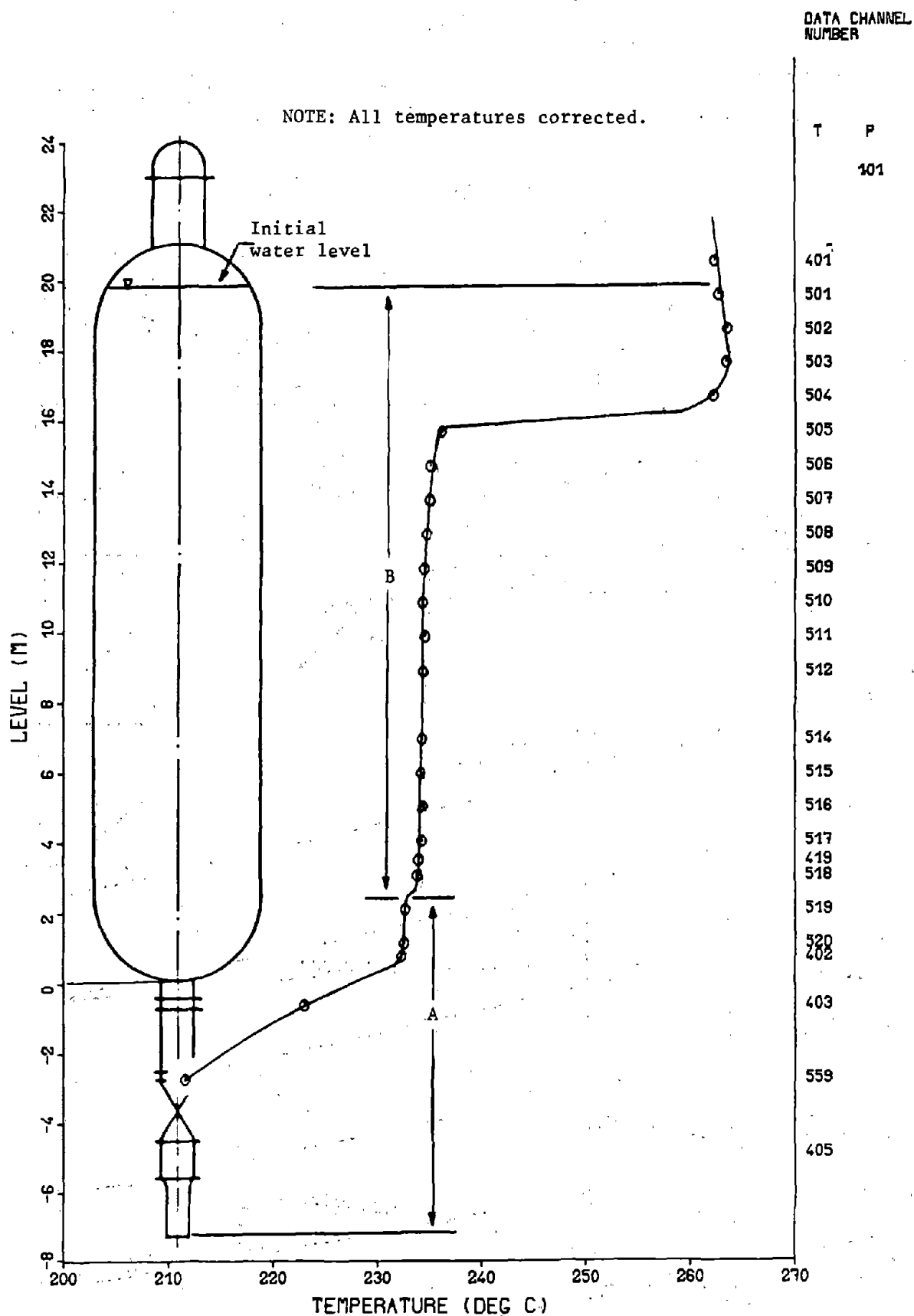
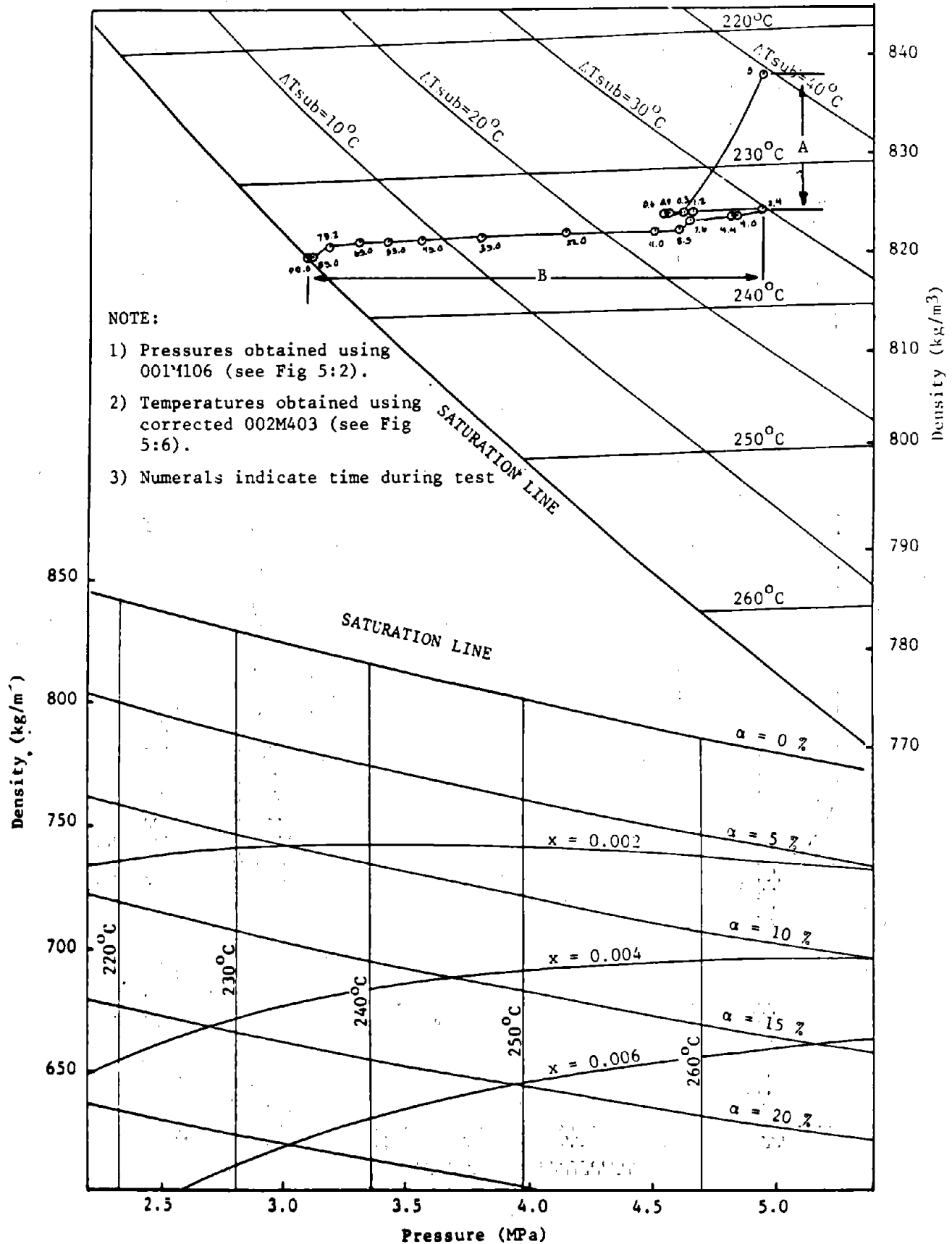


Fig 8:2b Category II test stagnation state history at the discharge pipe inlet



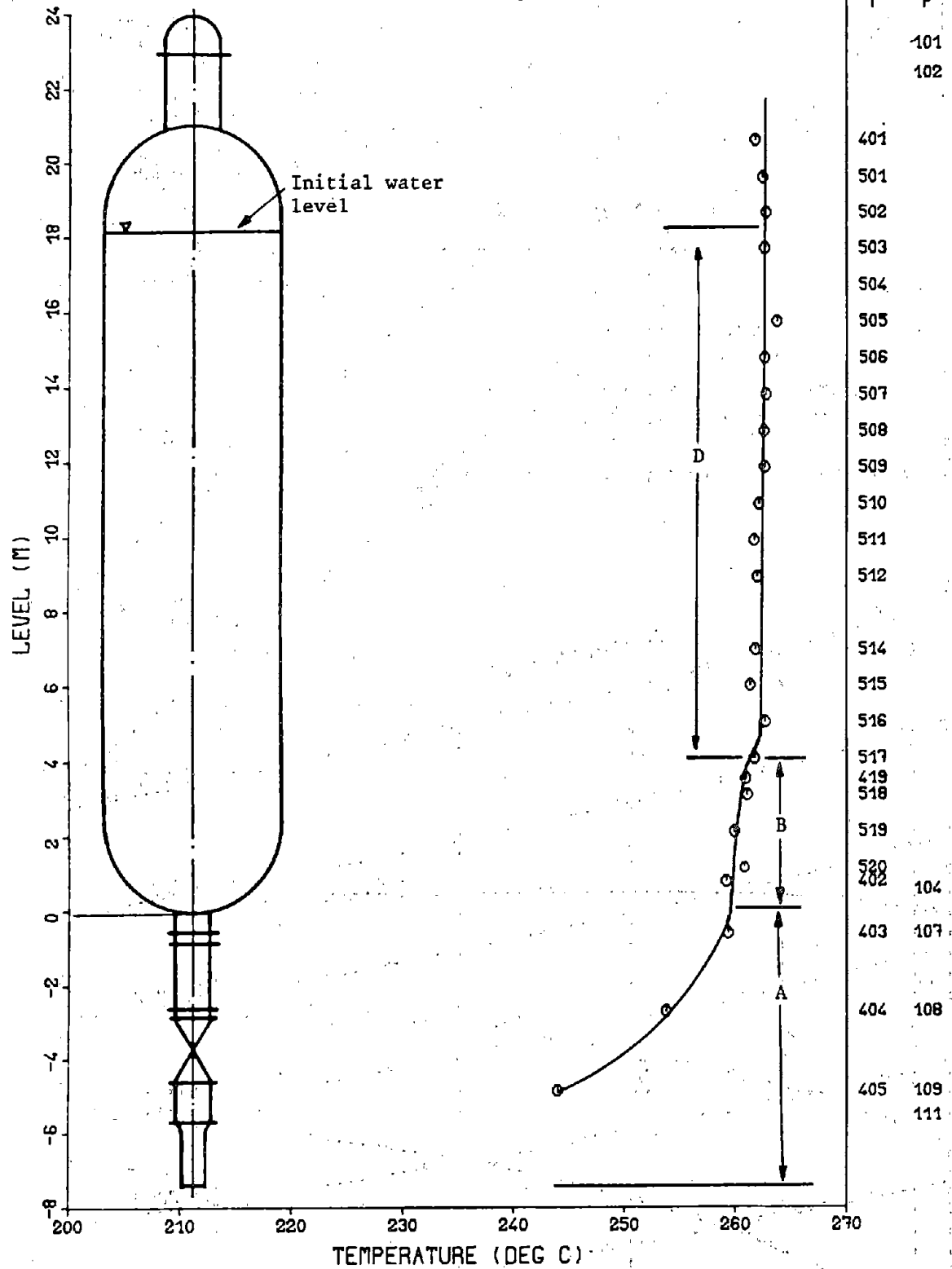
TEST 9

Fig 8:3a Vessel initial temperature profile -
Category III test

TIME (S) = -1.00

DATA CHANNEL
NUMBER

NOTE: All temperatures corrected.



NOTE:

- 1) Pressure obtained using 001M106 (see Fig 5:2)
- 2) Temperatures obtained using corrected 002M403 (see Fig 5:6)
- 3) Densities obtained using 007M246 (see Fig 5:2)
- 4) Numerals indicate time during test

The graph displays Density (kg/m^3) on the Y-axis (ranging from 770 to 840) against Pressure (MPa) on the X-axis (ranging from 2.5 to 5.0). It includes isotherms for temperatures from 220°C to 260°C and a SATURATION LINE. Data points are plotted along these isotherms, with some labeled with numerals indicating time during the test. Key features include the critical point at approximately 3.9 MPa and 311°C, and the liquid-liquid phase transition region between 250°C and 260°C.

Fig 8:4 Comparison between the instrumentation ring I temperatures at all locations and the centerline temperature: Test 12

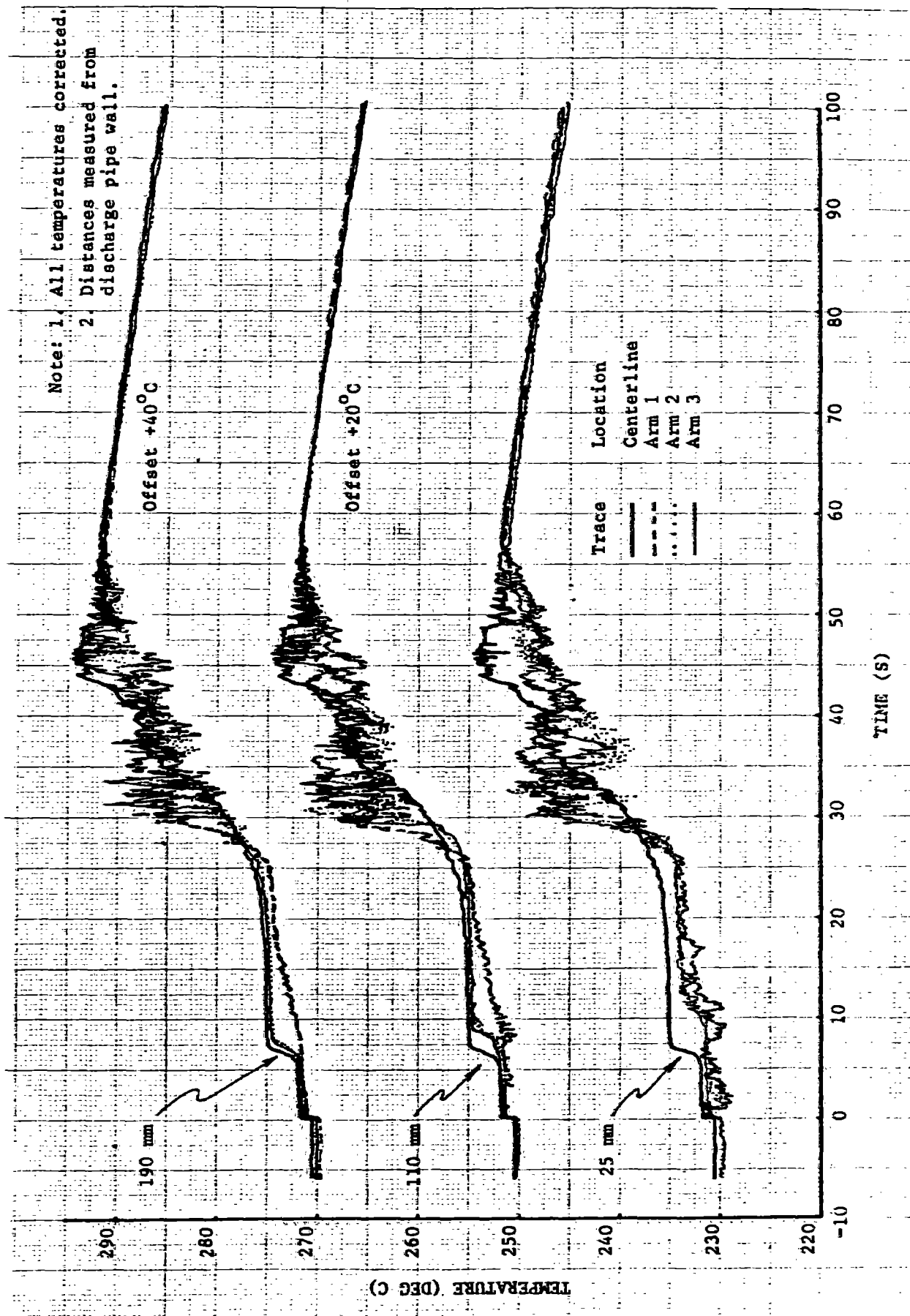


Fig 8:5 Comparison between the instrumentation ring II temperatures at all locations and the centerline temperature: Test 18

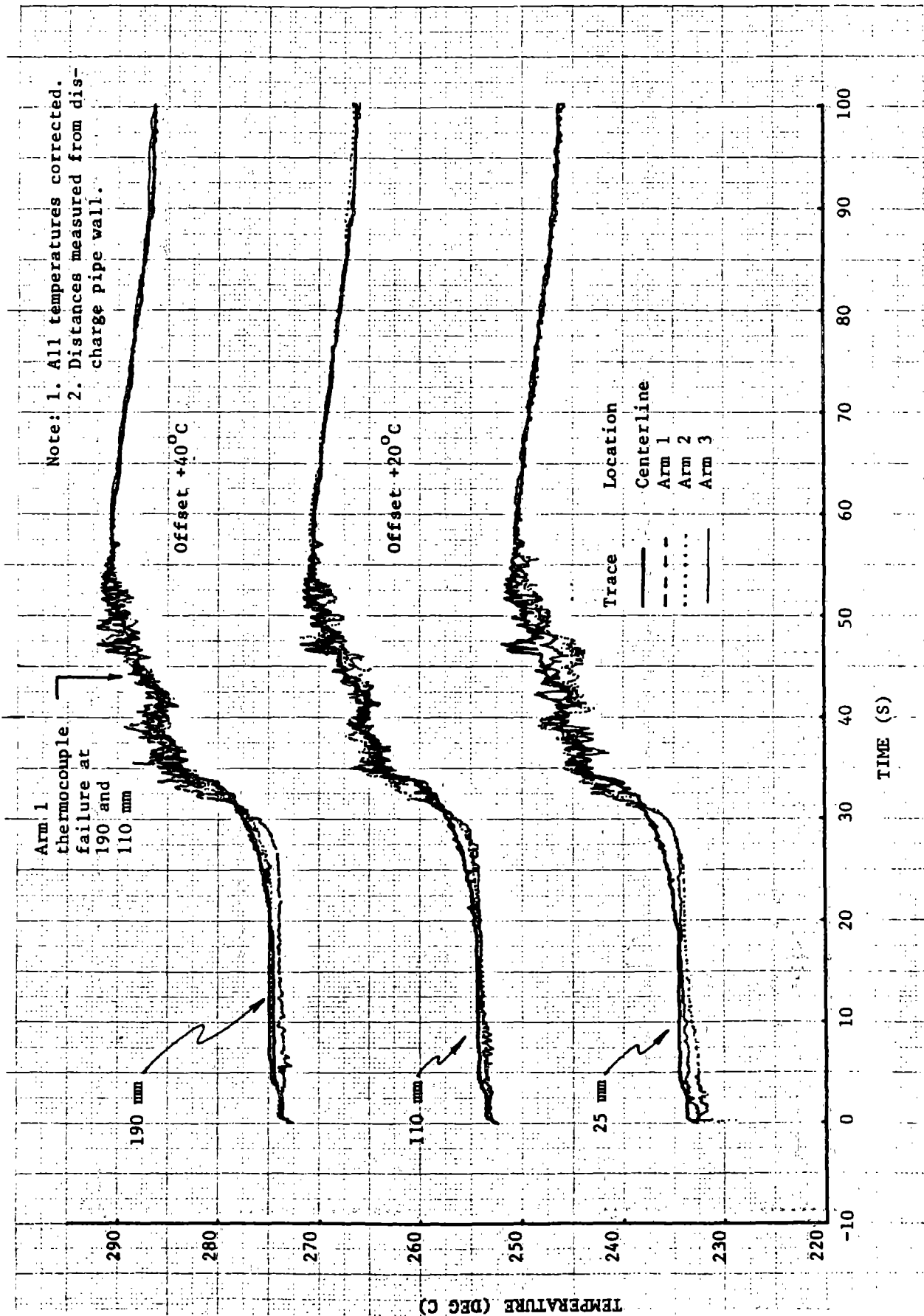


Fig 8:6 Comparison between the instrumentation ring II temperatures at all locations and the centerline temperature: Test 17

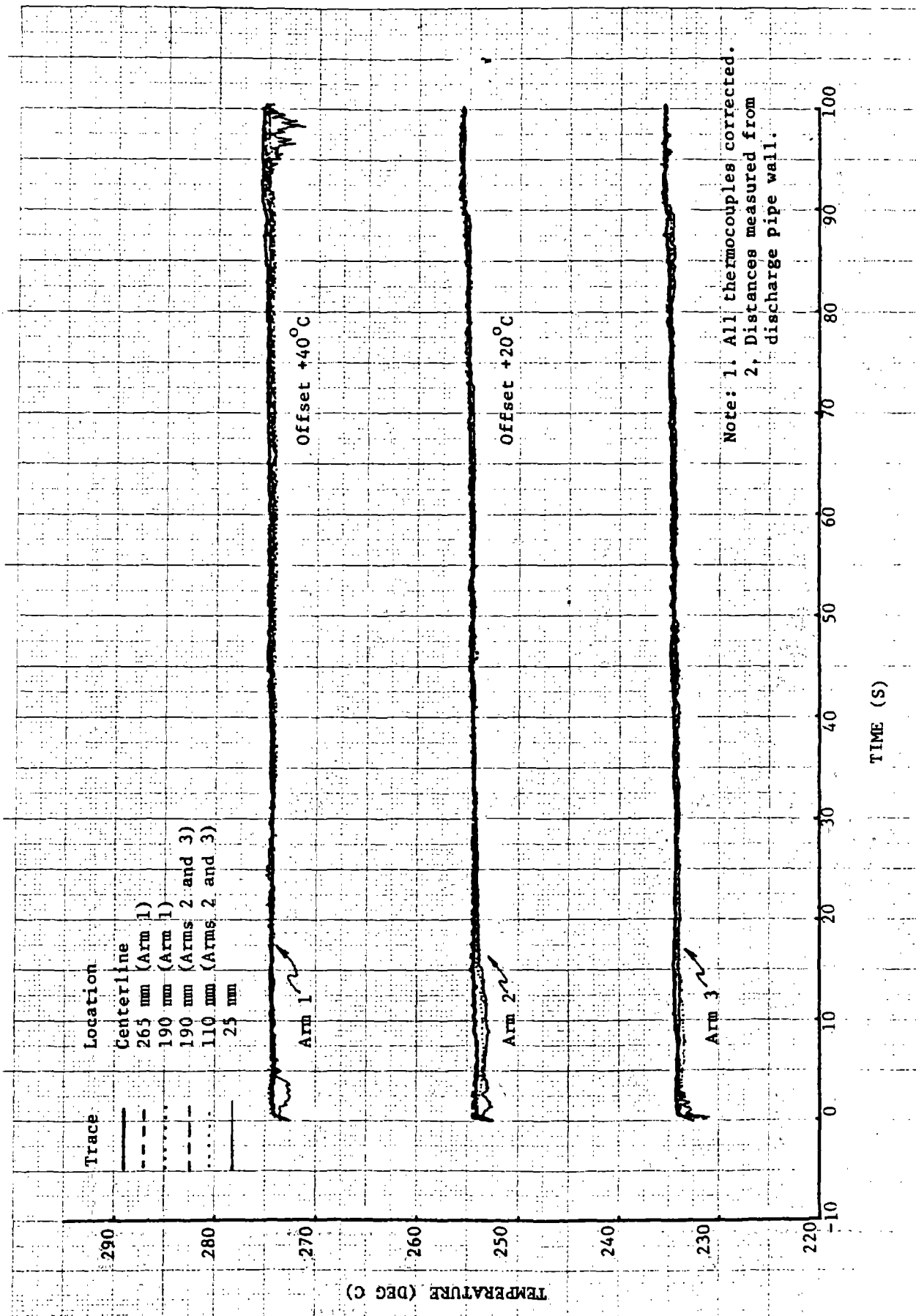


Fig 8:7 Comparison between the stagnation pressure at the vessel bottom and the stagnation pressure at the nozzle inlet plane: Test 22

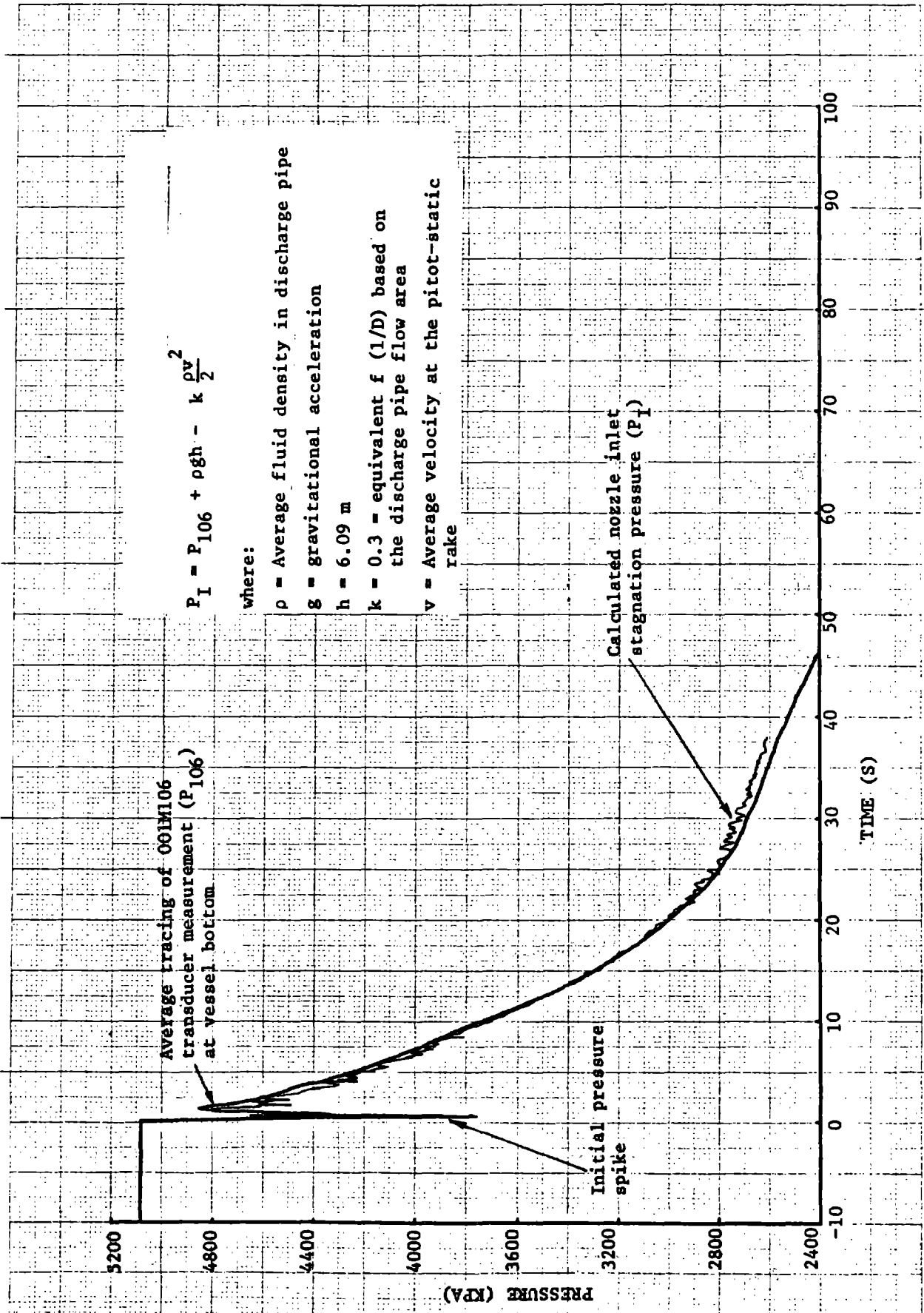


Fig 8:8 Fluid transit time from the instrumentation station to the nozzle inlet plane: Test 14

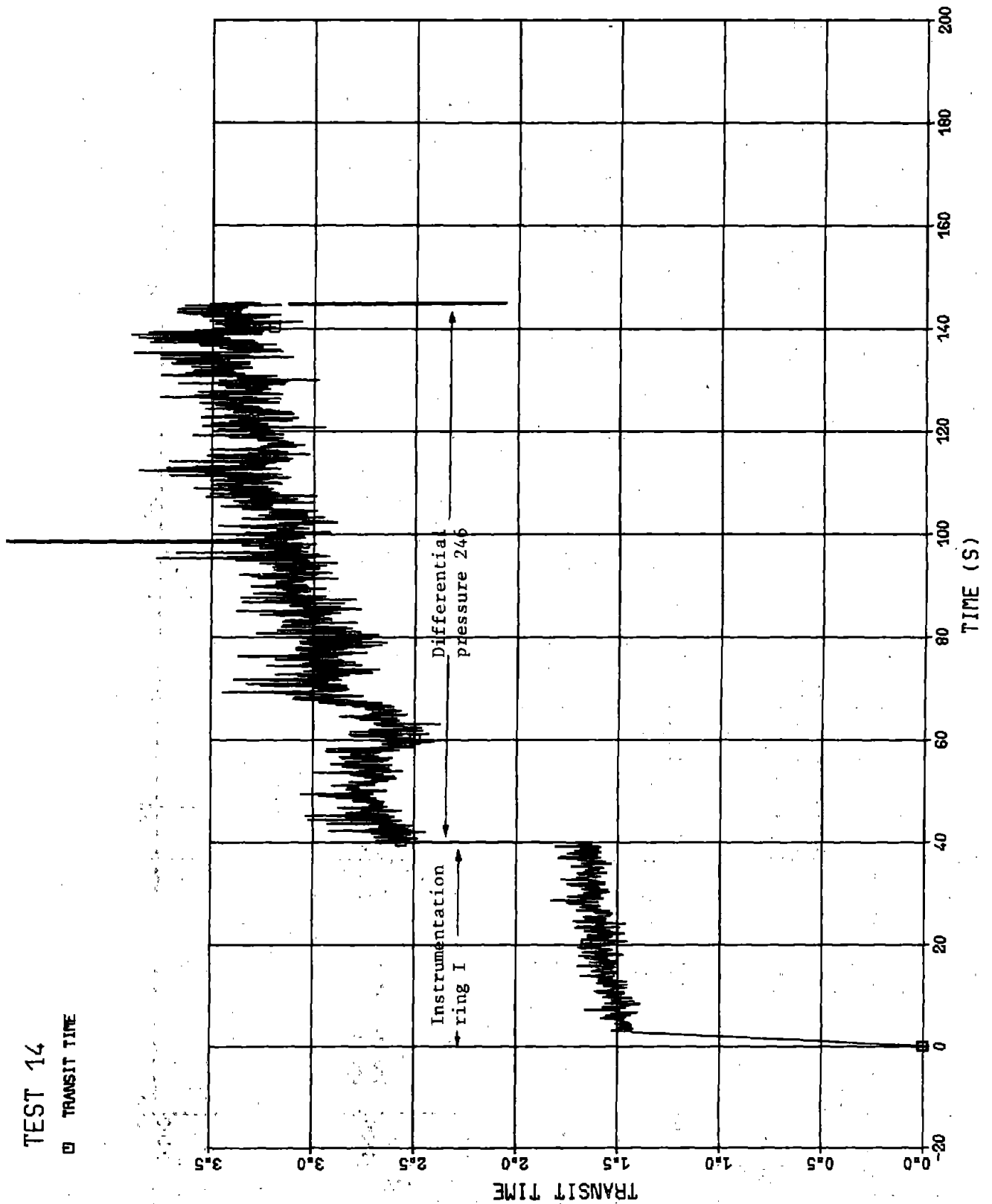


Fig 8:9 Fluid transit time from the instrumentation station to the nozzle inlet plane: Test 11

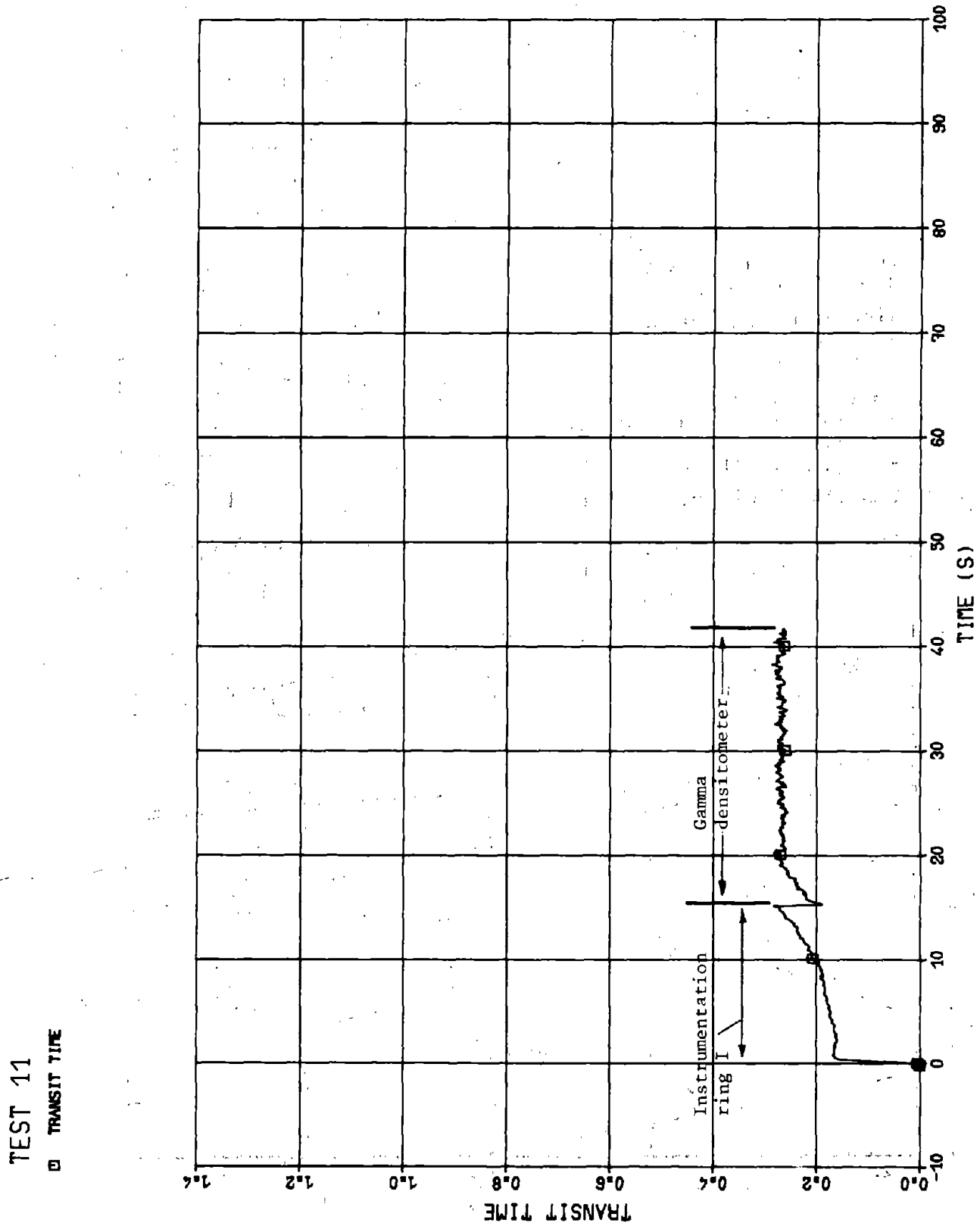


Fig 8:10 The test envelope for the nozzle with diameter 200 mm and L/D = 3.0: stagnation state history at the discharge pipe inlet

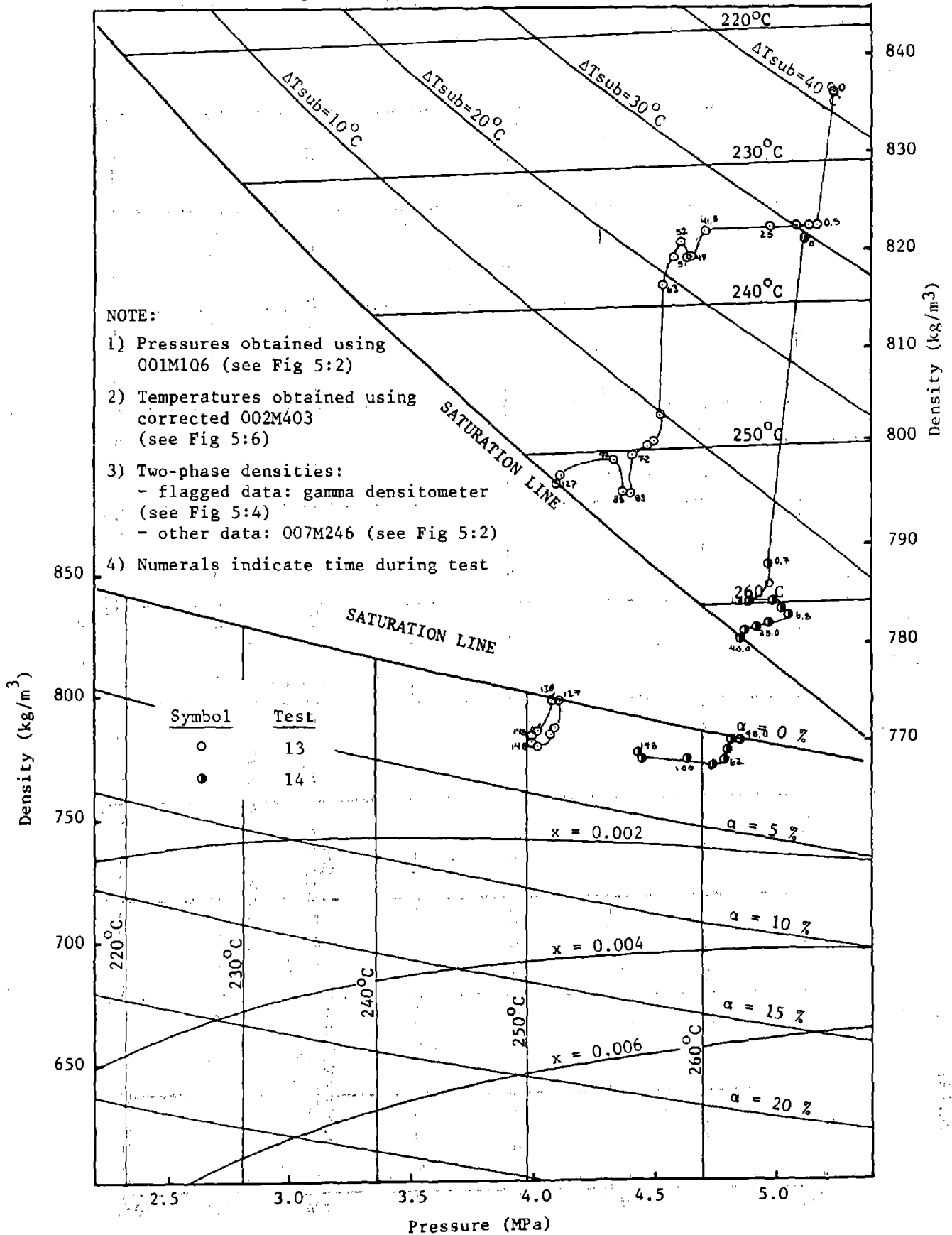


Fig 8:11 The test envelope for the nozzle with diameter 300 mm and L/D = 1.0: stagnation state history at the discharge pipe inlet

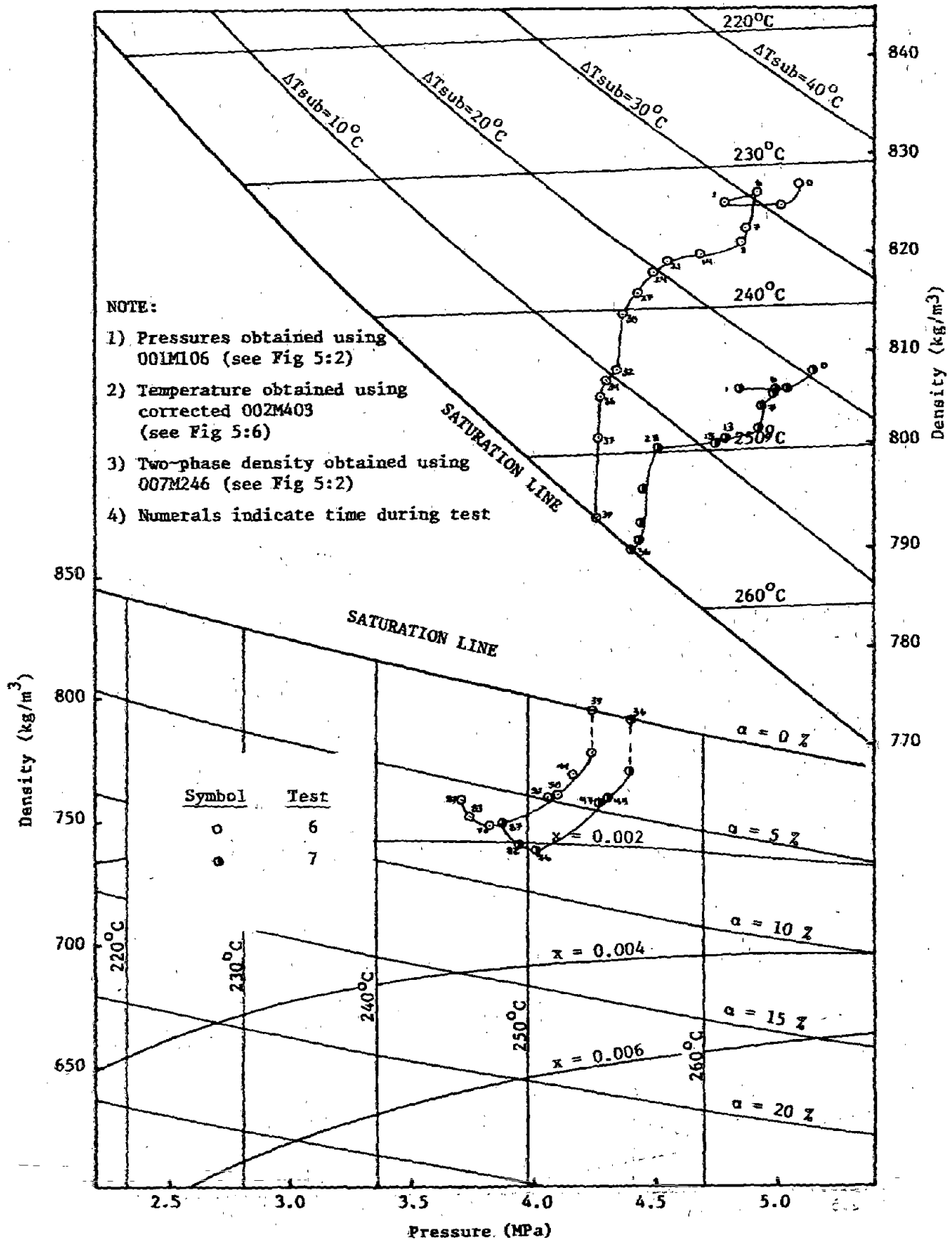


Fig 8:12 The test envelope for the nozzle with diameter 300 mm and L/D = 1.7: stagnation state history at the discharge pipe inlet

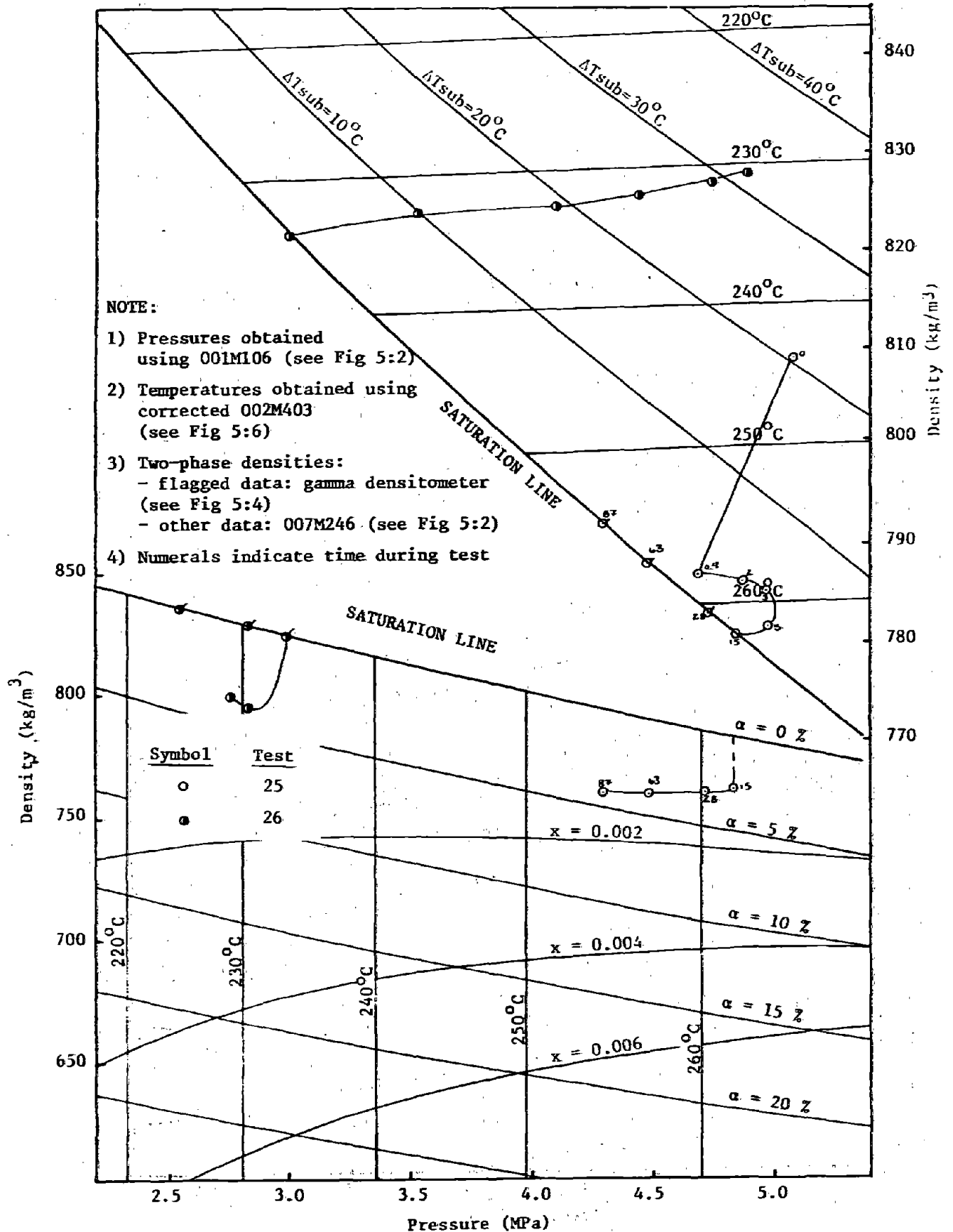


Fig 8:13 The test envelope for the nozzle with diameter 300 mm and L/D = 3.0: stagnation state history at the discharge pipe inlet

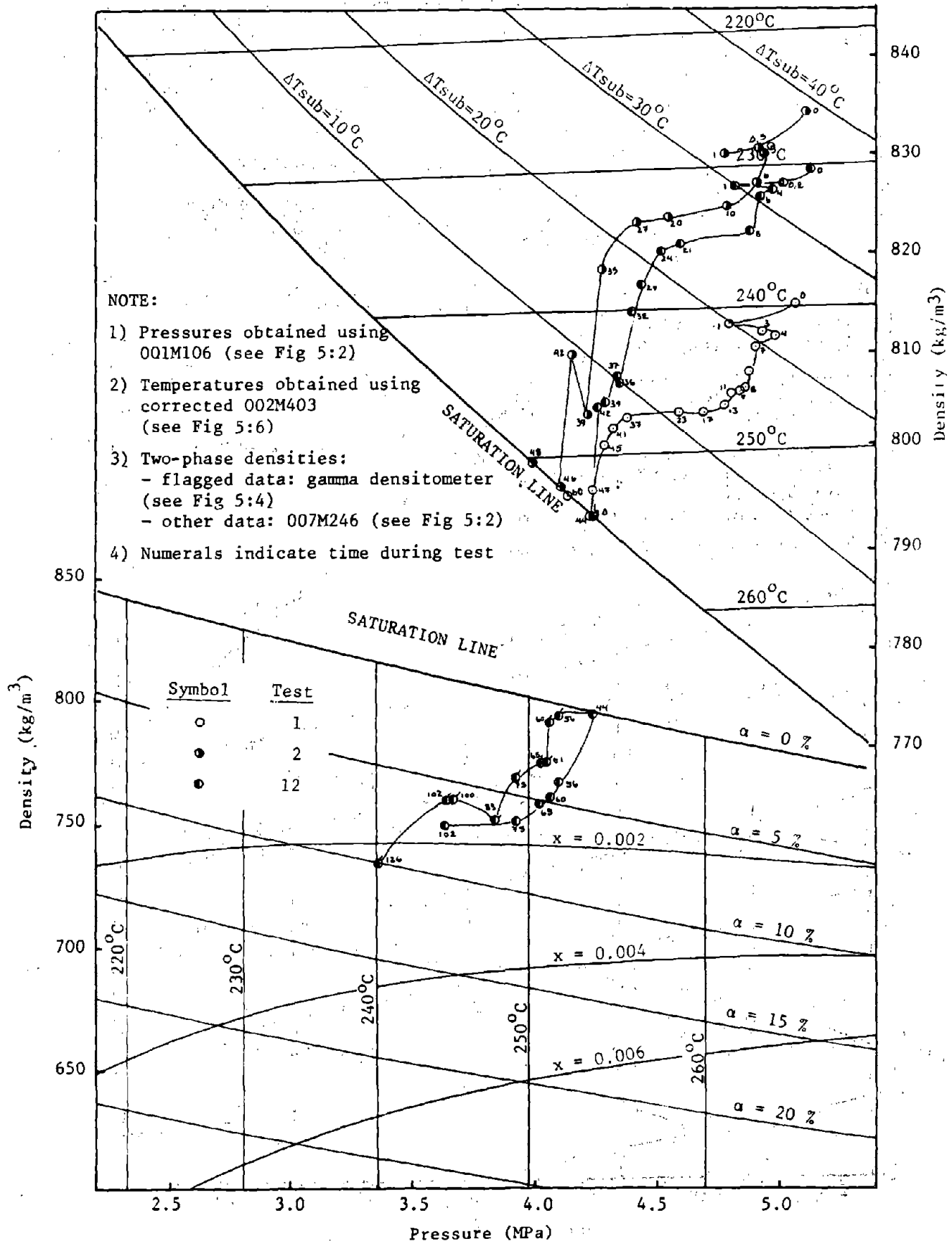


Fig 8:14 The test envelope for the nozzle with diameter 300 mm and L/D = 3.7: stagnation state history at the discharge pipe inlet

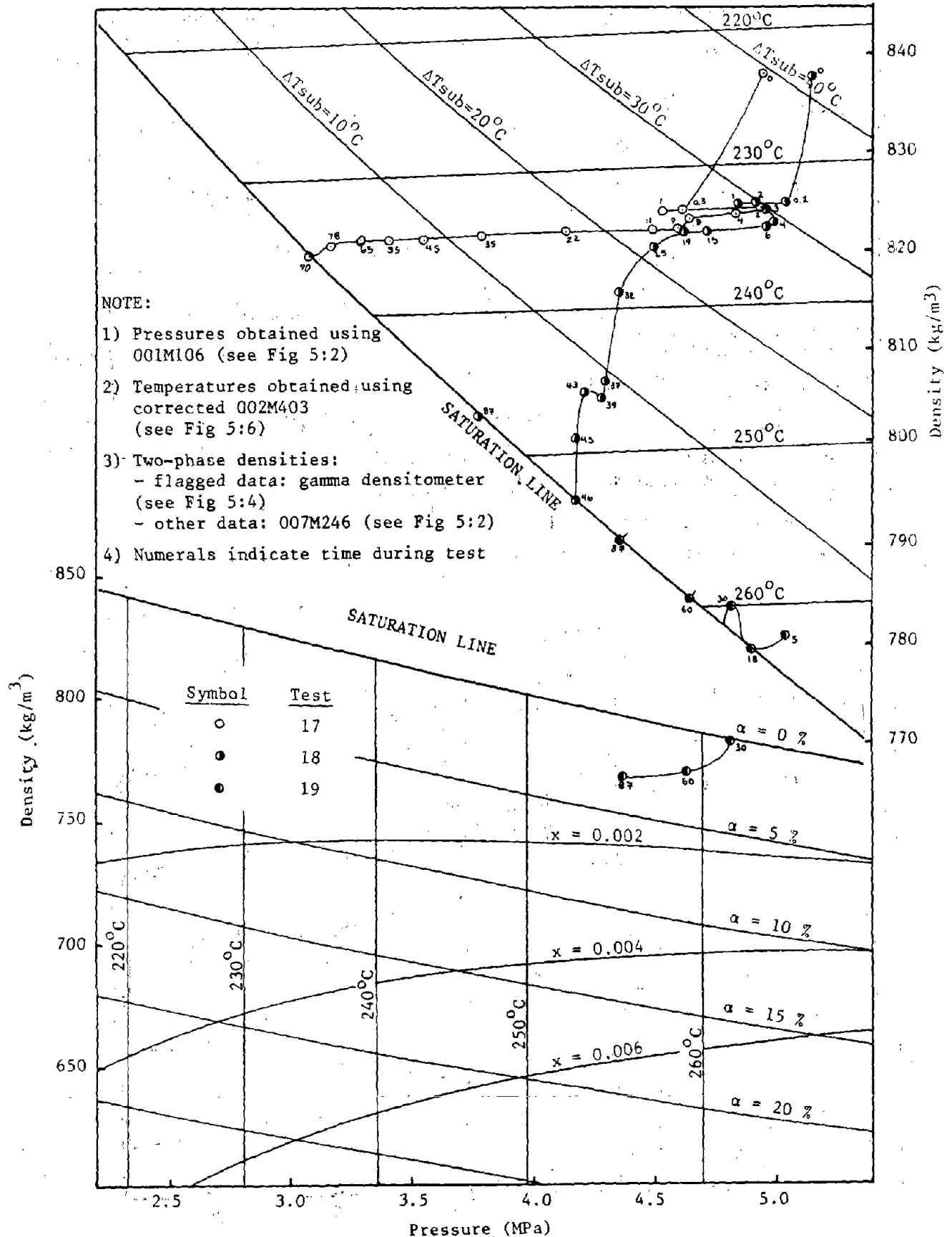


Fig 8:15 The test envelope for the nozzle with diameter 500 mm and L/D = 0.3: stagnation state history at the discharge pipe inlet

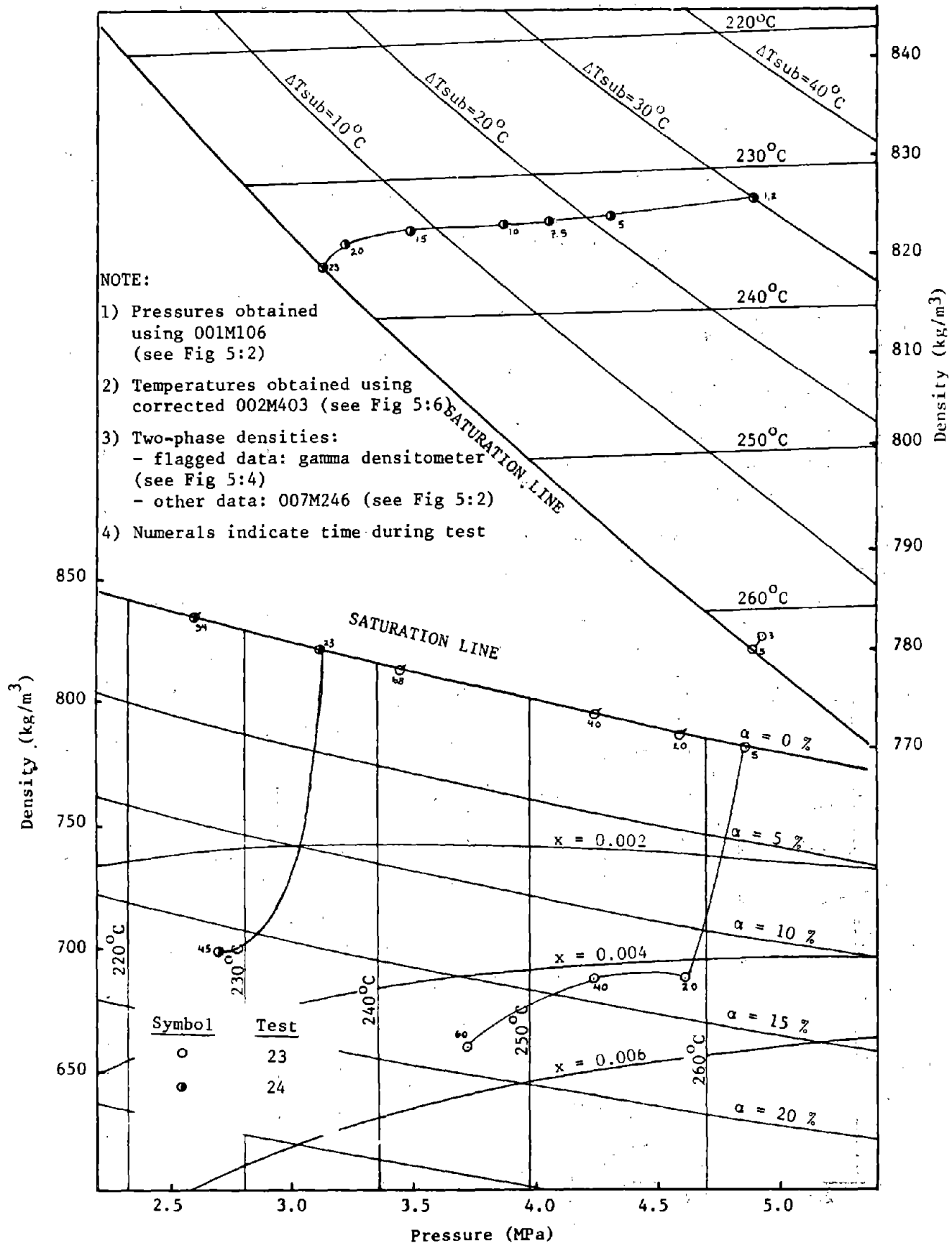


Fig 8:16 The 500 mm diameter L/D = 1.5 nozzle test envelope: stagnation state history at the discharge pipe inlet

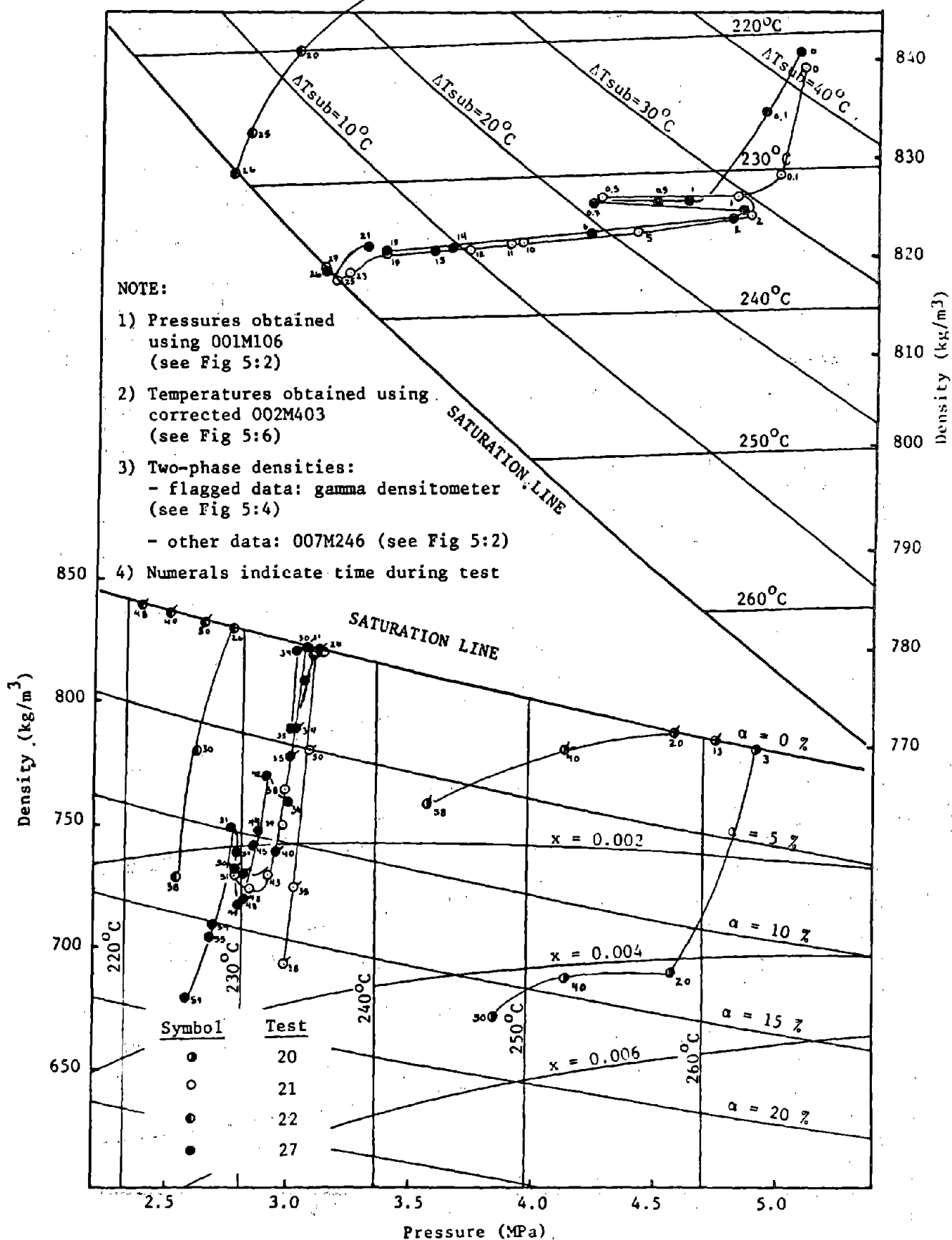


Fig 8:17 The test envelope for the nozzle with diameter 500 mm and L/D = 3.6: stagnation state history at the discharge pipe inlet

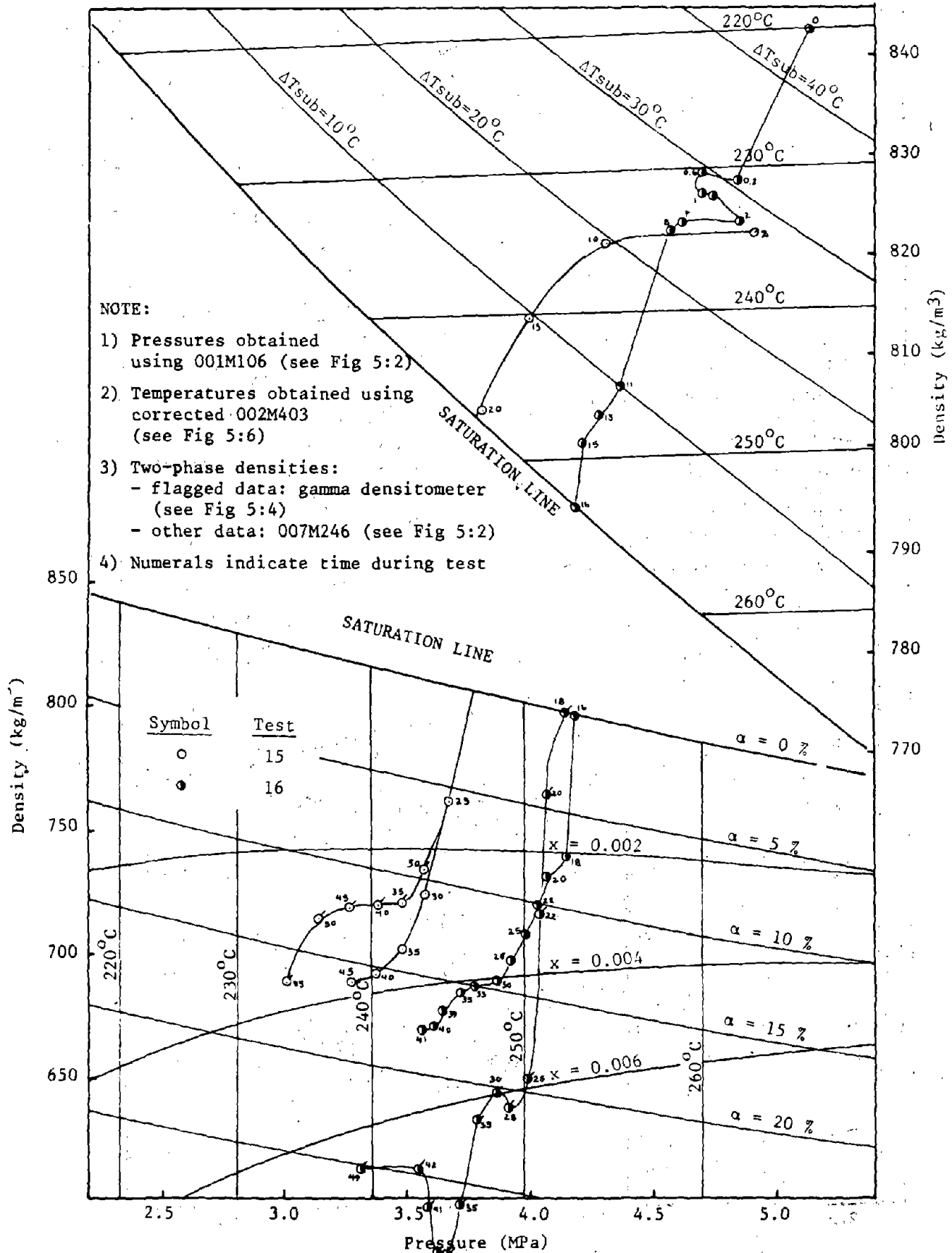


Fig 8:18 The test envelope for the nozzle with diameter 509 mm and L/D = 3.1: stagnation state history at the discharge pipe inlet

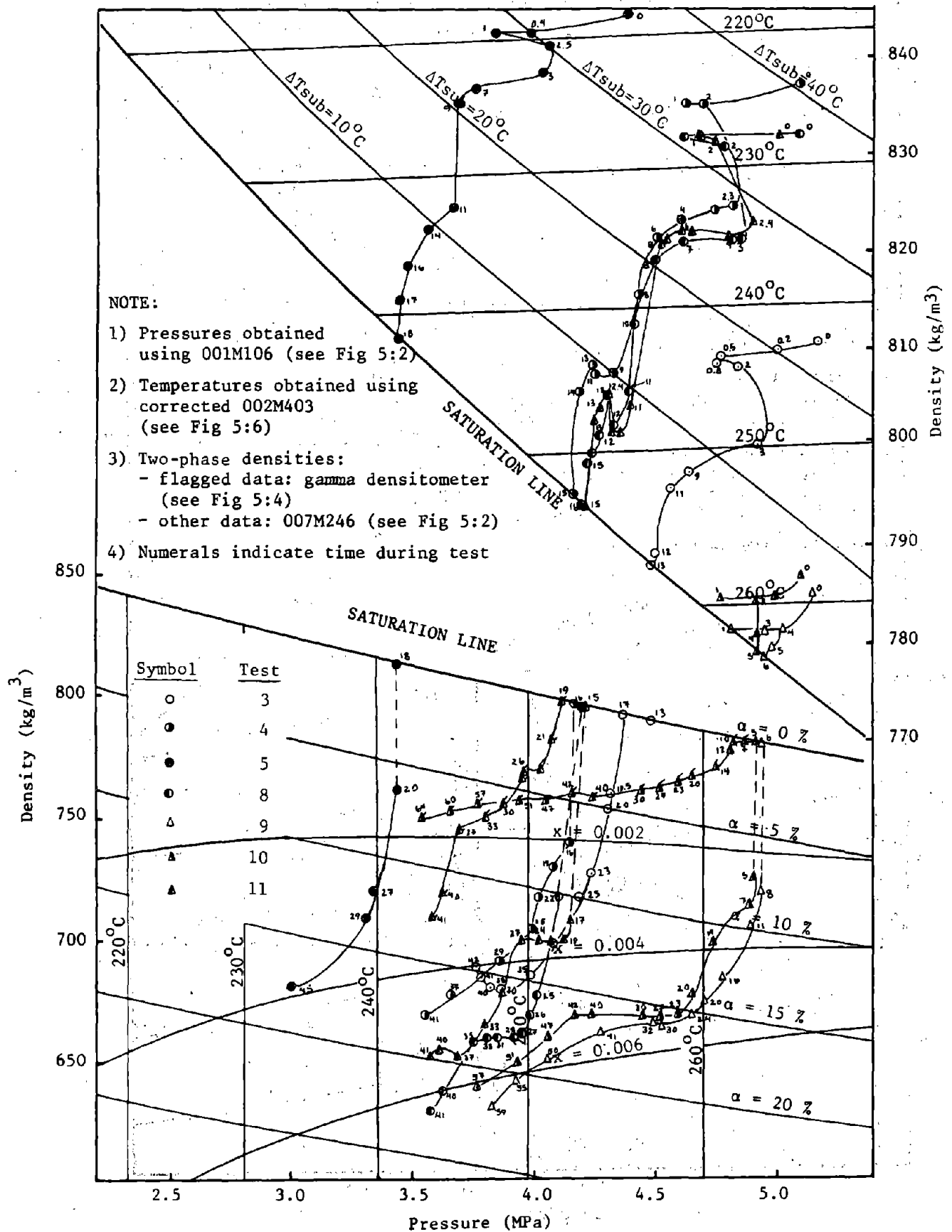


Fig 8:20 The test envelope for the nozzle with diameter 300 mm and $L/D = 1.0$: nozzle mass flux and inlet stagnation pressure

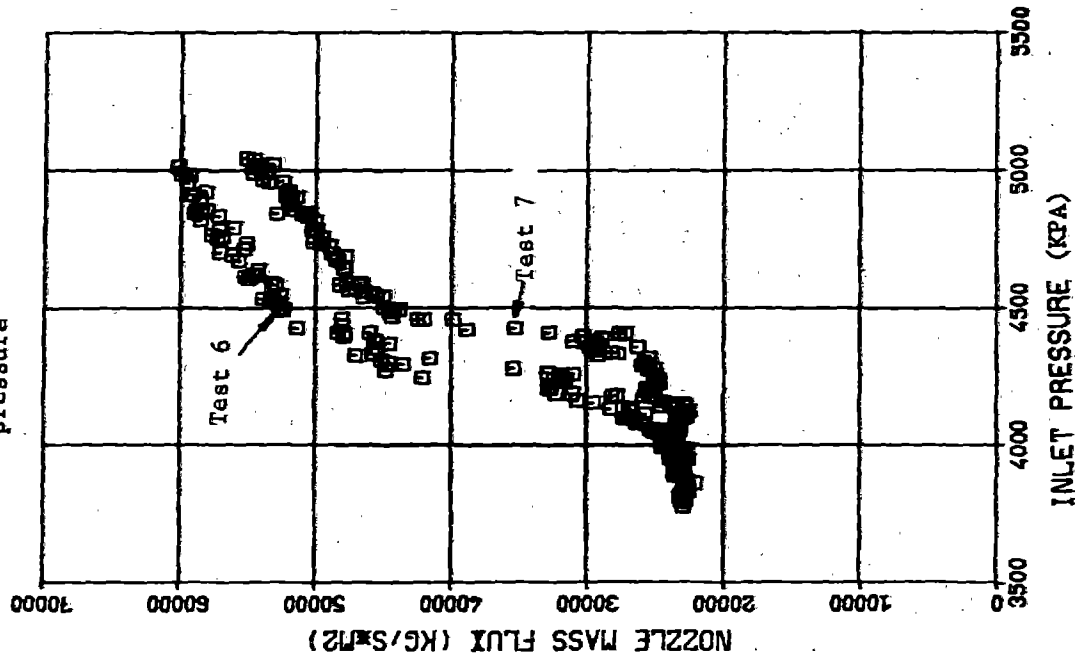


Fig 8:19 The test envelope for the nozzle with diameter 200 mm and $L/D = 3.0$: nozzle mass flux and inlet stagnation pressure

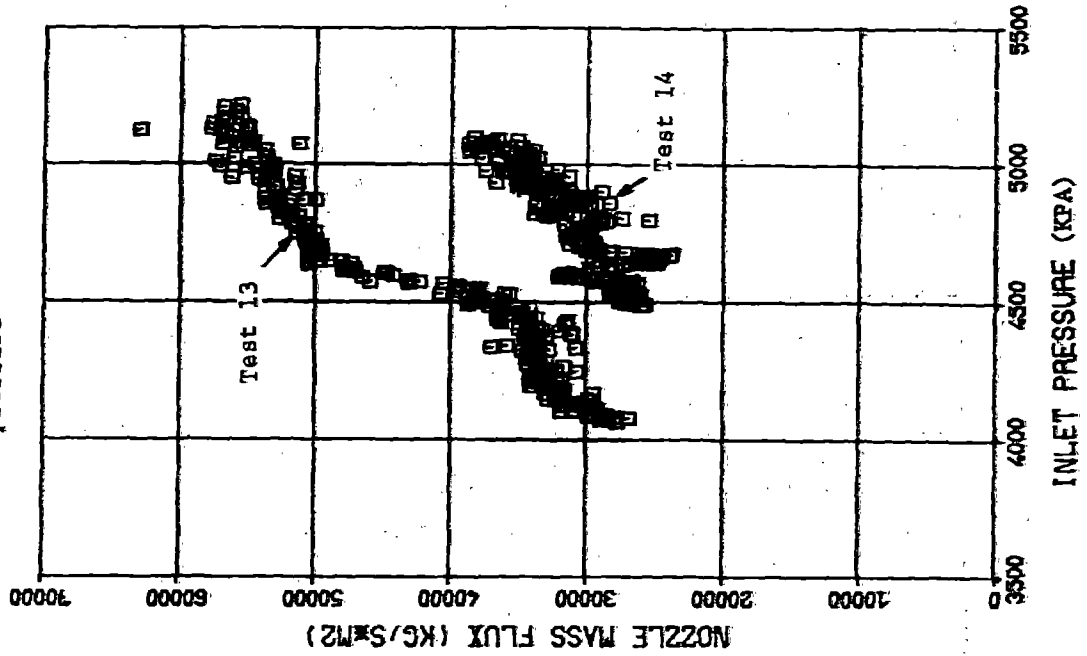


Fig 8:22 The test envelope for the nozzle with diameter 300 mm and $L/D = 3.0$: nozzle mass flux and inlet stagnation pressure

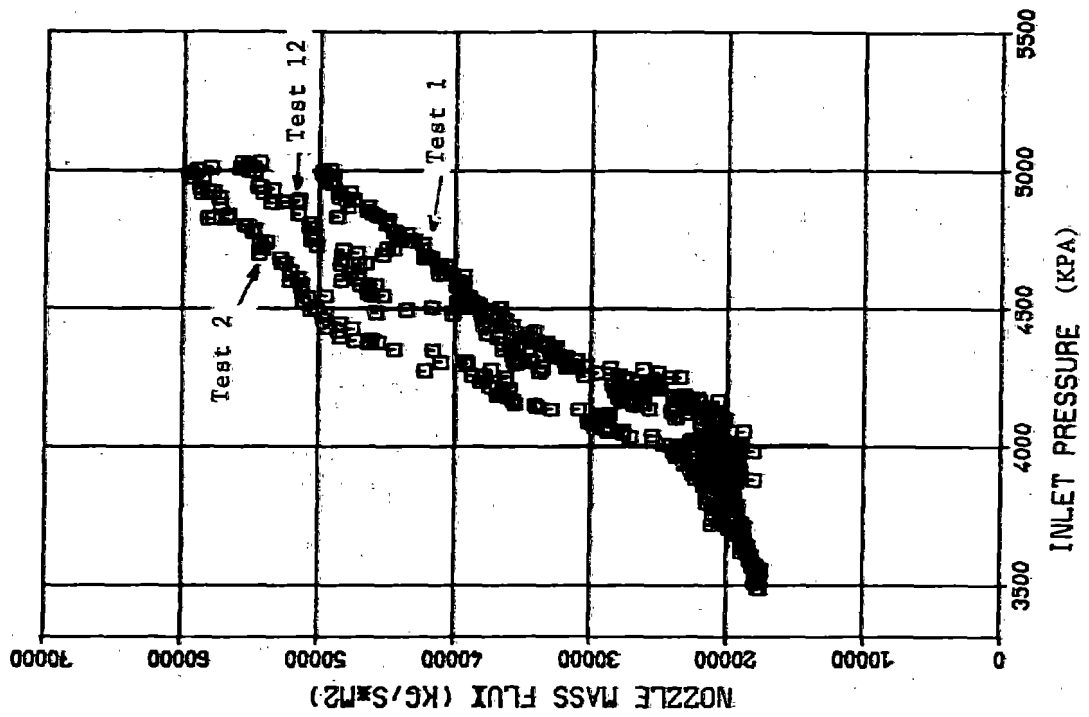


Fig 8:21 The test envelope for the nozzle with diameter 300 mm and $L/D = 1.7$: nozzle mass flux and inlet stagnation pressure

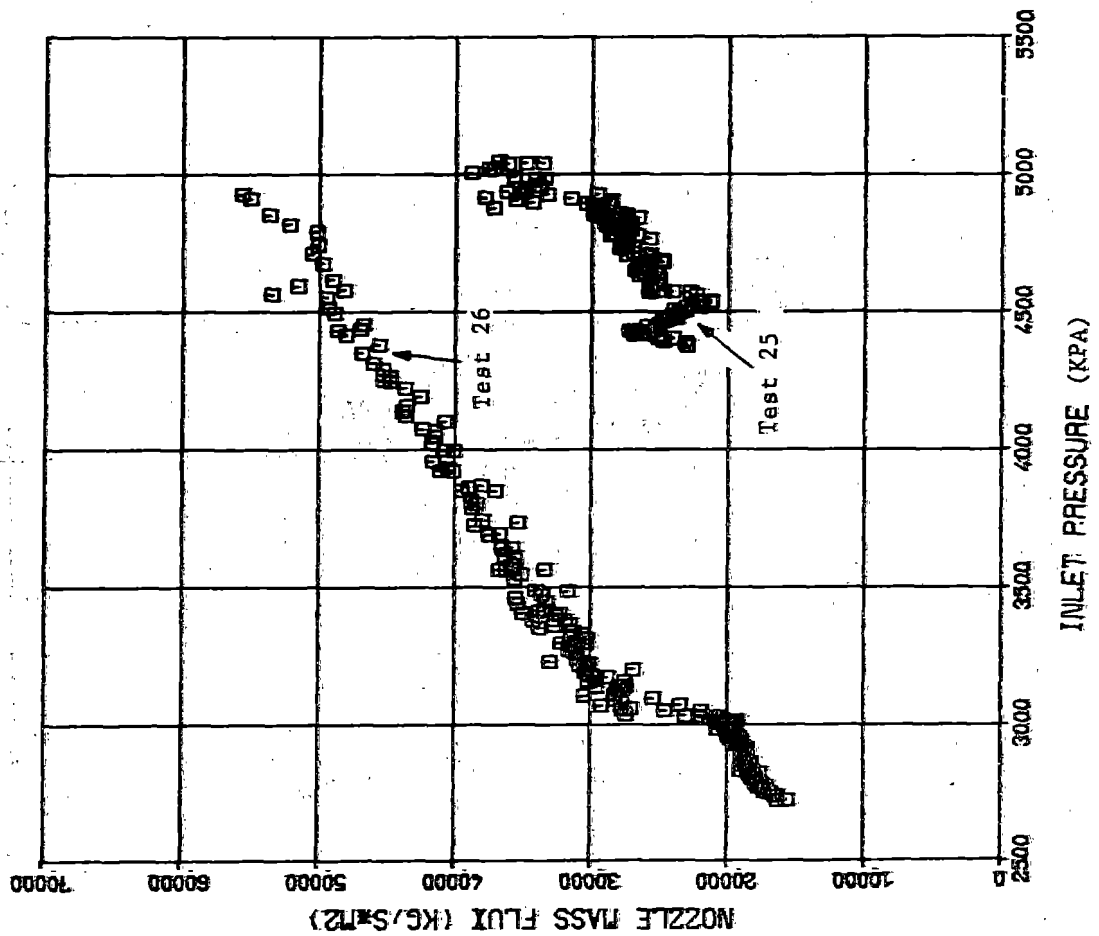


Fig 8:24 The test envelope for the nozzle with diameter 500 mm and $L/D = 0.3$: nozzle mass flux and inlet stagnation pressure

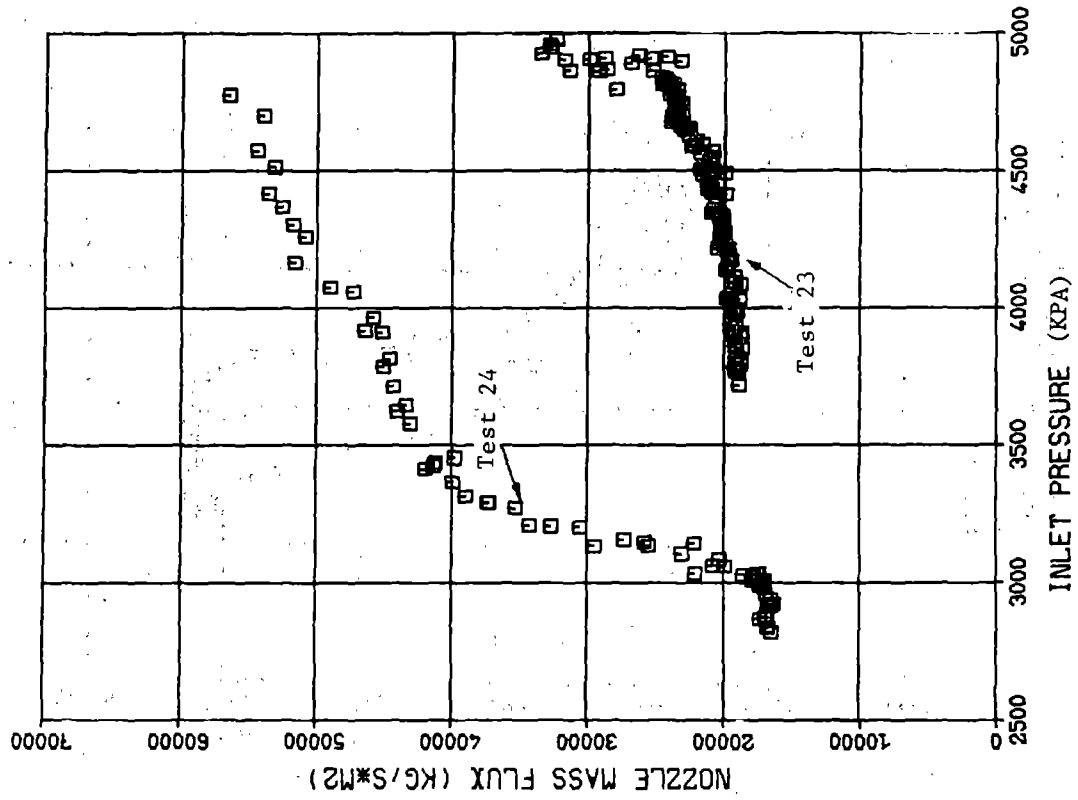


Fig 8:23 The test envelope for the nozzle with diameter 300 mm and $L/D = 3.7$: nozzle mass flux and inlet stagnation pressure

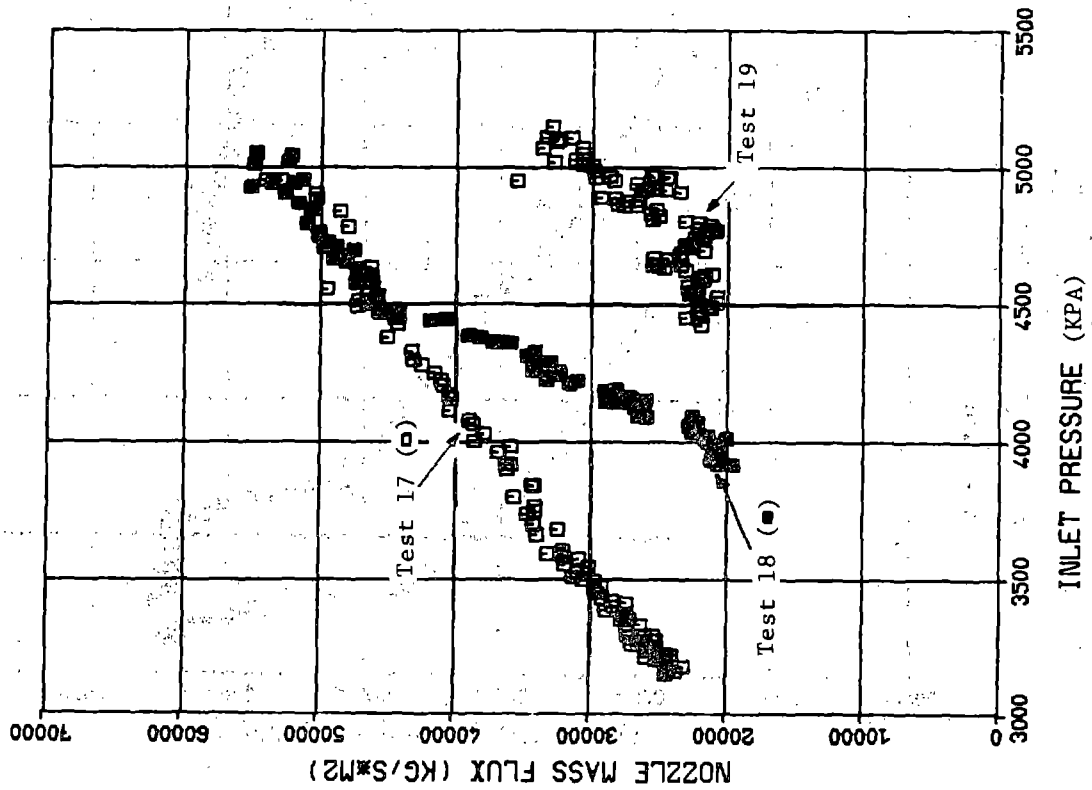


Fig 8:26 The test envelope for the nozzle with diameter 500 mm and $L/D = 3.6$: nozzle mass flux and inlet stagnation pressure

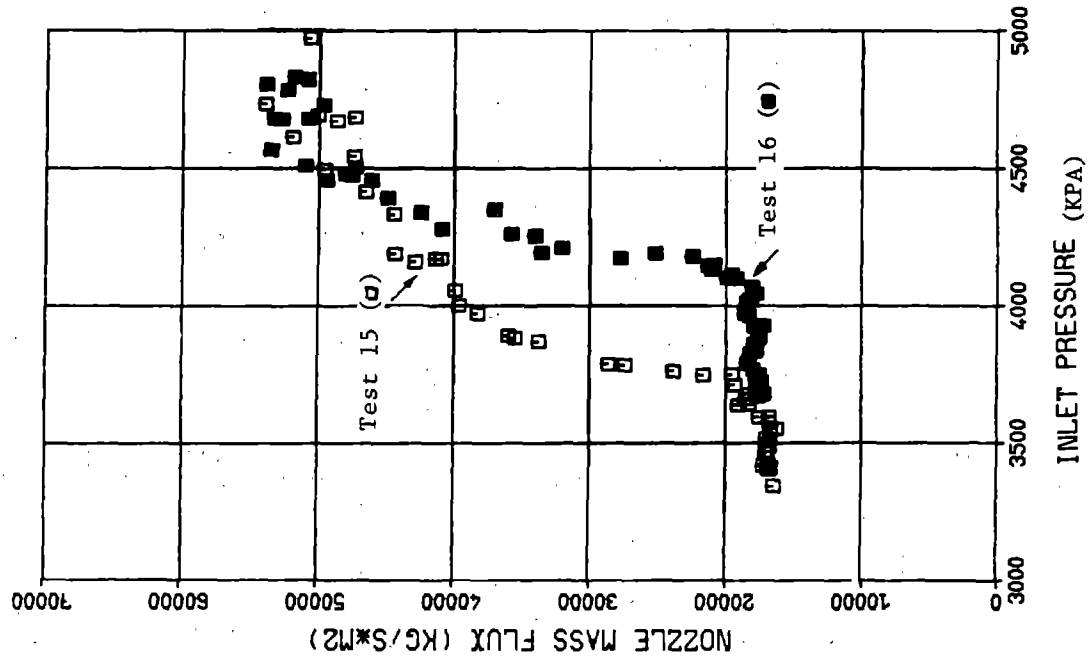


Fig 8:25 The test envelope for the nozzle with diameter 500 mm and $L/D = 1.5$: nozzle mass flux and inlet stagnation pressure

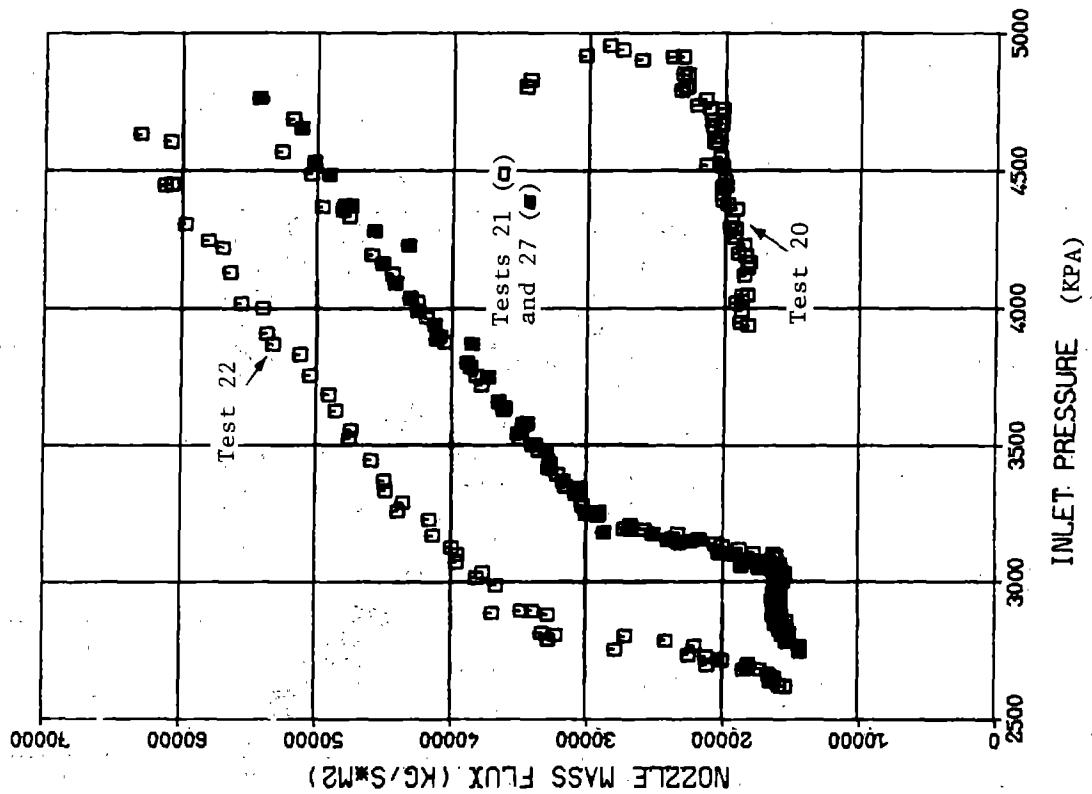


Fig 8:27 The test envelope for the nozzle with diameter 509 mm and $L/D = 3.1$: nozzle mass flux and inlet stagnation pressure

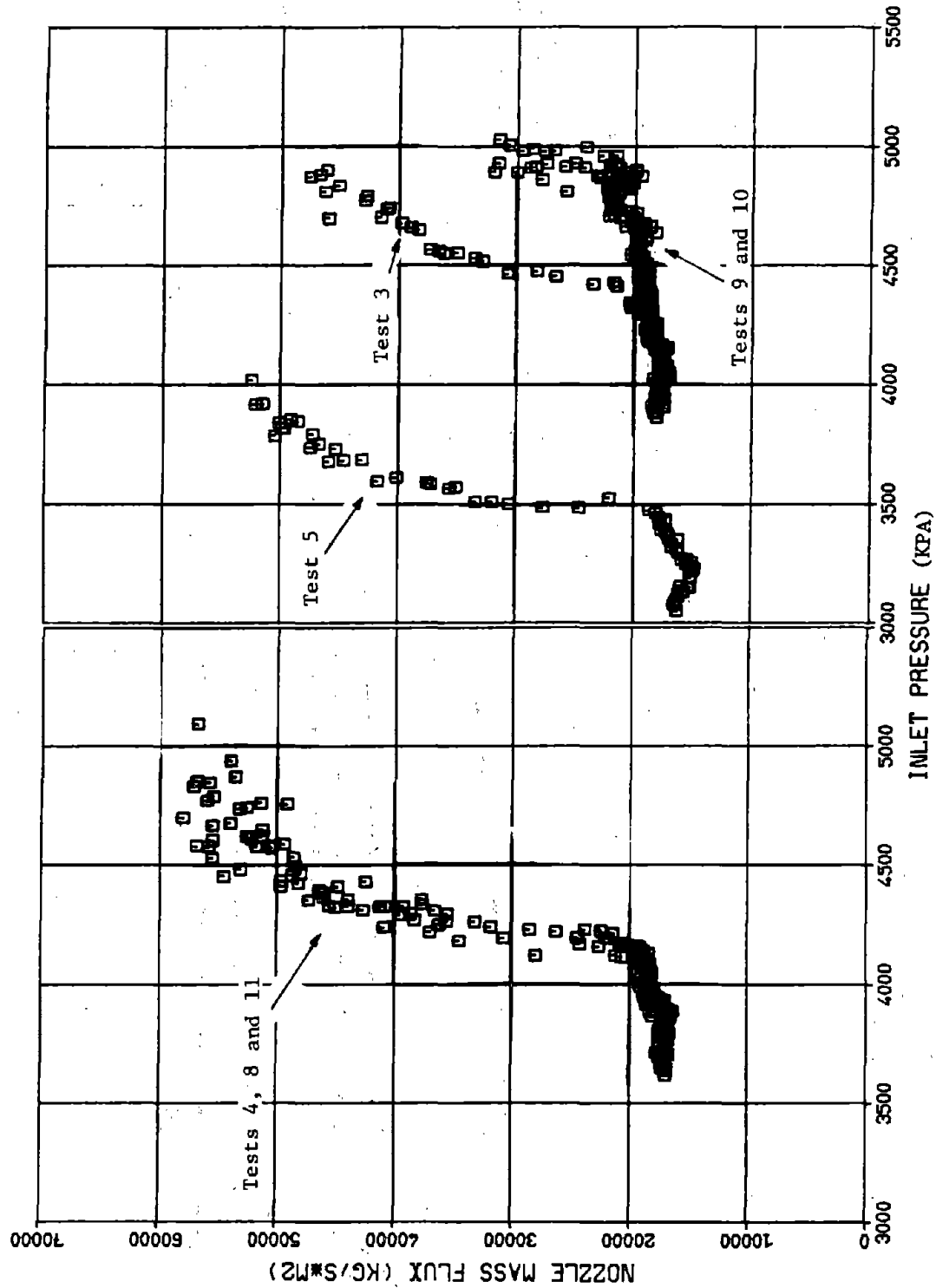


Fig 8:28 Ratio between nozzle wall static pressure and vessel bottom pressure along the nozzle length: Test 11

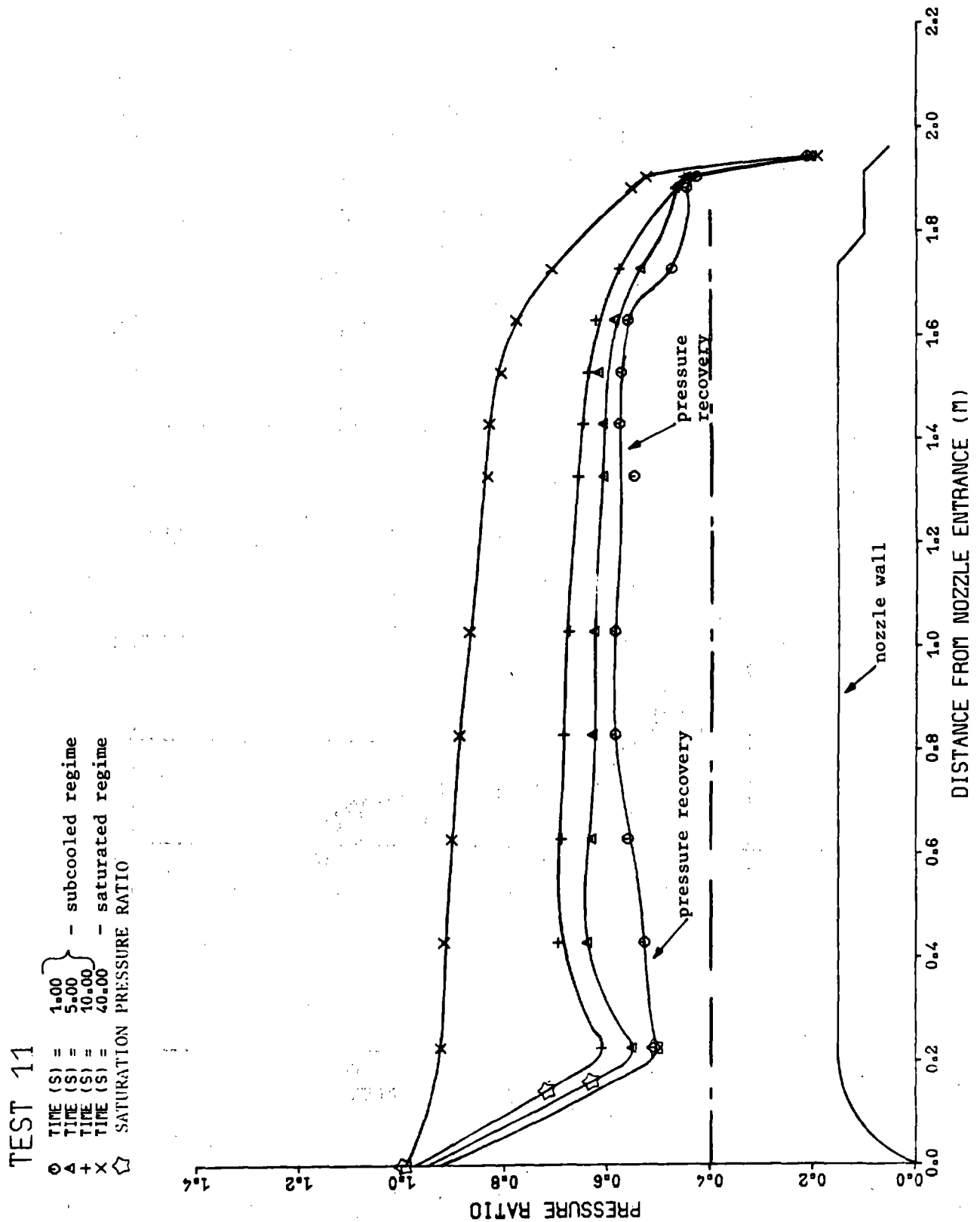


Fig 8:29 Ratio between nozzle wall pressure and discharge pipe saturation pressure along the nozzle length: Test 8

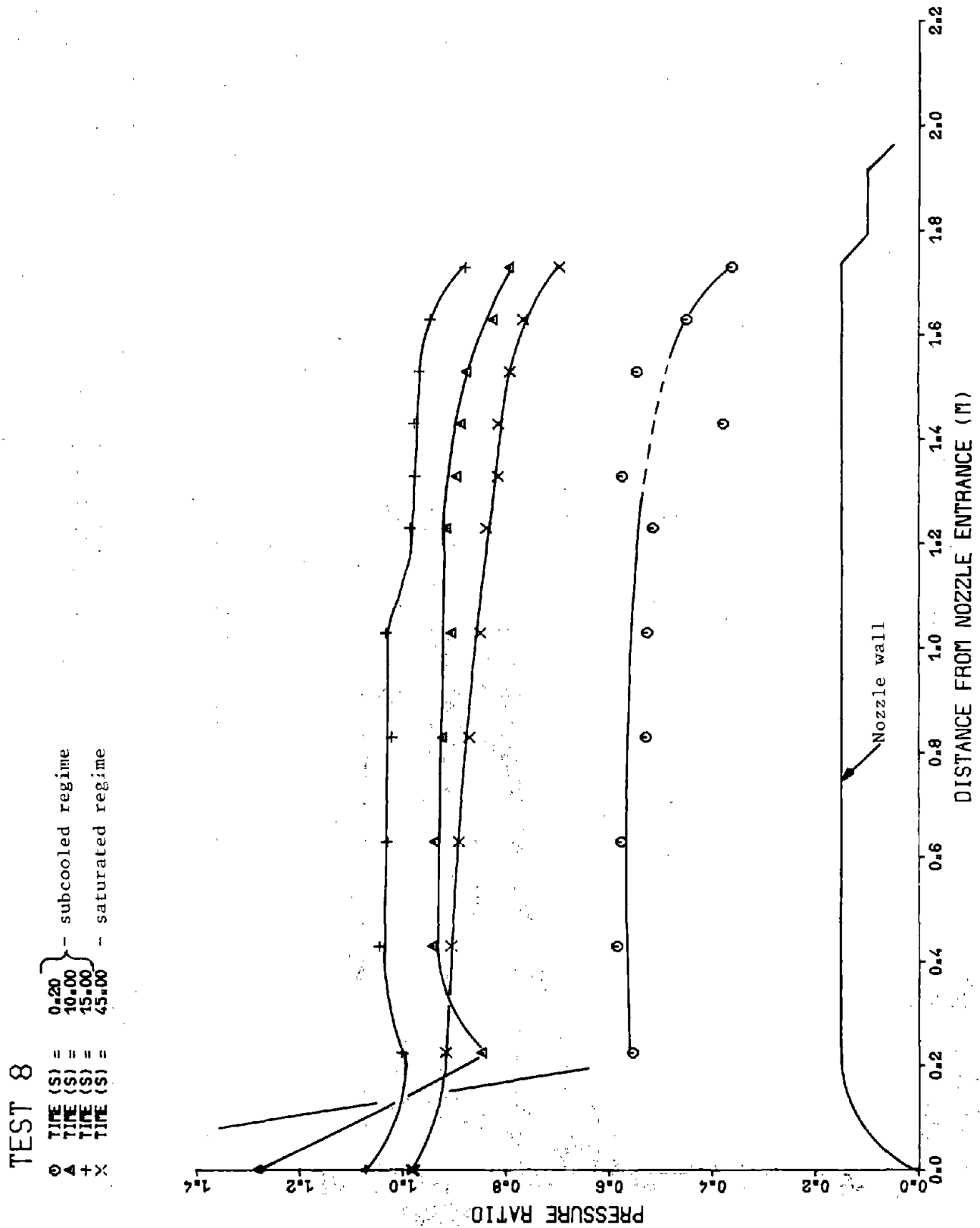


Fig 8:30 Ratio between nozzle wall static pressure and vessel bottom pressure along the nozzle length: Test 18

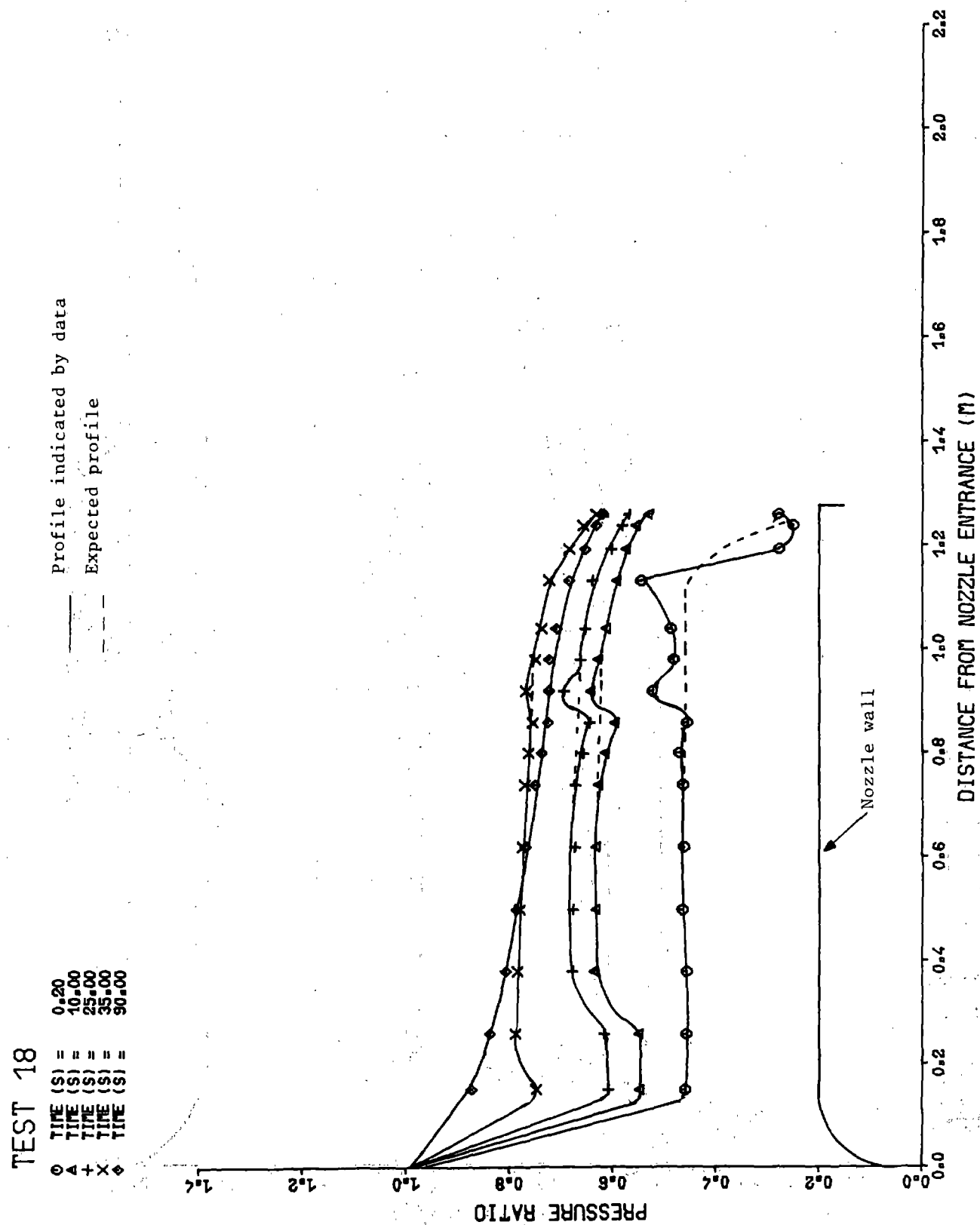


Fig 8:31 Steam dome pressure: Test 21

TEST 21

001M01 23.13 N ELEVATION (AVERAGED 20:1)
 001M03 23.13 N ELEVATION (AVERAGED 20:1) - Post-test value low by 20 kPa.

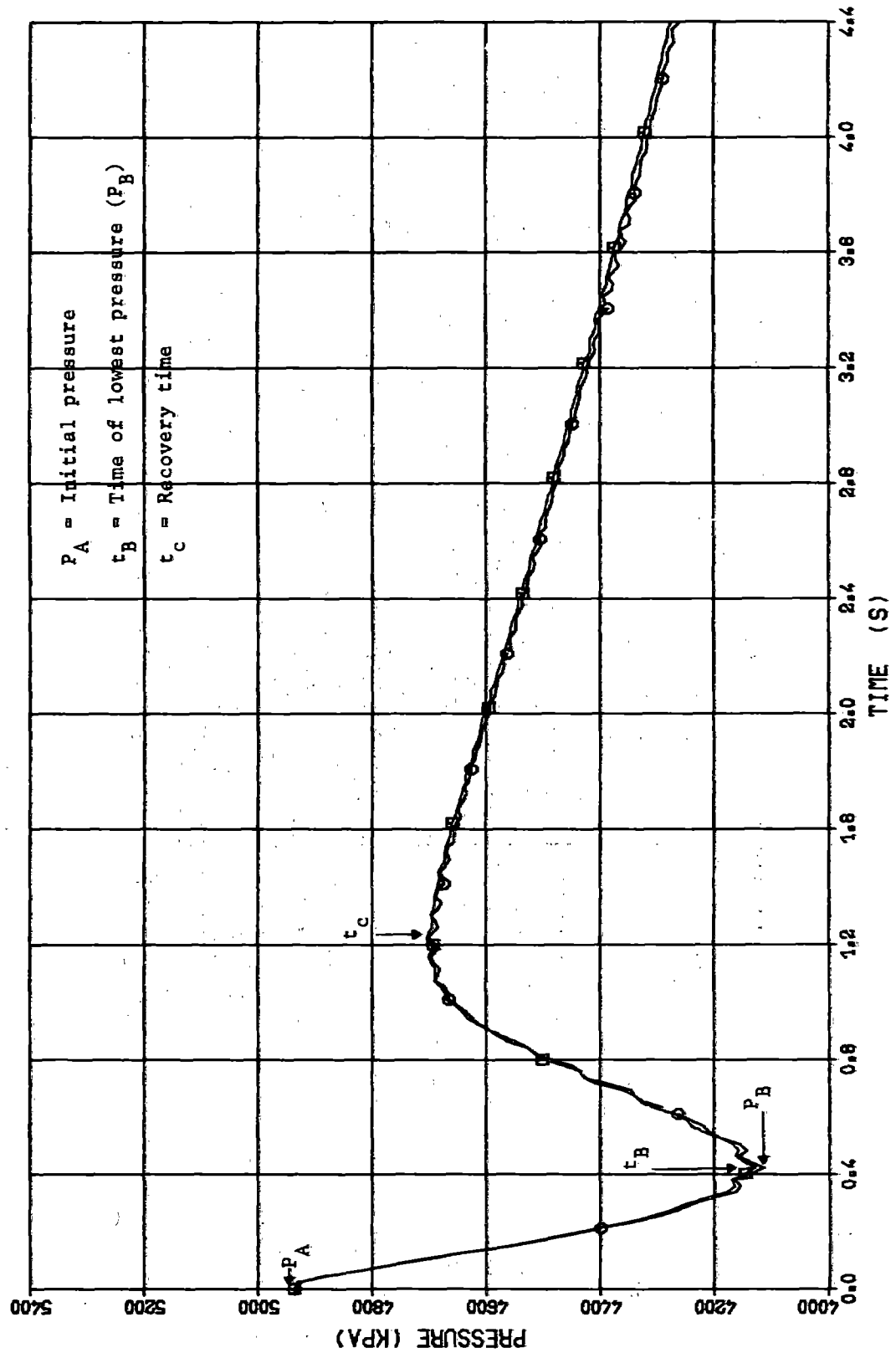


Fig 8:32 Steam dome pressure undershoot and recovery times

NOTE: Numerals indicate test numbers

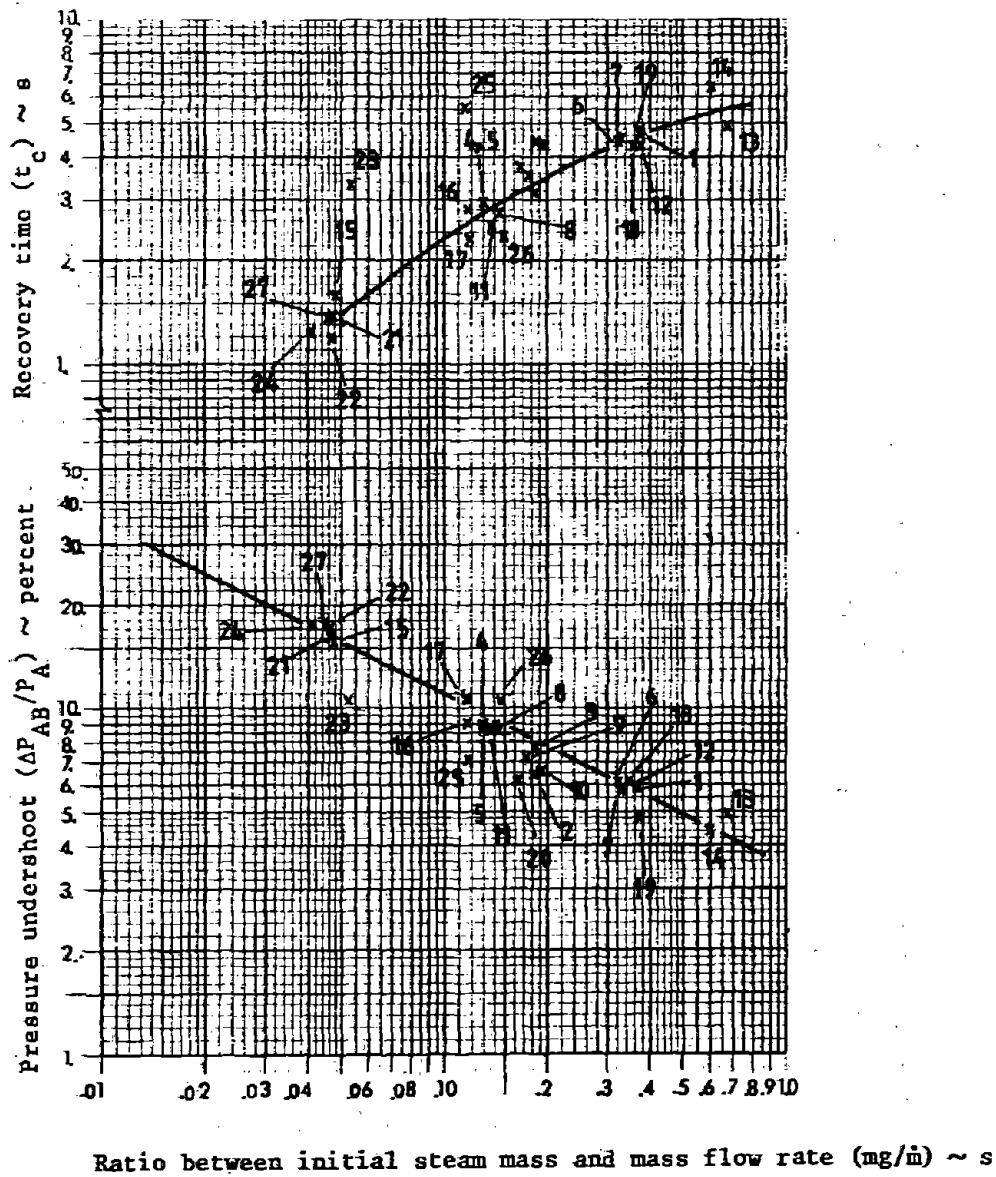
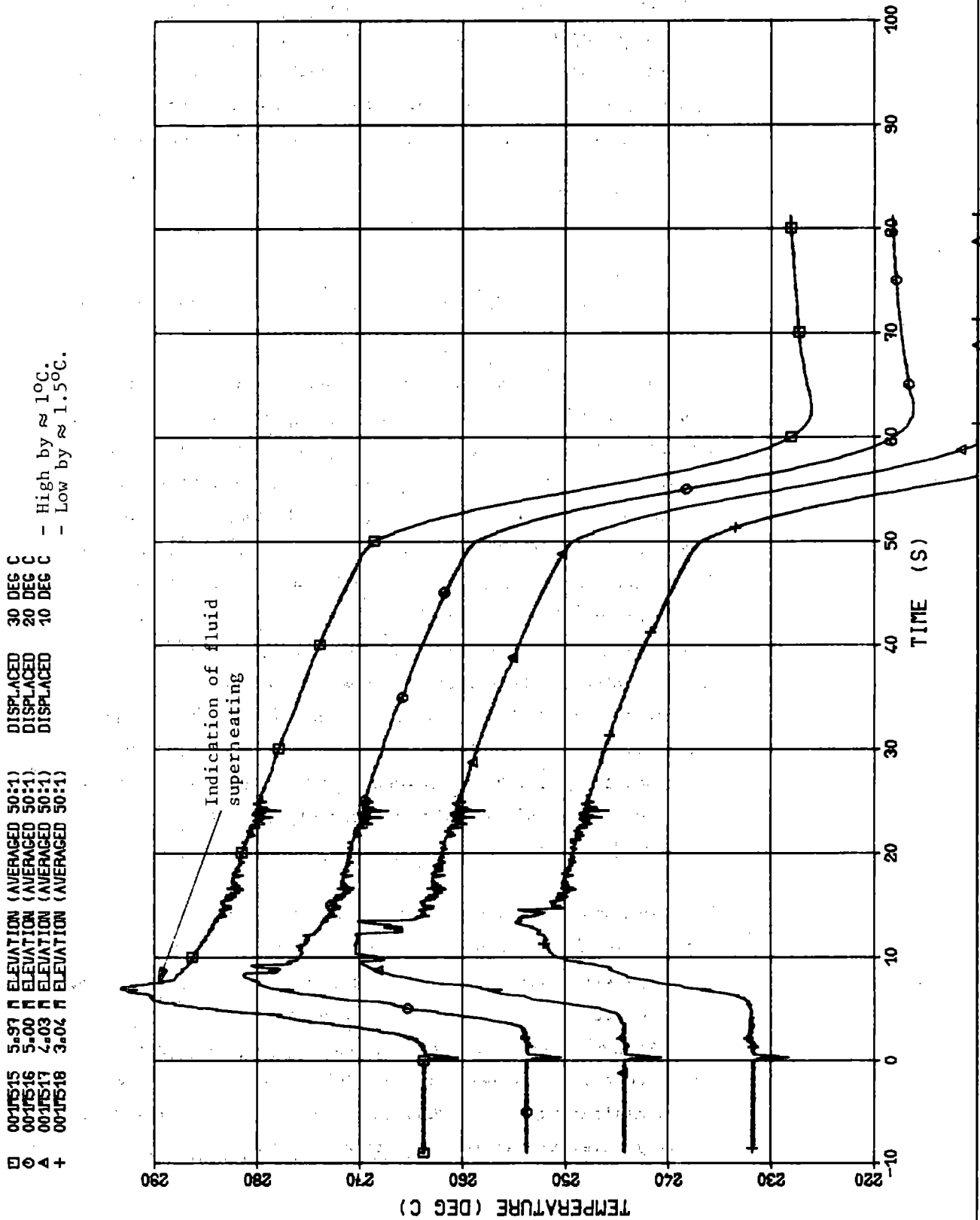


Fig 8:33 Fluid superheating at the vessel thermocouples

TEST 16



9 EFFECT OF PARAMETER VARIATIONS

The effect of nozzle inlet conditions on the nozzle mass fluxes with subcooled and saturated fluid is discussed. The differences between the mass fluxes calculated for each nozzle geometry at equal inlet conditions attributed to nozzle diameter, length and L/D (length to diameter) ratio are summarized. Finally, the effect of the air content in the vessel water on the nozzle mass fluxes is presented.

9.1 Effect of nozzle inlet conditions on nozzle mass flux

The data obtained during each test were divided into three periods (see Table 8:3). The data are presented and discussed for each of these periods separately because either the inlet stagnation conditions or the nozzle mass flux was evaluated differently in each period, depending on the efficacy of the test instrumentation (see Sections 8.1 and 8.2). Consequently, the error limits are different during each period. The subcooled period has the lowest uncertainty limit because the stagnation conditions are defined on the basis of an average temperature obtained at instrumentation ring I or II, using thermocouples with a probable error of less than $\pm 0.5^{\circ}\text{C}$. Further, the mass flux evaluated using the pitot-static method has error limits between ± 3 and $\pm 10\%$, since the fluid density is well known (see Section 7).

The saturated period has error limits that are significantly larger than in the subcooled period. The main contributor to large error limits was the instrumentation for measuring density. During the CFT program, the gamma densitometer was not available until Test 9. Thereafter, the gamma densitometer performance proved to be unpredictable and often misleading. The backup instrumentation for measuring density, i.e. DP 246 (differential pressure 007M246)

provided average fluid density data using a 1.9 m span in the vessel near the discharge pipe inlet. Thus the true density at the discharge pipe inlet was somewhat greater than the average density measured by DP 246. The densities determined using DP 246 and the gamma densitometer often disagreed substantially. However, the mass fluxes presented for the saturated period were determined using the mass inventory method and they have a probable error limit of $\pm 6\%$ (see Section 7).

The transition period contains the data with the largest uncertainty limit. First, the fluid conditions often change quite rapidly as the flow changes from a fully subcooled state to a fully saturated condition. Thus, the mass inventory method cannot be used to evaluate the mass flux history. In addition, the density must be evaluated using either the gamma densitometer or DP 246 since the presence of possible regions of two-phase flow remove the option of evaluating the density on the basis of the average temperature.

Figure 9:1 shows the variation in universal stagnation quality with the nozzle mass flux for Test 27. Plots of this type (these variables plus the inlet pressure are shown for all tests in Appendix E) have been obtained by selecting points at random from the nozzle inlet stagnation data with a specified time interval (see Appendices B, C and D). Consequently, the oscillations observed in the inlet stagnation conditions and mass flow histories are represented by the point scatter in Fig 9:1 (and Plots E:1 through E:27). Perhaps a more accurate method of presenting the data would have been to use a data splining technique (see Section 6). However, the splining technique used to evaluate the steam dome depressurization data in the CFT program was not well suited to the quality/mass flux evaluation. No other technique was currently available.

The divisions between the subcooled, transition and saturated periods are shown in Fig 9:1 (and on the plots in Appendix E). The data are discussed with respect to each of these time periods in the following paragraphs:

Subcooled period

Test 27 was conducted with a subcooled period extending from x 's (qualities) of -0.0017 to -0.0001 . Thereafter the transition period began (see Fig 9:1). The subcooled x /mass flux curve is clearly defined with the exception of a low point at $x = -0.001$.

Of special interest is the change in the slope between nozzle mass flux and quality which occurs at $x = 0.0001$. The slope change is also apparent for other nozzle geometries, cf Plots E:17 and E:21.

Because the change of slope in the curve of mass flux versus x occurs near the time at which the transition period begins in Fig 9:1, it is tempting to assume that the change in calculational procedure is the primary cause. However, closer examination of Plots E:17 and E:21 show that during these tests slope changes occur at qualities over -0.0004 , well before the transition period begins. In fact, Test 17 never reached saturation. Thus it is believed that the slope change is real.

The change in calculational technique which occurs in going from the subcooled to the transition period does have an effect on the relationship between mass flux and x , as might be expected.

One of the more evident examples of the effect of a different calculational method is seen in Plot E:3, where a step change in density (see Plot C:3) at 12.6 s has caused a shift in the plot of mass flux versus x . Thus some of the figures shown in Appendix E have a slope change which is caused partly by the calculational technique and partly by the actual relationship between mass flux and quality.

The fact that the slope change occurs at $x = 0.0004$ for Test 17 (see Plot E:17) when testing a 300 mm diameter nozzle, and at $x = 0.0001$ for a 500 mm diameter nozzle in Test 21, suggests that the slope change is related to the nozzle diameter. However, data recorded in other tests do not support such an observation.

Figure 9:2 compares the mass flux and x data for Tests 4 and 5. Although the inlet pressure during Test 4 was usually 700 to 800 kPa above the Test 5 pressure at equal qualities, the mass fluxes of the two tests are usually within 1000 to 1500 $\text{kg/m}^2\text{s}$ at the same qualities. Thus the inlet stagnation pressure does not seem to exert a large influence on the relationship between mass flux and quality.

Transition period

The transition period (see Fig 9:1) lies between the subcooled and saturated periods. Because the fluid conditions are partly subcooled and partly saturated (possibly containing steam) in combination with a mass flux that is changing rapidly with time, and possibly also with density, the variation in mass flux with quality is prone to large uncertainties. Thus the nozzle critical flow behaviour in the transition zone has not been studied in detail.

Saturated period

The saturated flow behaviour shows a relatively constant mass flux over the range of quality and pressure existing upstream of the test nozzle in any given test (see Fig 9:1). However, the saturated mass flux decreases with decreasing pressure at equal qualities for the same nozzle geometry (see Fig 9:2).

9.2 Effects of other parameters

The data presented in Appendix E have been examined to determine the effect of nozzle length, diameter and L/D (length to diameter) ratio on the critical mass flux through the nine test nozzles. The data are shown in Figs 9:3 to 9:5 in the form of graphs of the nozzle mass flux as a function of the nozzle length at subcooled x (Figs 9:3 and 9:4) and at $x = 0$ (Fig 9:5). The saturated data at qualities above zero have not been shown since the two-phase density measurement equipment provided inconsistent and often conflicting results. Future studies concerning the saturated critical flow through the nozzles will yield the most consistent results if they consider only one of the two-phase density measurements (it must also be remembered that the DP 246 behaviour changed after Test 14 when the hole in the moderator tank bottom was enlarged).

Figure 9:3 shows that the nozzle mass fluxes for the 200, 300 and 500 mm diameter nozzles lie within calculational error limits for lengths greater than 500 mm. However, the mass fluxes obtained using the 300 and 500 mm diameter nozzles with lengths less than 500 mm increase sharply. The fact that the 300 mm diameter nozzle shows a larger mass flux than the 500 mm diameter nozzle, even though the straight section of the former nozzle is over 120 mm longer than that in the larger diameter nozzle, implies that nozzle diameter is an important variable for short nozzles. However, as the length increases, the effect of diameter seems to be less than the 3 to 10 percent error limit which is applicable to the pitot-static method.

The behaviour shown in Fig 9:3 is also apparent at other subcooled qualities, so that at equal qualities the nozzle with a diameter of 300 mm and $L/D = 1.0$ shows greater mass fluxes than the nozzle with a diameter of 500 mm and $L/D = 0.3$. Thus the nozzle mass fluxes seem to be dependent on the L/D ratio, but this dependence is only noticeable when L/D is less than 1.5.

Fig 9:4 shows how the mass flux depends on length for the 500 mm diameter nozzle at several subcooled stagnation qualities. The graphs of critical mass flux versus length are similar for $x = -0.0014$, -0.0001 and -0.0005 .

The nozzle diameter assumes a new importance at a zero fluid quality. Data (see Table 8:3) obtained at the time when all thermocouples showed saturation in the discharge pipe provide a basis for comparison of the nozzle test geometries. When the mass flux through the nozzle is plotted against nozzle length at inlet pressures between 4.8 and 4.9 MPa (see Fig 9:5), it is found that the relationship is similar to that observed at subcooled qualities for the 500 mm diameter nozzle. Data are not available from Table 8:3 for the 300 mm diameter nozzle over the same pressure range. However, it is clear that the mass flux is dependent on nozzle diameter since the nozzle mass flux increased with decreasing diameter.

Figure 9:6 shows the nozzle mass flux as a function of the quality for Tests 21 and 27. These tests were conducted with the same nozzle and the same inlet conditions except that the measured water O_2 (oxygen) content in Test 27 was a factor of 140 less than in Test 21 (see Table 4:2). The water O_2 content is a direct measure of the water air content since no other source of O_2 was available which could have a significant effect on the O_2 gas measurement. The data plotted in Fig 9:6 show that the decrease in water air content did not result in any detectable difference in the nozzle mass flux under equal inlet stagnation conditions. The degree of liquid superheating in the vessel seems to be related to the content of dissolved air. The indicated superheatings were in general larger in Test 27 than in Test 21.

Fig 9:1 Nozzle mass flux and the universal stagnation quality:
Test 27

NOZZLE MASS FLUX VS. UNIVERSAL QUALITY

□ TEST 27

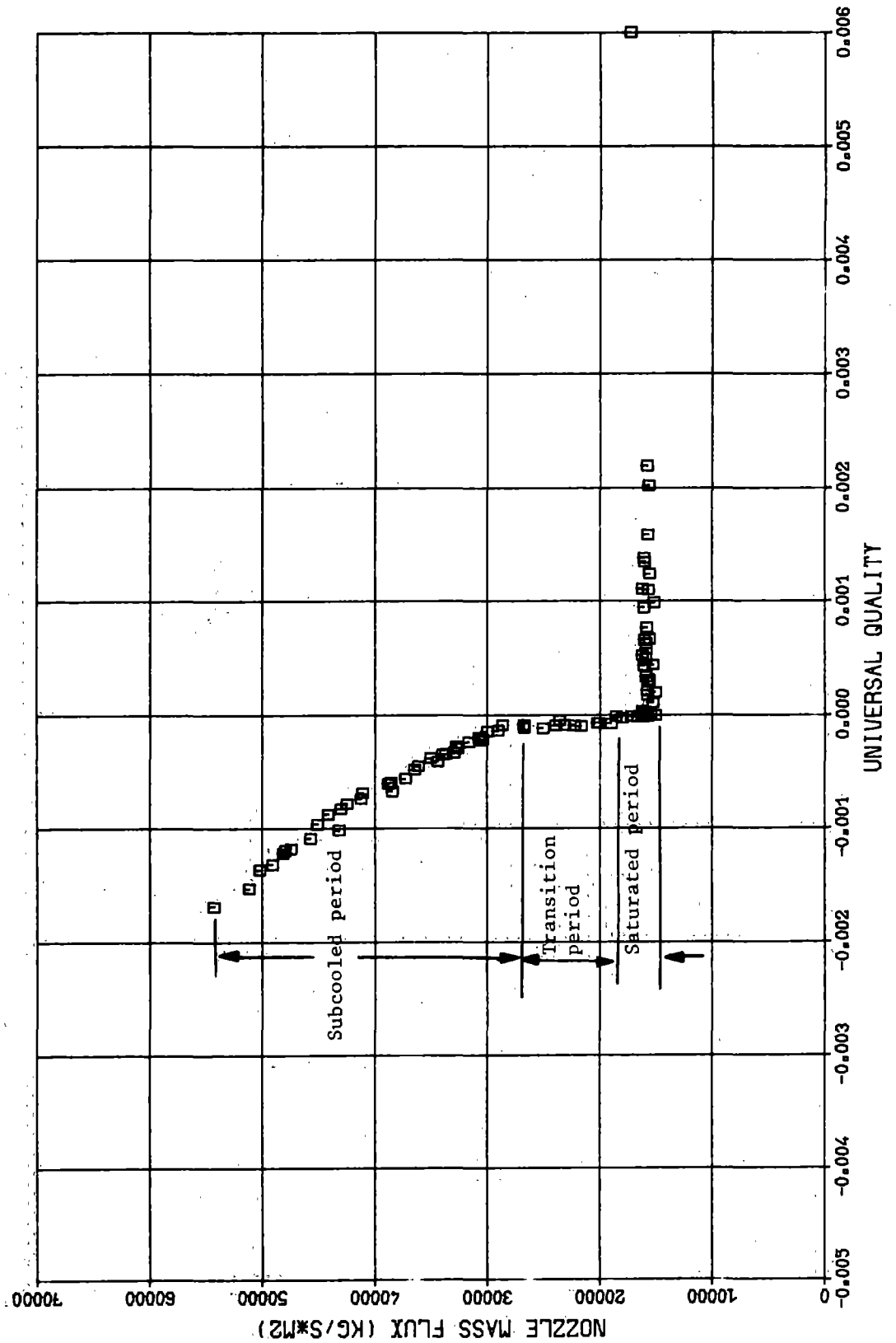


Fig 9:2 Comparison between the nozzle mass flux and the universal stagnation quality for Tests 4 and 5

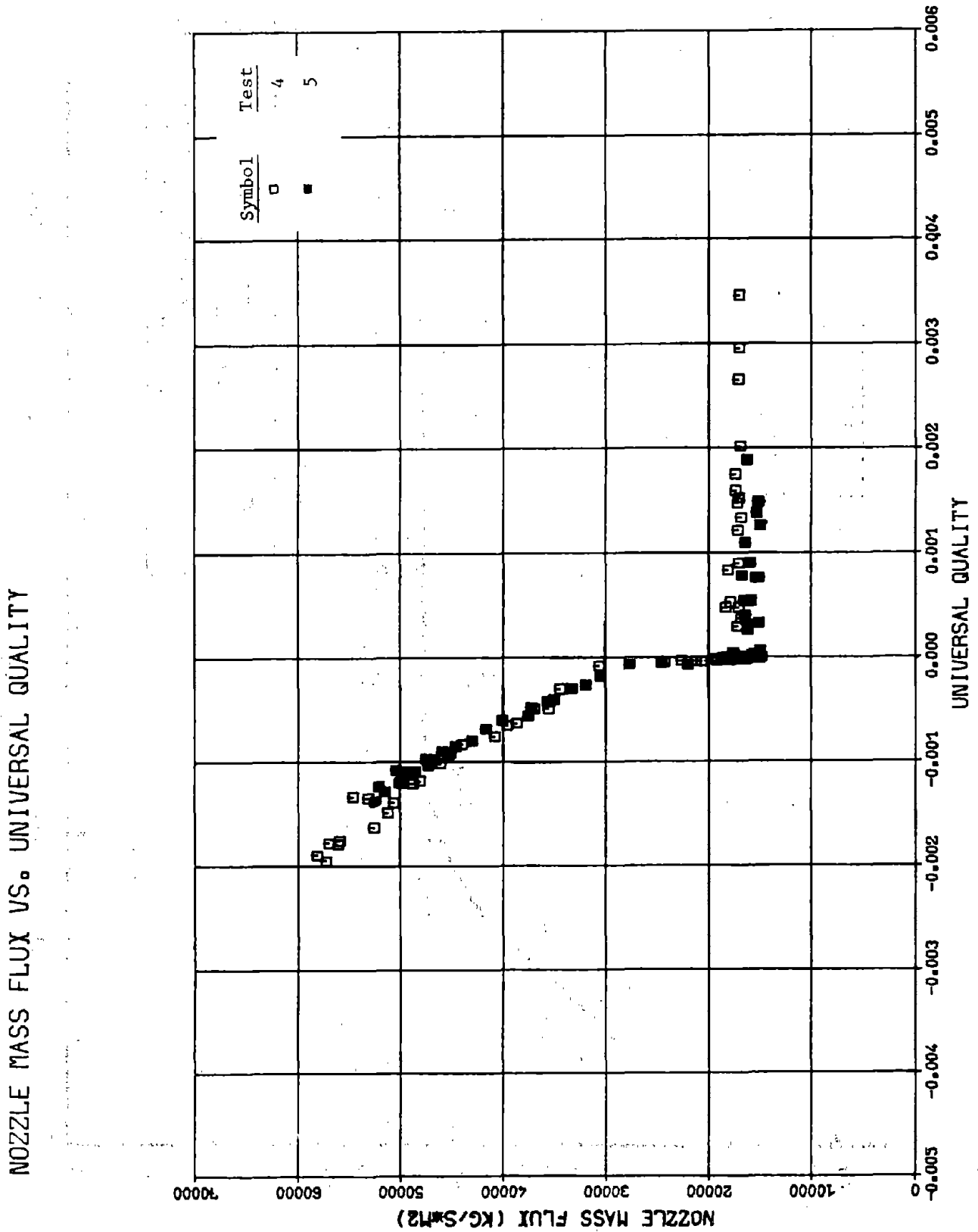


Fig 9:3 Variation of nozzle mass flux with nozzle length at a universal stagnation quality of -0.001

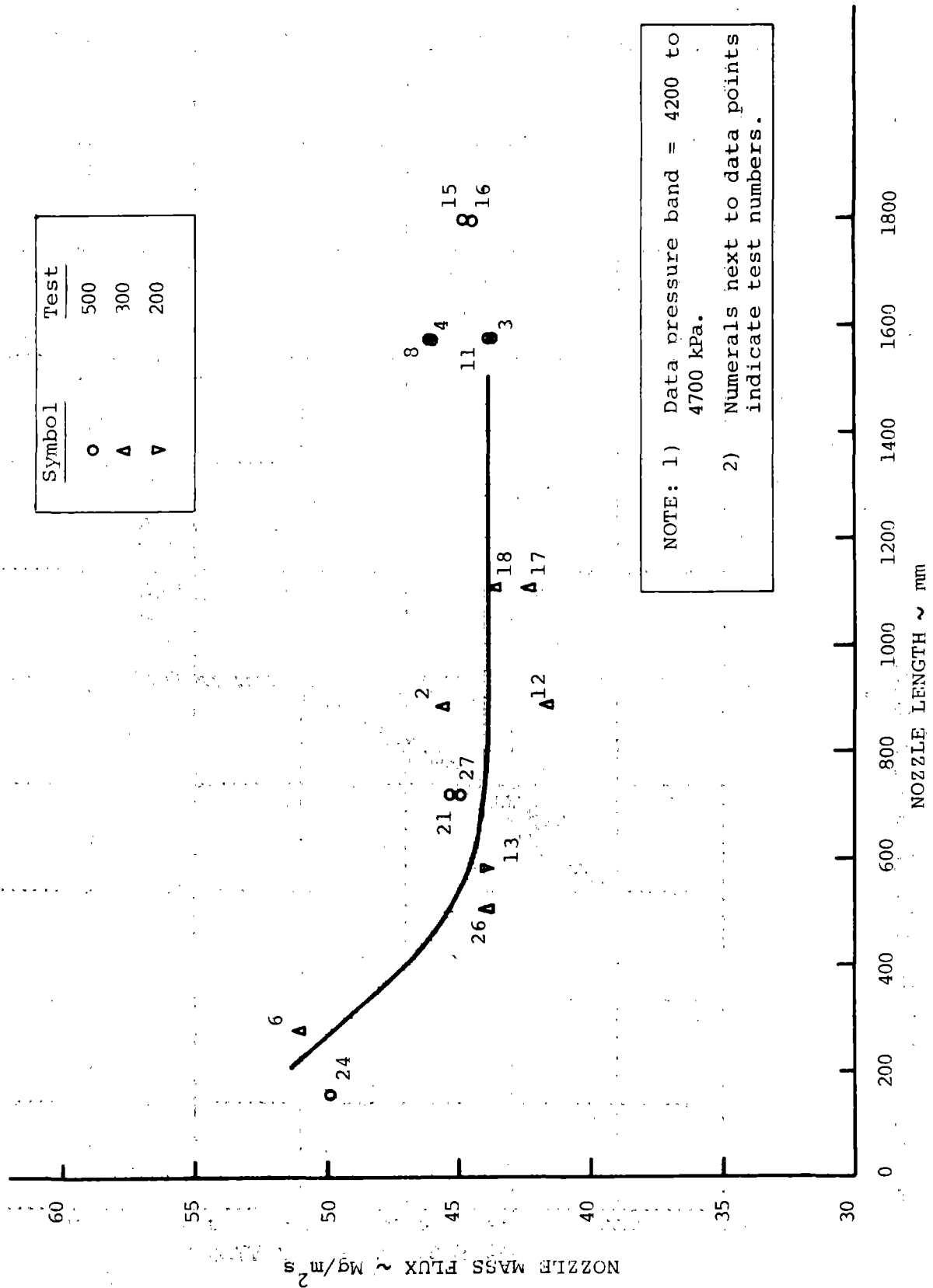


Fig 9:4 Variation of nozzle mass flux with nozzle length at universal stagnation qualities of -0.0005, -0.0010 and -0.0014: 500 mm diameter nozzles

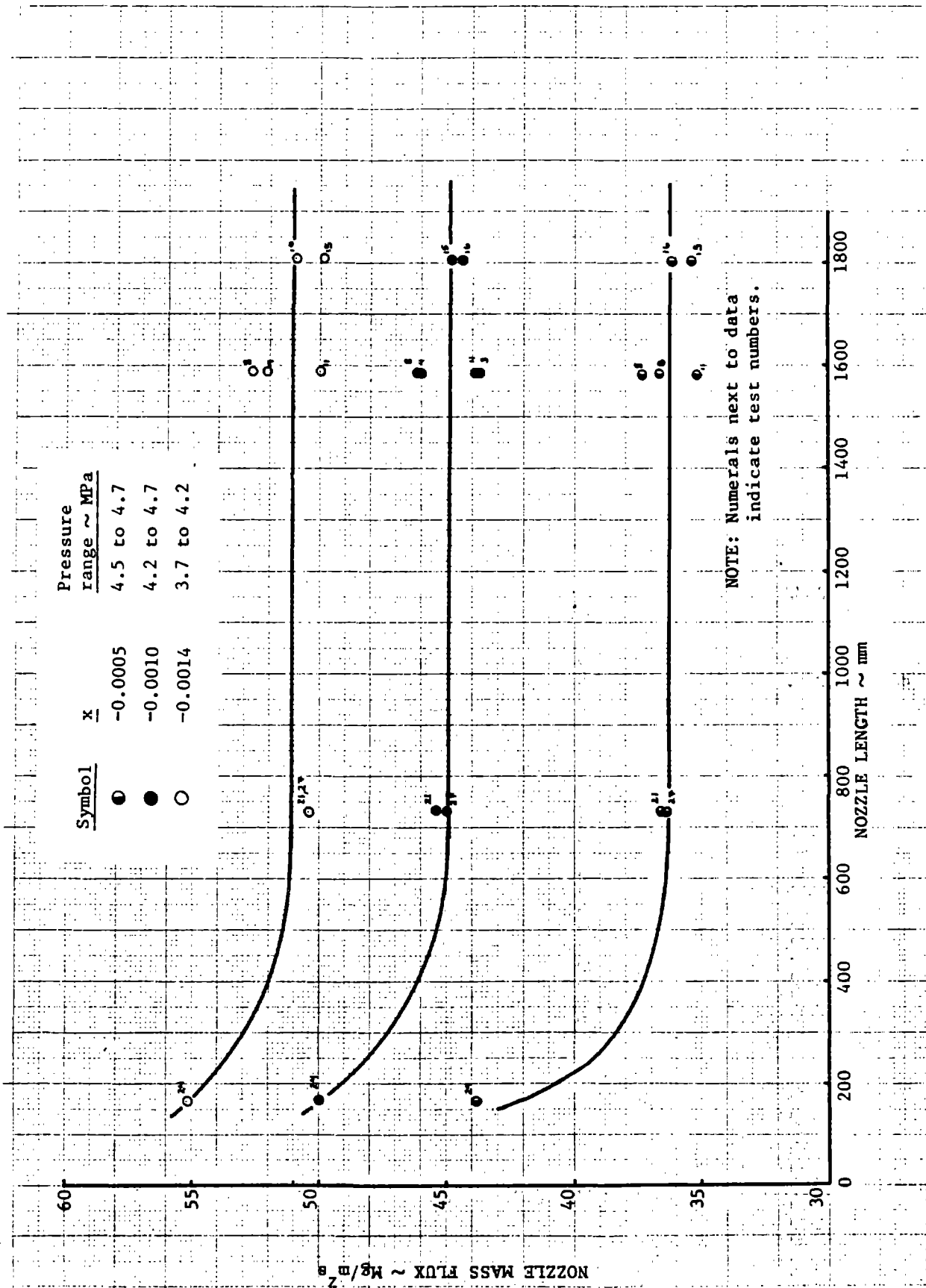


Fig 9:5 Variation of nozzle mass flux with nozzle length at zero quality

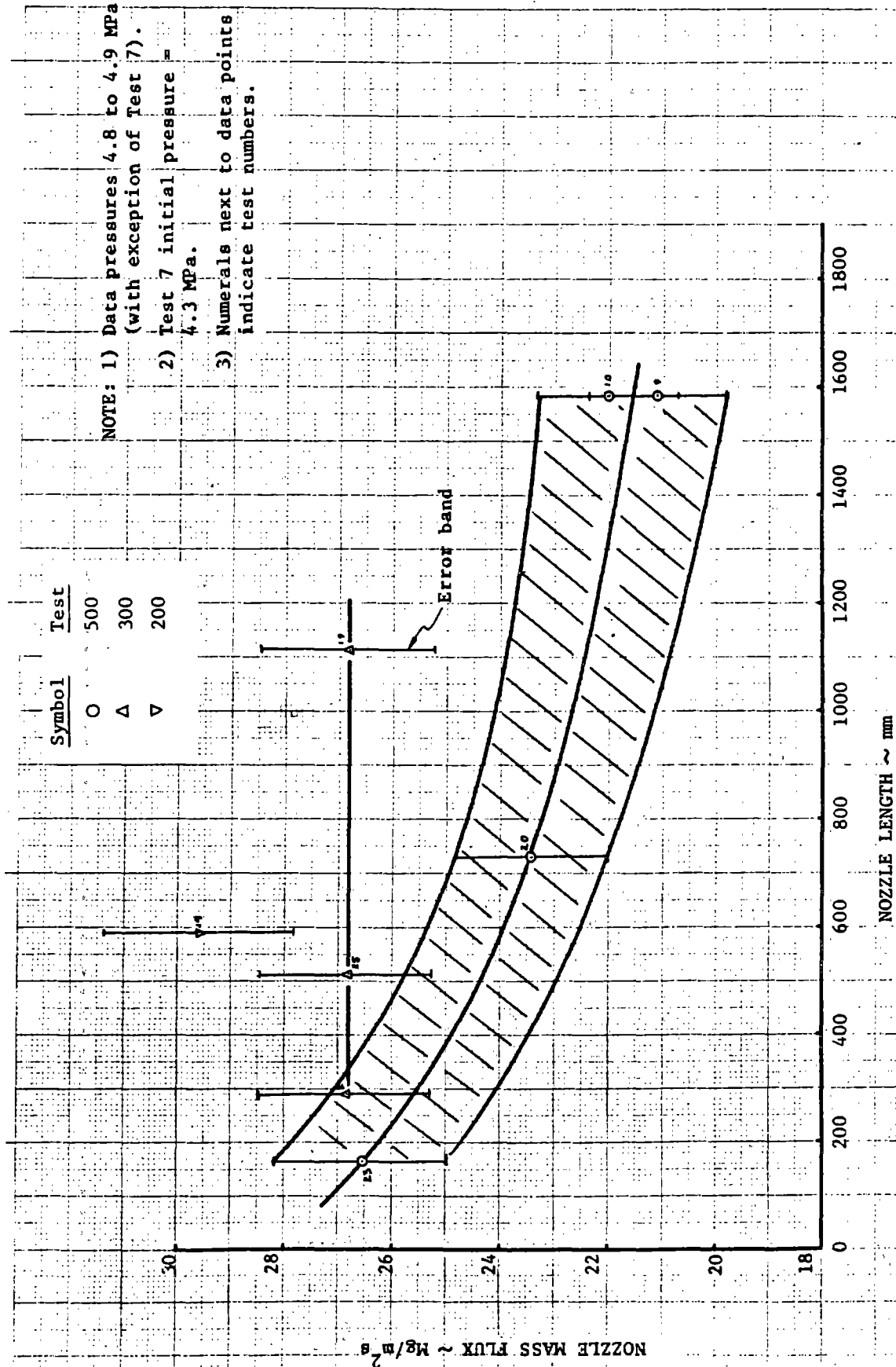
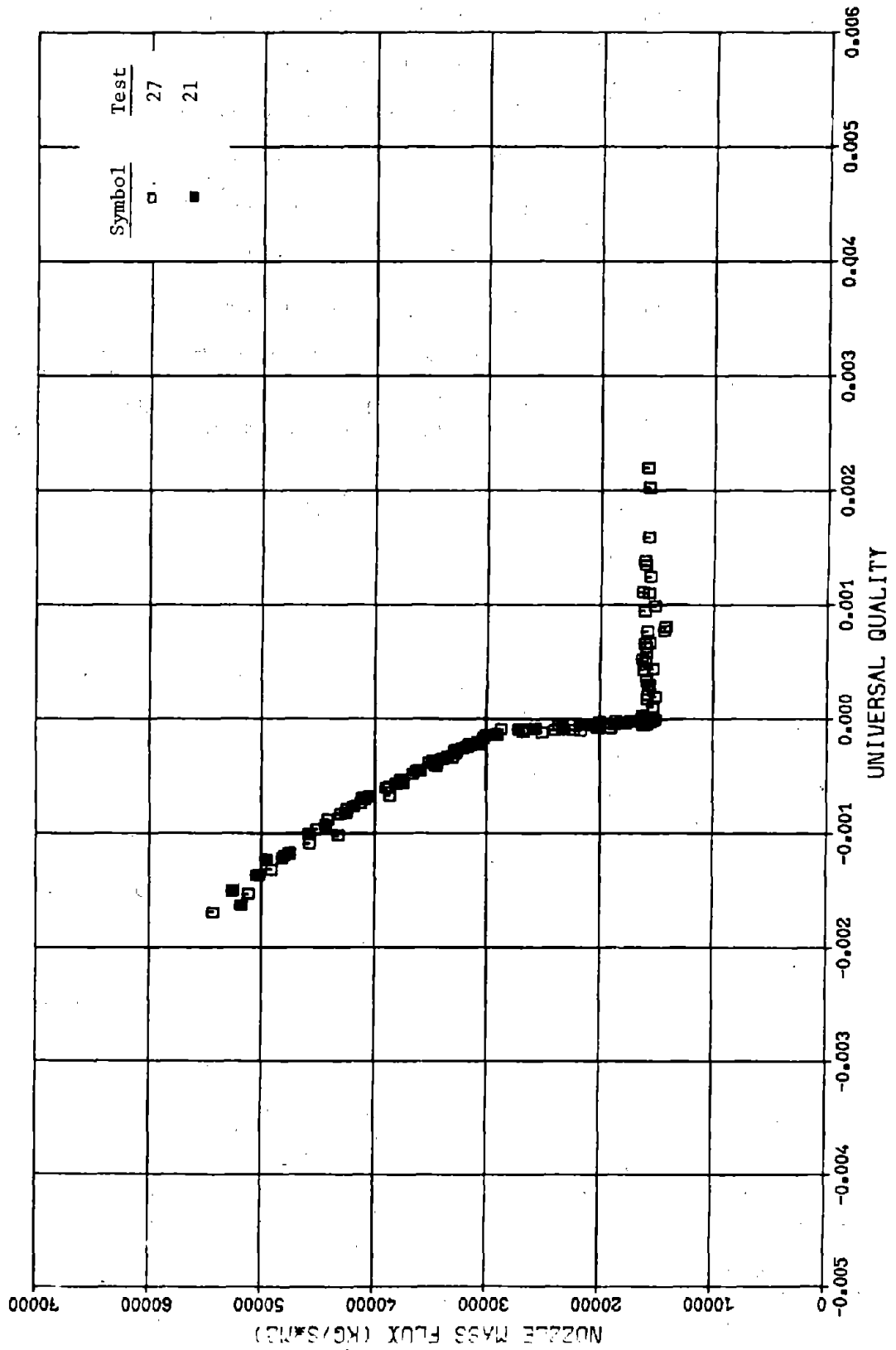


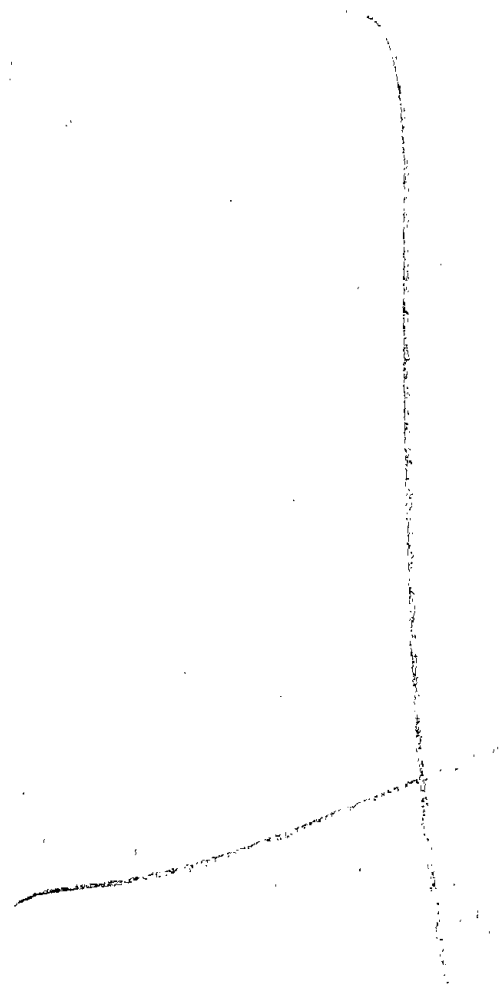
Fig 9:6 Comparison between the nozzle mass flux and the universal stagnation quality for Tests 21 and 27

NOZZLE MASS FLUX VS. UNIVERSAL QUALITY



10 REFERENCES

1. MXC-101: "Description of the Test Facility"
2. MXC-102: "Measurement System Description"
3. MXC-105: "Evaluation methods"
4. MXC-104: "Data Accuracy"
5. MXC-208: "Results from Test 8"
6. MXC-211: "Results from Test 11"
7. MXC-212: "Results from Test 12"
8. MXC-210: "Results from Test 10"
9. MXC-227: "Results from Test 27"
10. R Garcia, F.D. Hammitt and M.J. Robinson,
"Acoustic Noise from a Cavitating Venturi",
The University of Michigan College of En-
gineering, Technical Report No. 1, June 1964



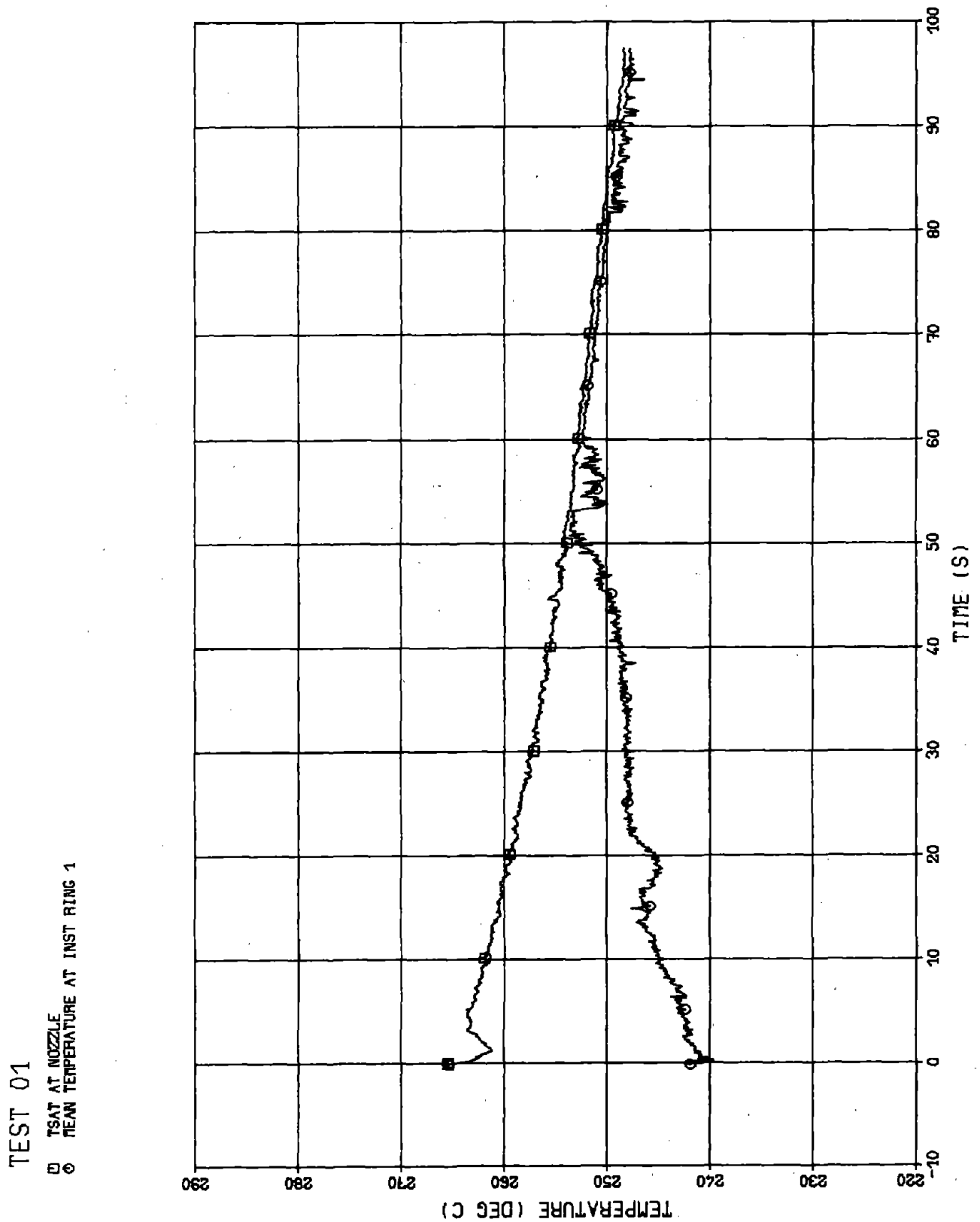
APPENDIX A

Comparisons between the saturation temperature and the
average temperature measured at the test instrumentation
rings: Tests 1 through 27

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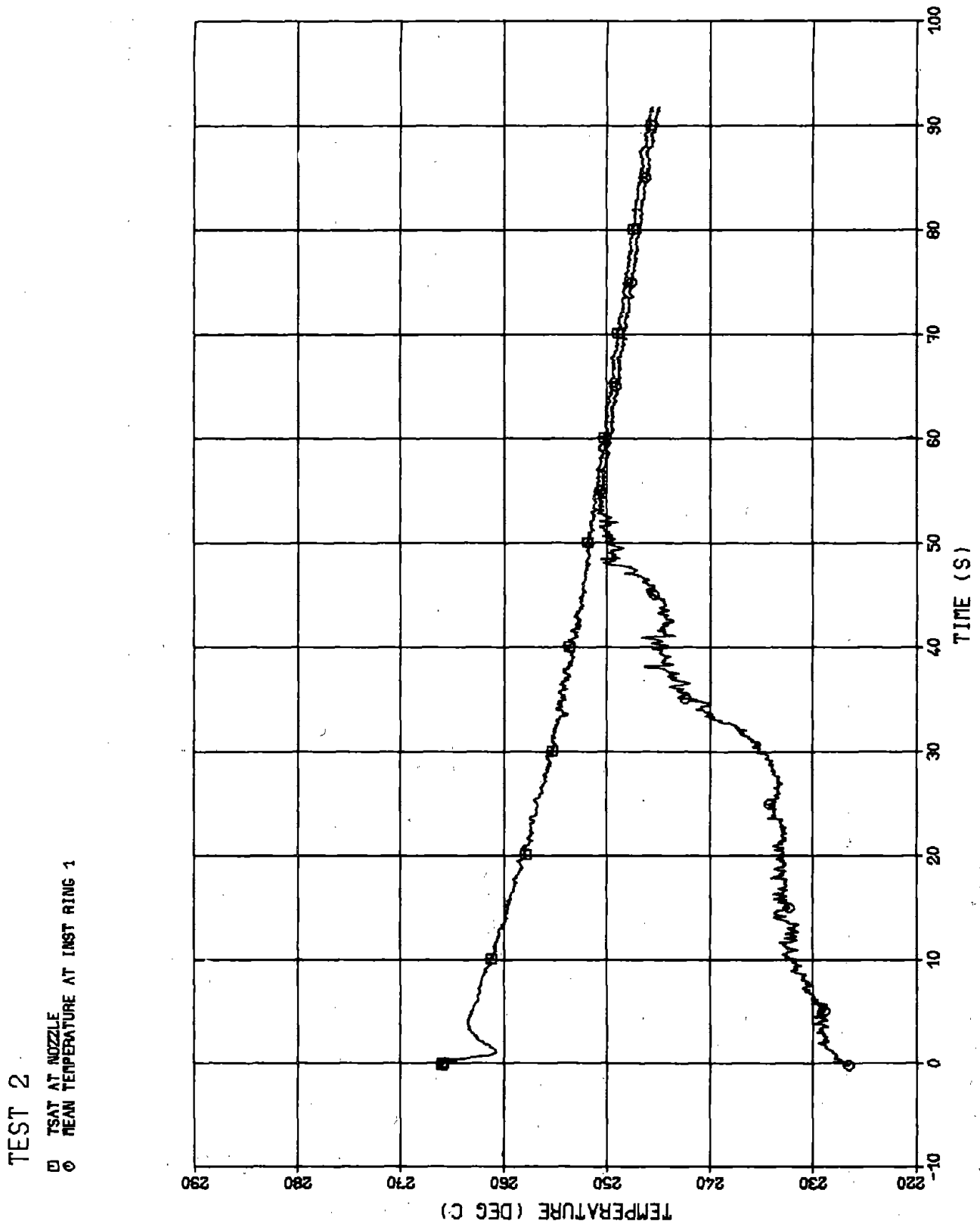


Plot A:1 - Test 1 Comparison between the average temperature at instrumentation ring I and the saturation temperature at the nozzle inlet plane



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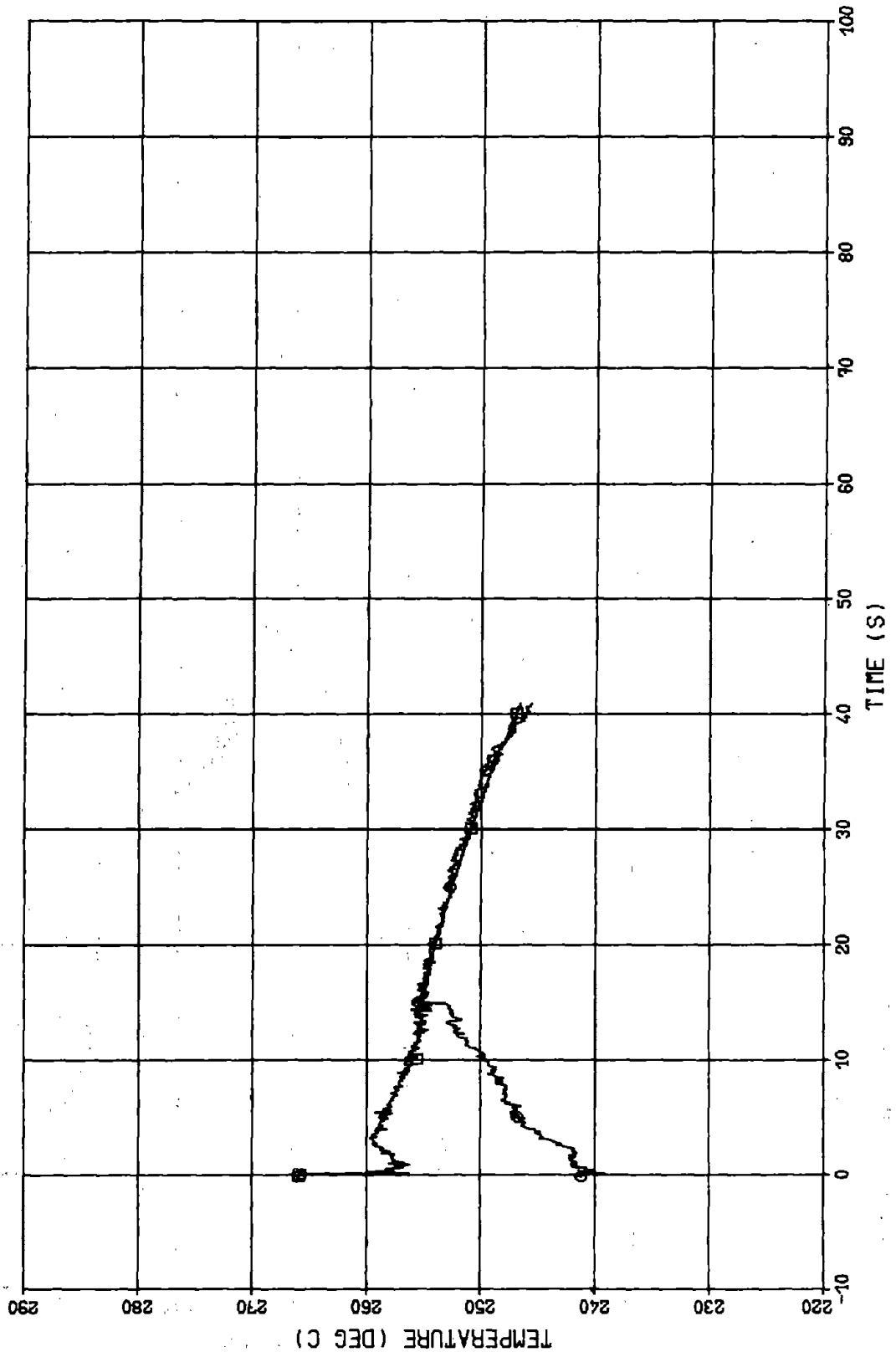
Plot A:2 - Test 2 Comparison between the average temperature at instrumentation ring 1 and the saturation temperature at the nozzle inlet plane



Plot A:3 - Test 3

TEST 3

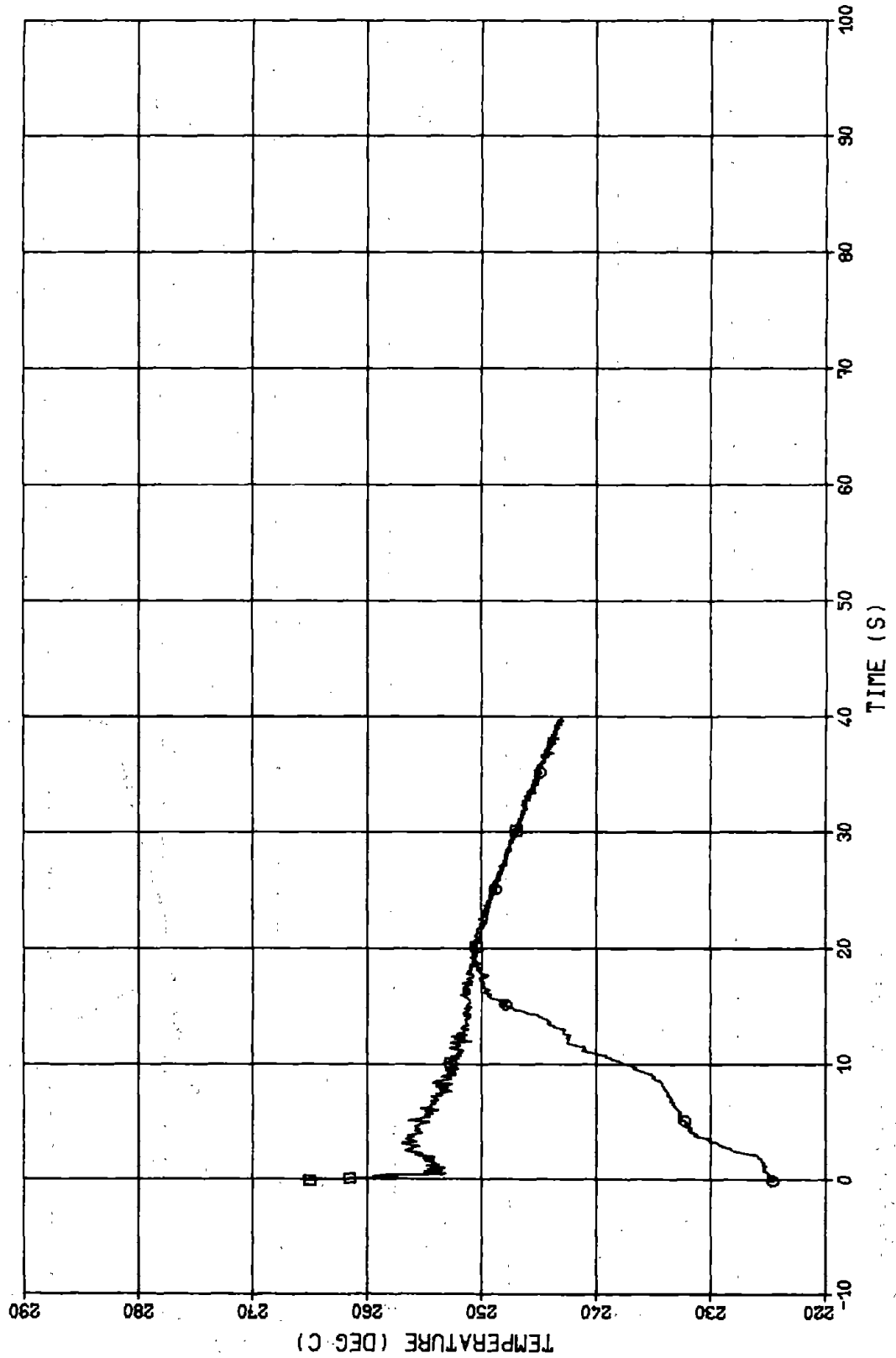
□ TSAT AT RING 1
○ MEAN TEMPERATURE AT INST RING 1



Plot A:4 - Test 4

TEST 4

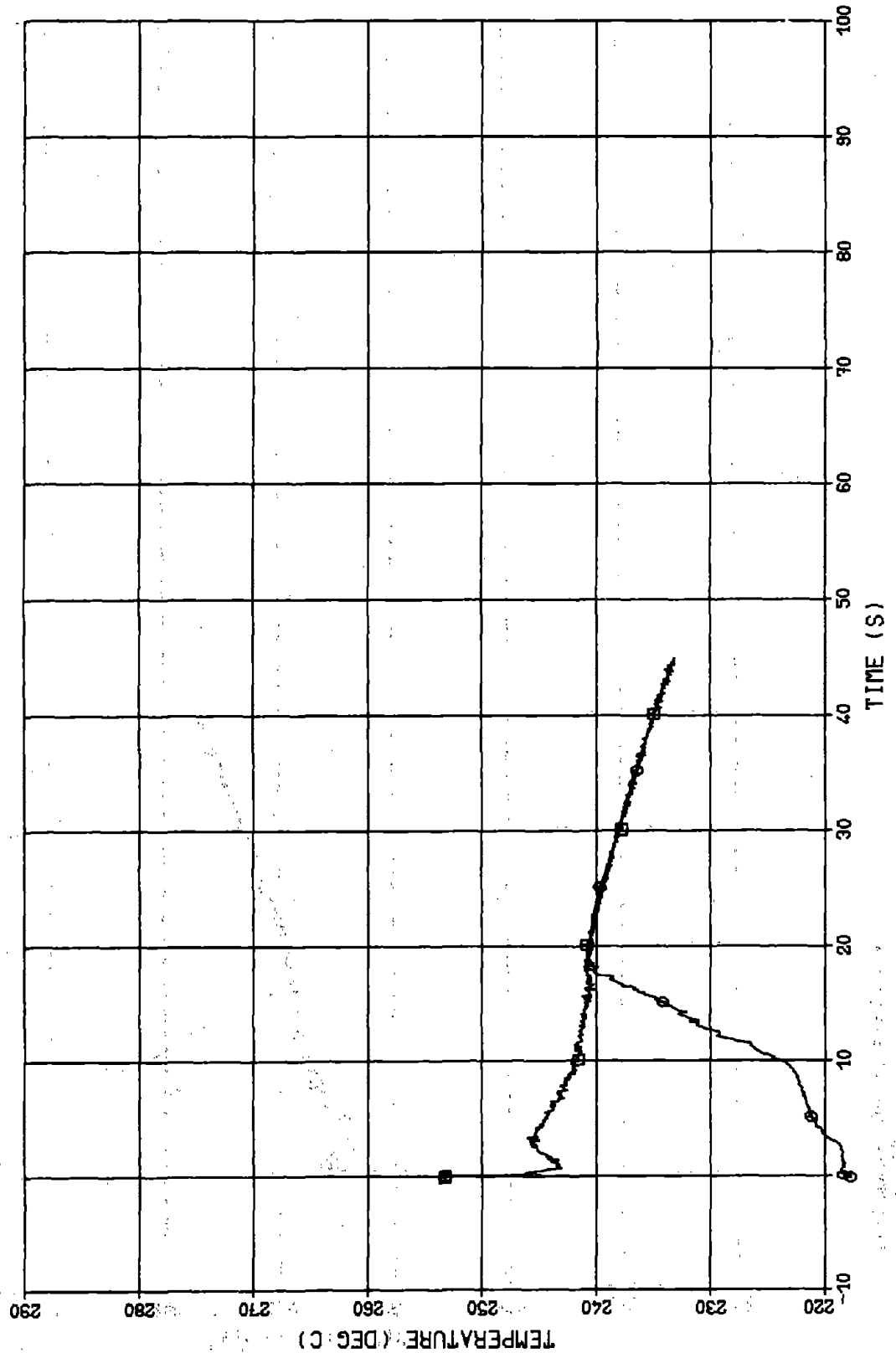
□ TSAT AT RING 1
○ MEAN TEMPERATURE AT INST RING 1



Plot A:5 - Test 5

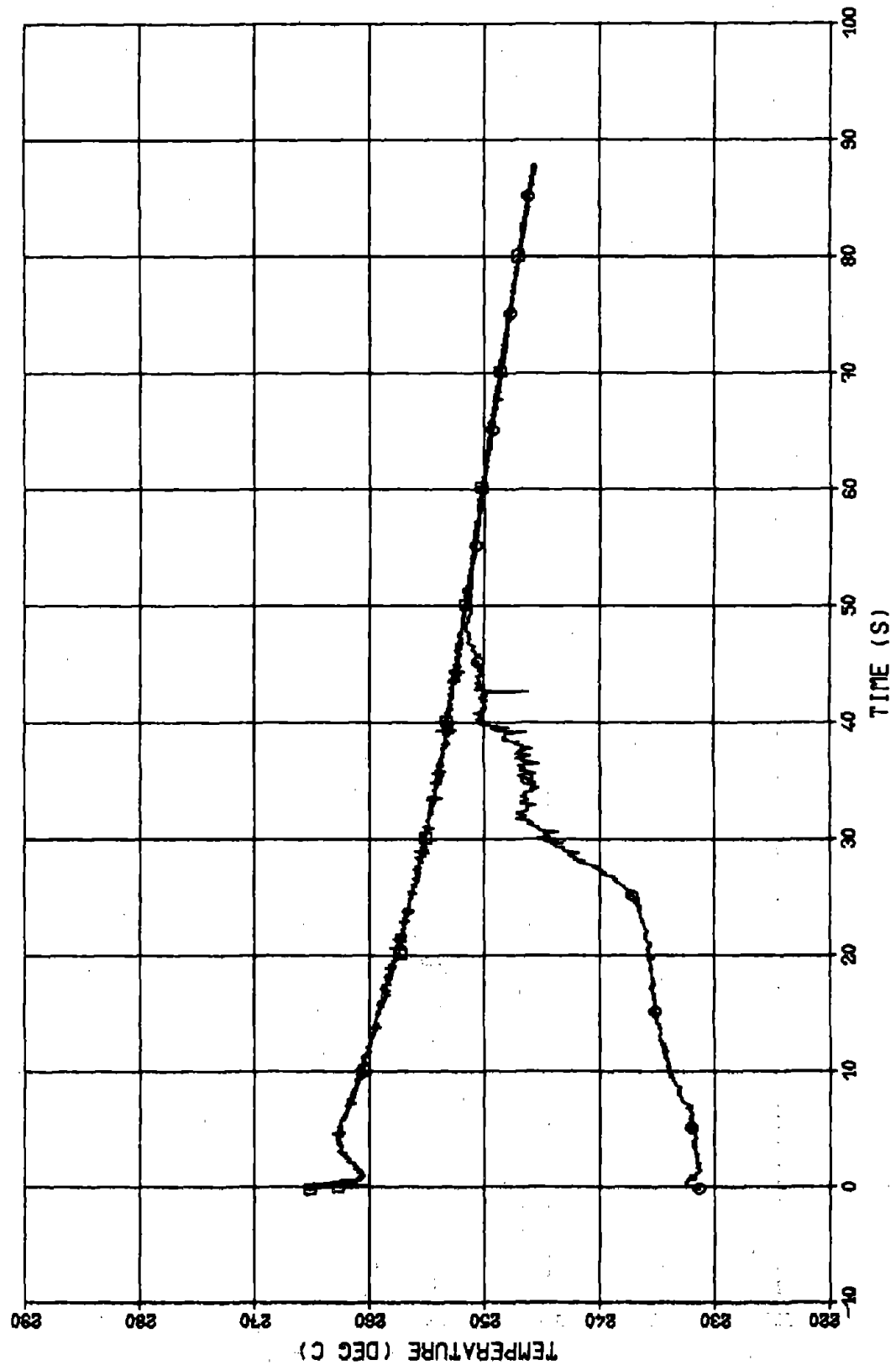
TEST 5

□ TSAT AT RING 1
○ MEAN TEMPERATURE AT INST RING 1

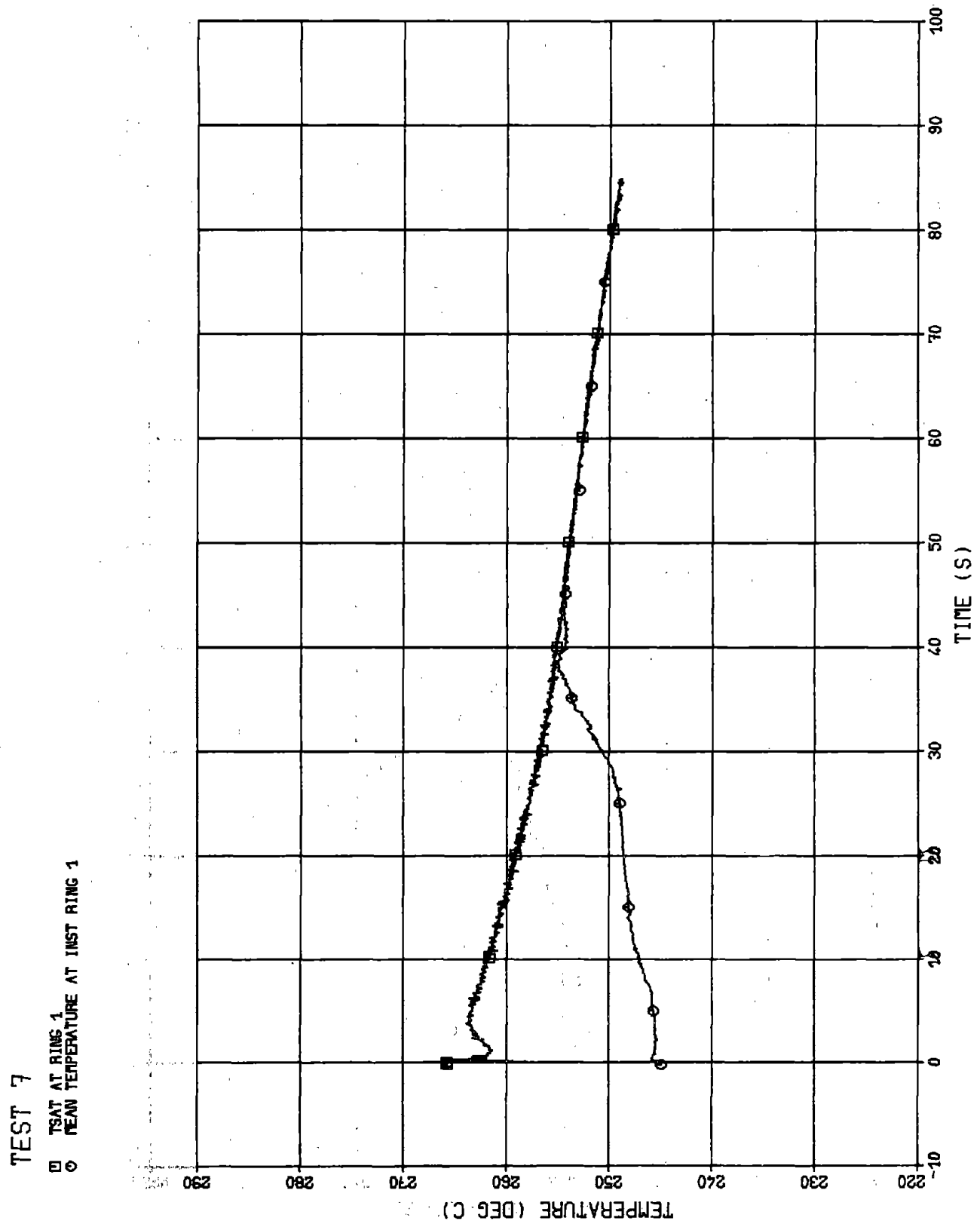


Plot A:6 - Test 6

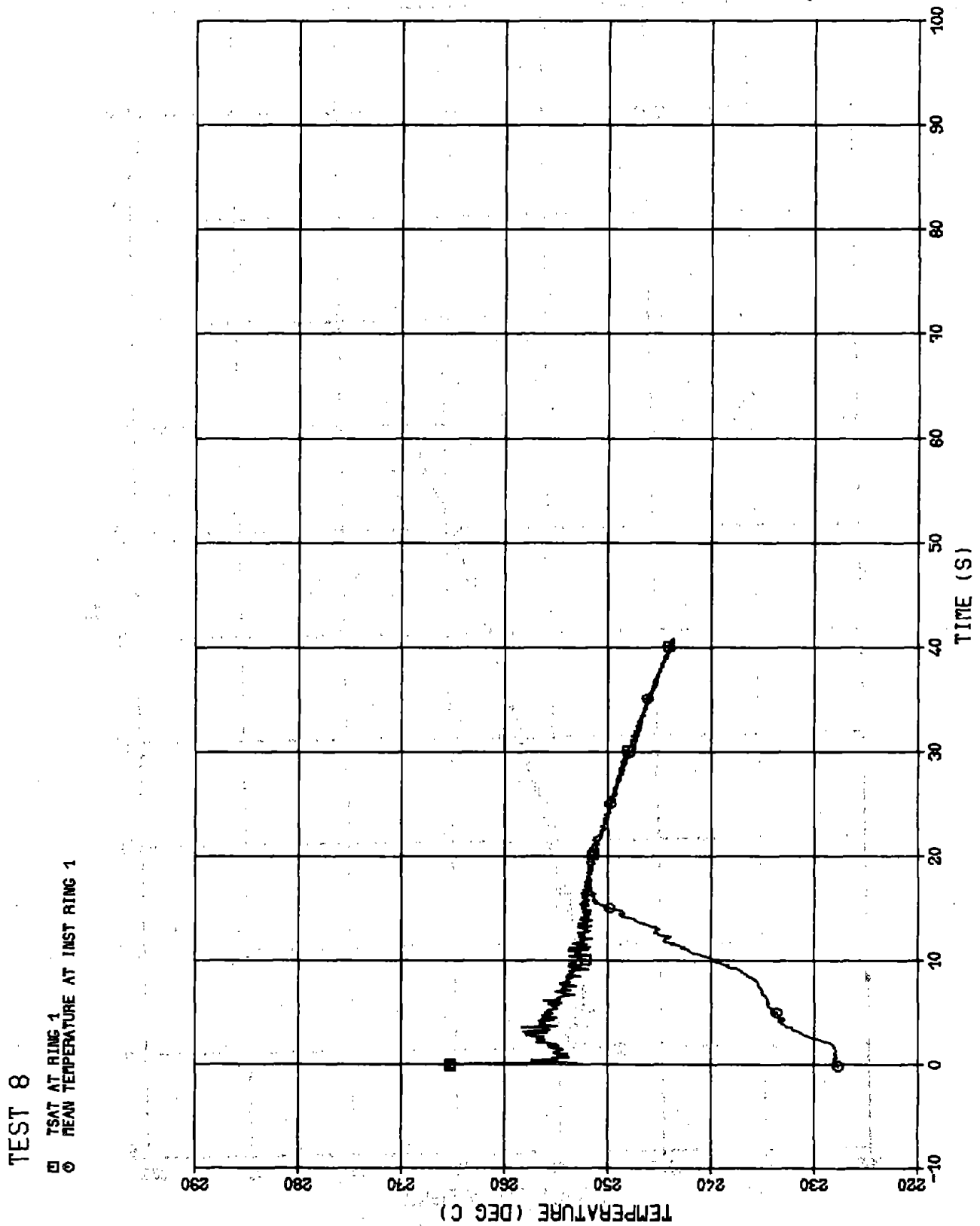
TEST 6
□ TSAT AT RING 1
○ MEAN TEMPERATURE AT INST RING 1



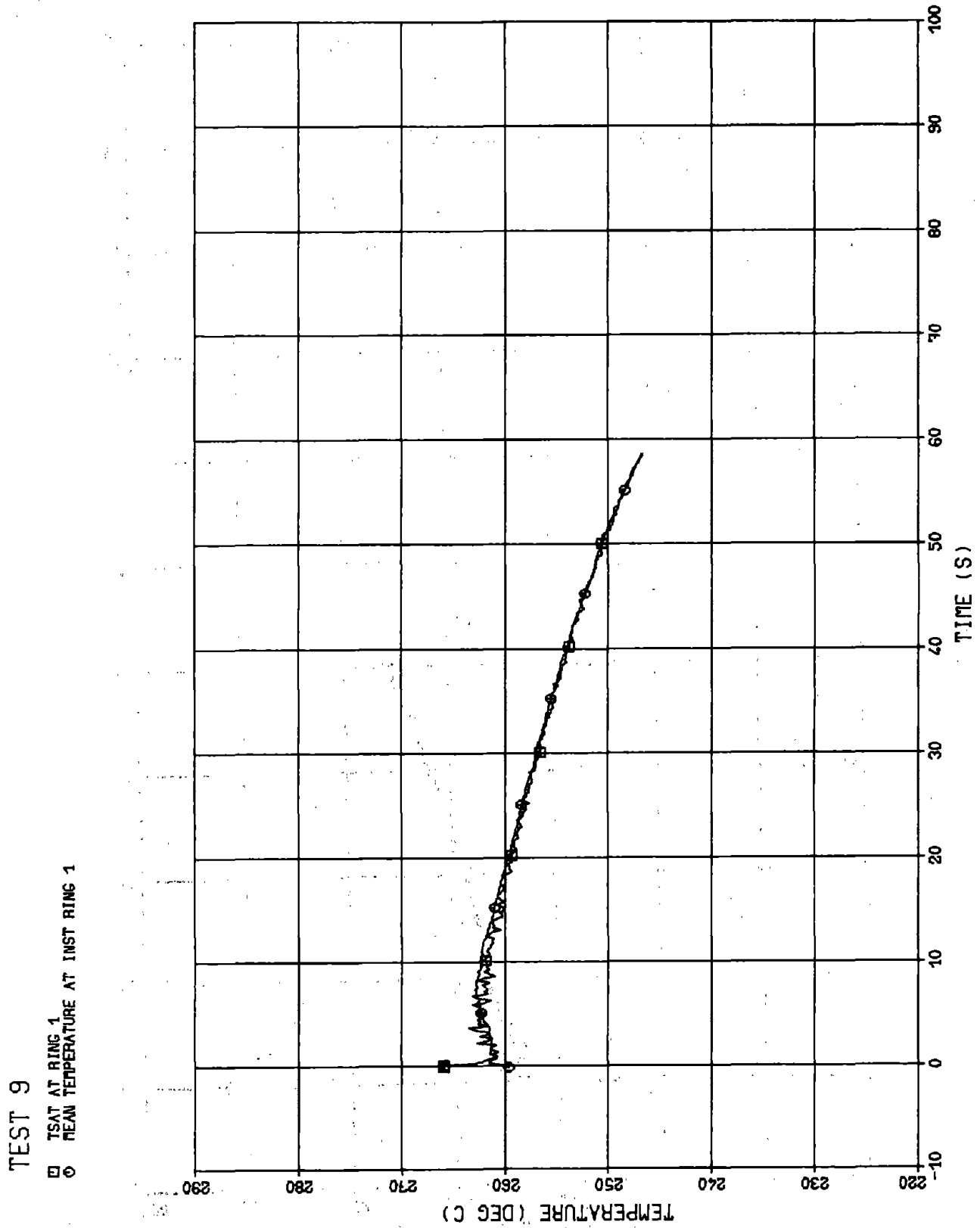
Plot A:7 - Test 7



Plot A:8 - Test 8



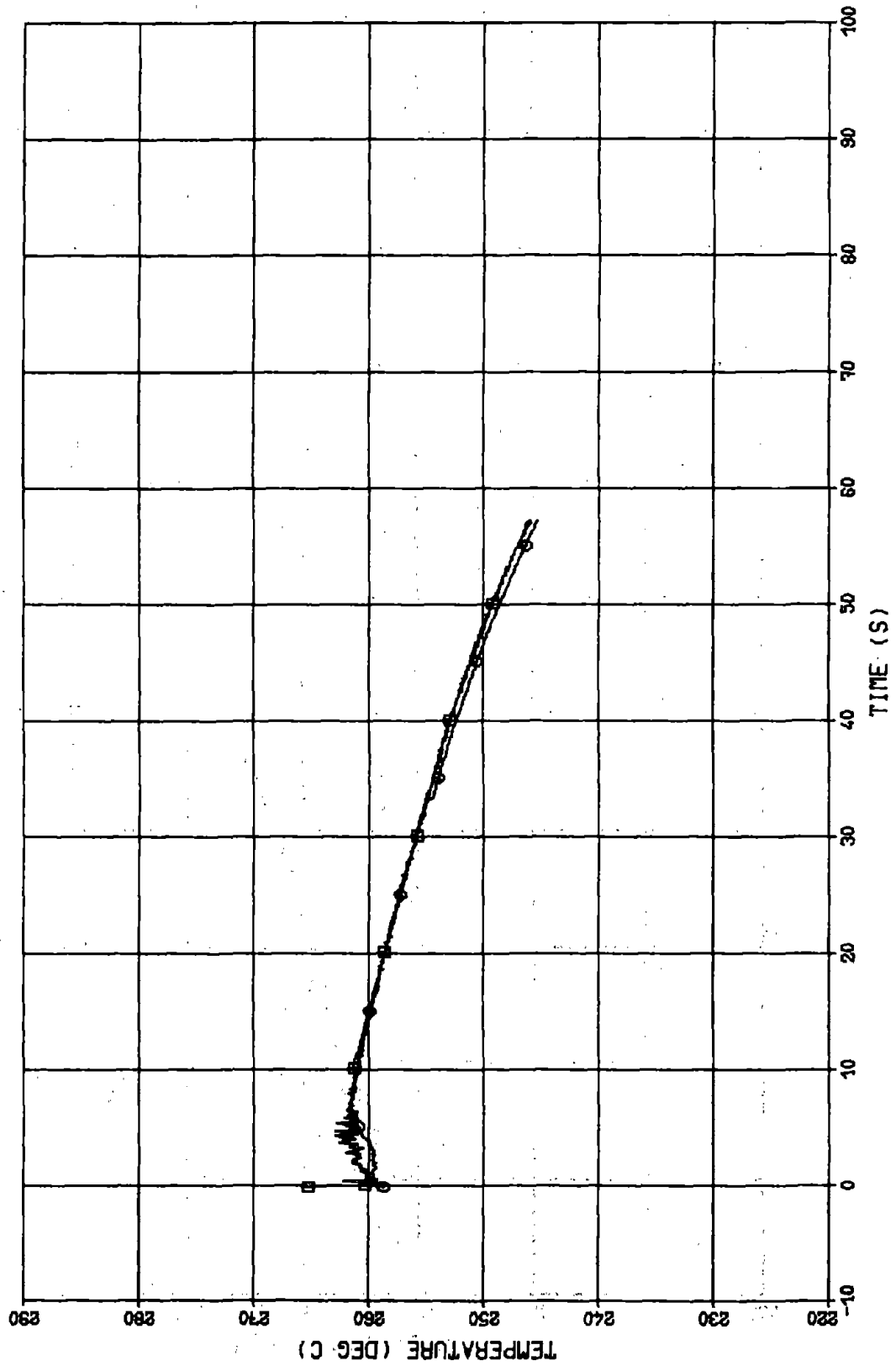
Plot A:9 - Test 9



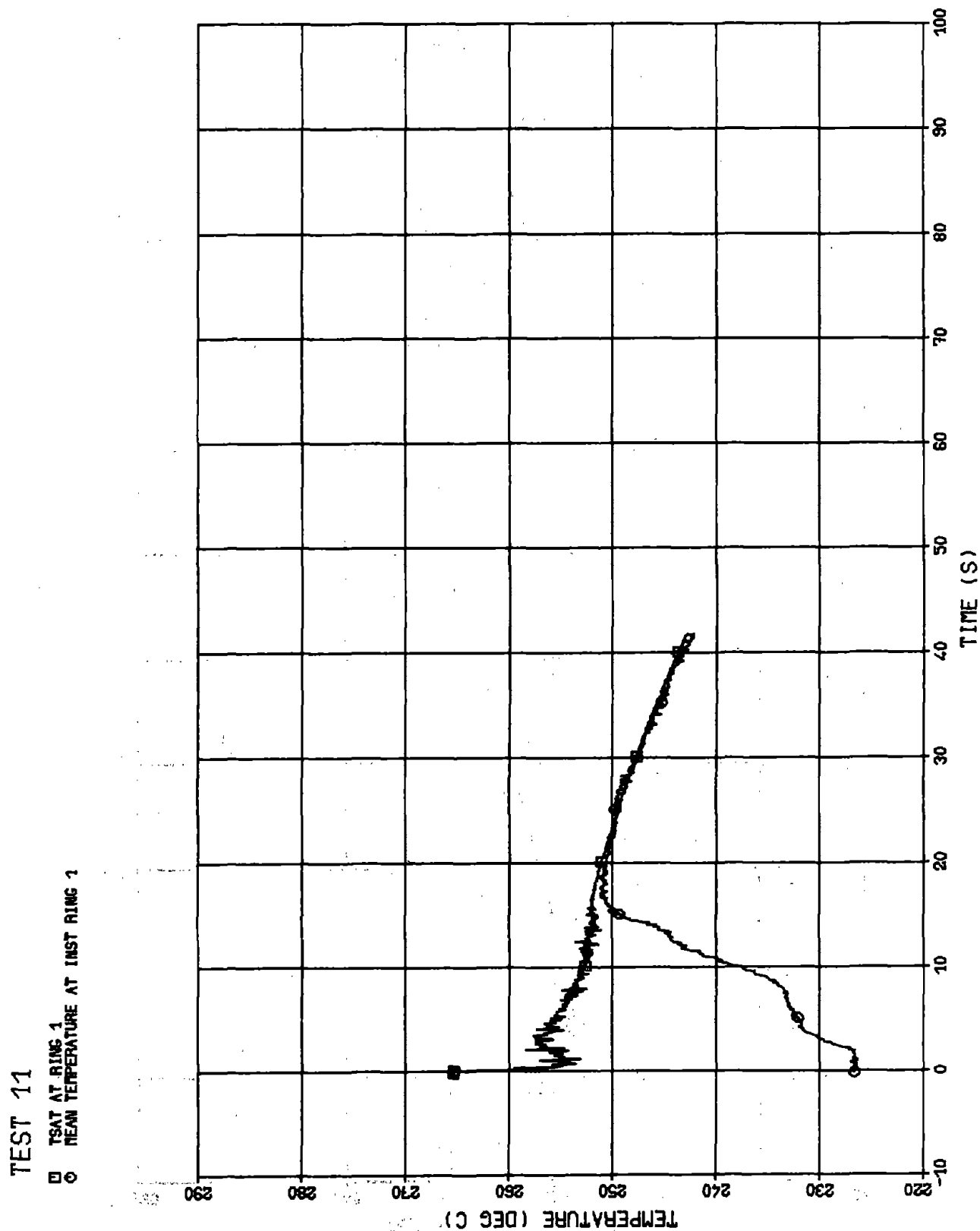
Plot A:10 - Test 10

TEST 10

□ TSAT AT RING 1
○ MEAN TEMPERATURE AT INST RING 1



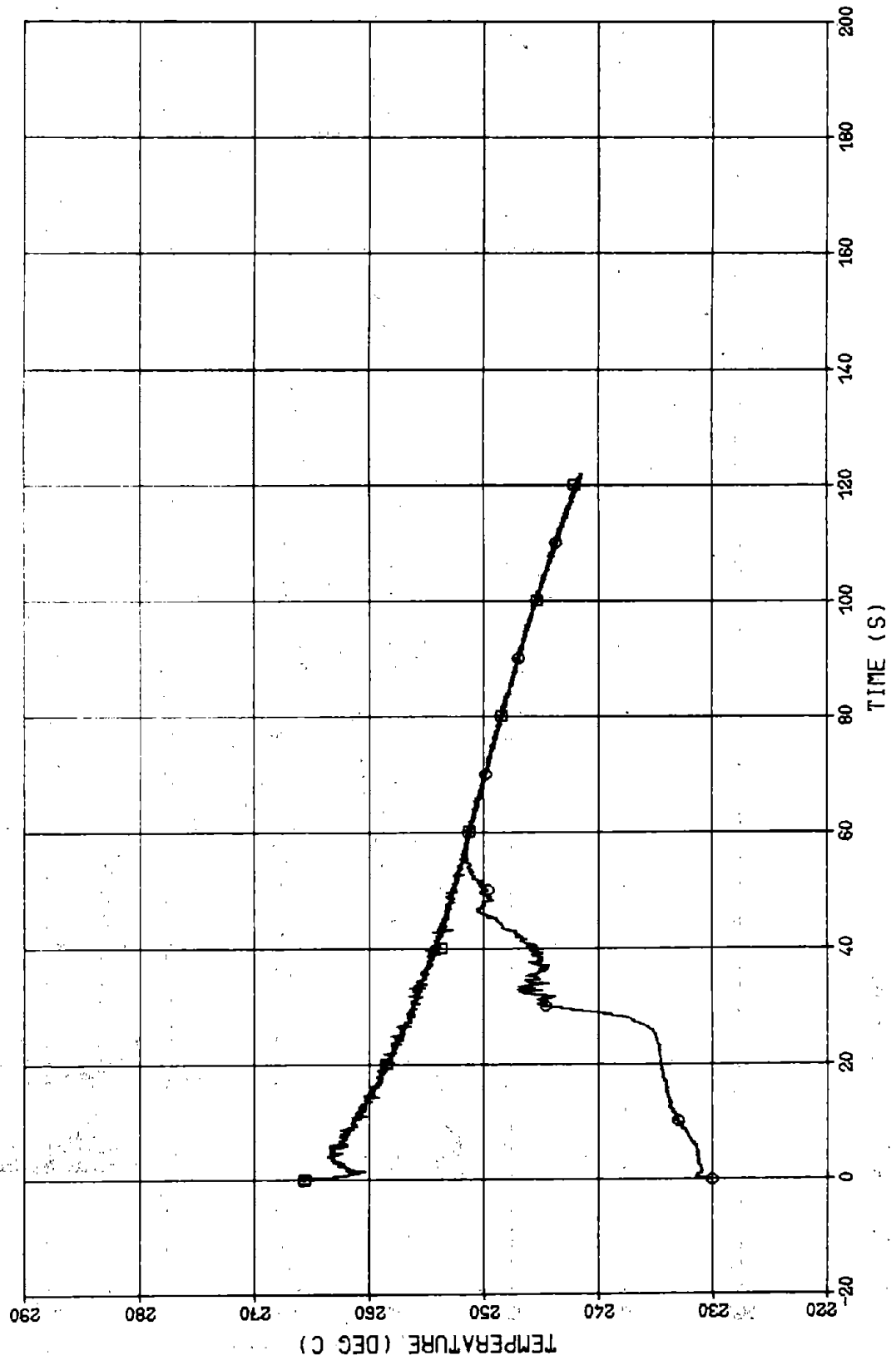
Plot A:11 - Test 11



Plot A:12 - Test 12

TEST 12

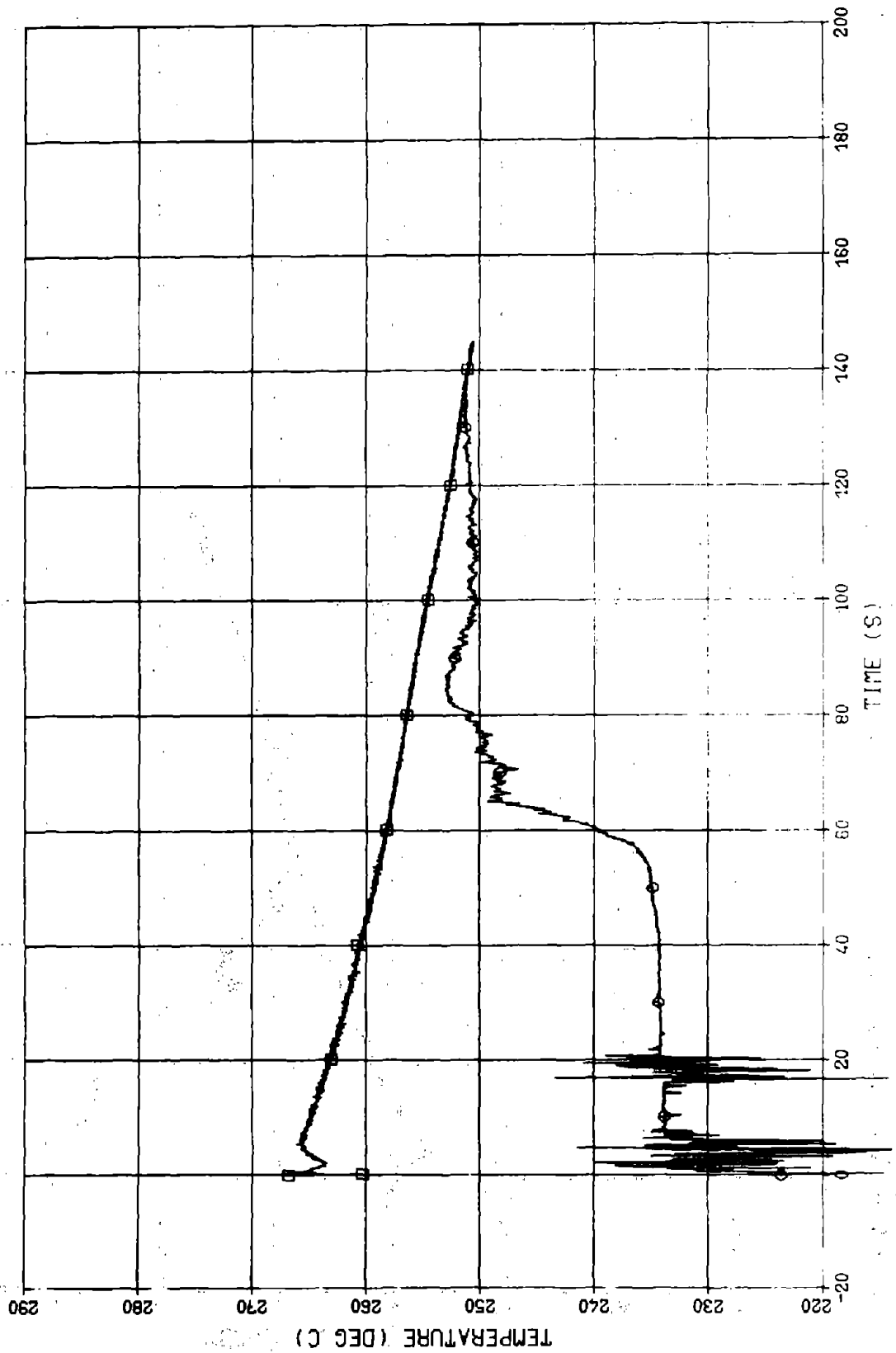
□ TSAT AT RING 1
○ MEAN TEMPERATURE AT INST RING 1



Plot A:13 - Test 13

TEST 13

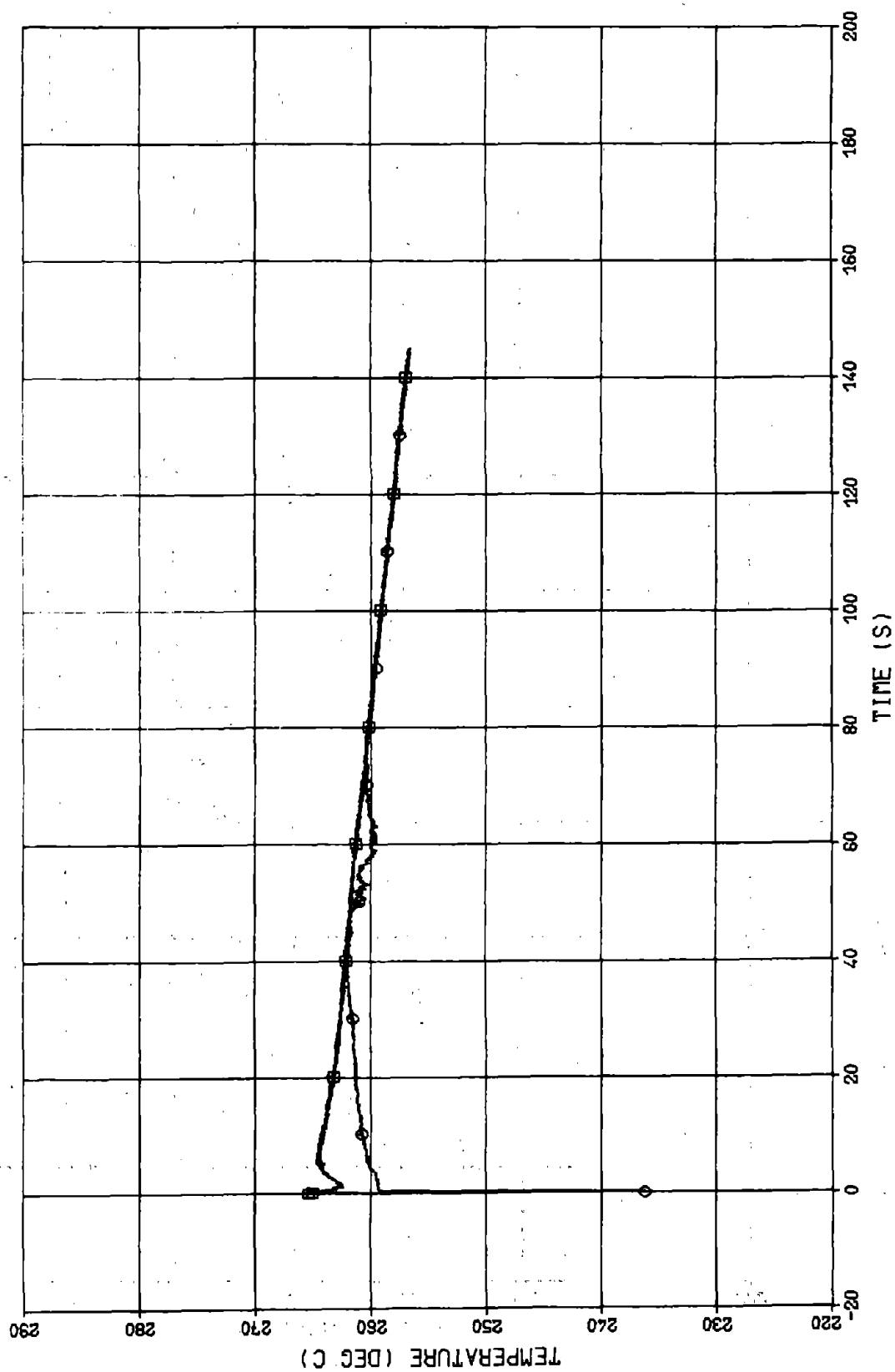
□ TSAT AT RING 1
○ MEAN TEMPERATURE AT INST RING 1



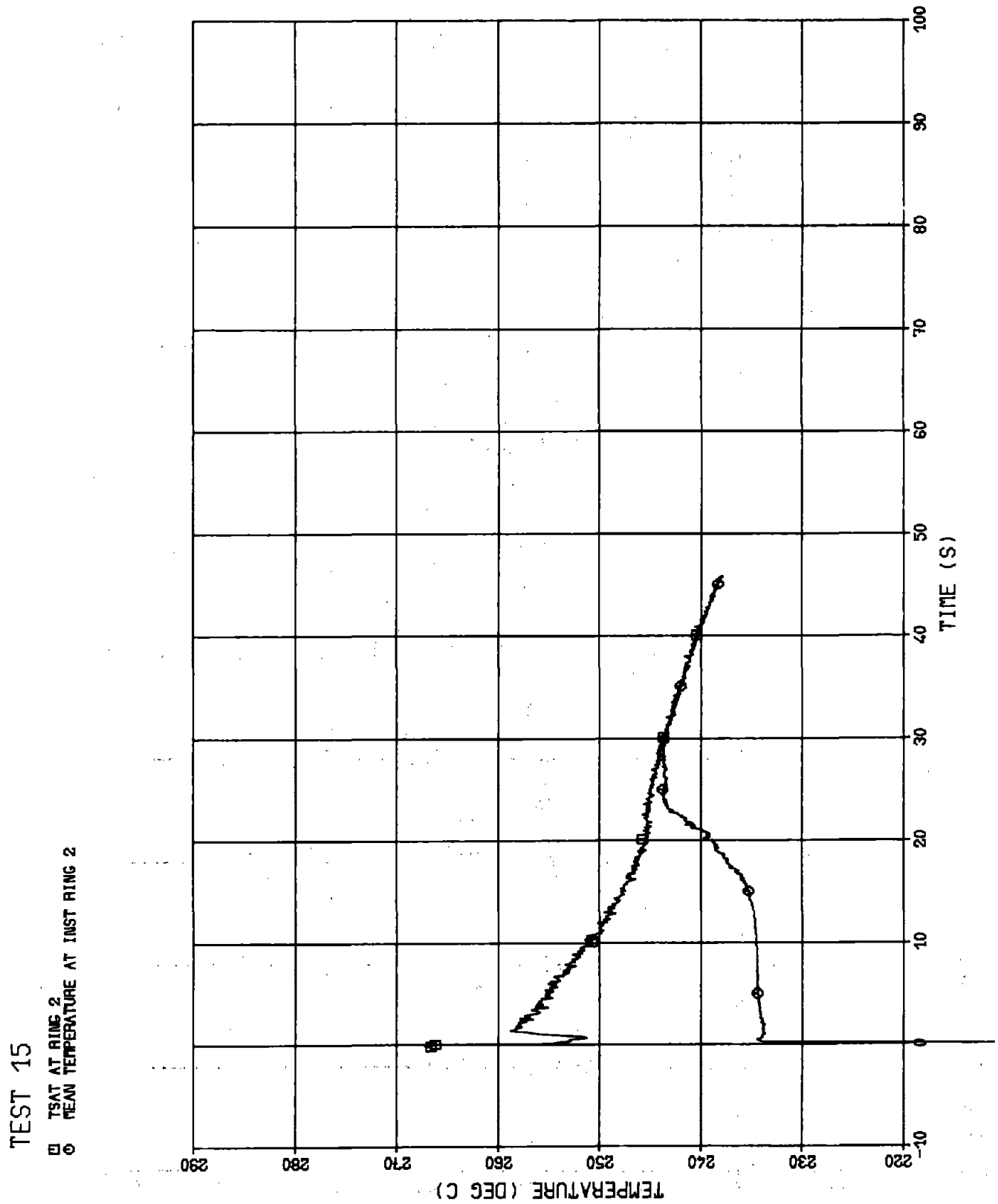
Plot A:14 - Test 14

TEST 14

□ TSAT AT RING 1
○ MEAN TEMPERATURE AT INST RING 1

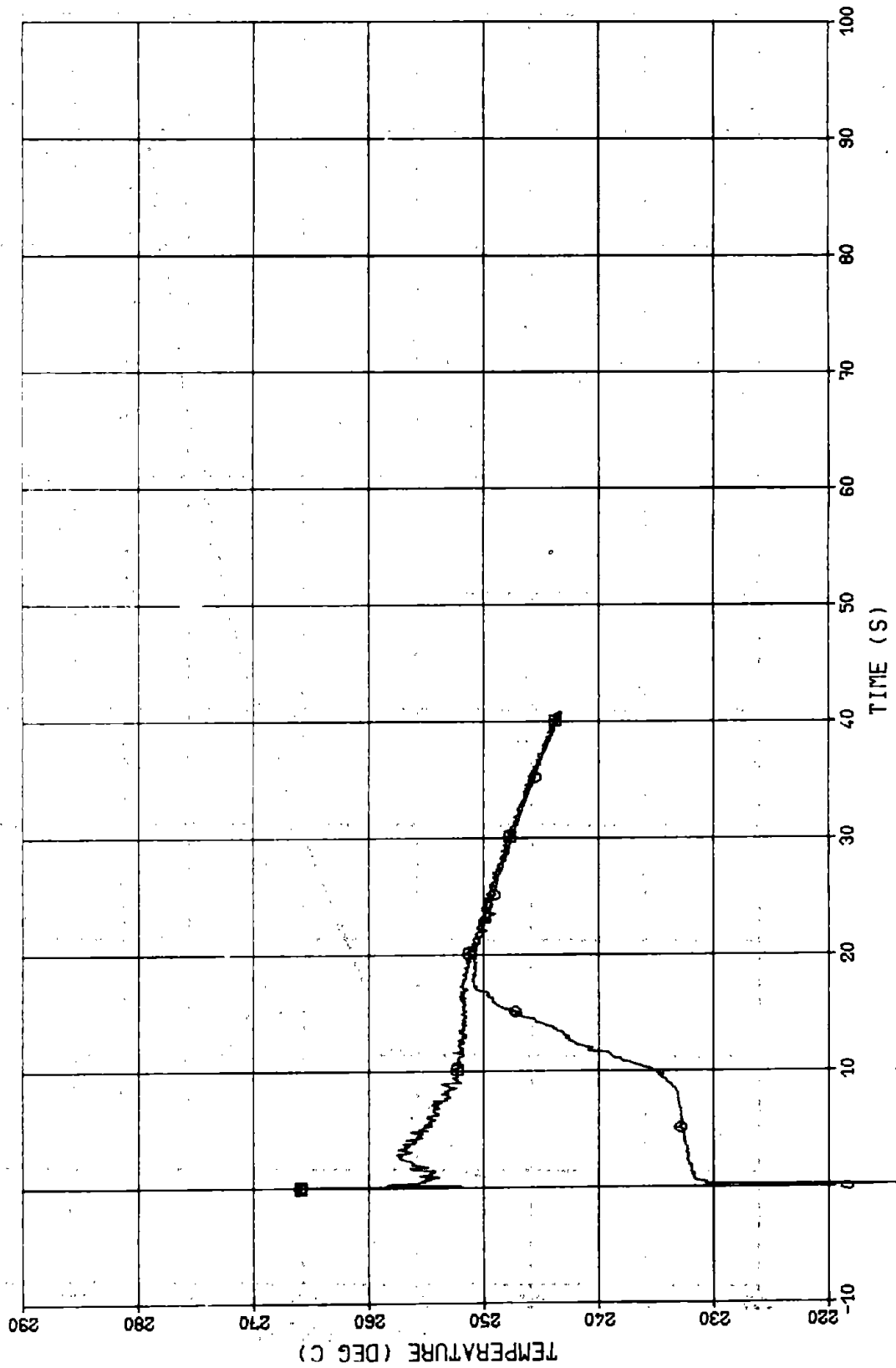


Plot A:15 - Test 15

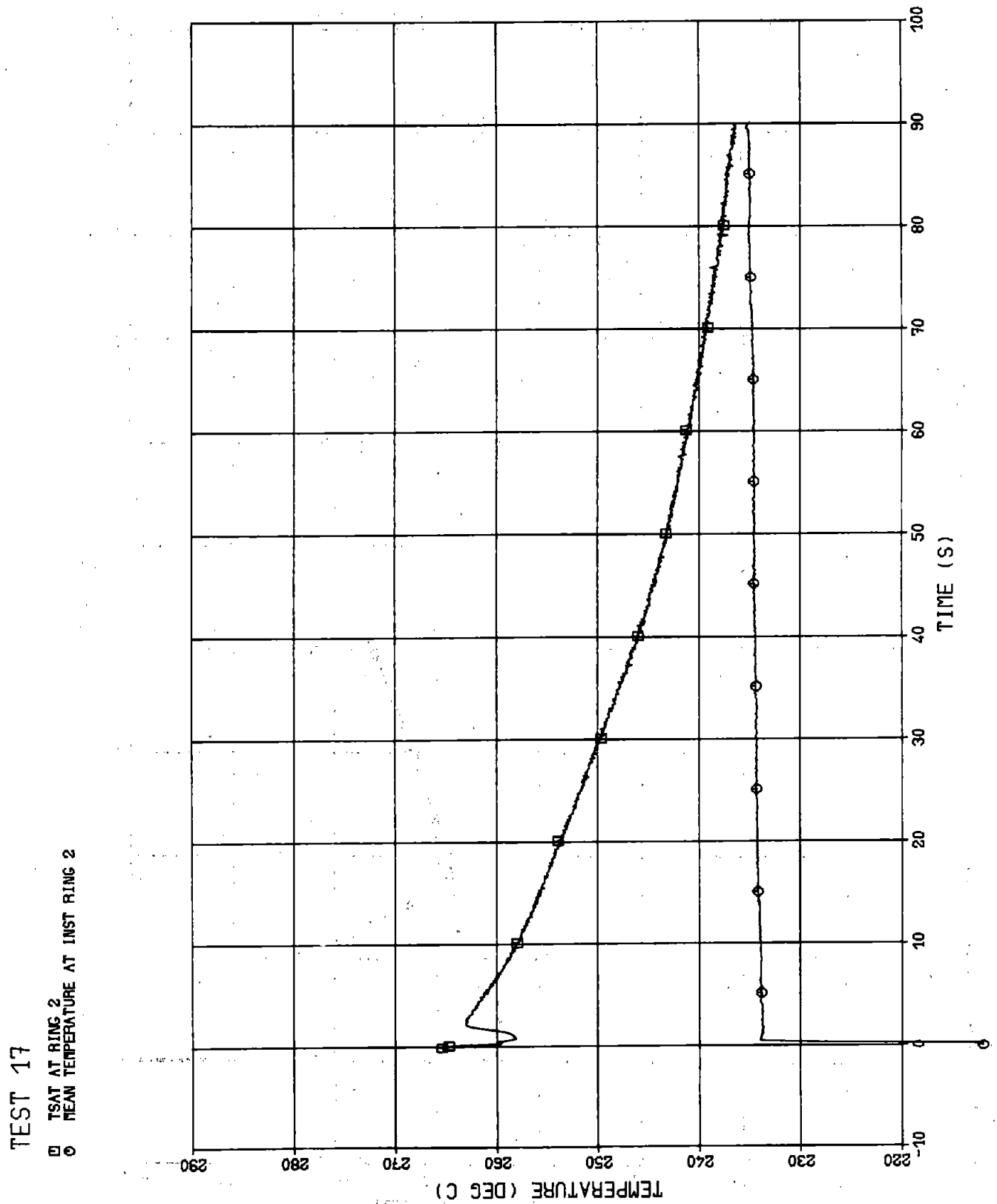


Plot A:16 - Test 16

TEST 16
□ TSAT AT RING 2
○ MEAN TEMPERATURE AT INST RING 2



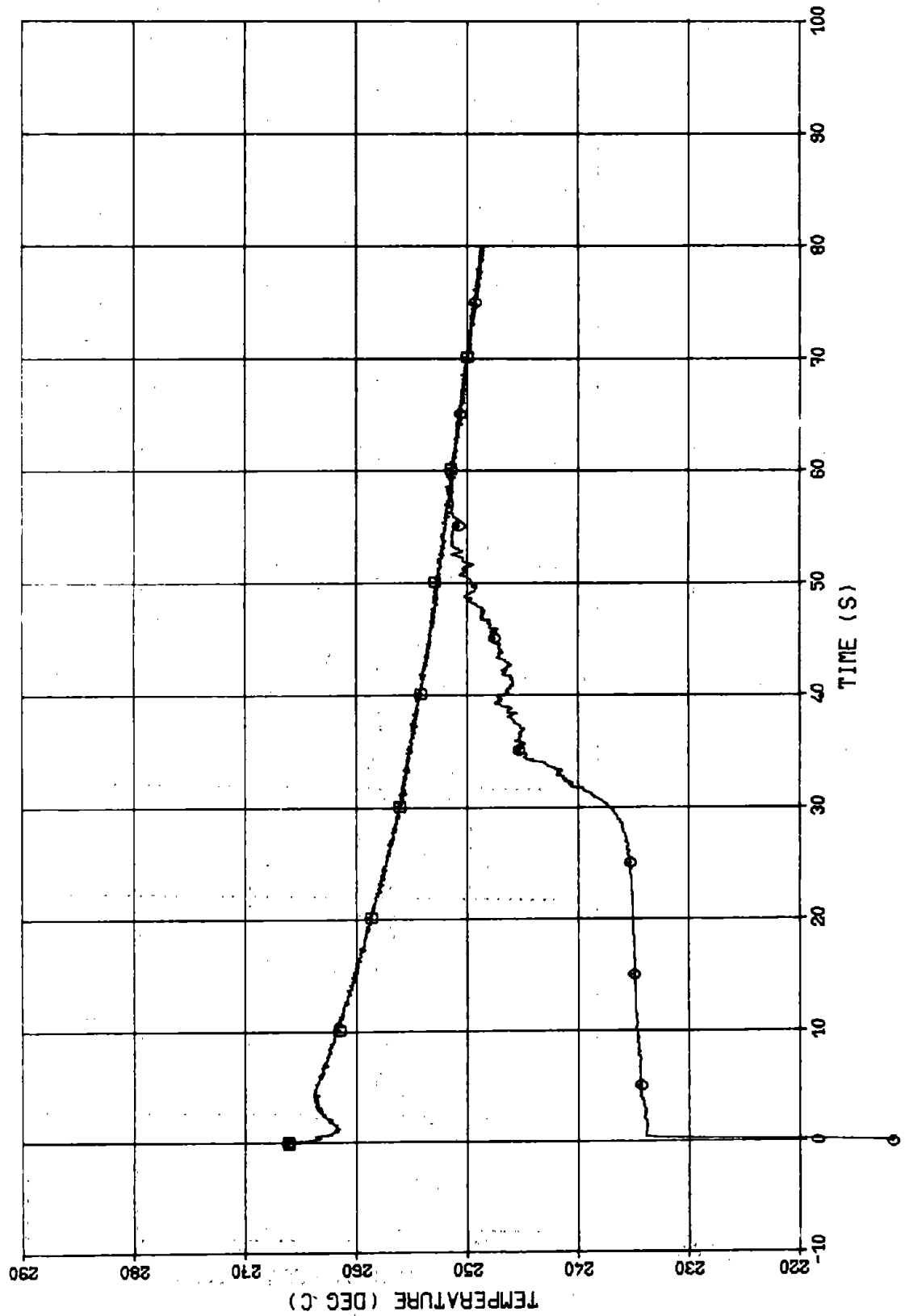
Plot A:17 - Test 17



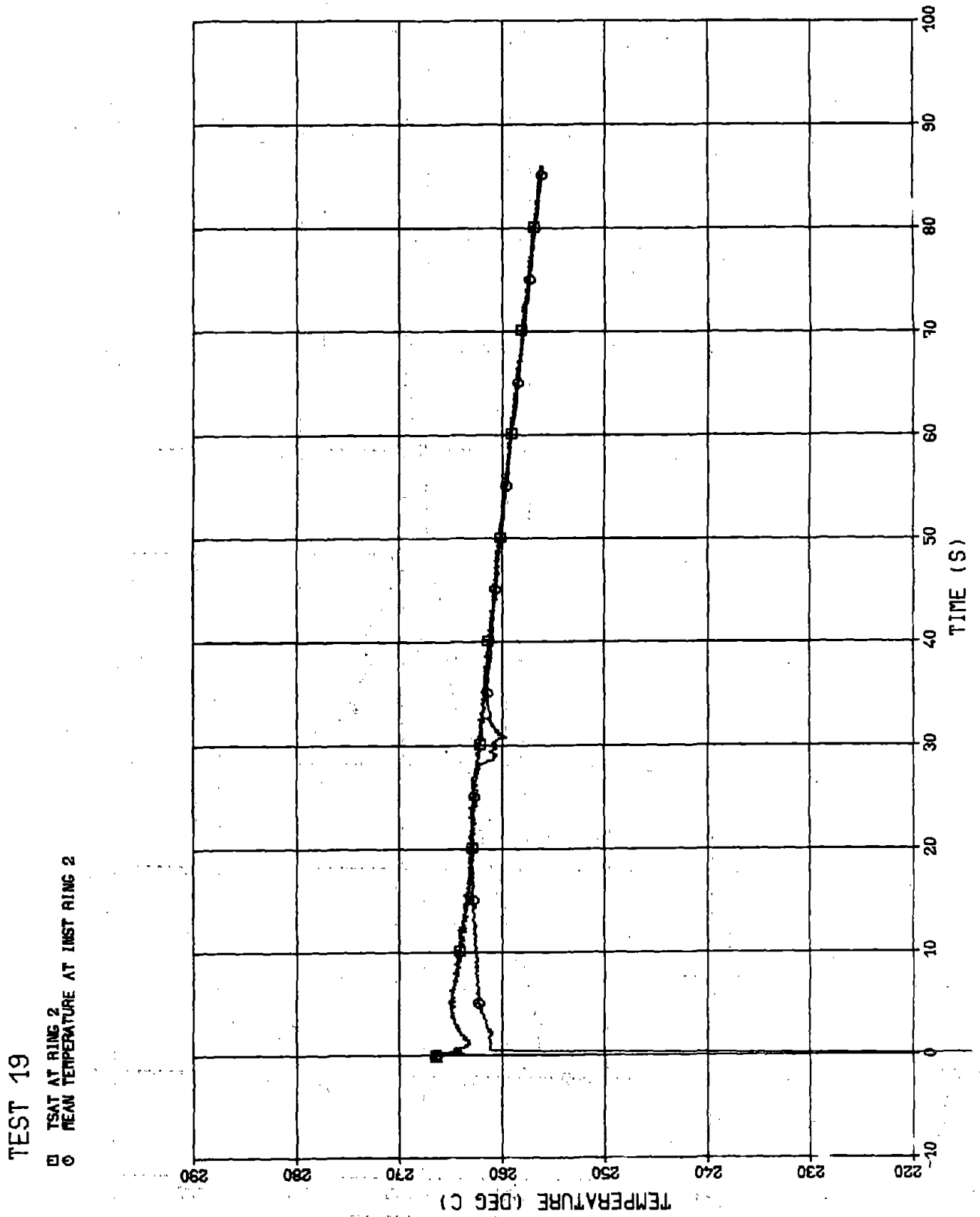
Plot A:18 - Test 18

TEST 18

□ TSAT AT RING 2
○ MEAN TEMPERATURE AT INST RING 2



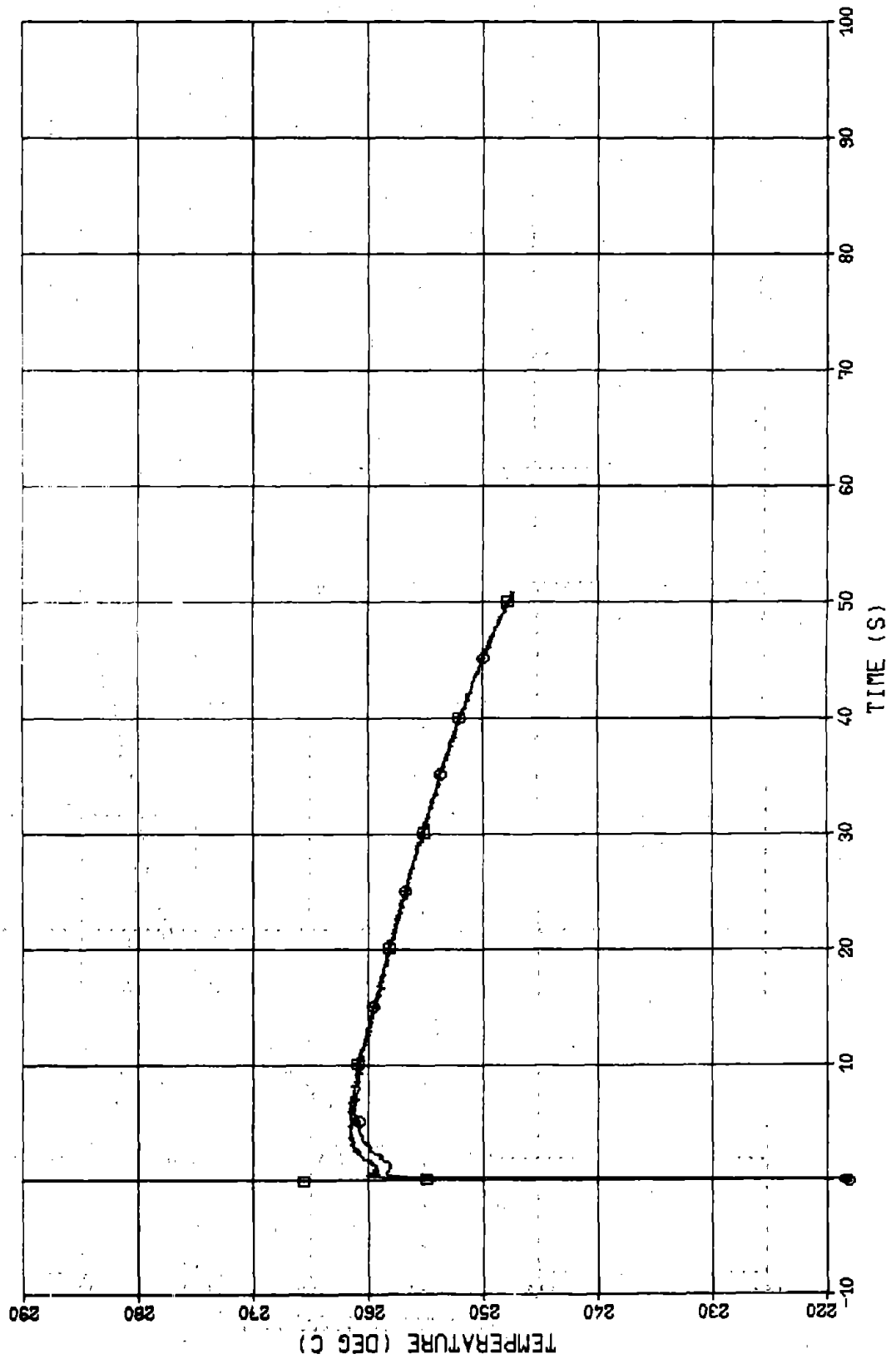
Plot A:19 - Test 19



Plot A:20 - Test 20

TEST 20

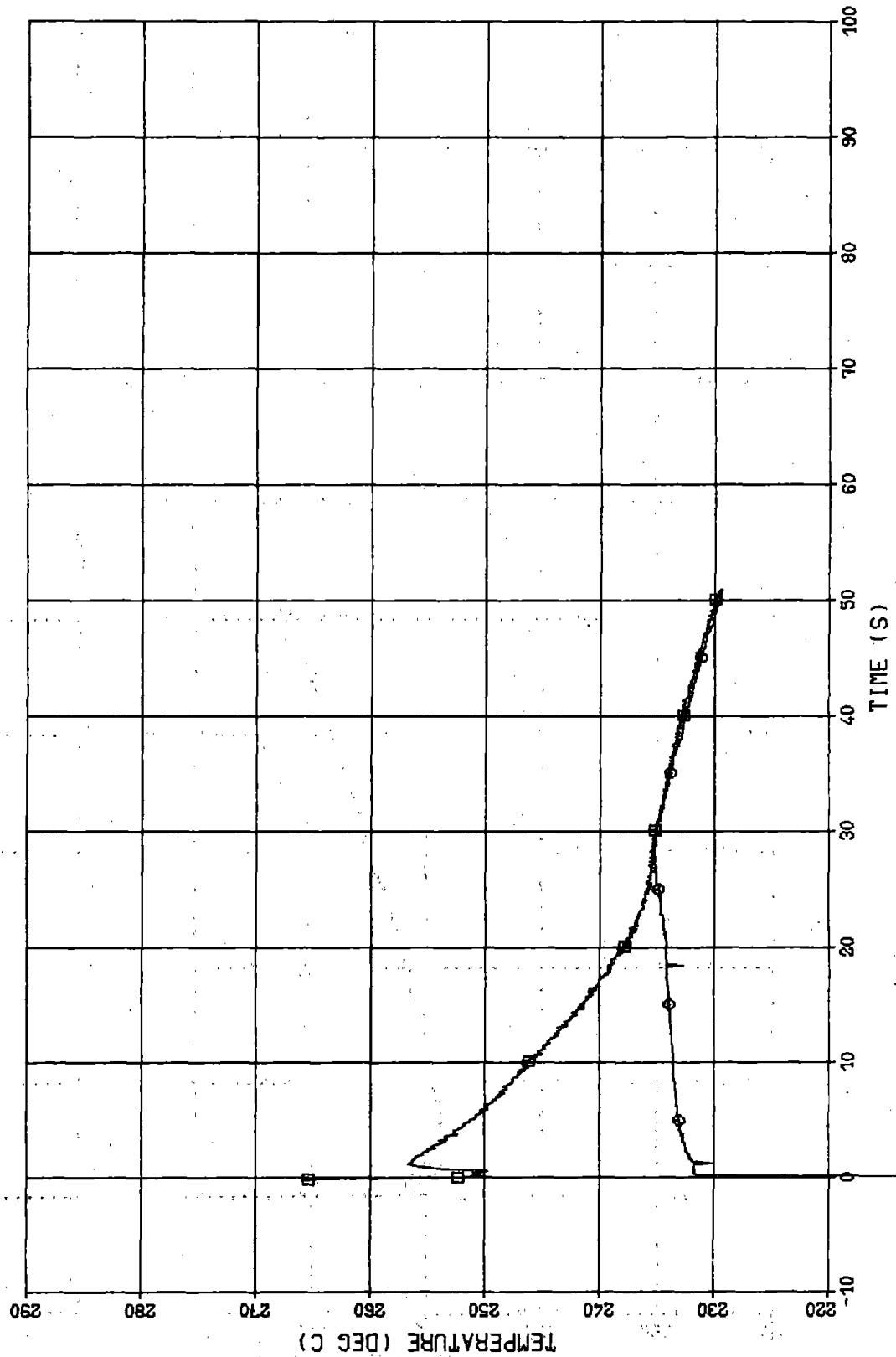
□ TSAT AT RING 2
○ MEAN TEMPERATURE AT INST RING 2



Plot A:21 - Test 21

TEST 21

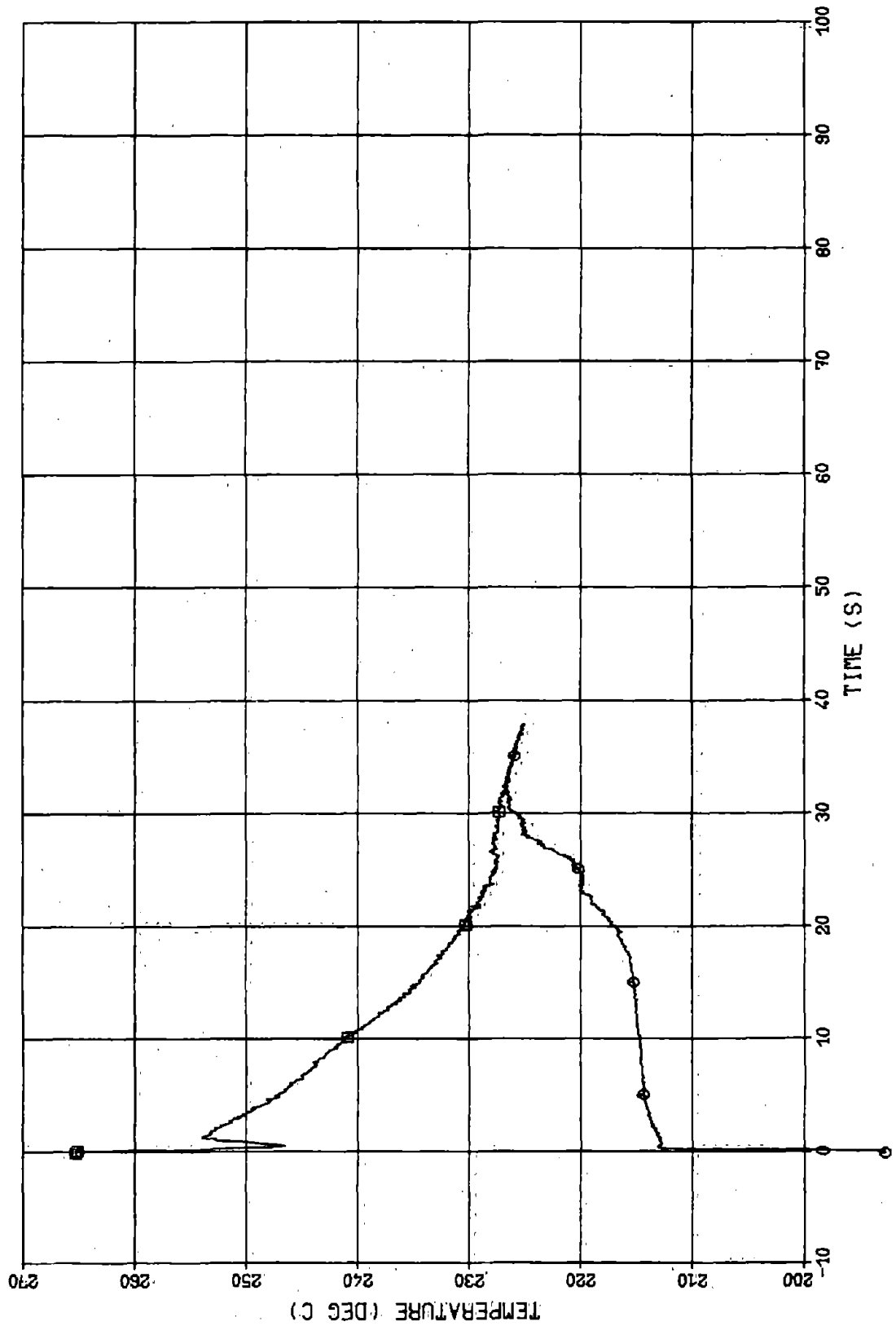
□ TSAT AT RING 2
○ NEW TEMPERATURE AT INST RING 2



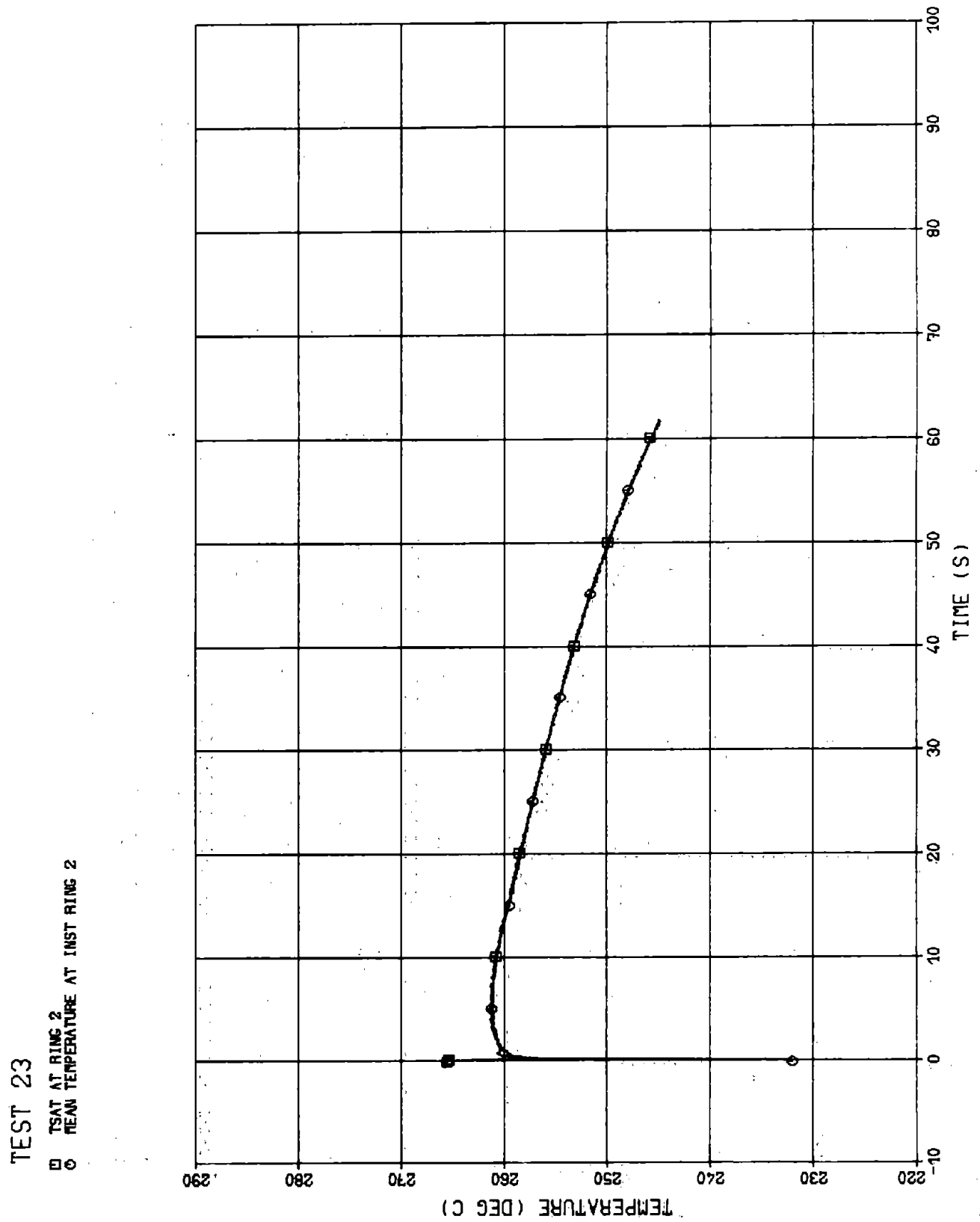
Plot A:22 - Test 22

TEST 22

□ TSAT AT RING 2
○ MEAN TEMPERATURE AT INST RING 2



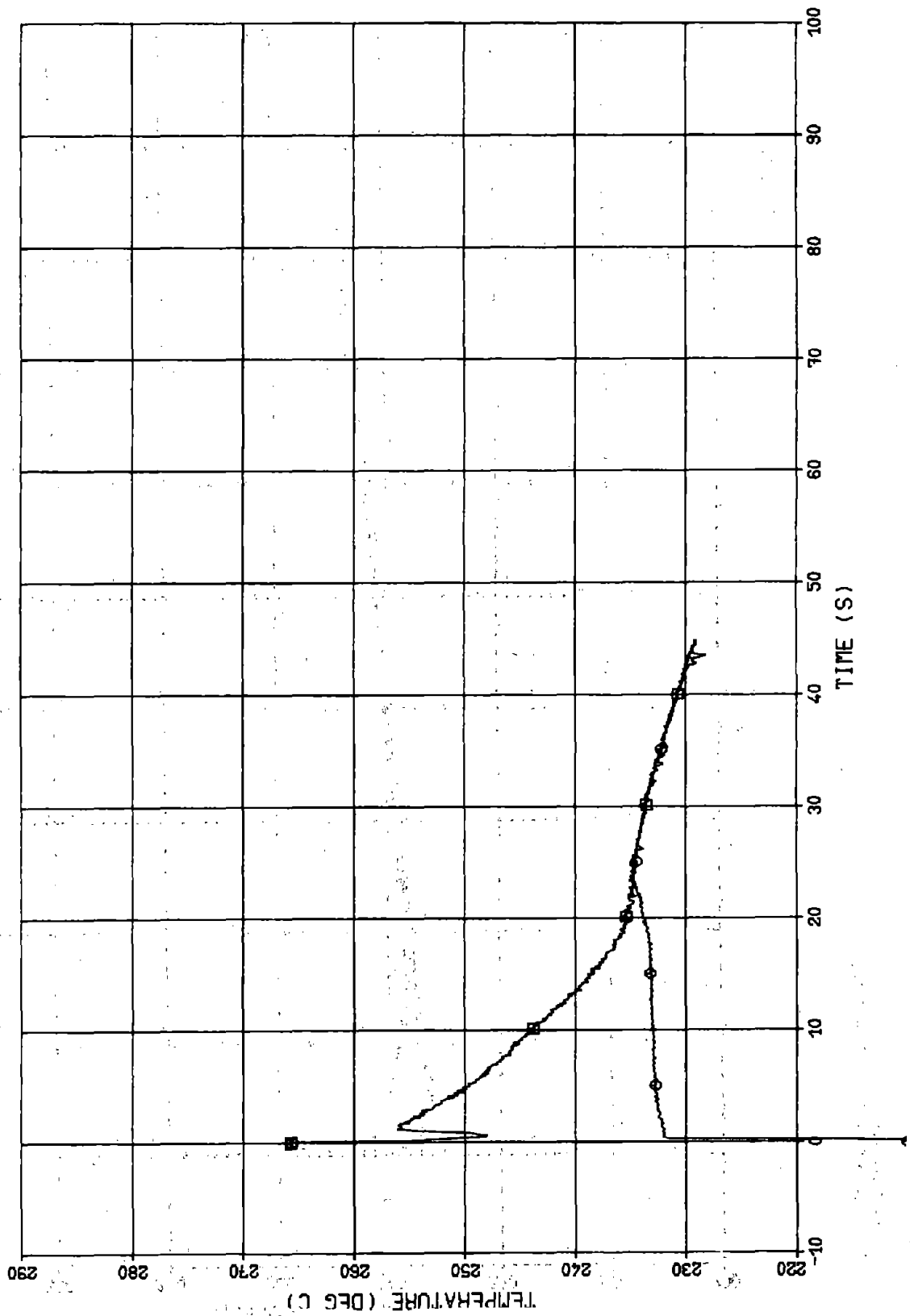
Plot A:23 - Test 23



Plot A:24 - Test 24

TEST 24

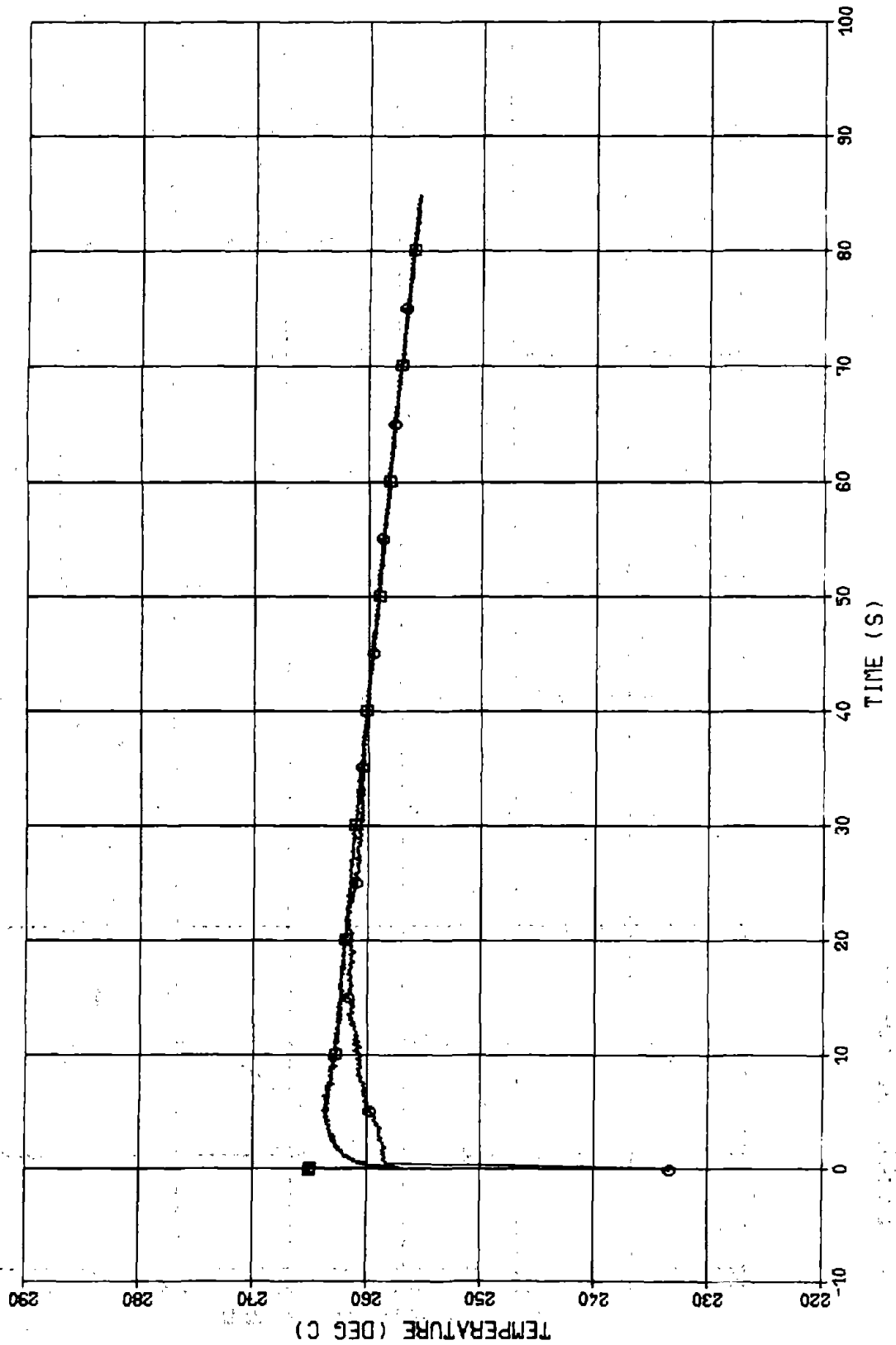
□ TSAT AT RING 2
○ MEAN TEMPERATURE AT INST RING 2



Plot A:25 - Test 25

TEST 25

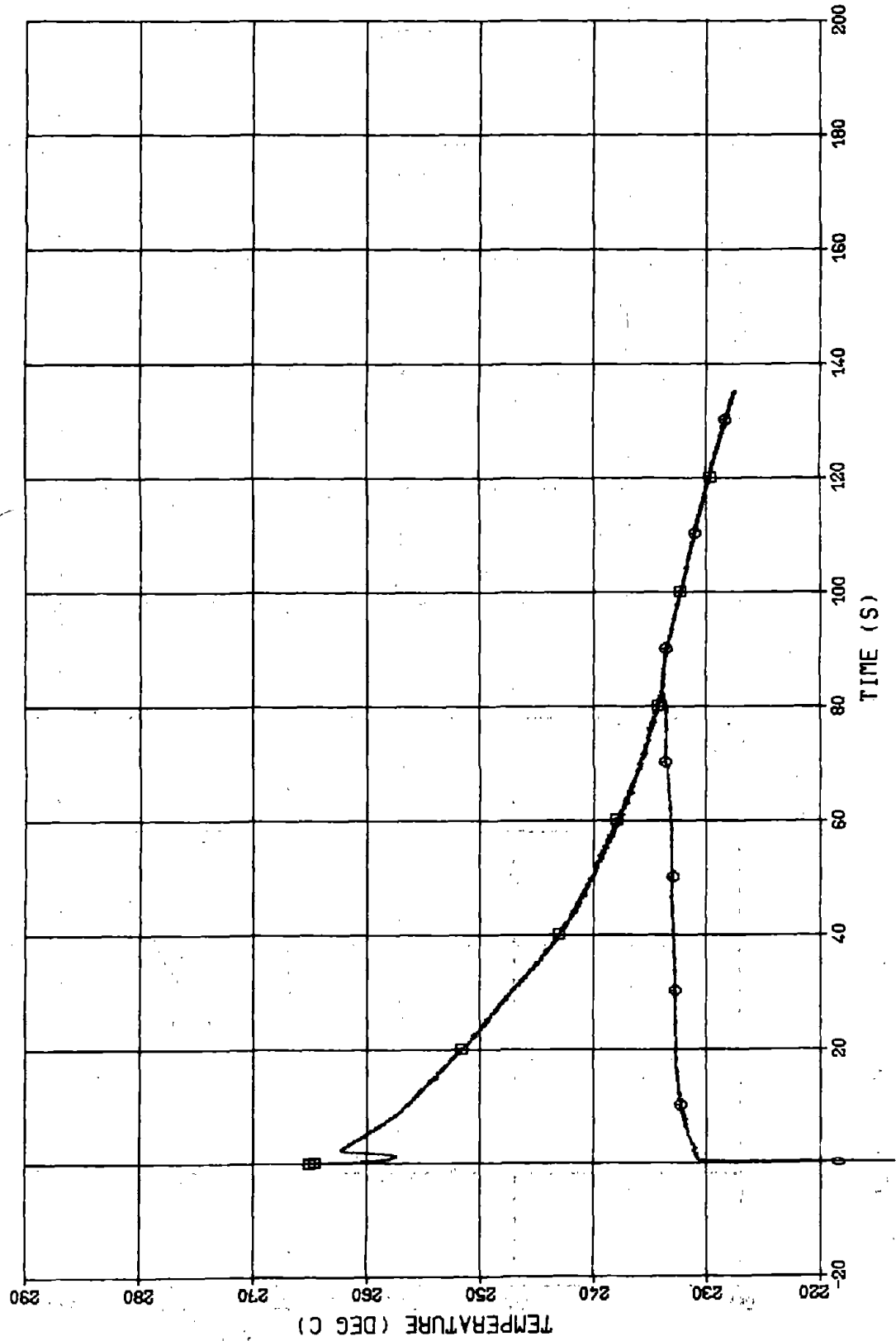
□ TSAT AT RING 2
○ MEAN TEMPERATURE AT INST RING 2



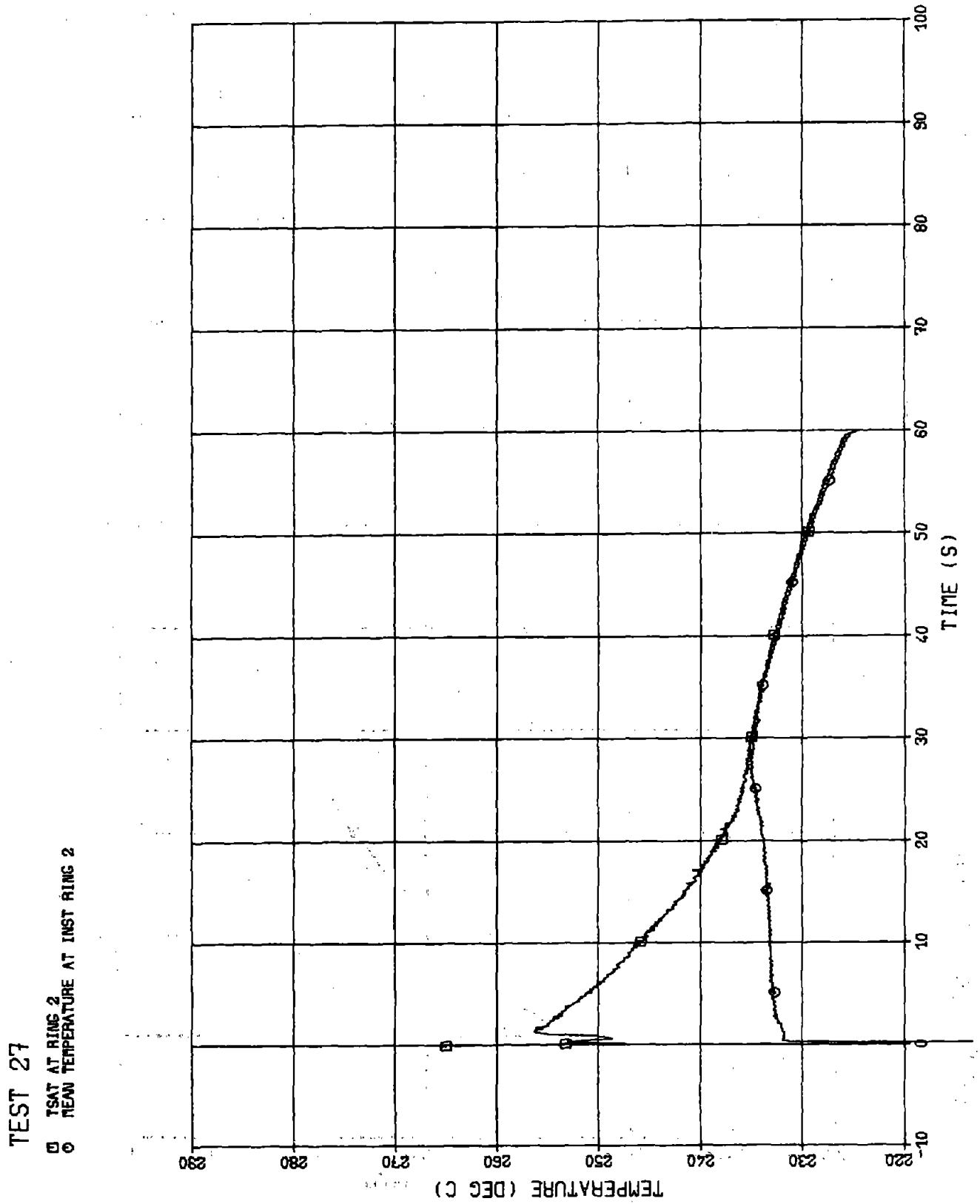
Plot A:26 - Test 26

TEST 26

□ TSAT AT RING 2
○ MEAN TEMPERATURE AT INST RING 2



Plot A:27 - Test 27

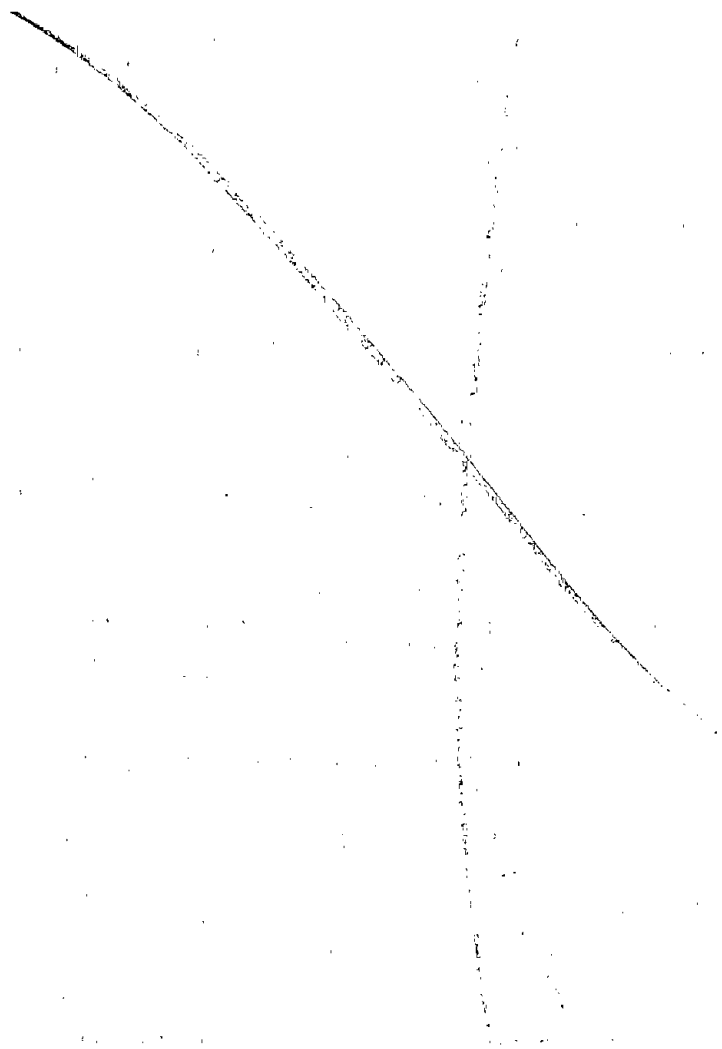




APPENDIX B

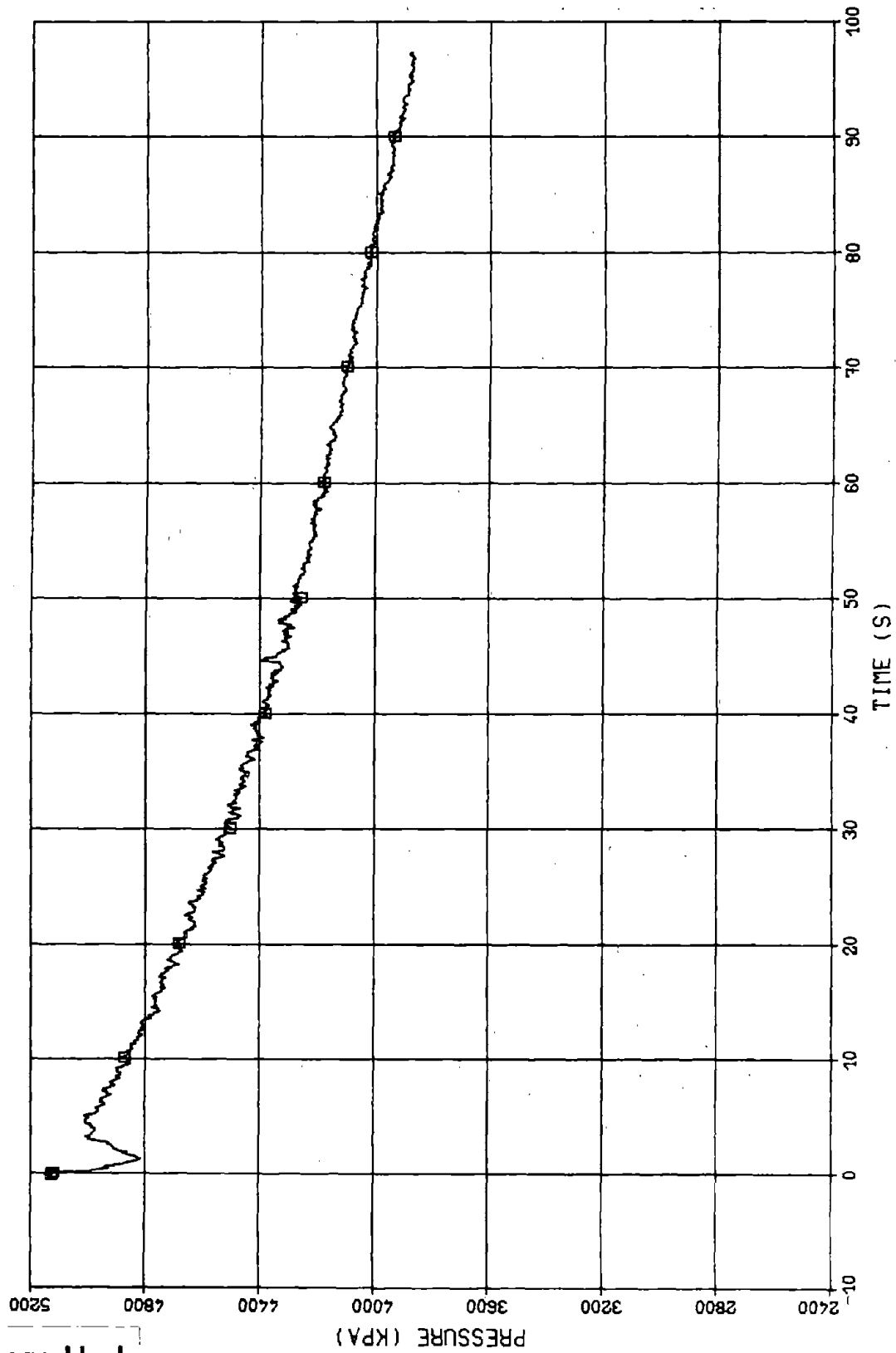
Stagnation pressures at the nozzle inlet plane: Tests 1
through 27

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Plot B:1 - Test 1

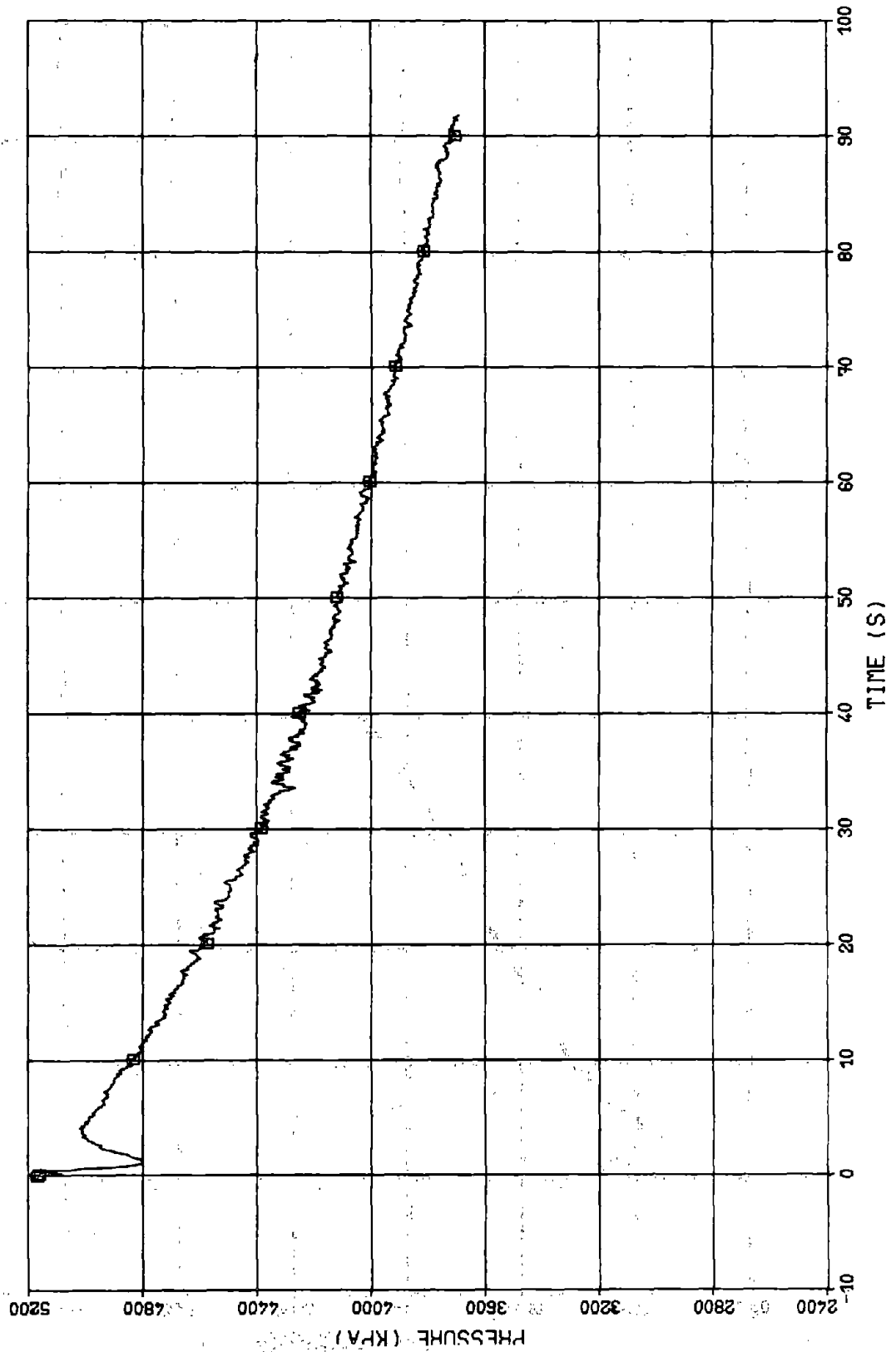
TEST 01
INLET PRESSURE



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Plot B:2 - Test 2

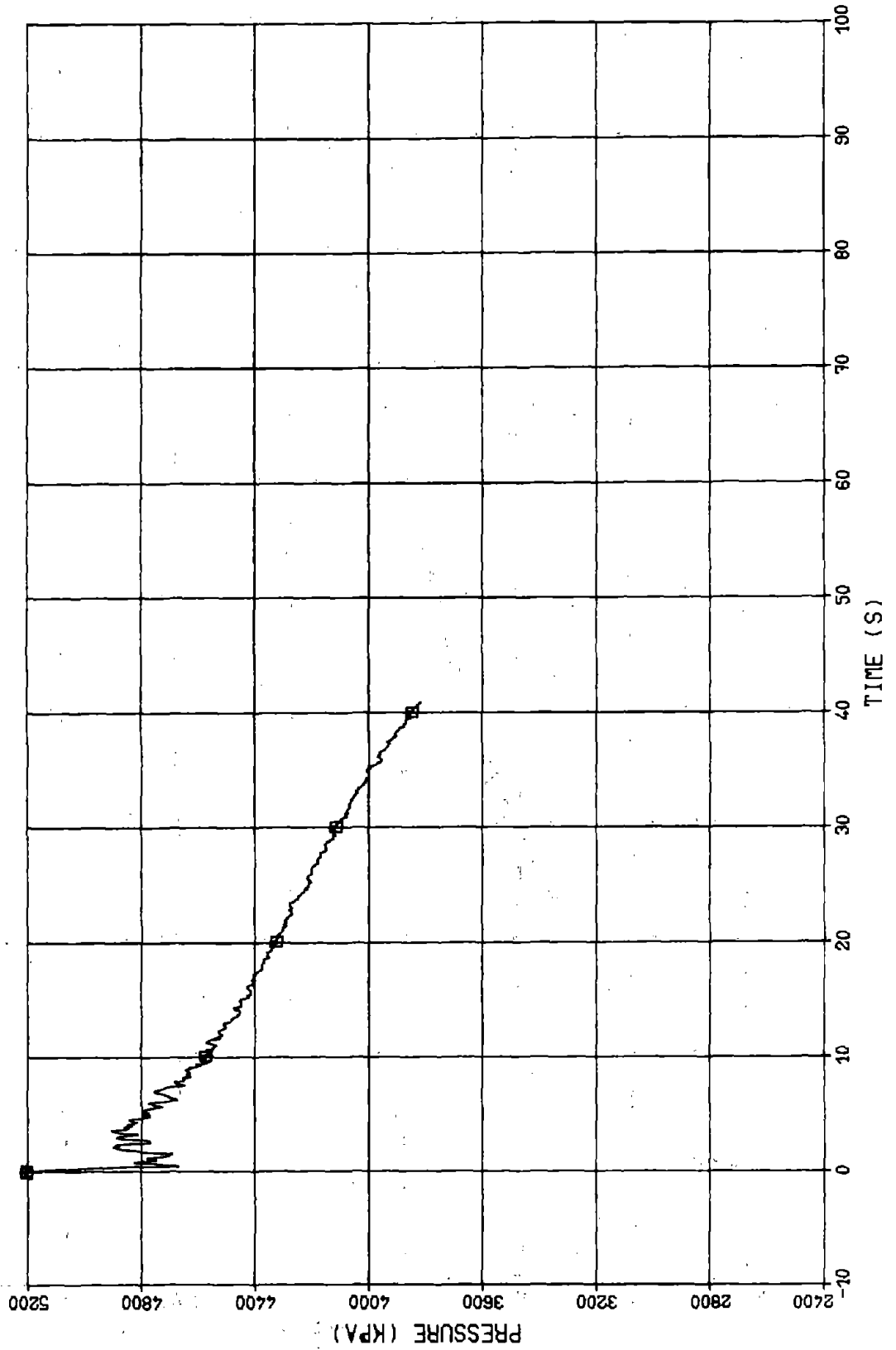
TEST 2
INLET PRESSURE



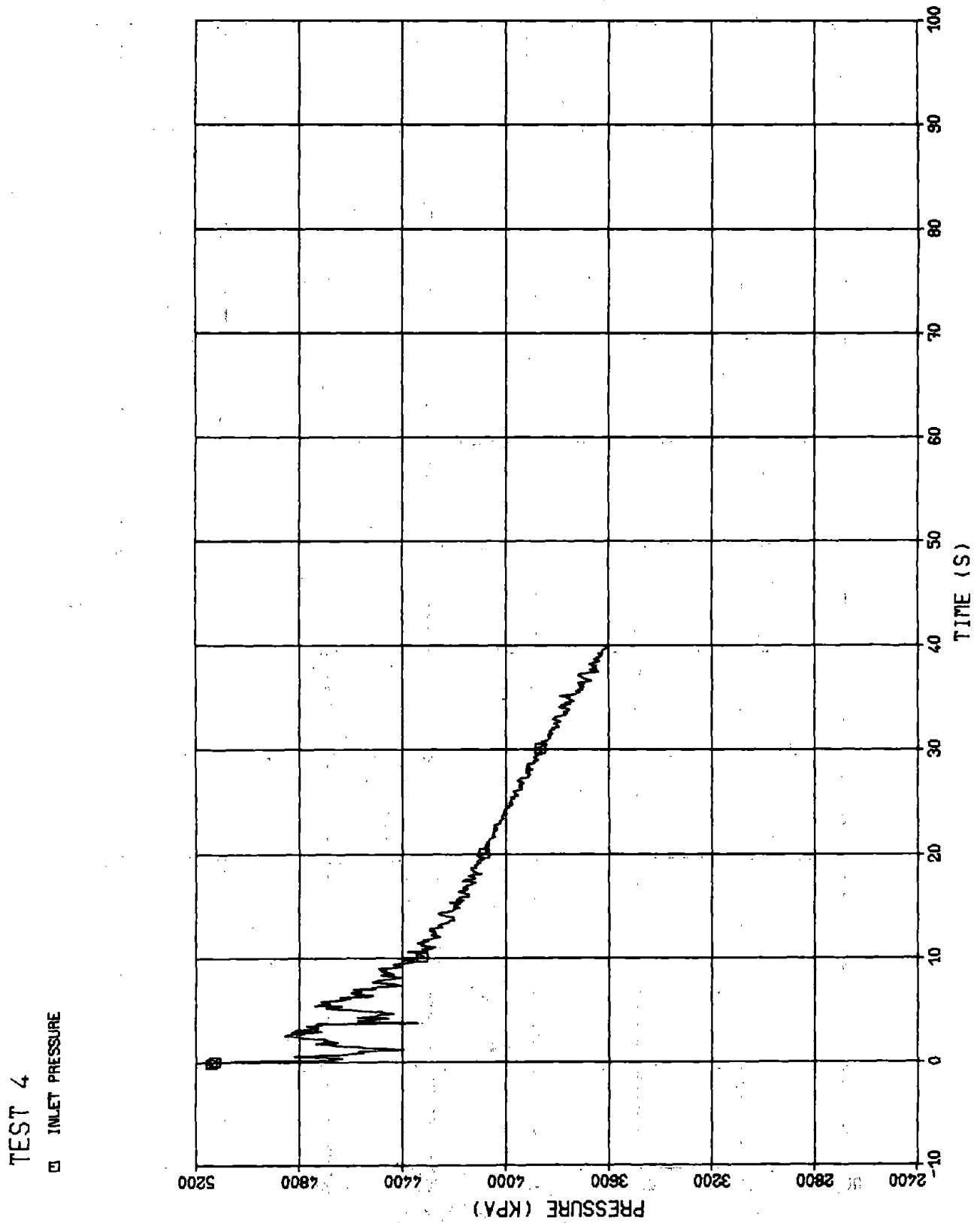
Plot B:3 - Test 3

TEST 3

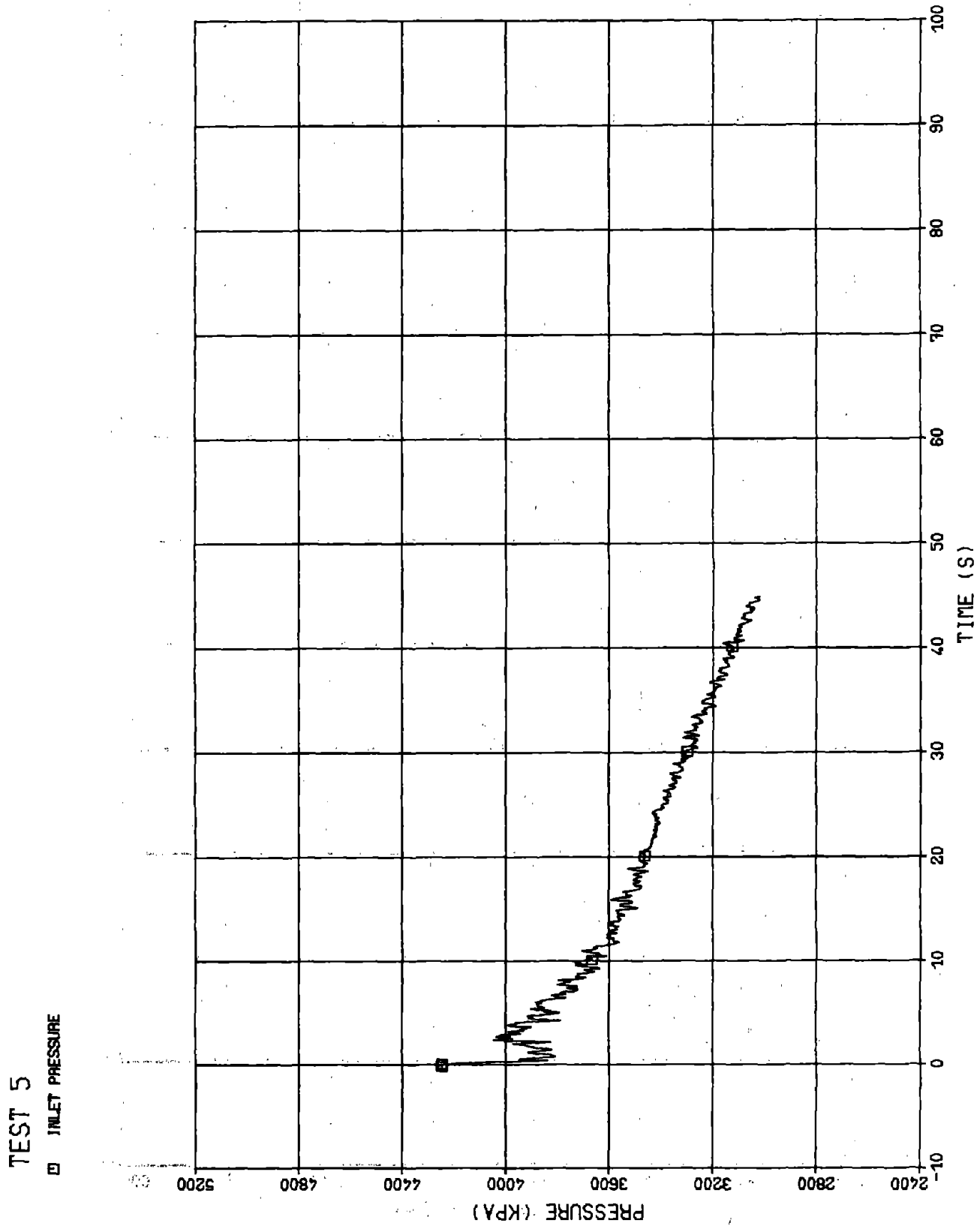
INLET PRESSURE



Plot B:4 - Test 4



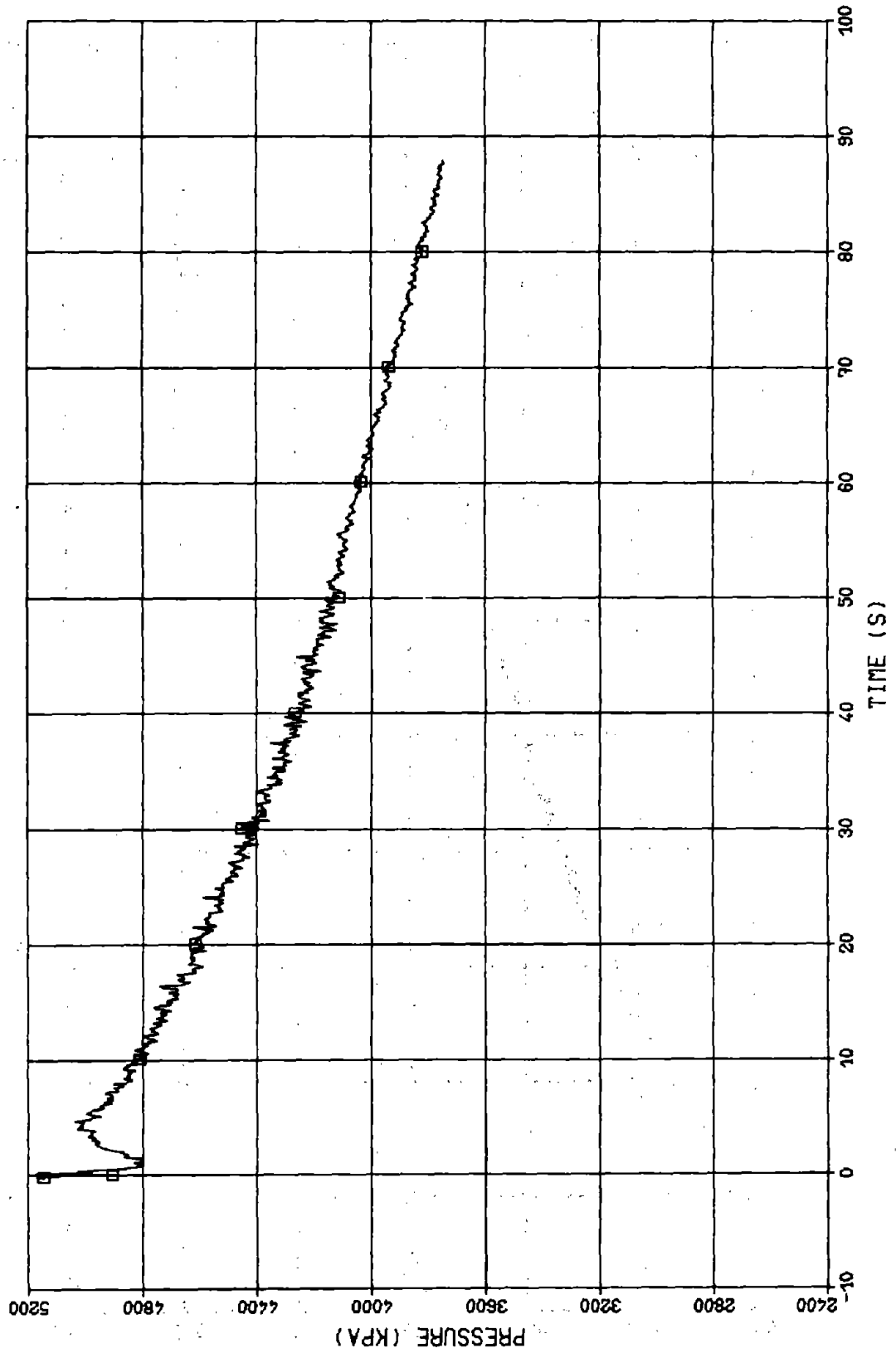
Plot B:5 - Test 5



Plot B:6 - Test 6

TEST 6

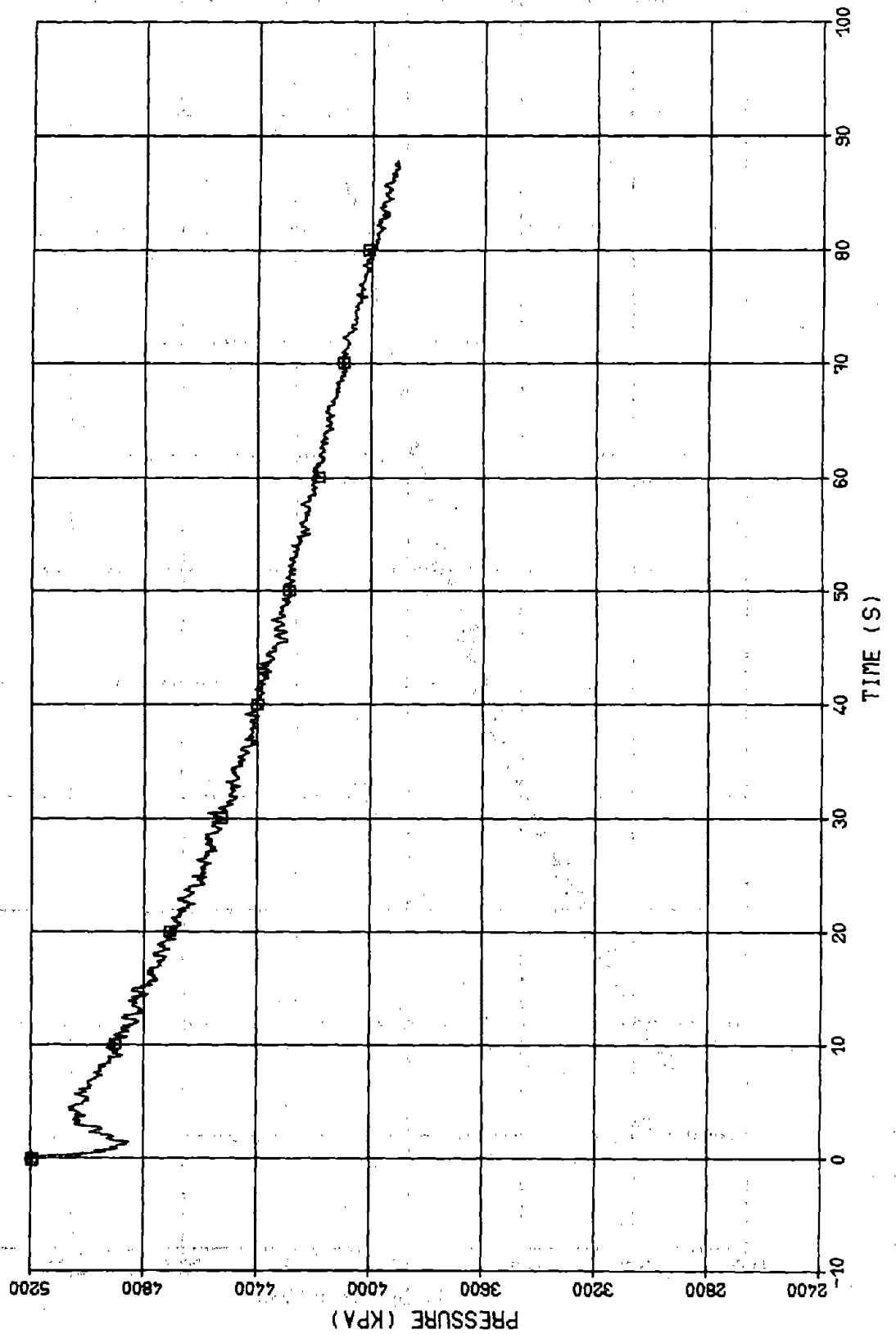
□ INLET PRESSURE



Plot B:7 - Test 7

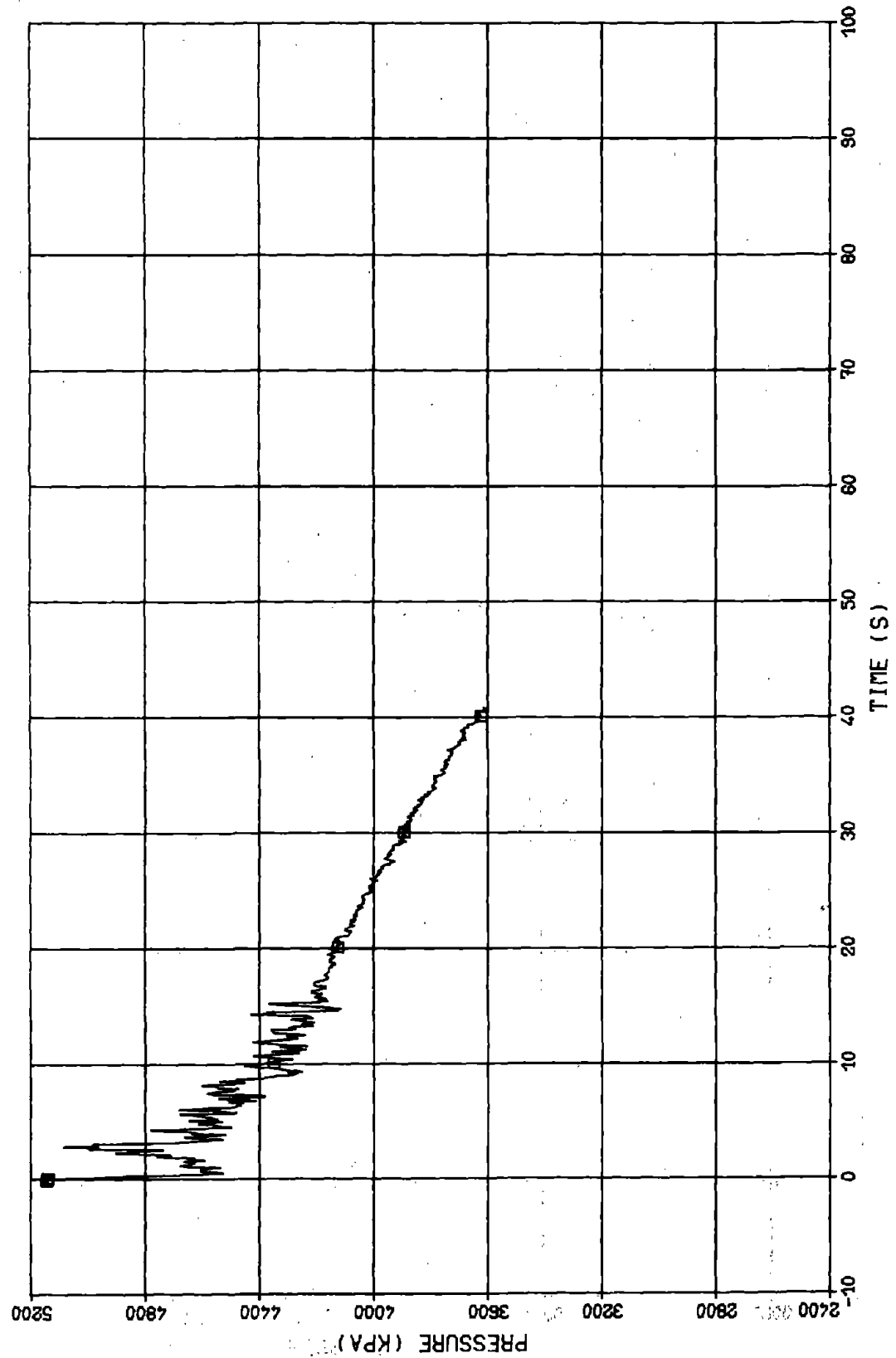
TEST 7

□ INLET PRESSURE



Plot B:8 - Test 8

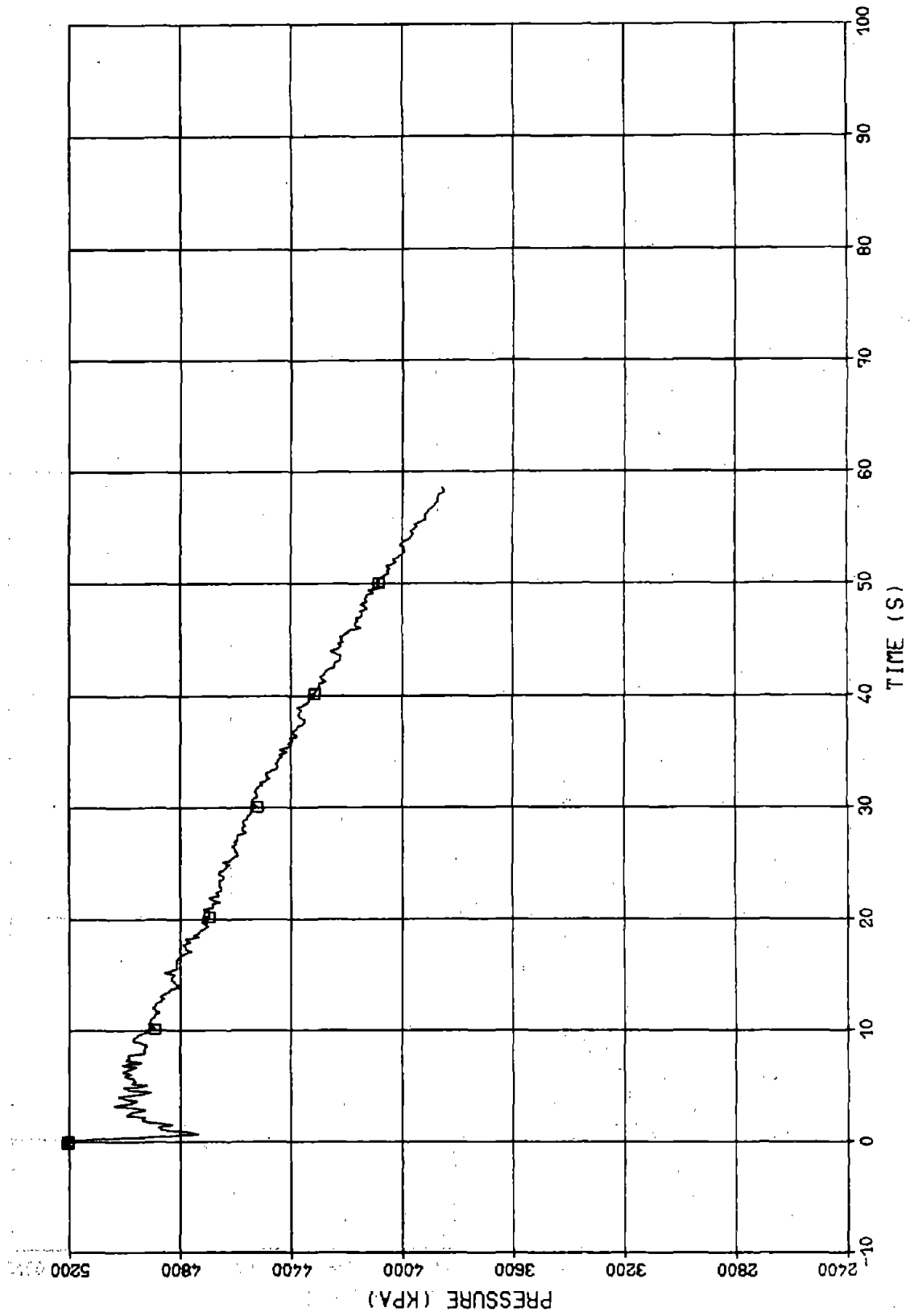
TEST 8
INLET PRESSURE



Plot B:9 - Test 9

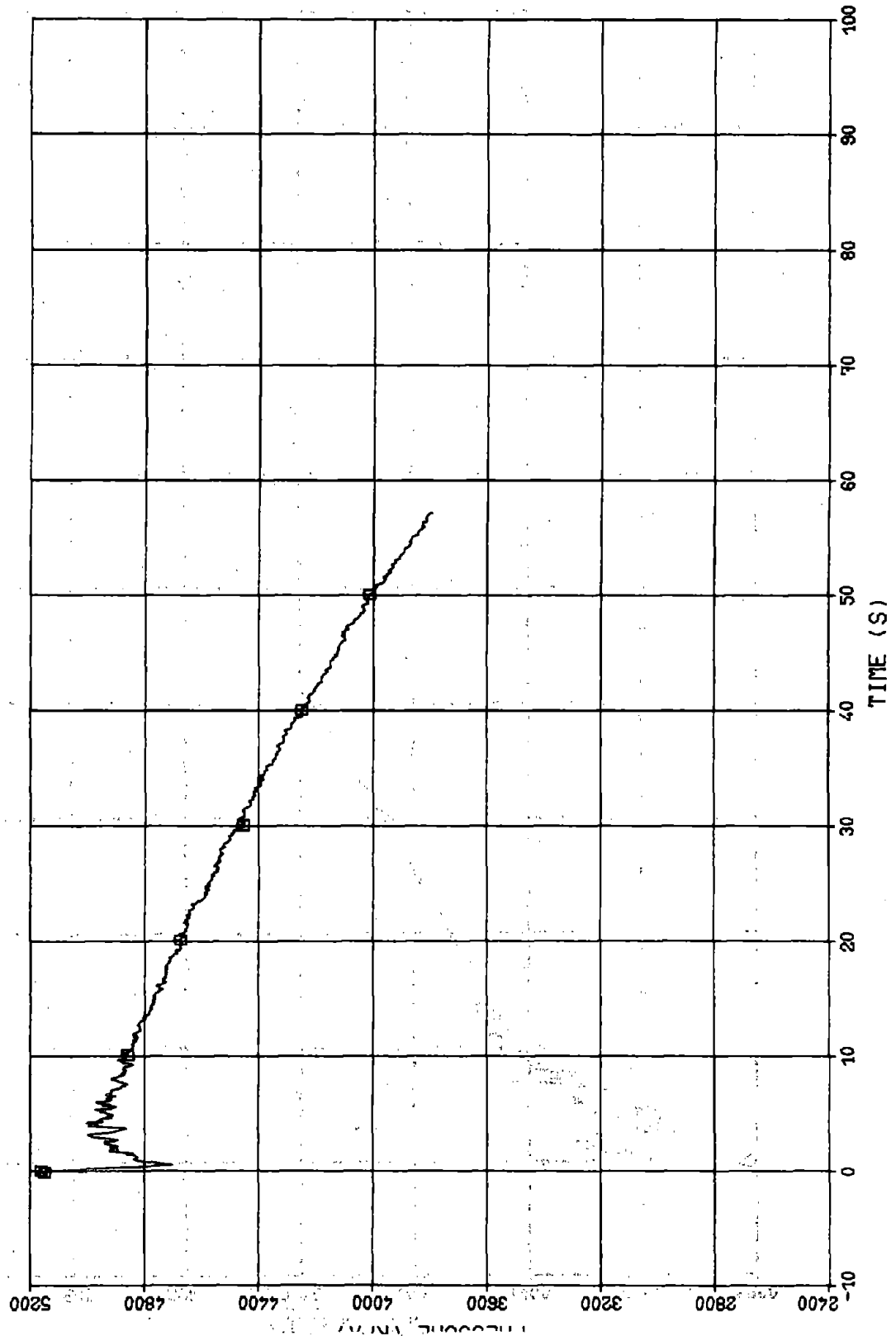
TEST 9

INLET PRESSURE

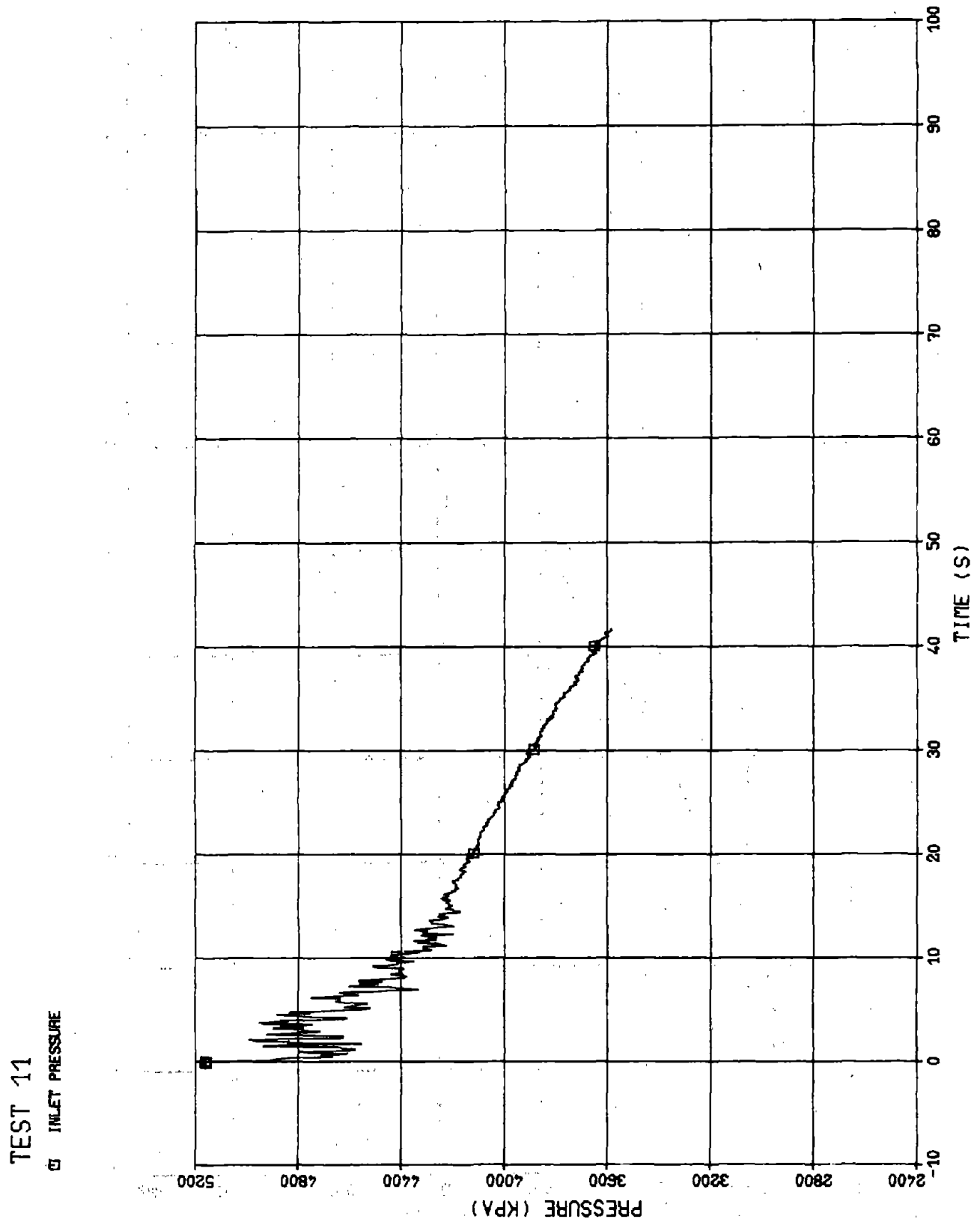


Plot B:10 - Test 10

TEST 10
INLET PRESSURE

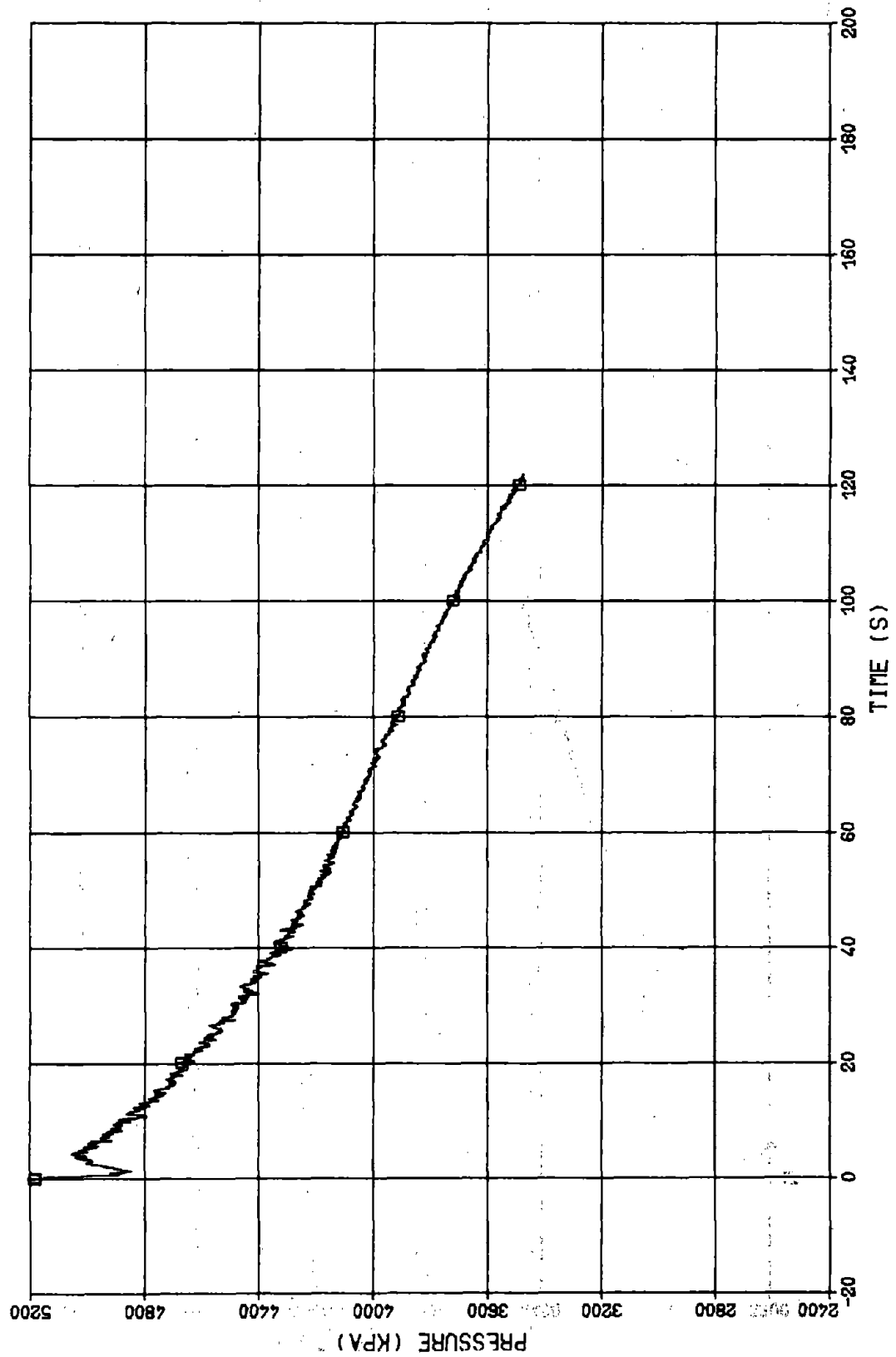


Plot B:11 - Test 11



Plot B:12 - Test 12

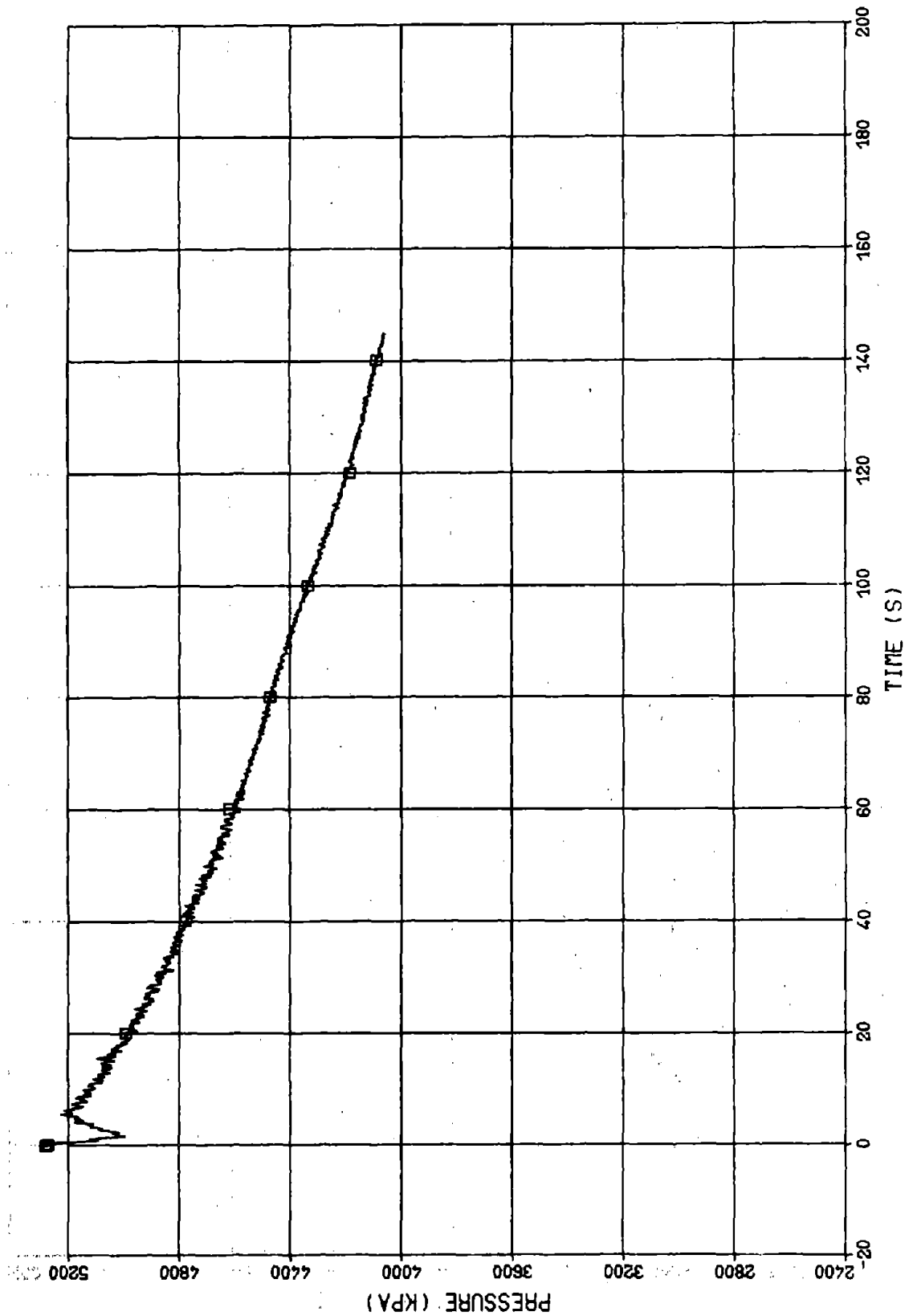
TEST 12
□ INLET PRESSURE



Plot B:13 - Test 13

TEST 13

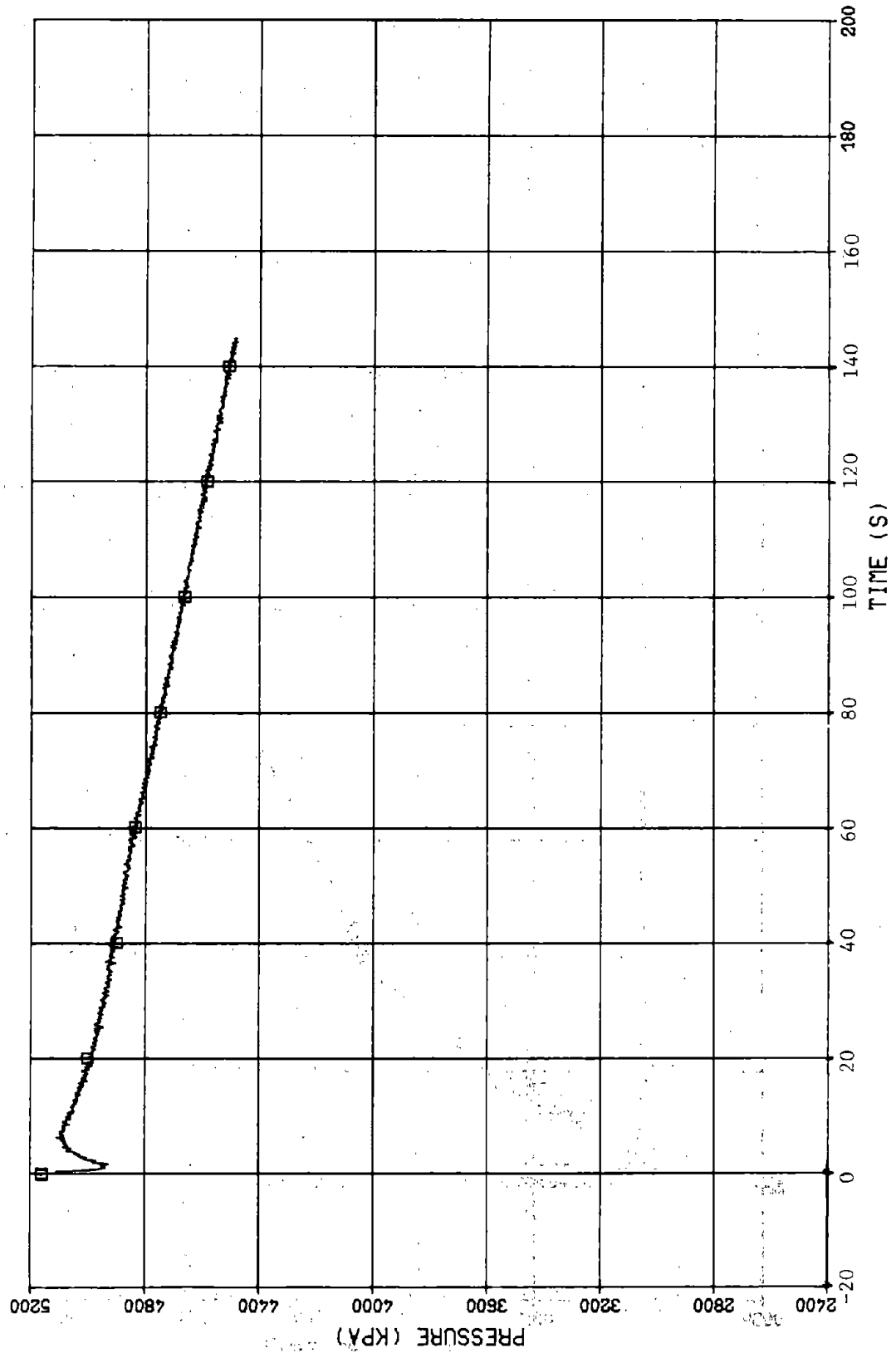
□ INLET PRESSURE



Plot B:14 - Test 14

TEST 14

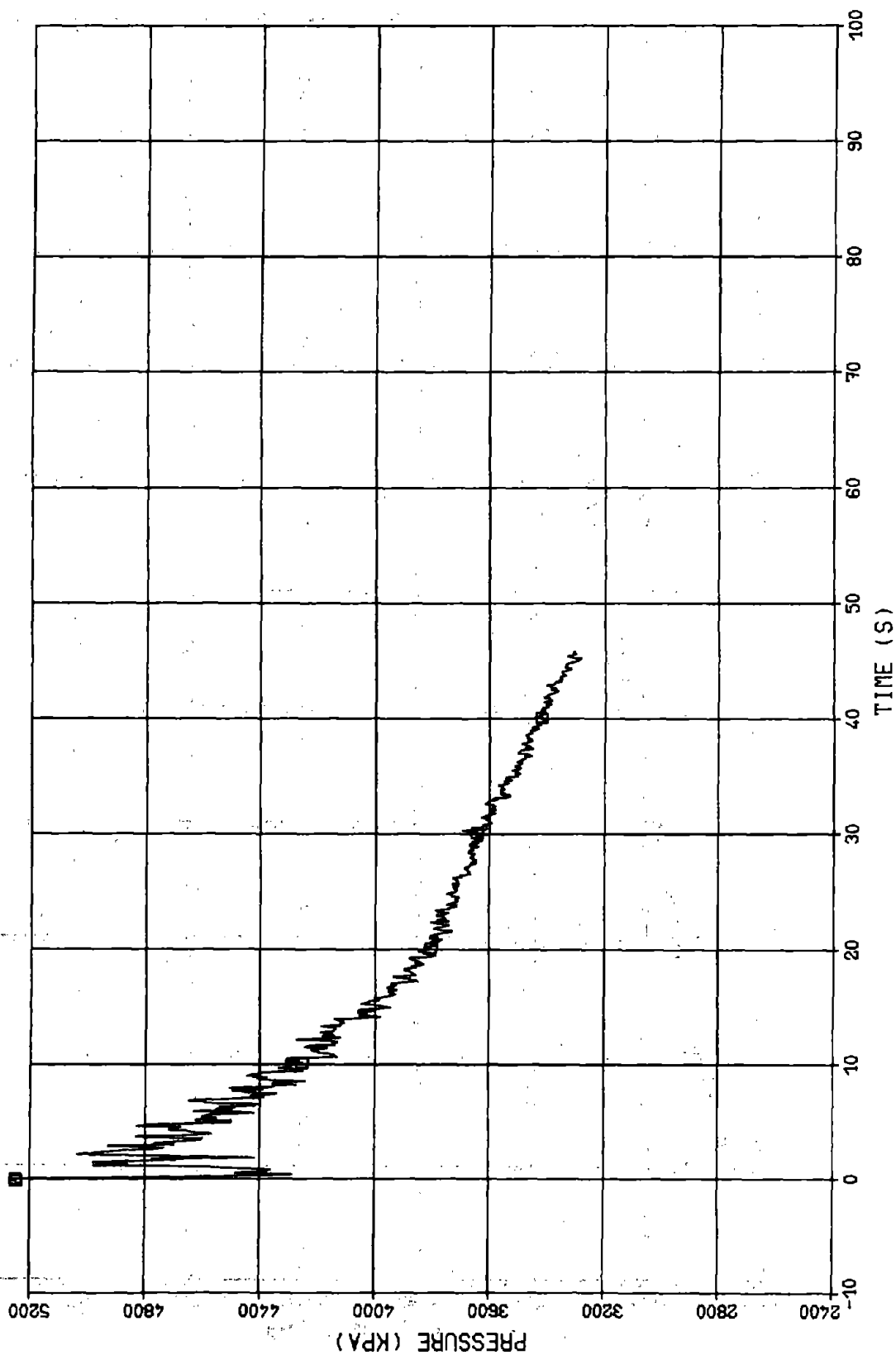
□ INLET PRESSURE



Plot B:15 - Test 15

TEST 15

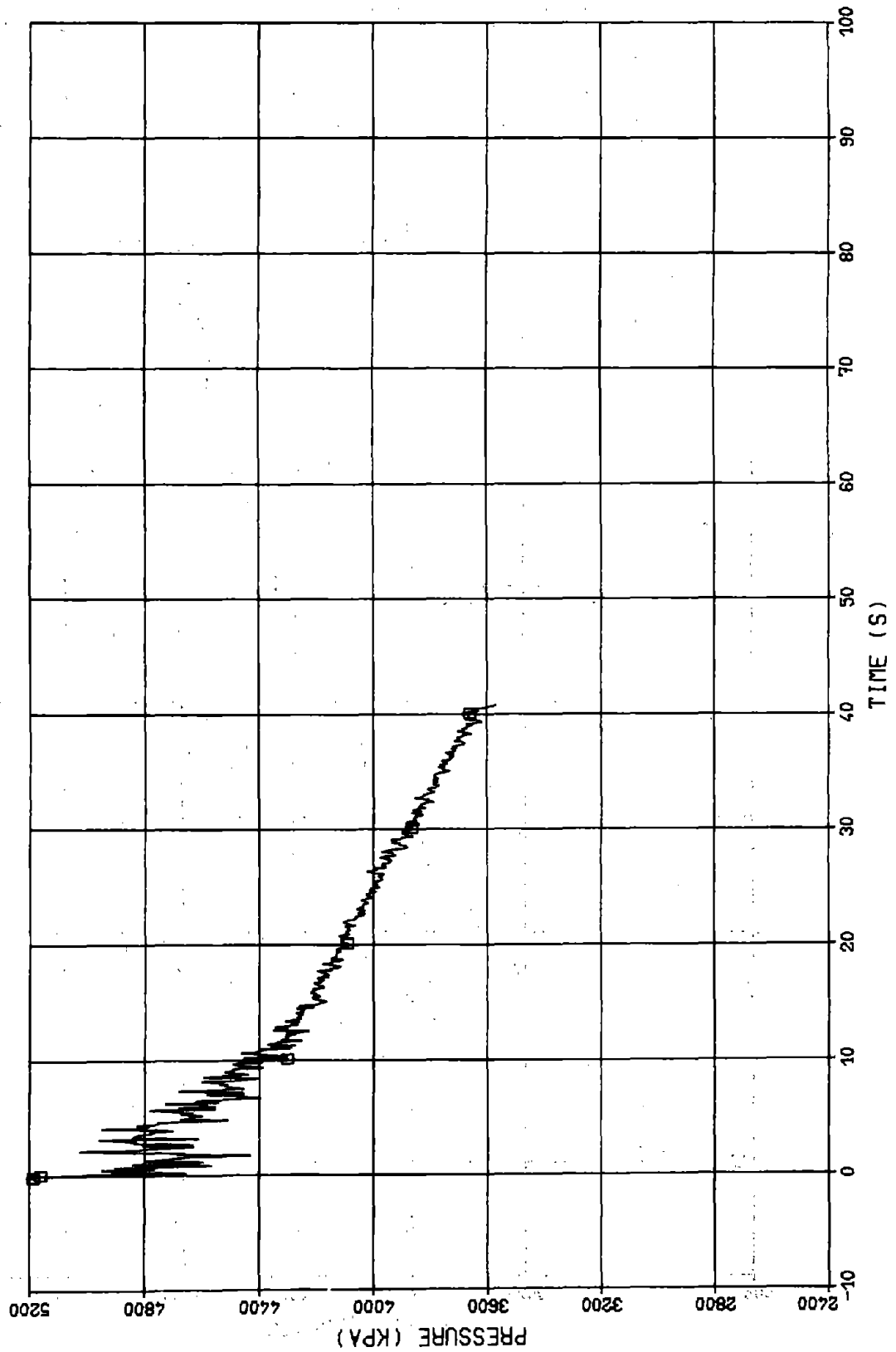
□ INLET PRESSURE



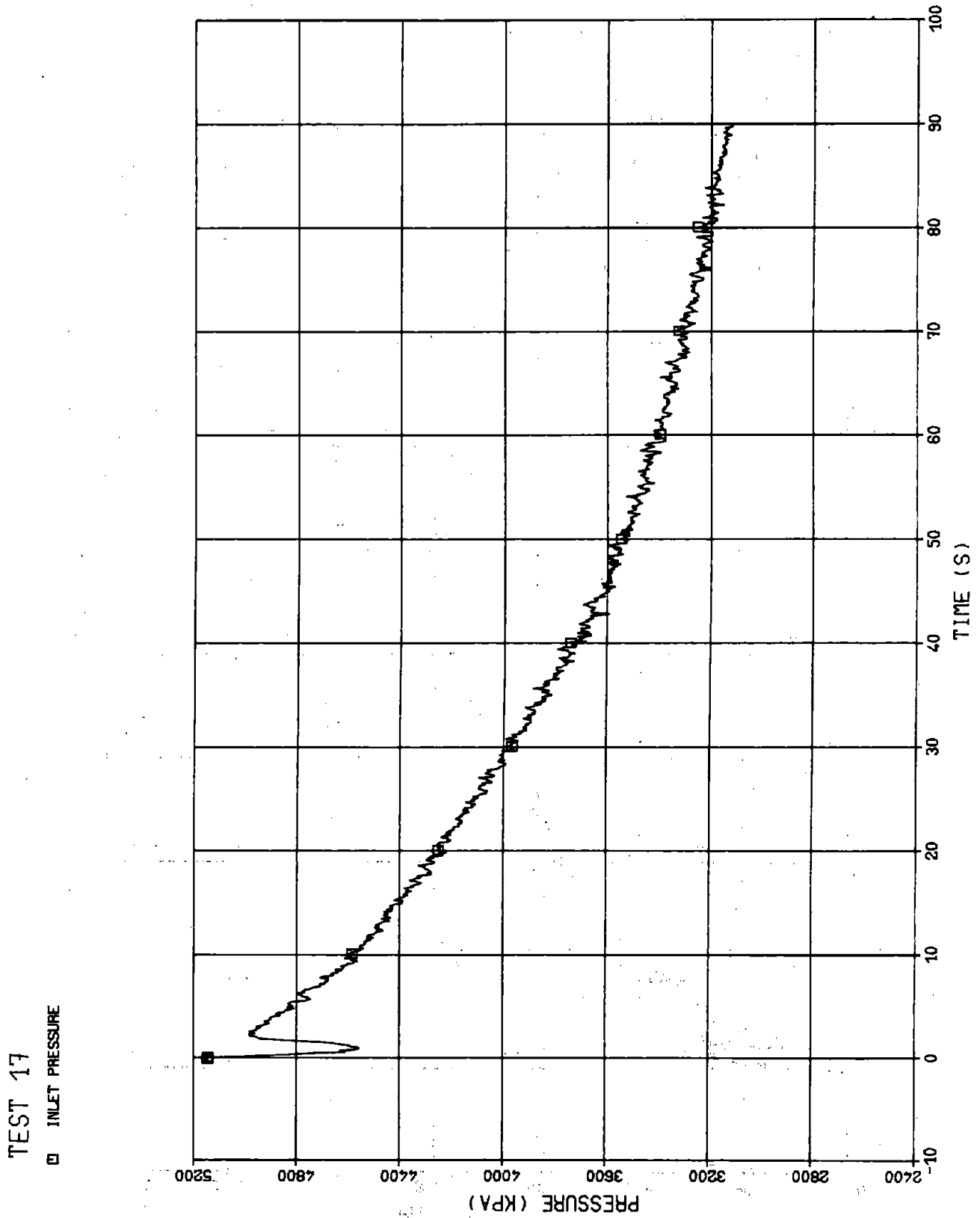
Plot B:16 - Test 16

TEST 16

INLET PRESSURE



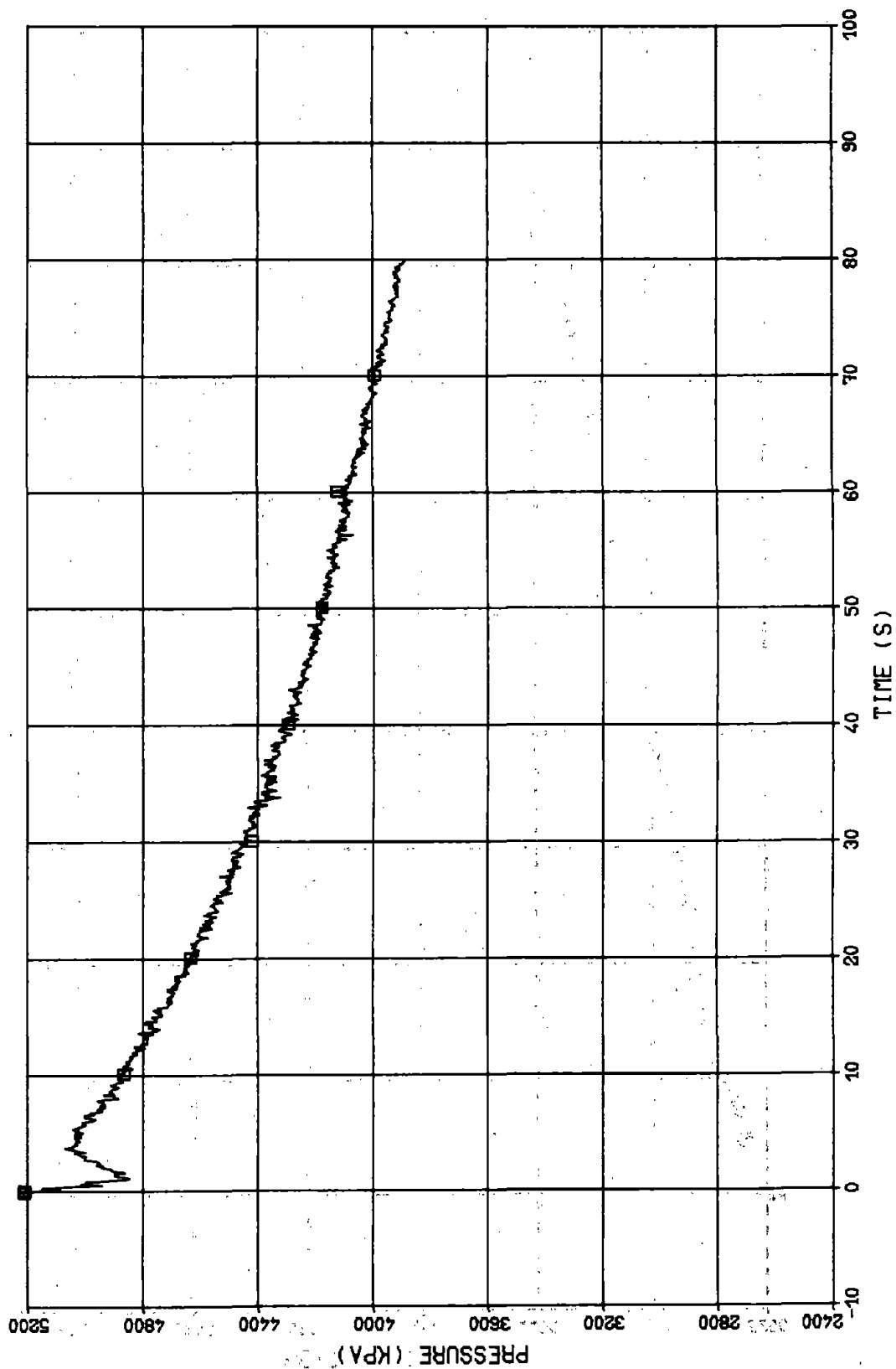
Plot B:17 - Test 17



Plot B:18 - Test 18

TEST 18

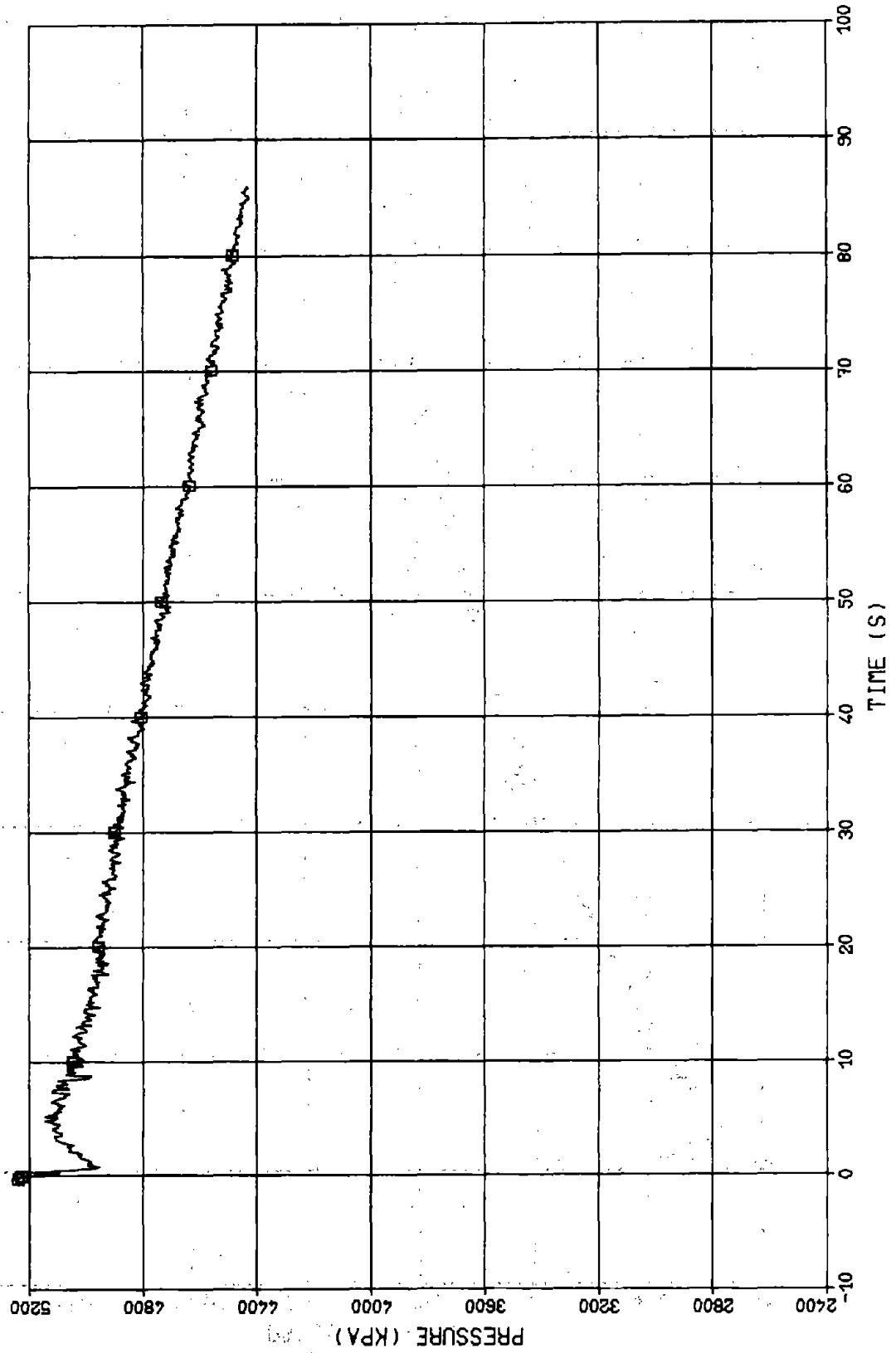
□ INLET PRESSURE



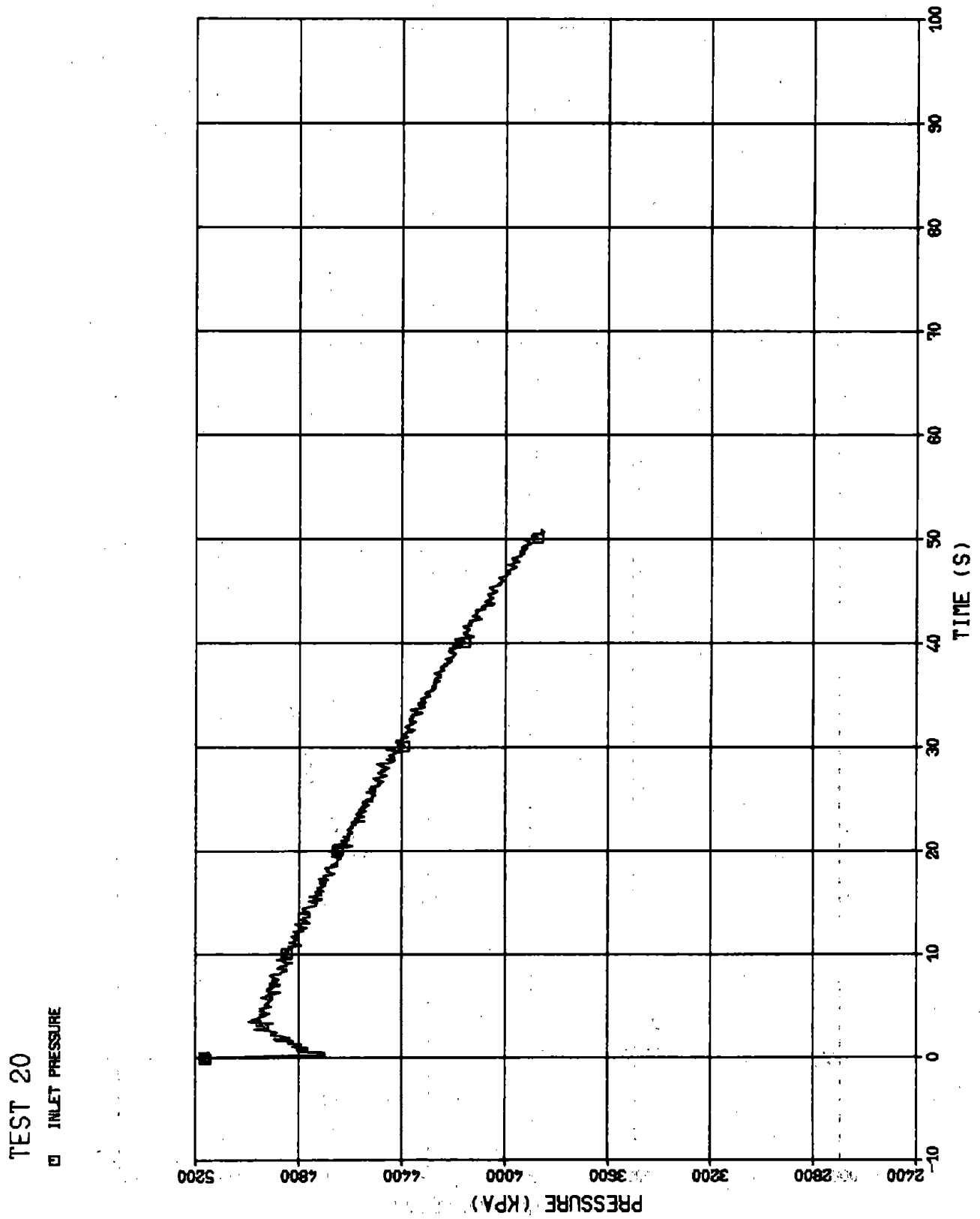
Plot B:19 - Test 19

TEST 19

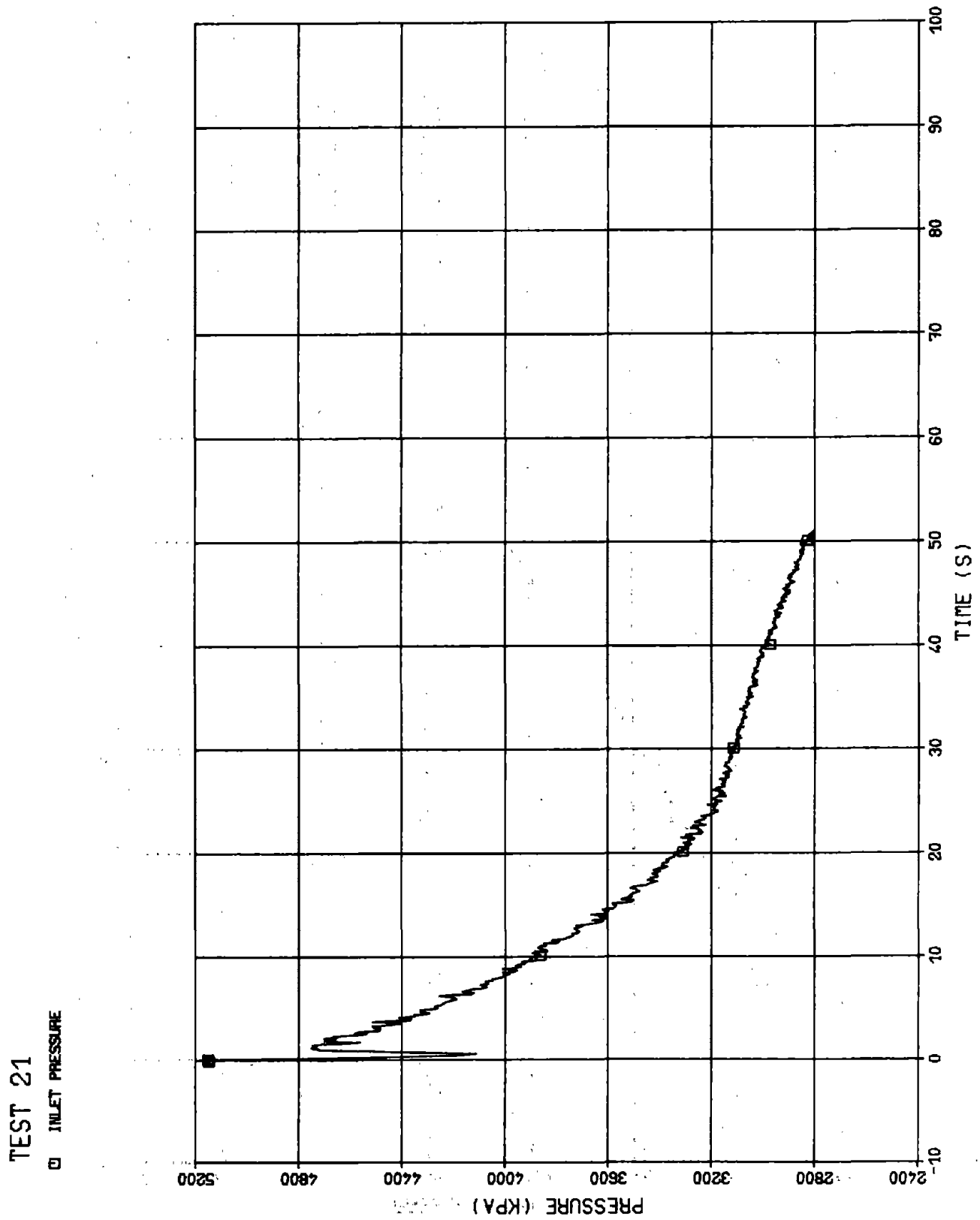
□ INLET PRESSURE



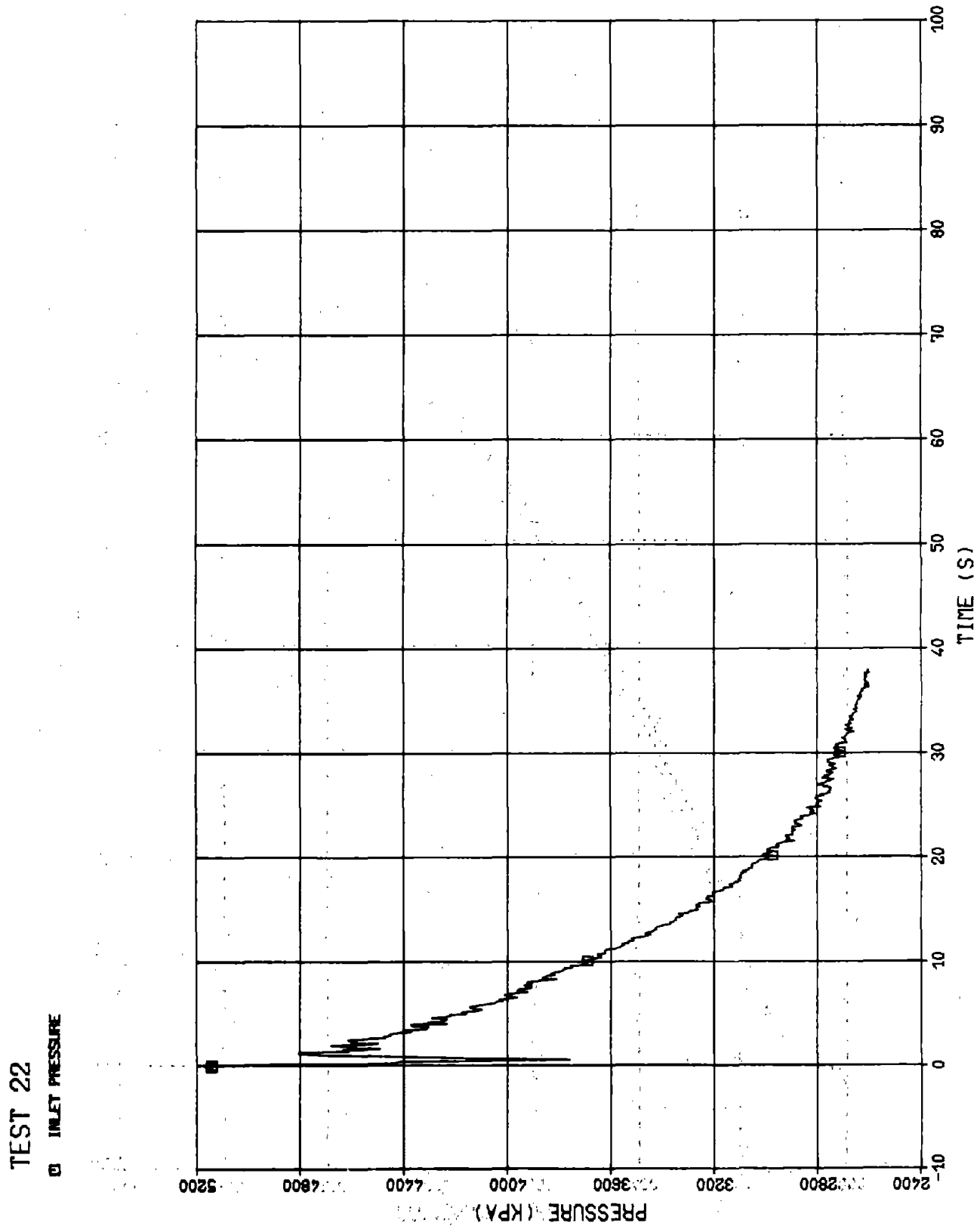
Plot B:20 - Test 20



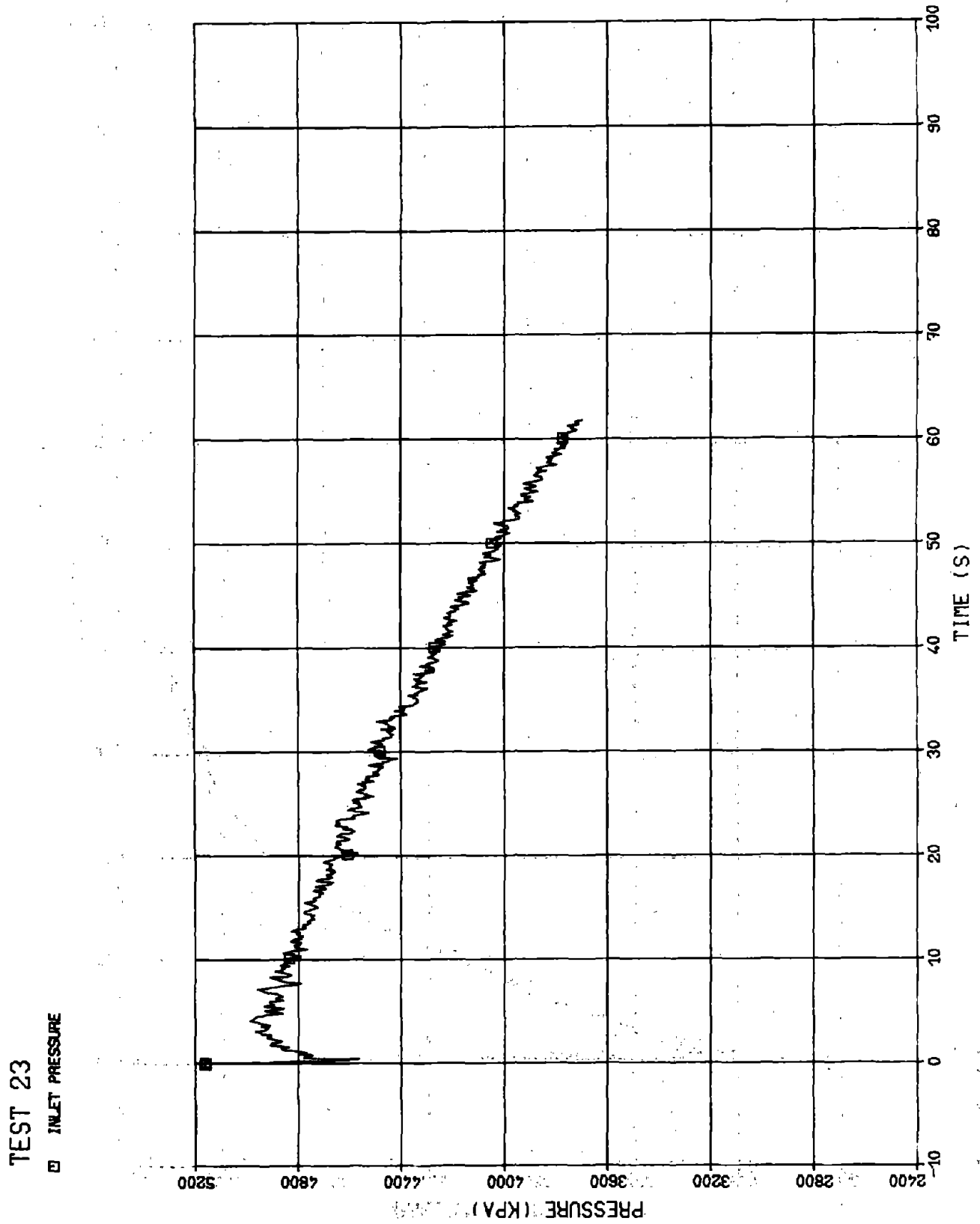
Plot B:21 - Test 21



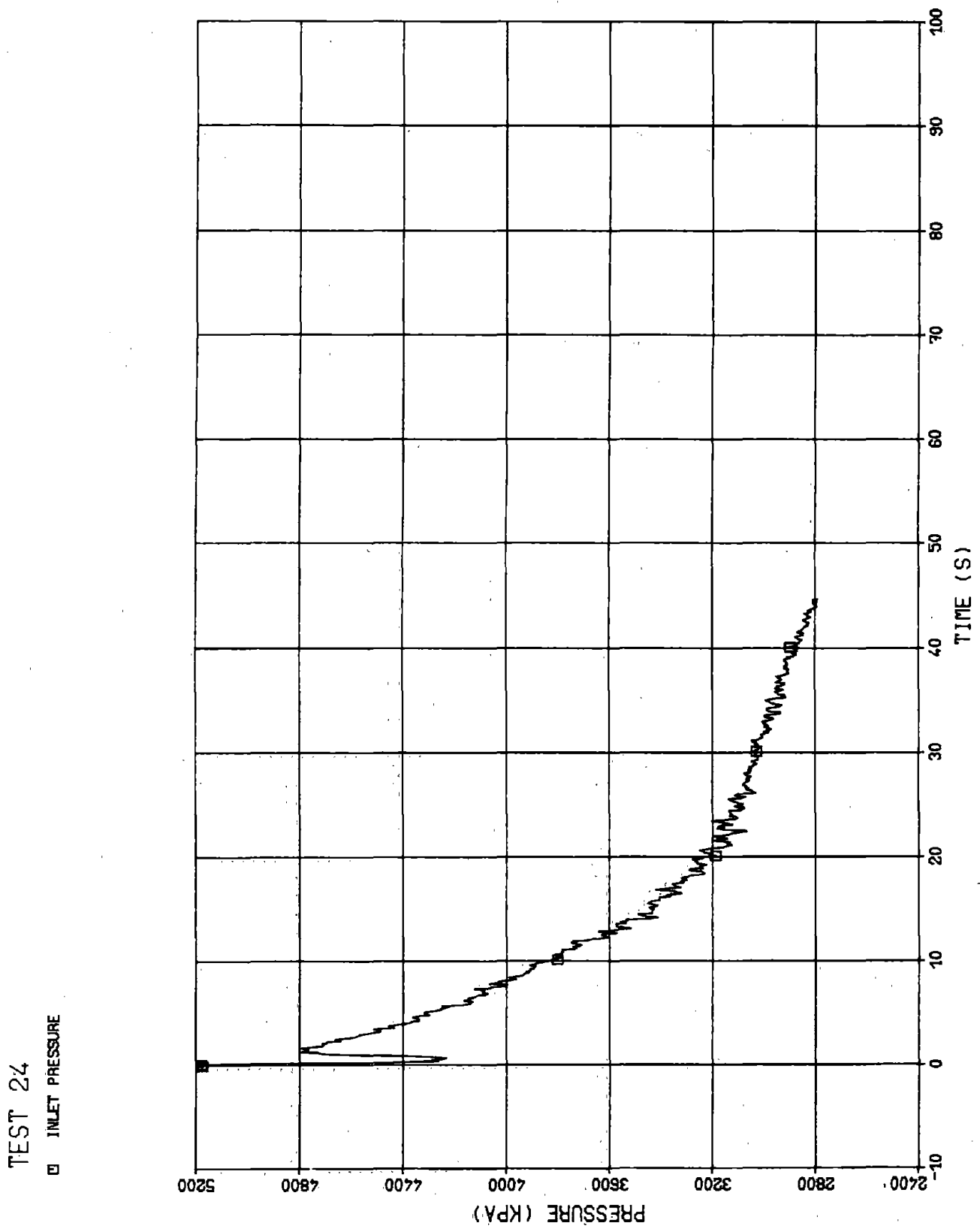
Plot B:22 - Test 22



Plot B:23 - Test 23



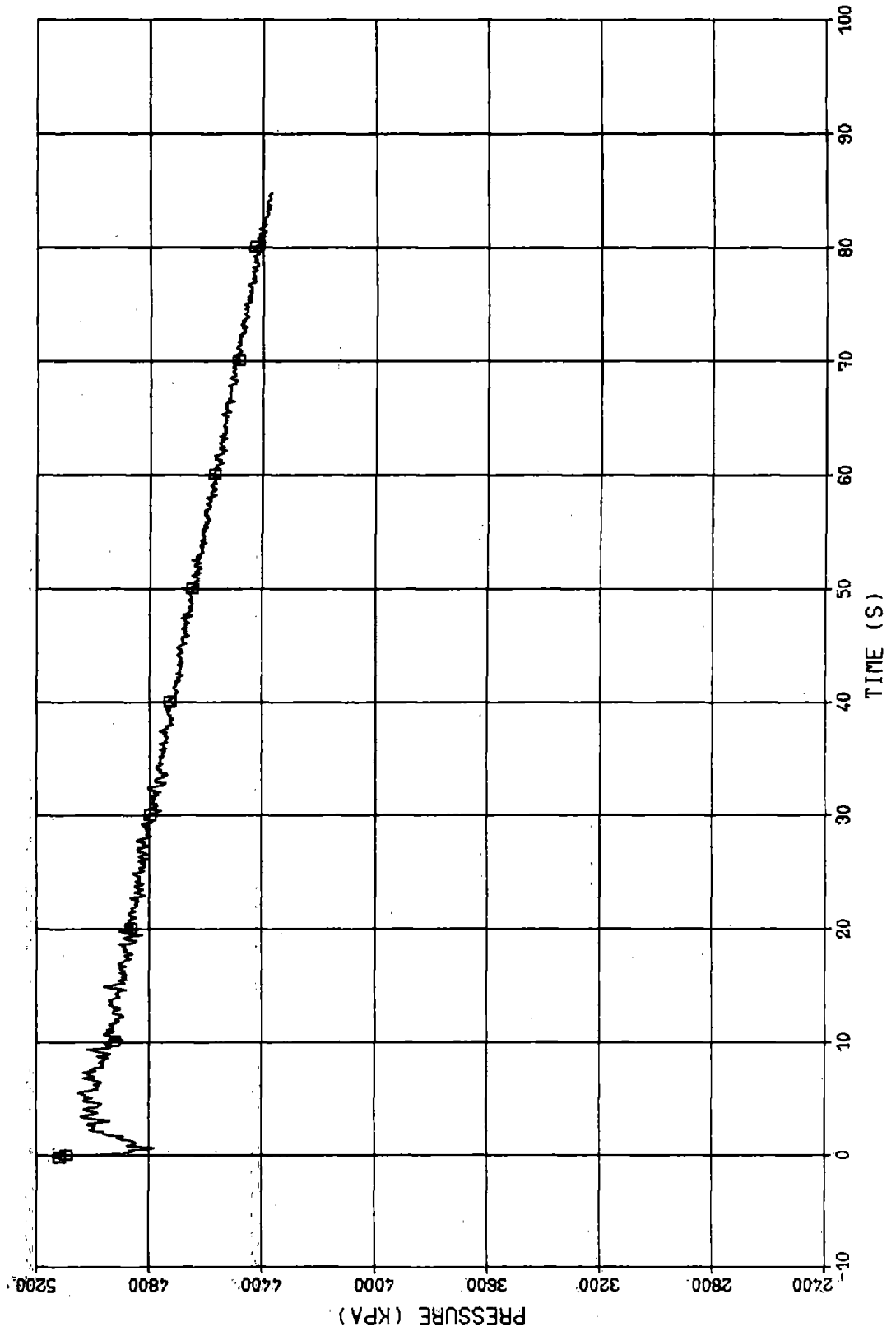
Plot B:24 - Test 24



Plot B:25 - Test 25

TEST 25

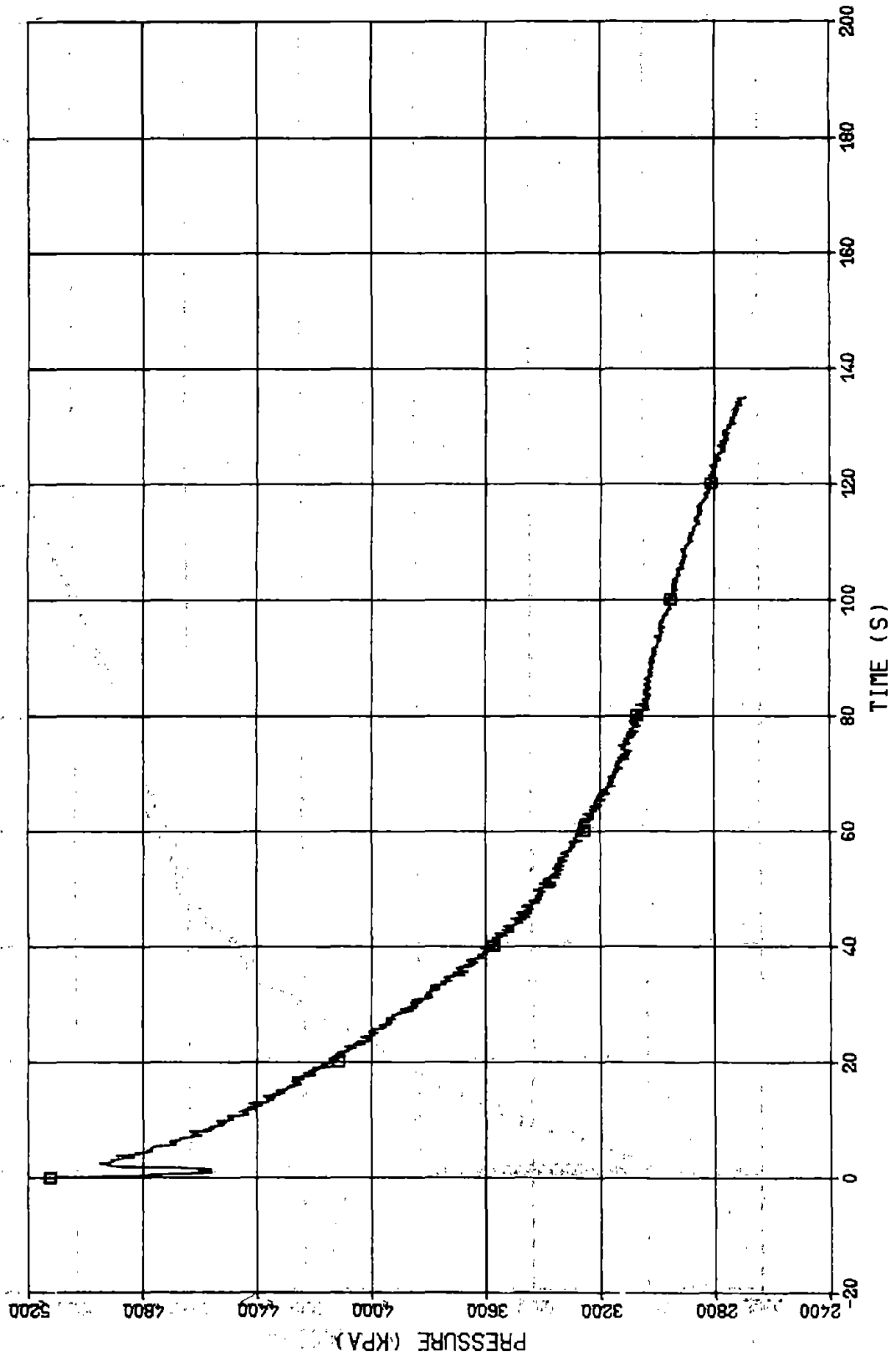
□ INLET PRESSURE



Plot B:26 - Test 26

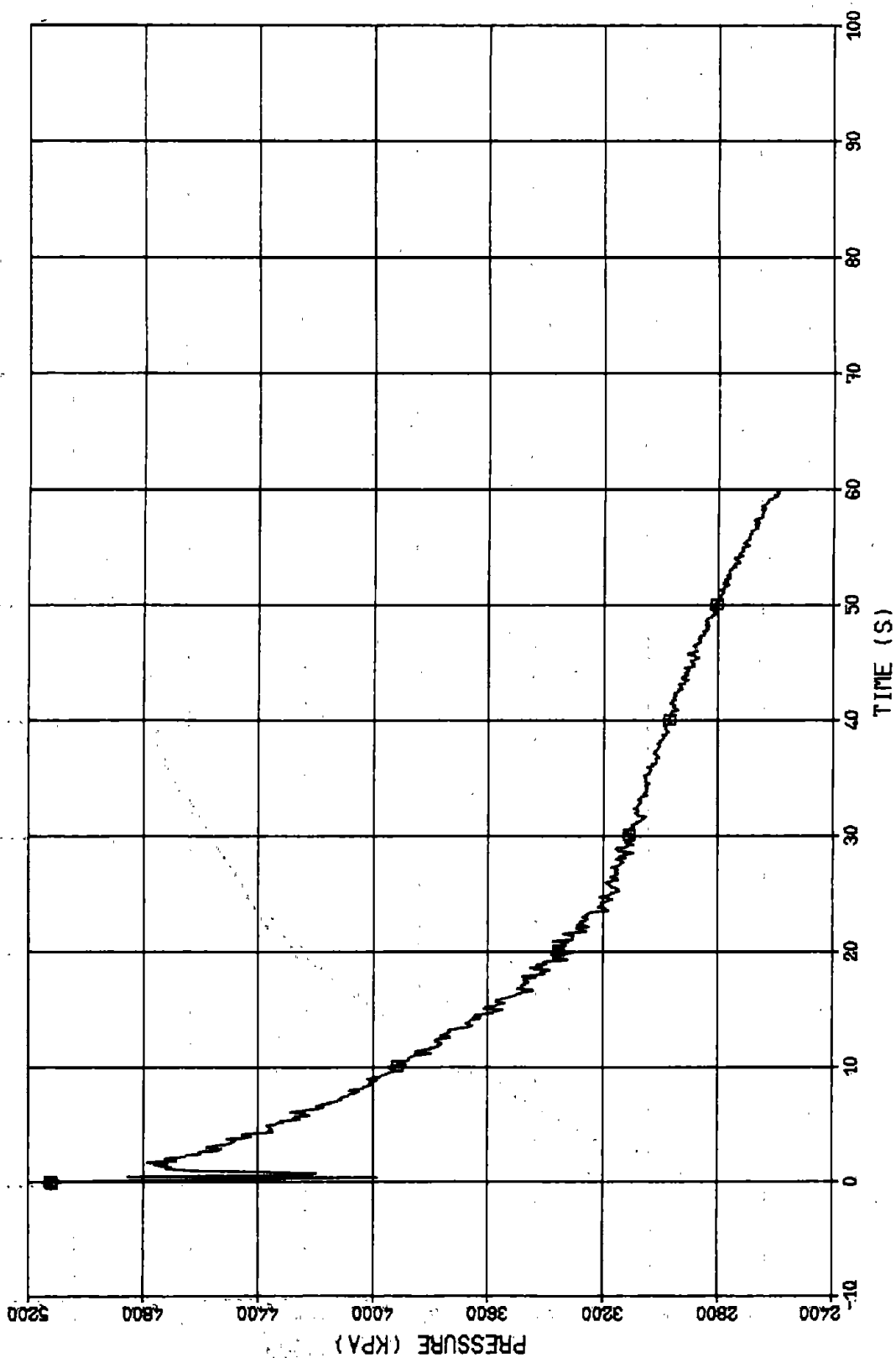
TEST 26

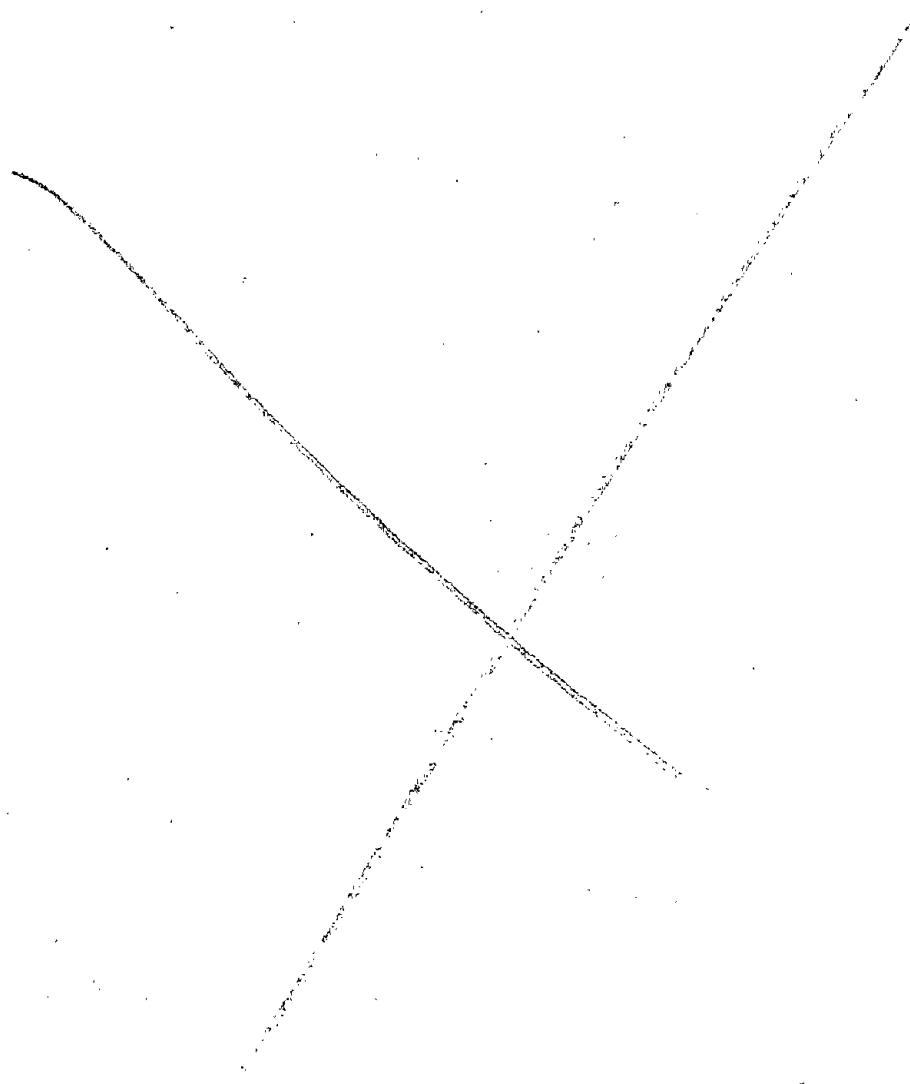
□ INLET PRESSURE



Plot B:27 - Test 27

TEST 27
INLET PRESSURE

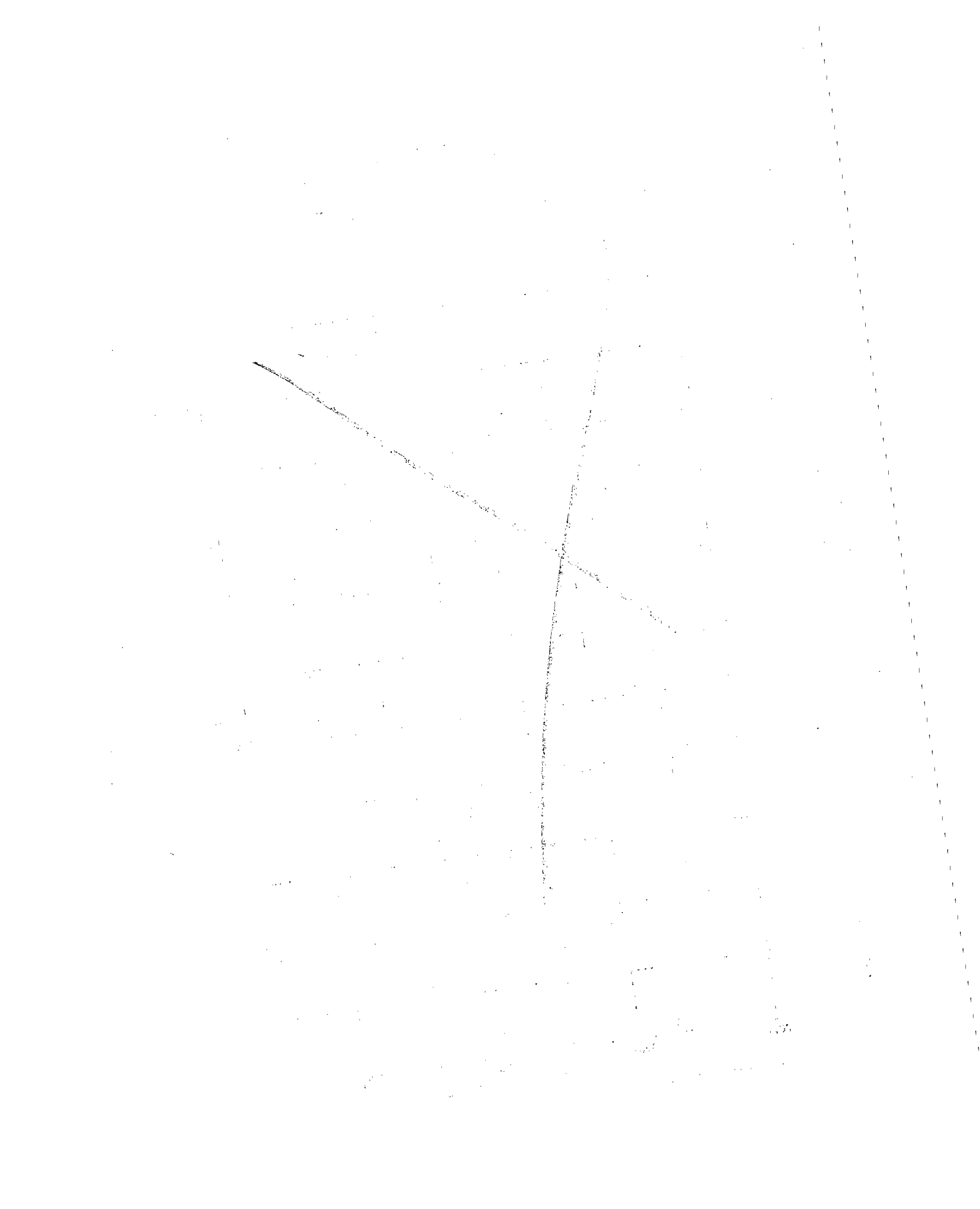




APPENDIX C

Density based on the stagnation conditions at the nozzle
inlet plane: Tests 1 through 27

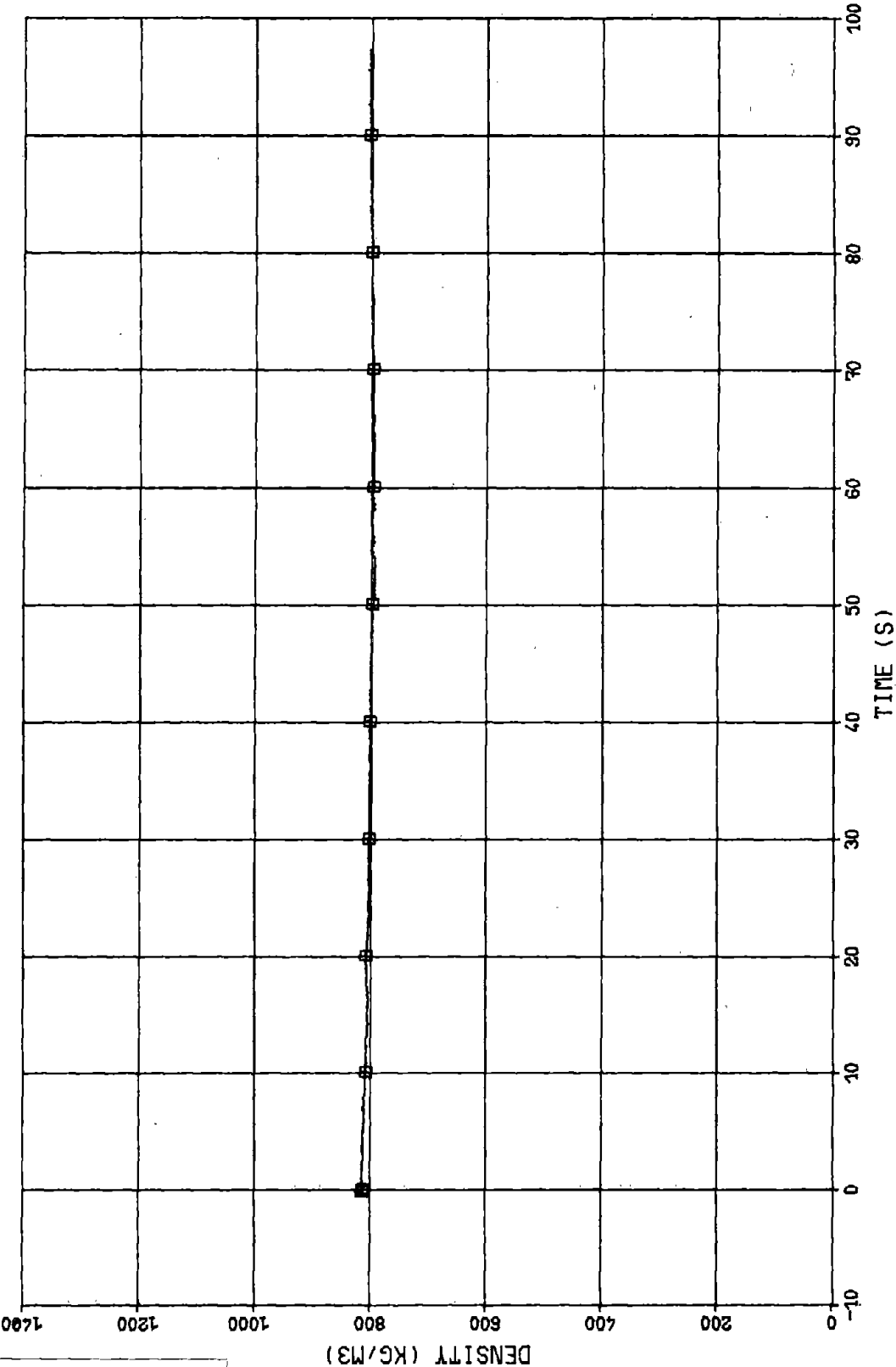
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Plot C:1 - Test 1

TEST 01

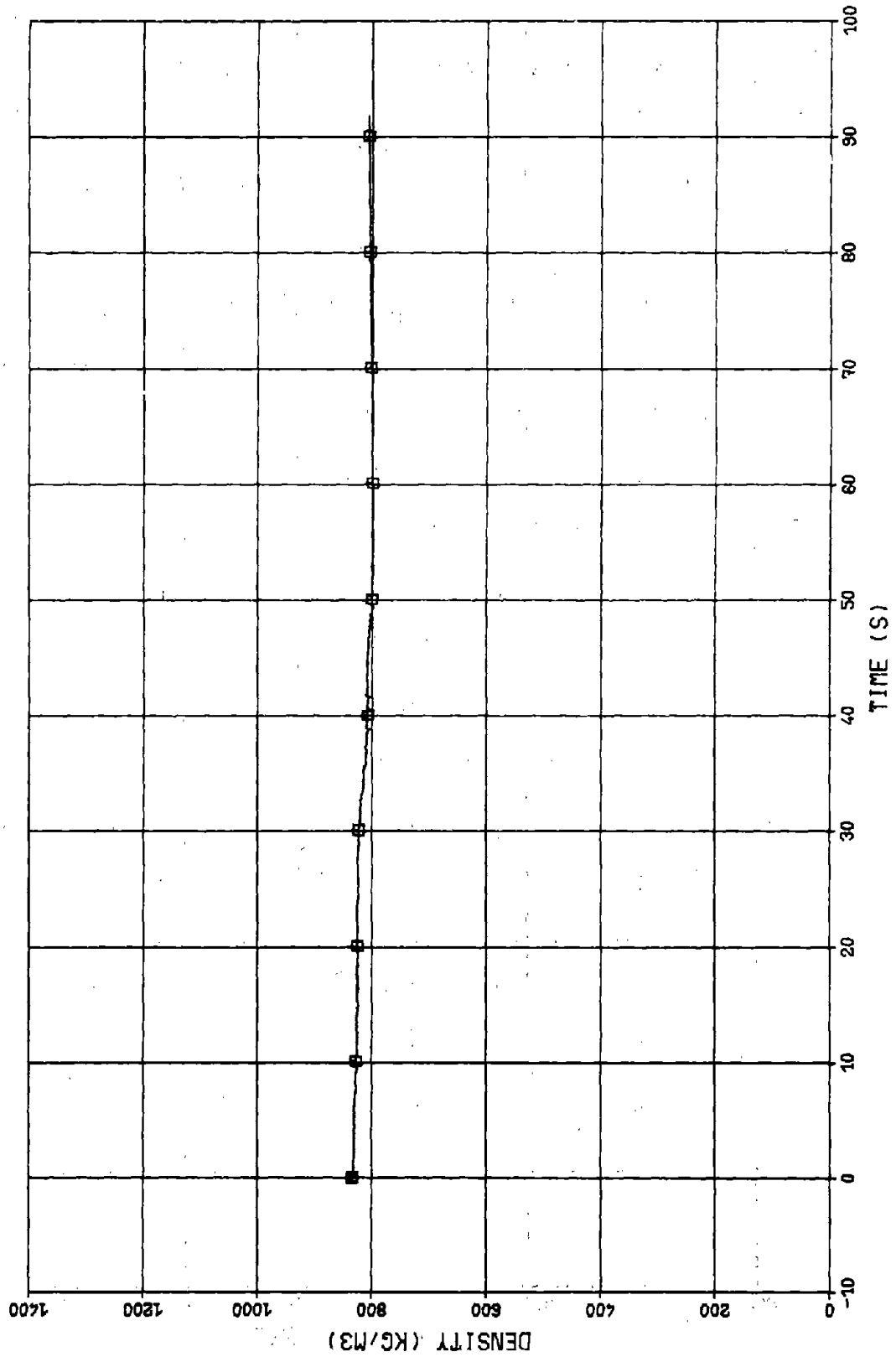
INLET DENSITY



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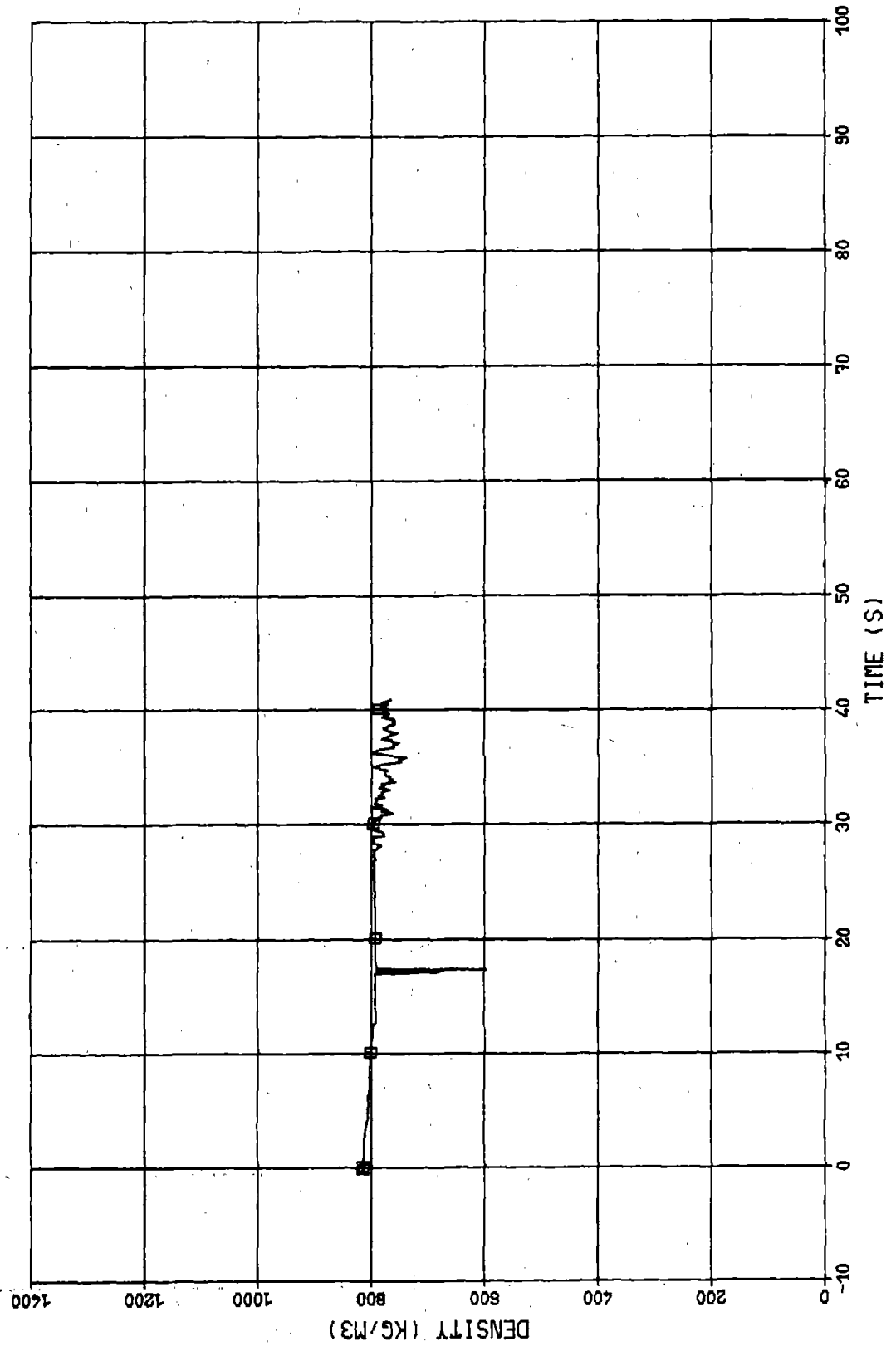
Plot C:2 - Test 2

TEST 2
INLET DENSITY



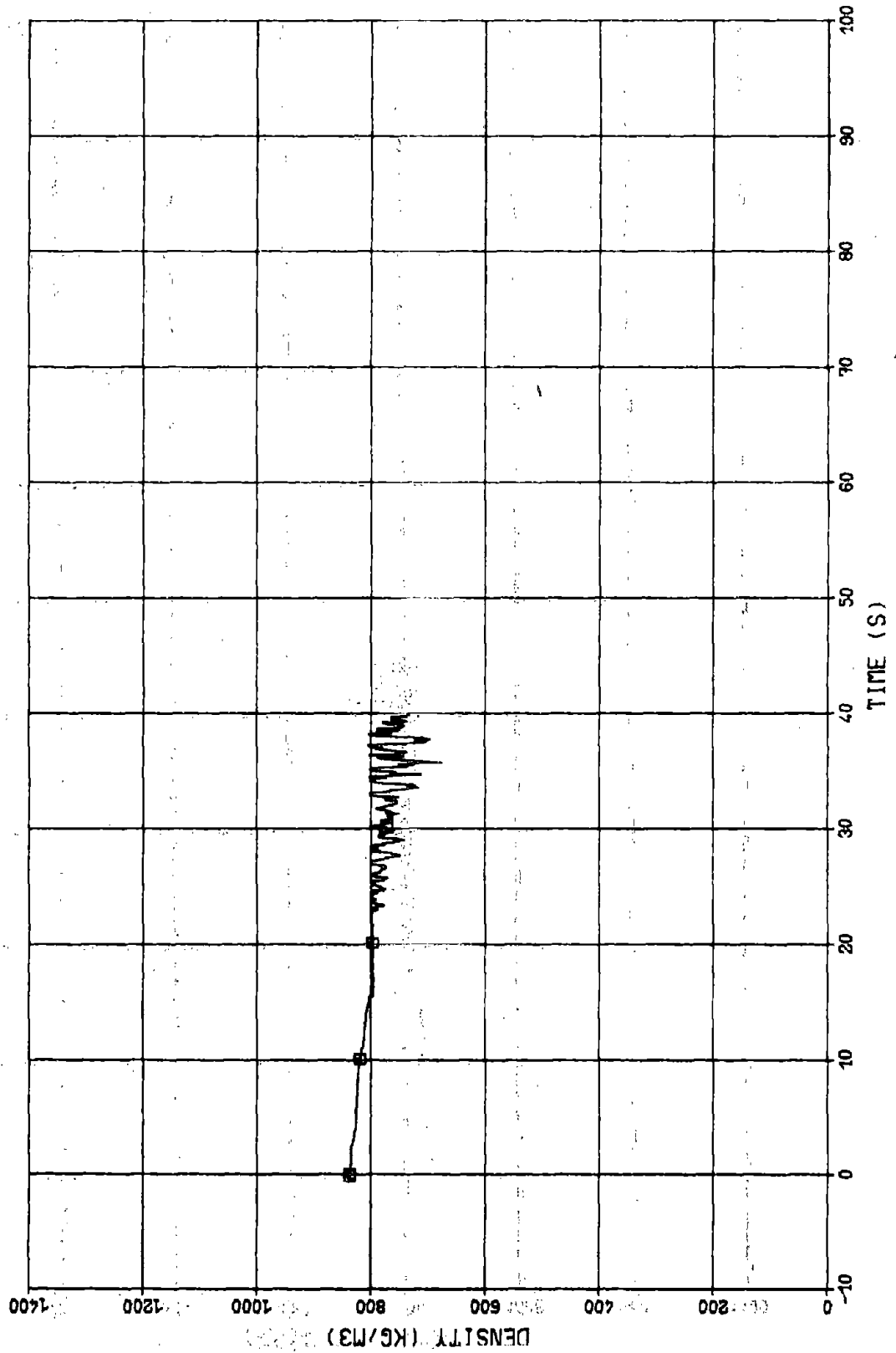
Plot C:3 - Test 3

TEST 3
INLET DENSITY

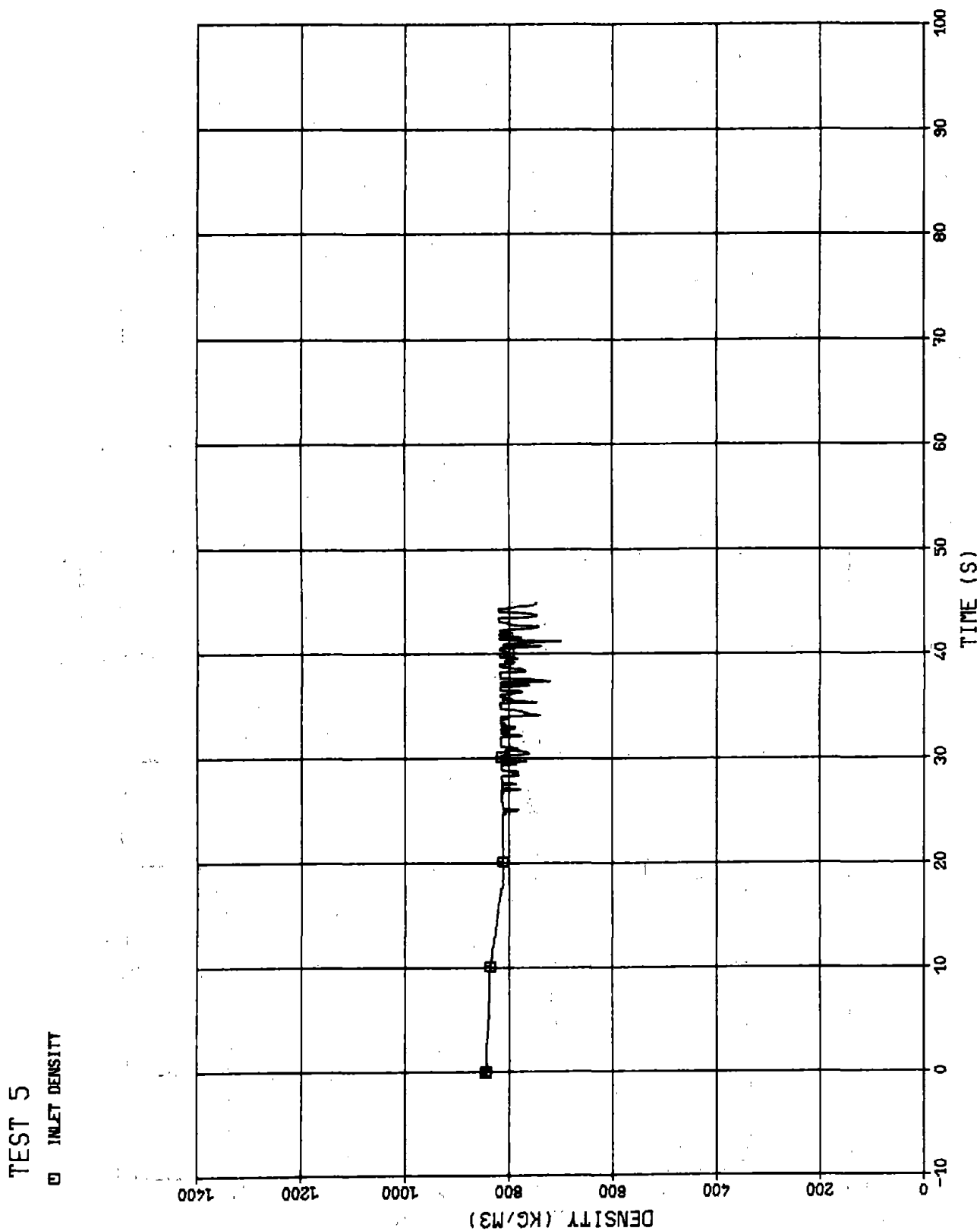


Plot C:4 - Test 4

TEST 4
INLET DENSITY

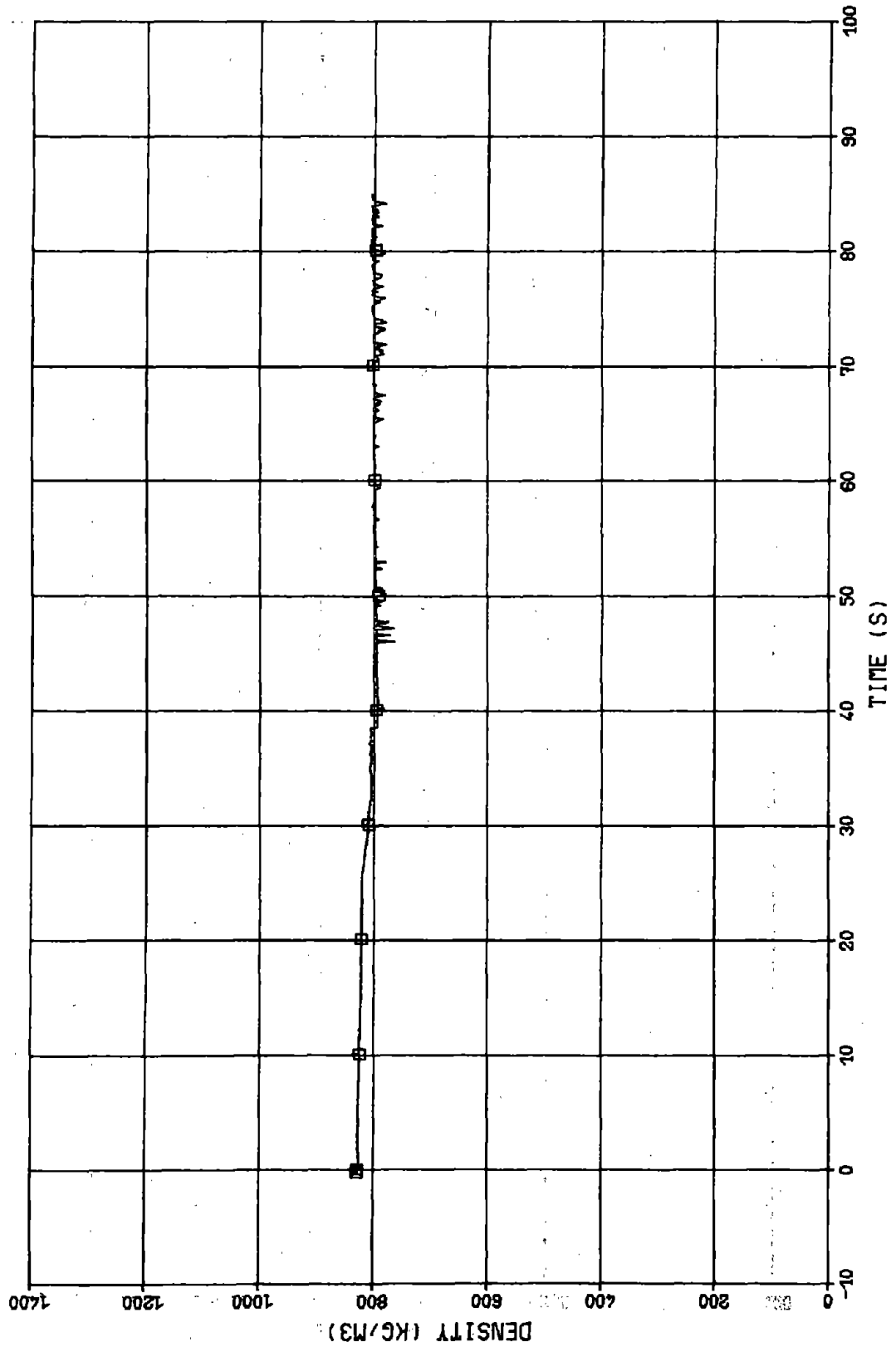


Plot C:5 - Test 5

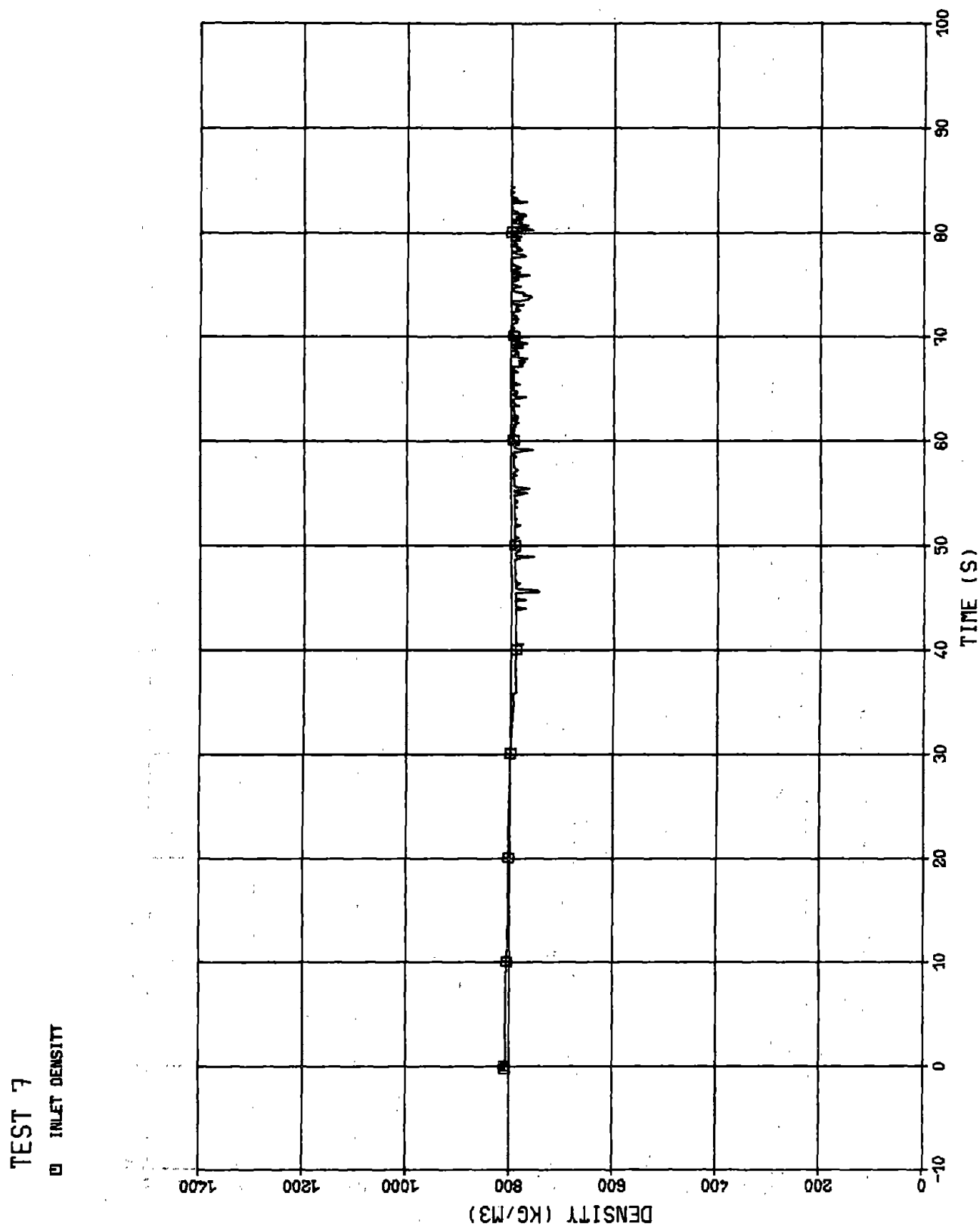


Plot C:6 - Test 6

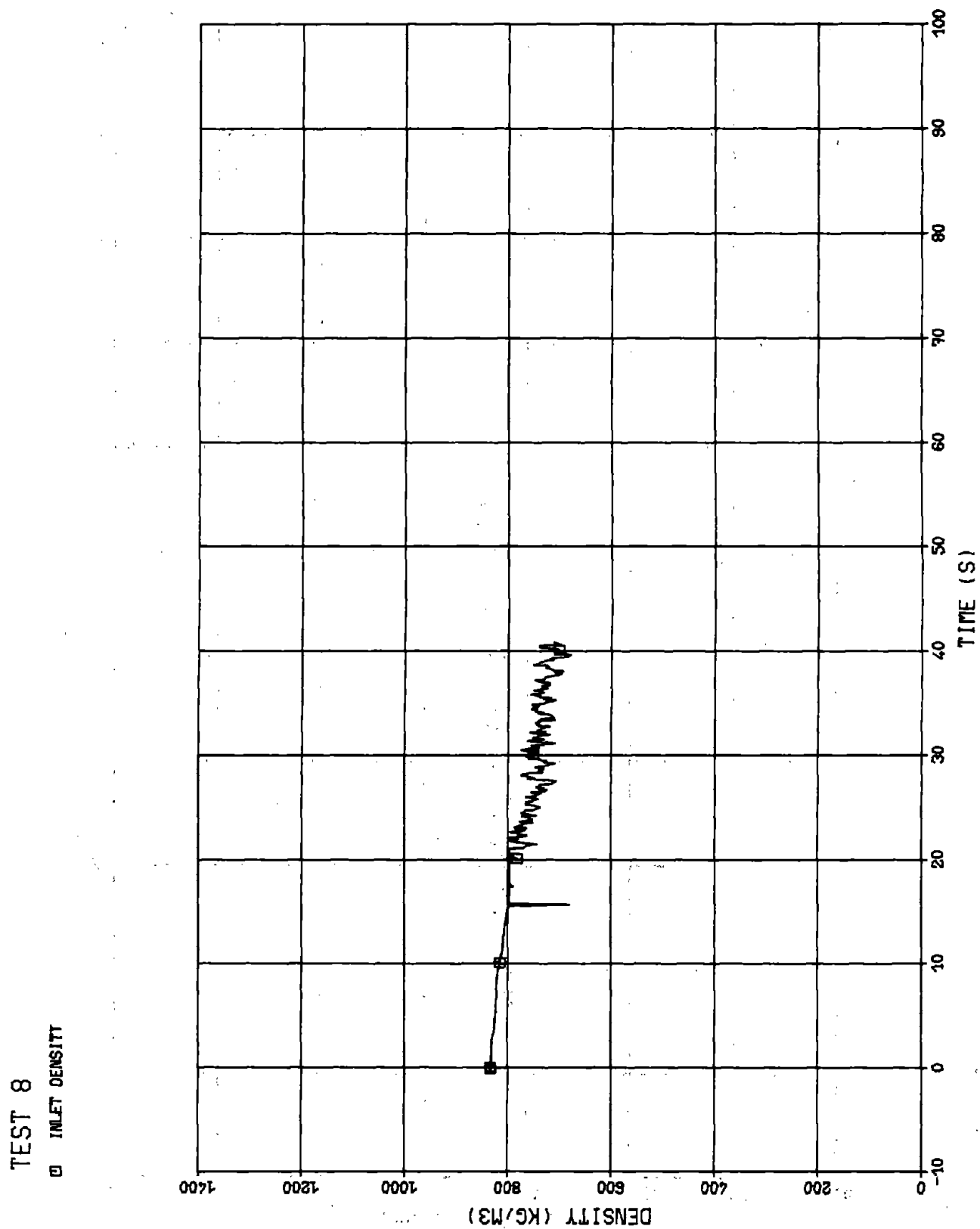
TEST 6.
□ INLET DENSITY



Plot C:7 - Test 7



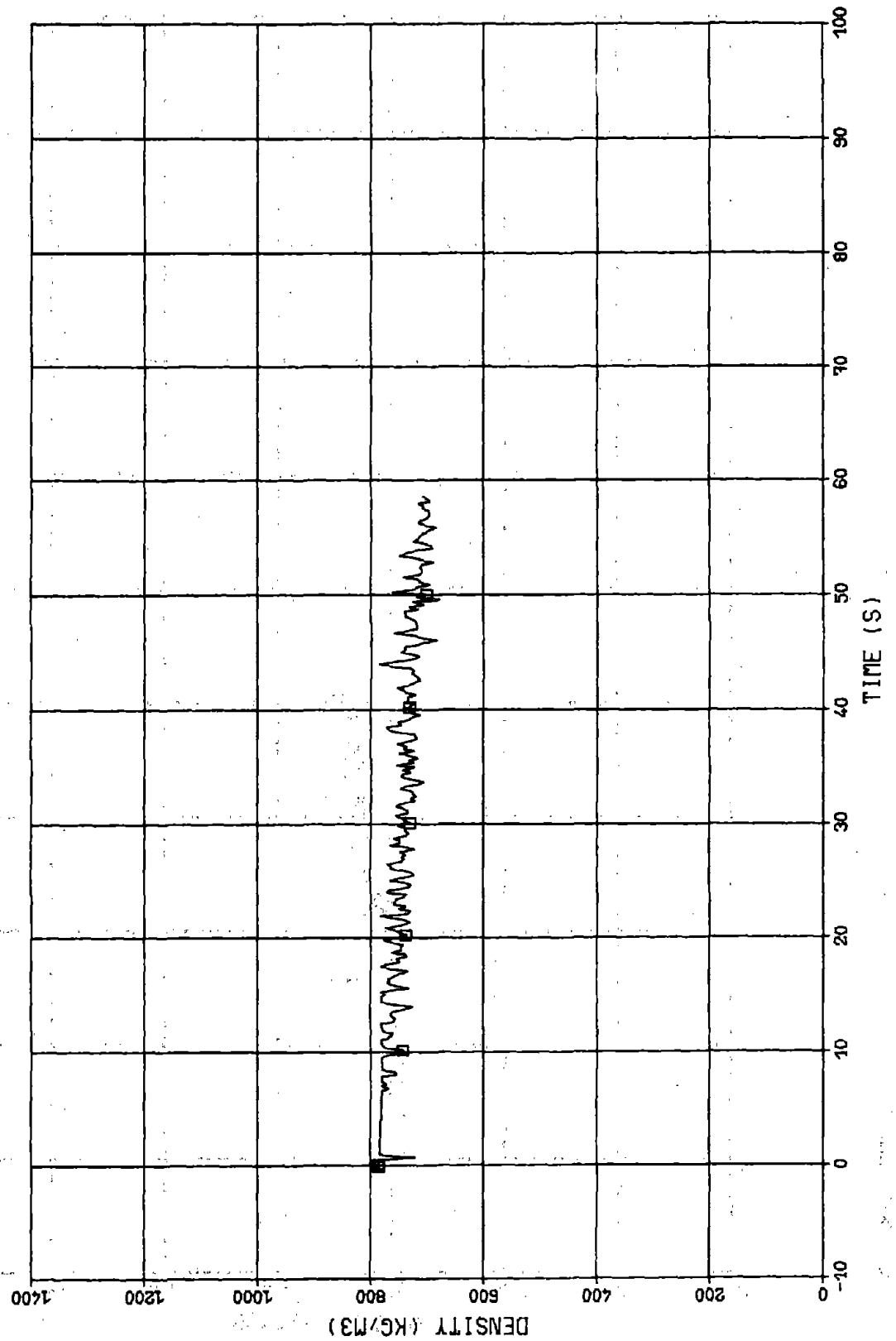
Plot C:8 - Test 8



Plot C:9 - Test 9

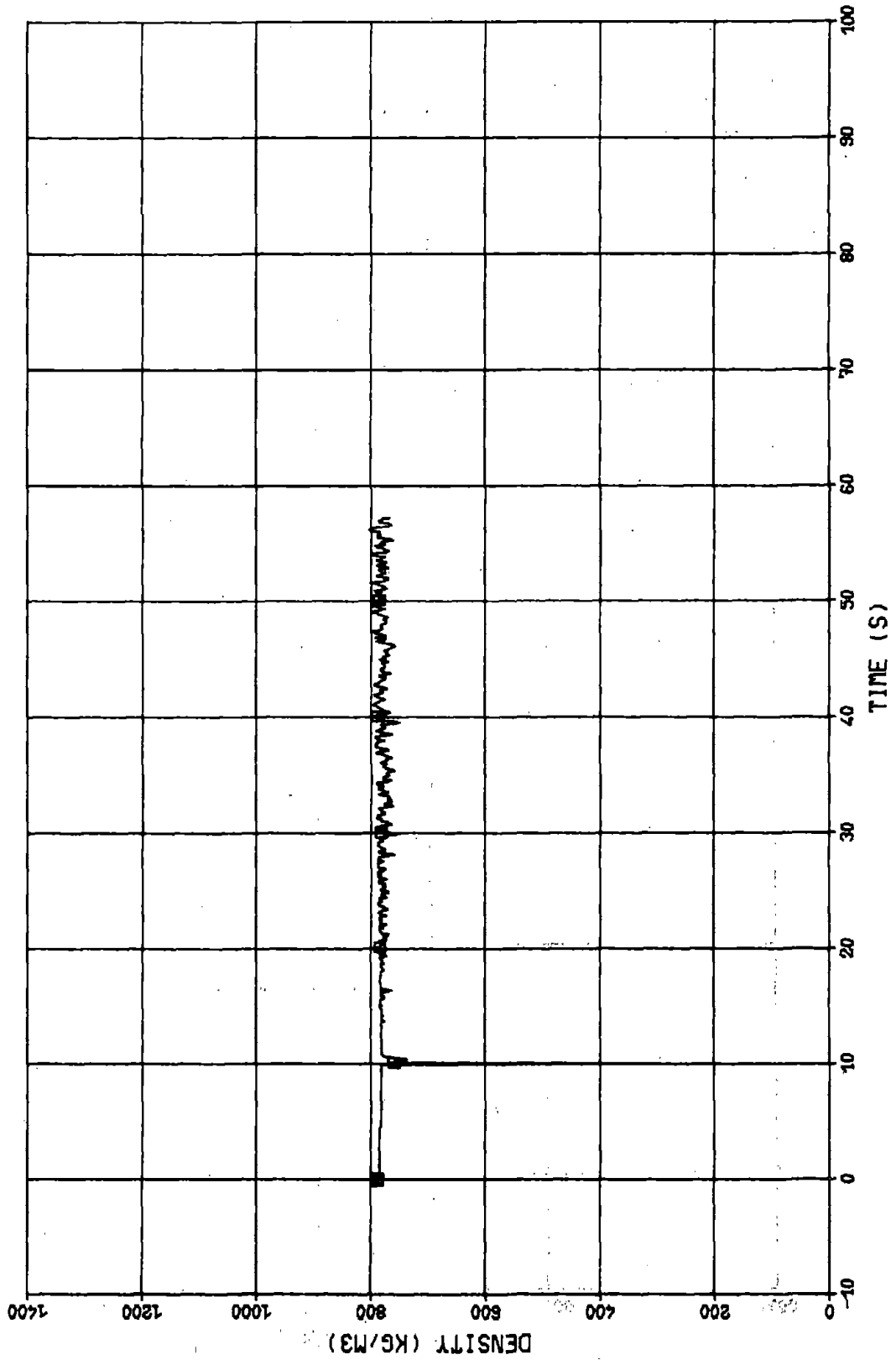
TEST 9

INLET DENSITY



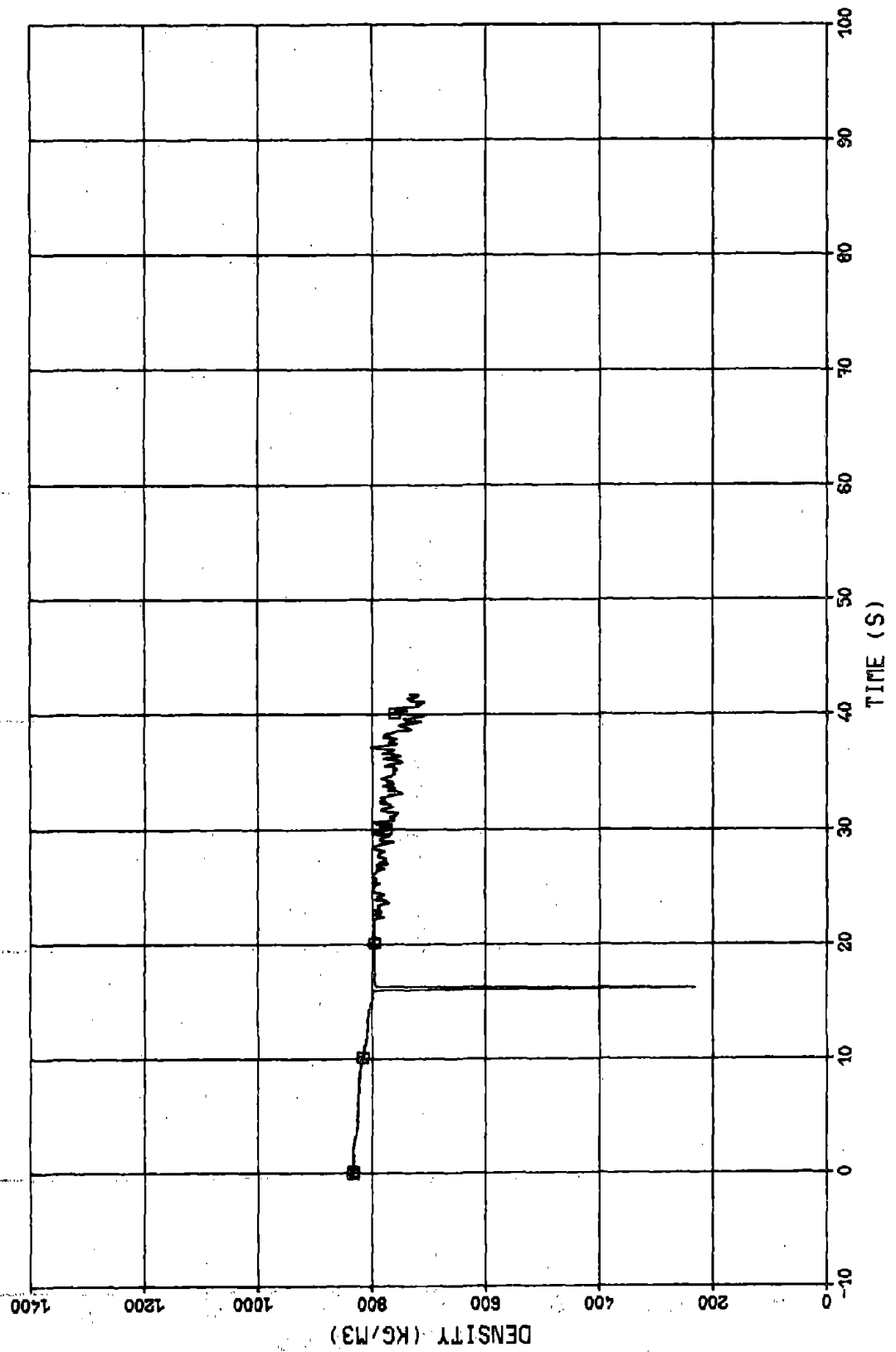
Plot C:10 - Test 10

TEST 10
INLET DENSITY



Plot C:11 - Test 11

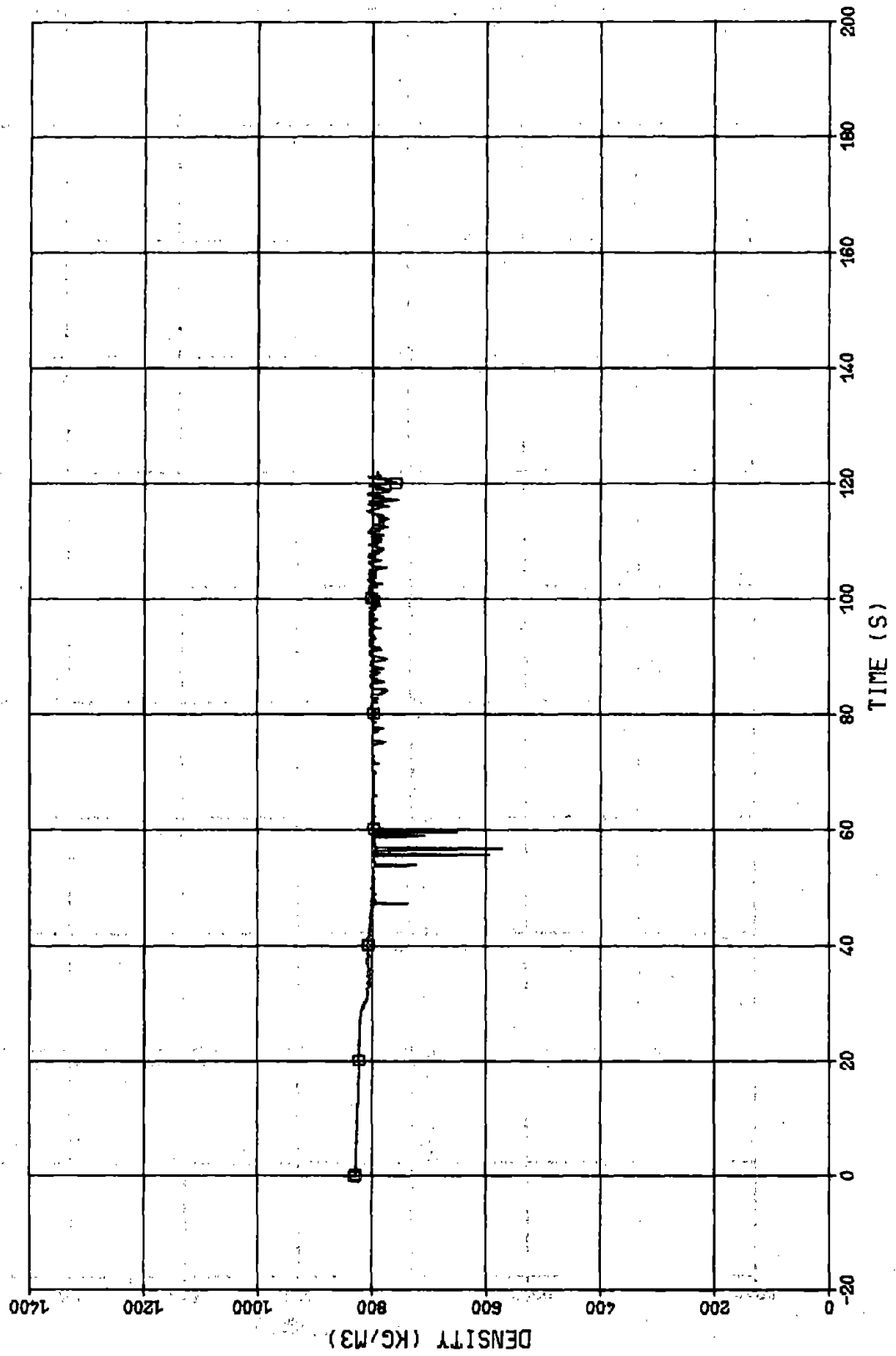
TEST 11
□ INLET DENSITY



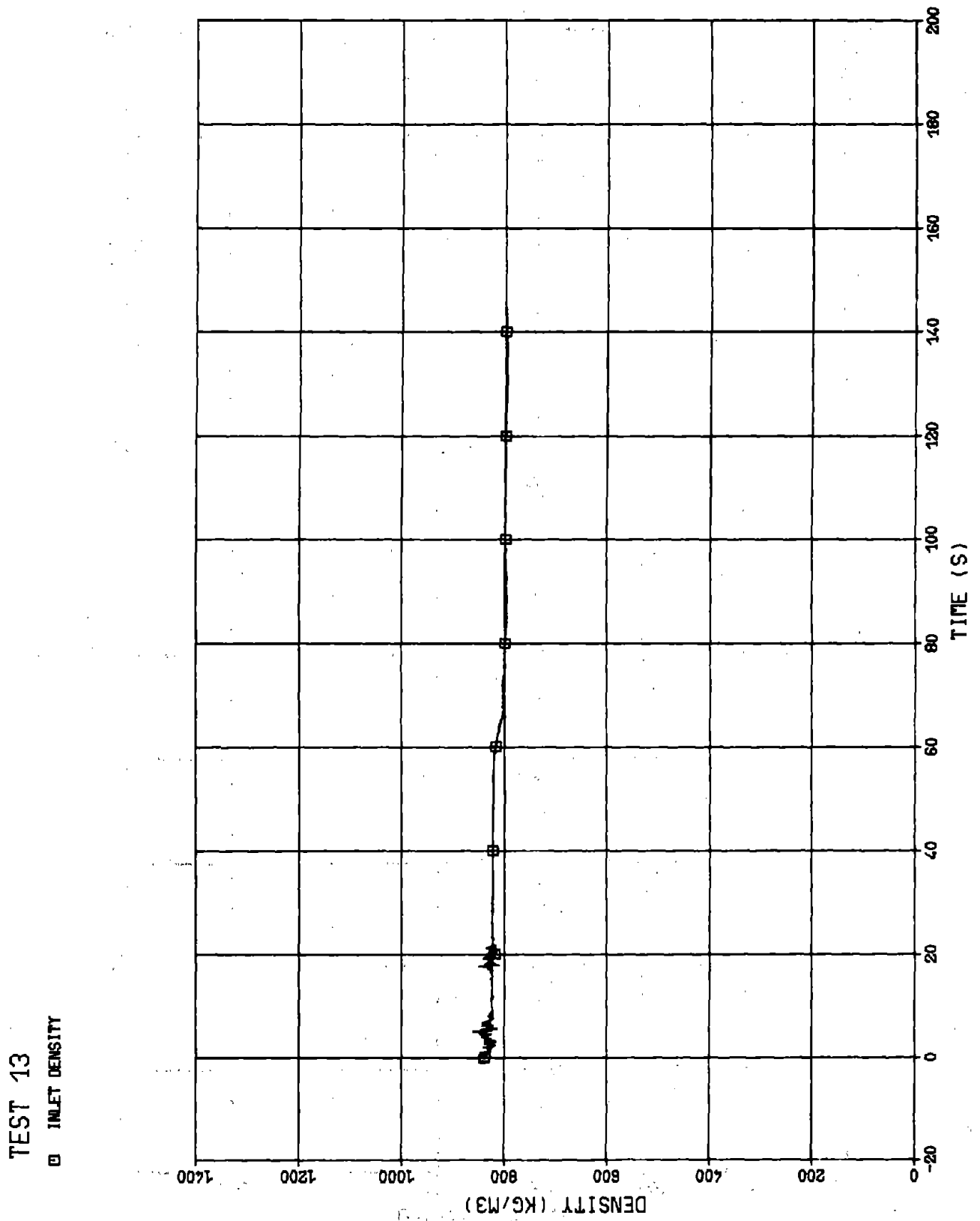
Plot C:12 - Test 12

TEST 12

□ INLET DENSITY

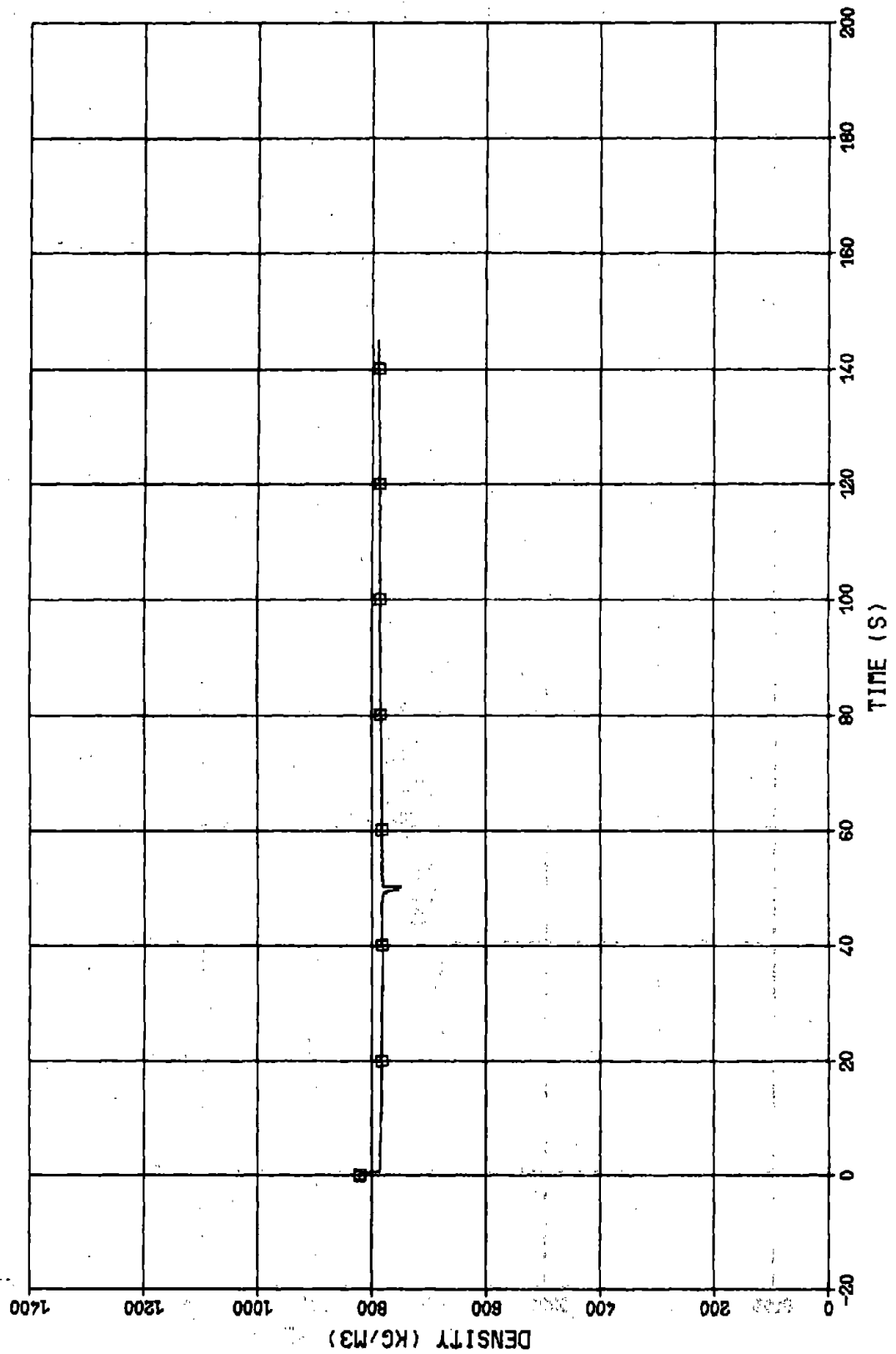


Plot C:13 - Test 13

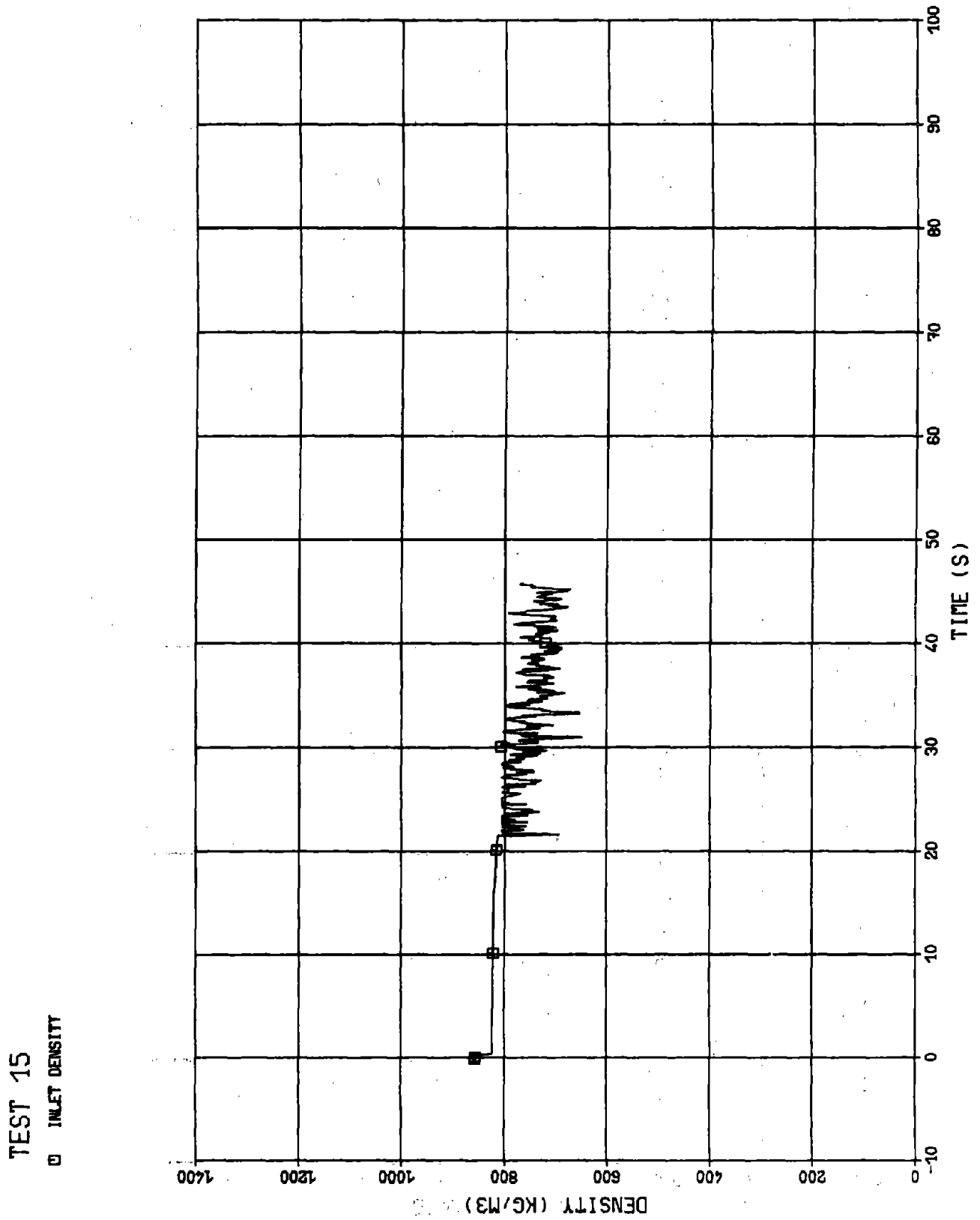


Plot C:14 - Test 14

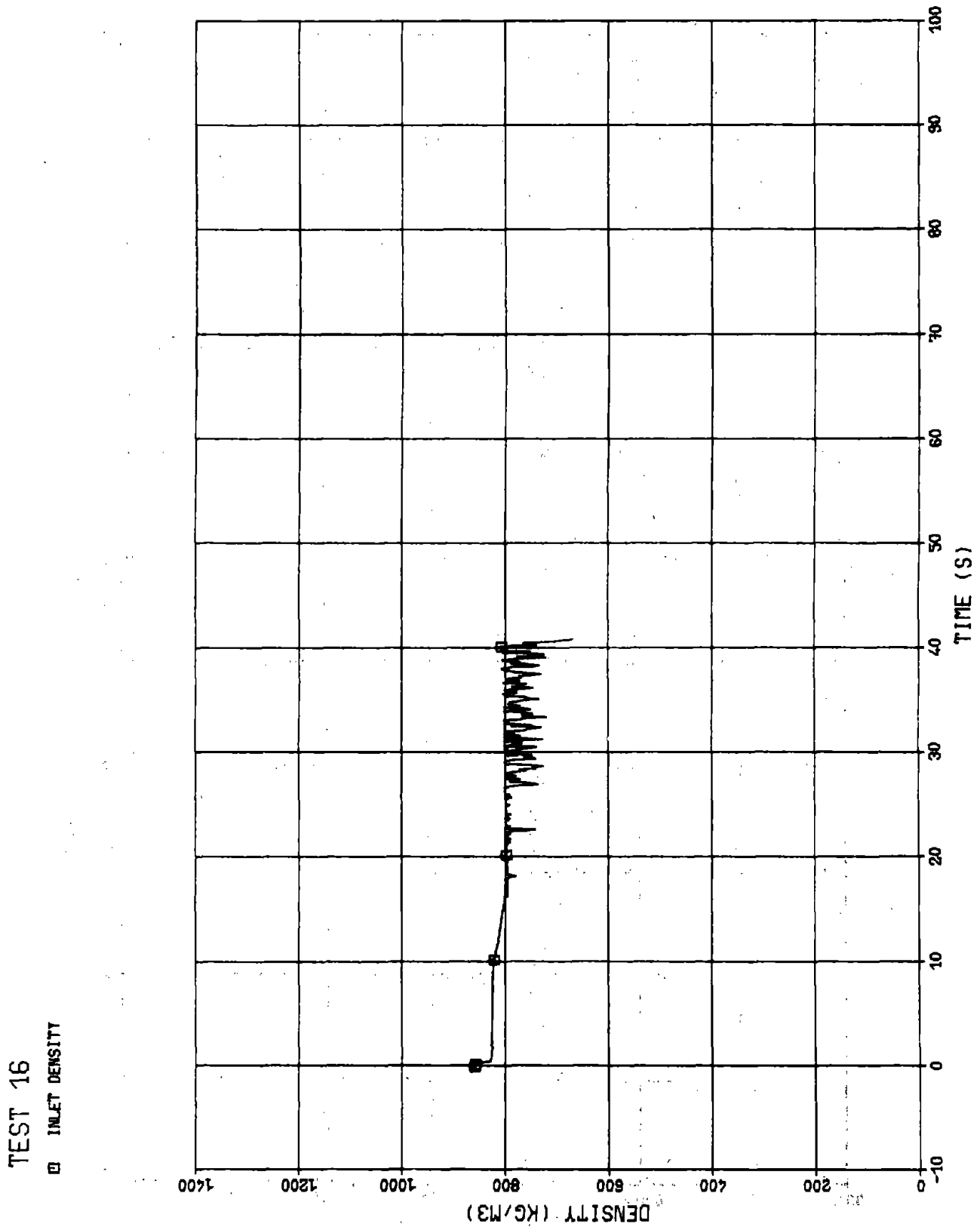
TEST 14
□ INLET DENSITY



Plot C:15 - Test 15

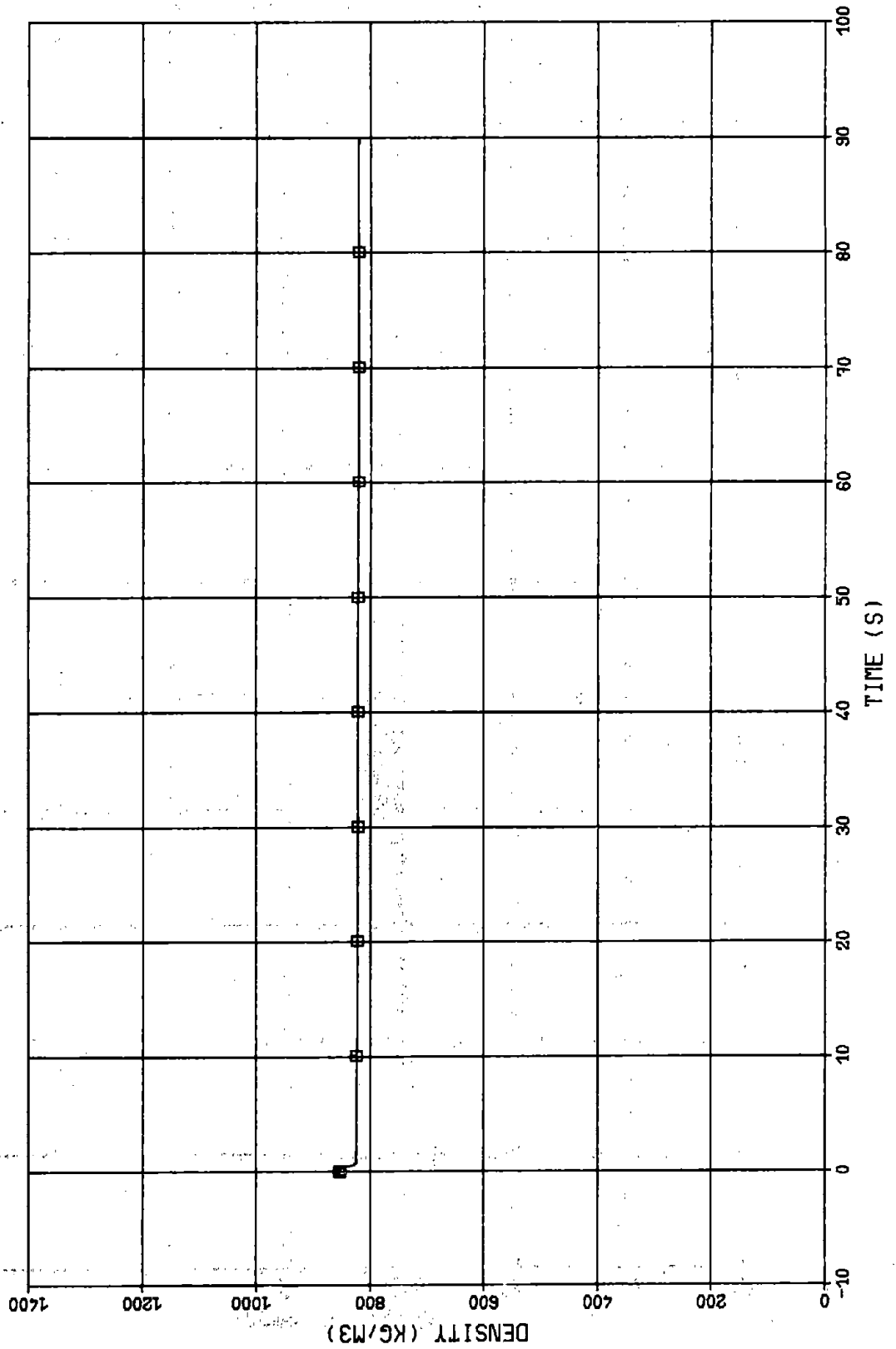


Plot C:16 - Test 16



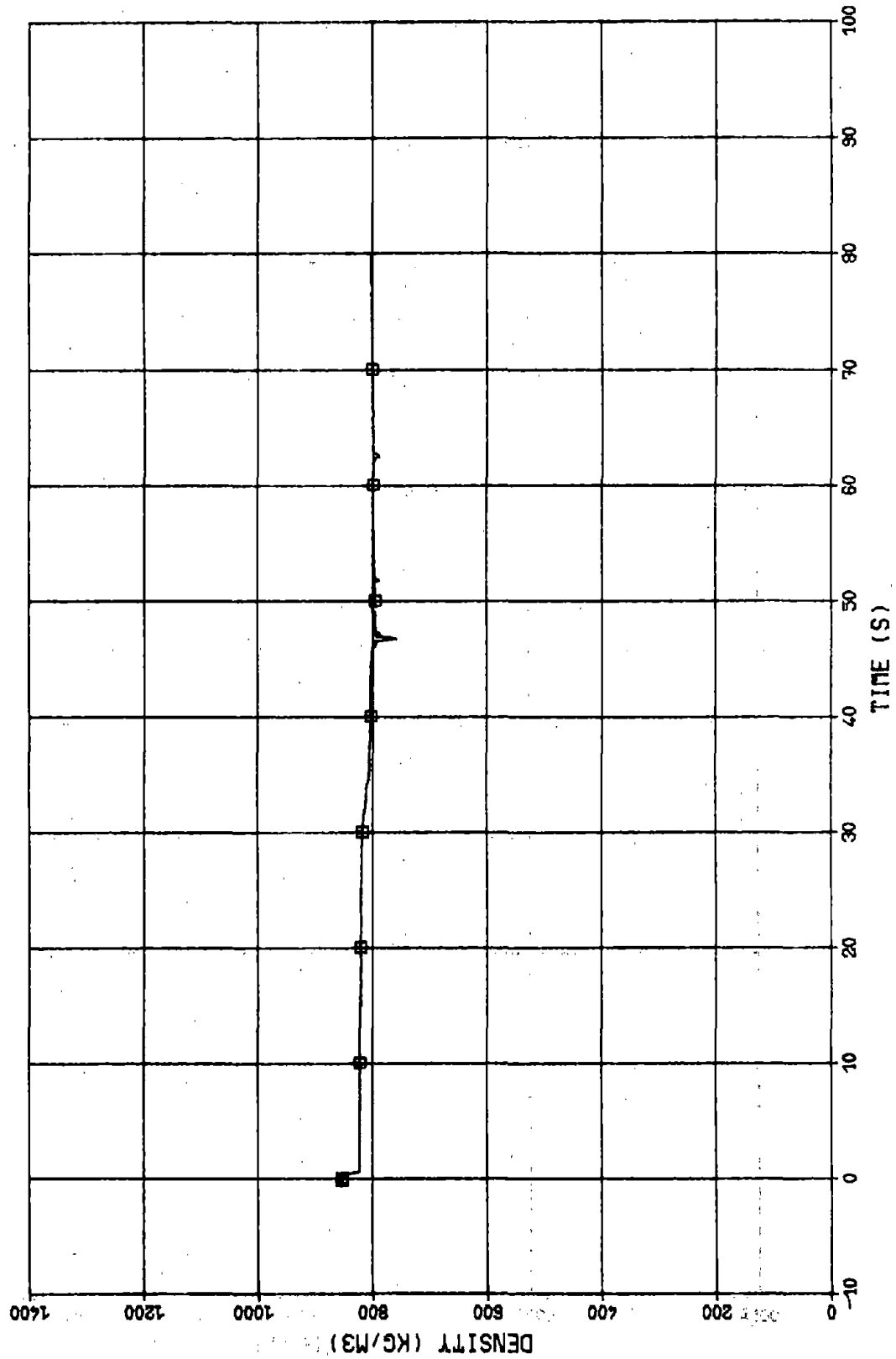
Plot C:17 - Test 17

TEST 17
□ INLET DENSITY



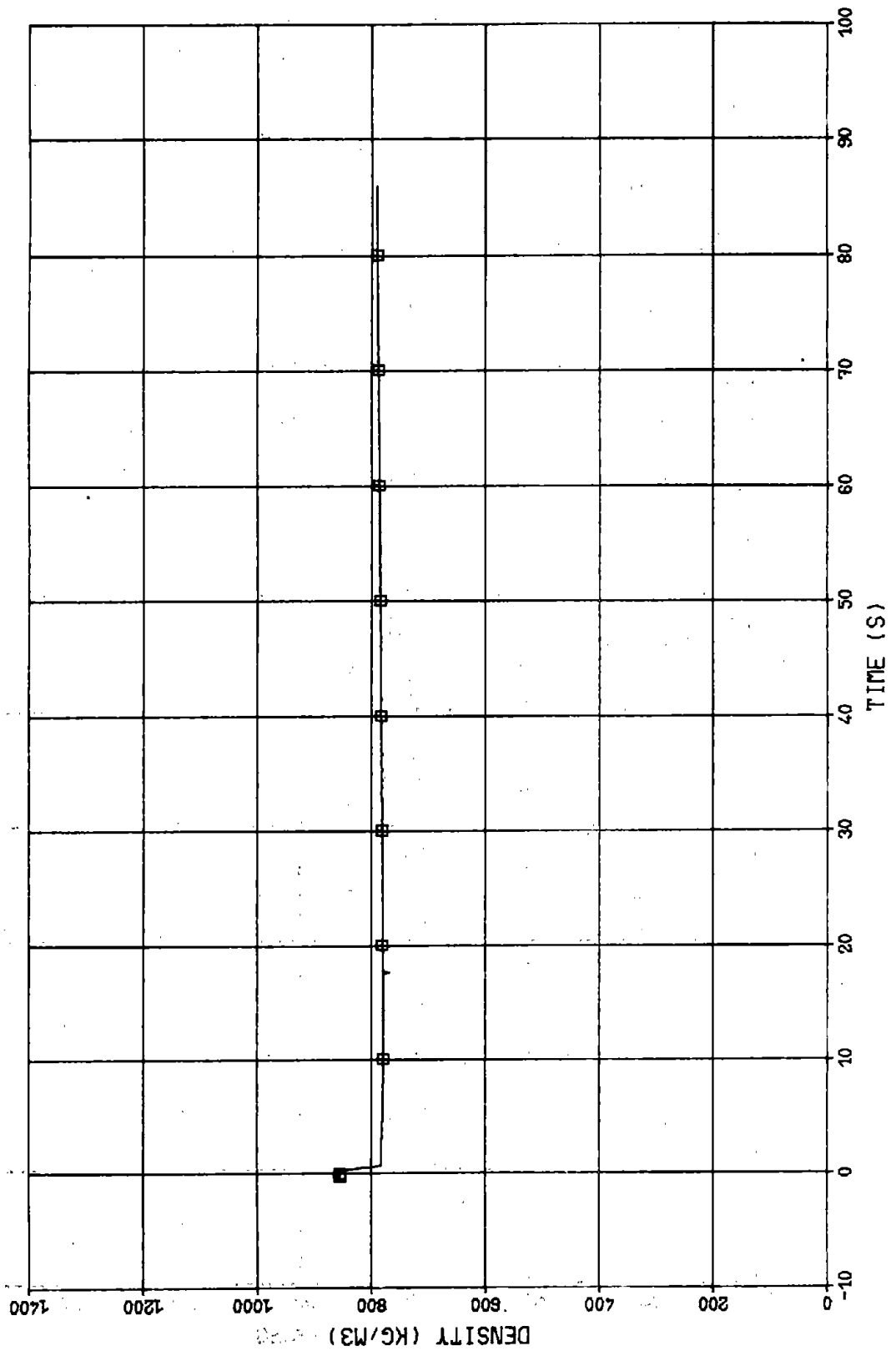
Plot C:18 - Test 18

TEST 18
INLET DENSITY



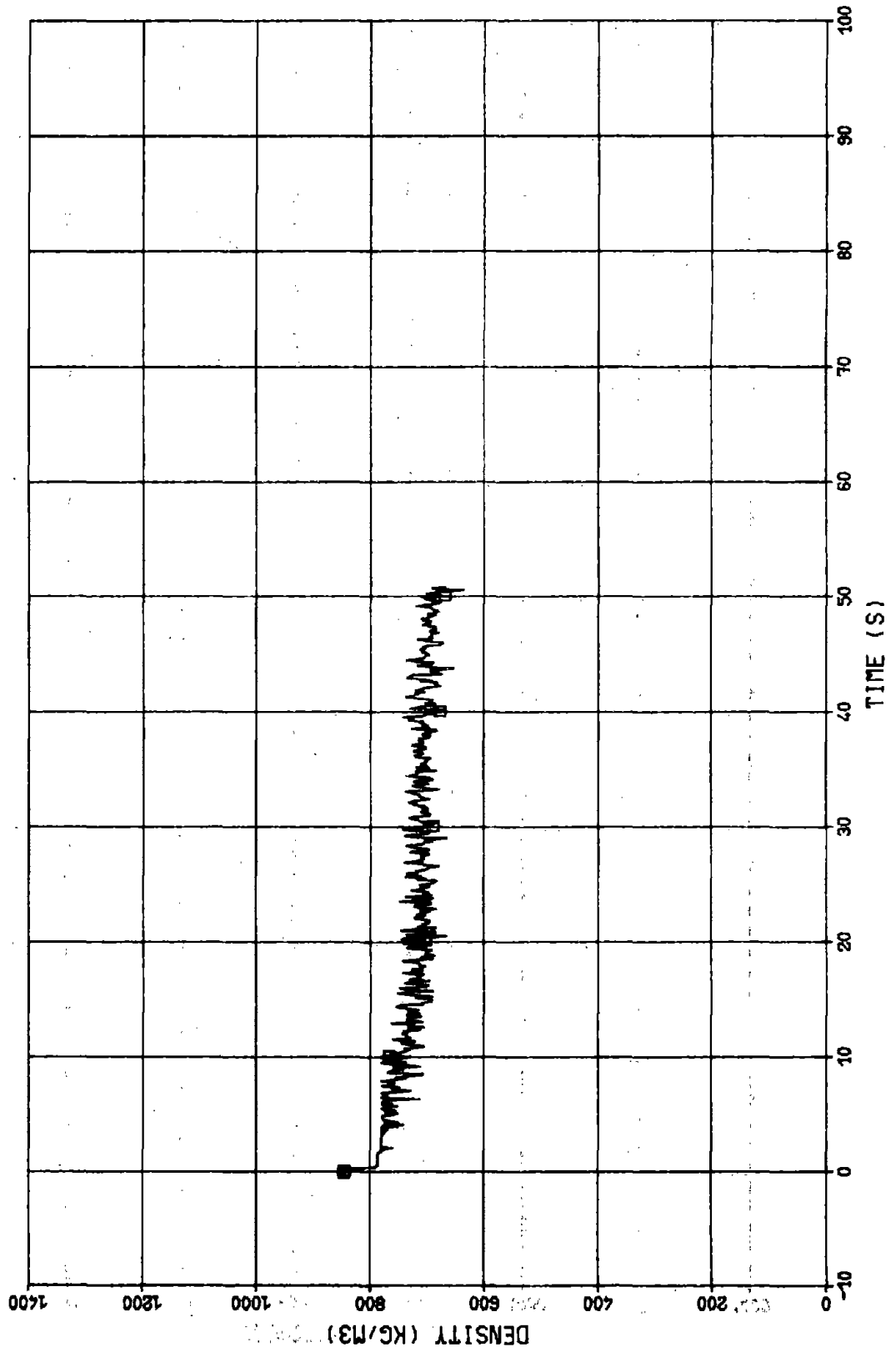
Plot C:19 - Test 19

TEST 19
INLET DENSITY



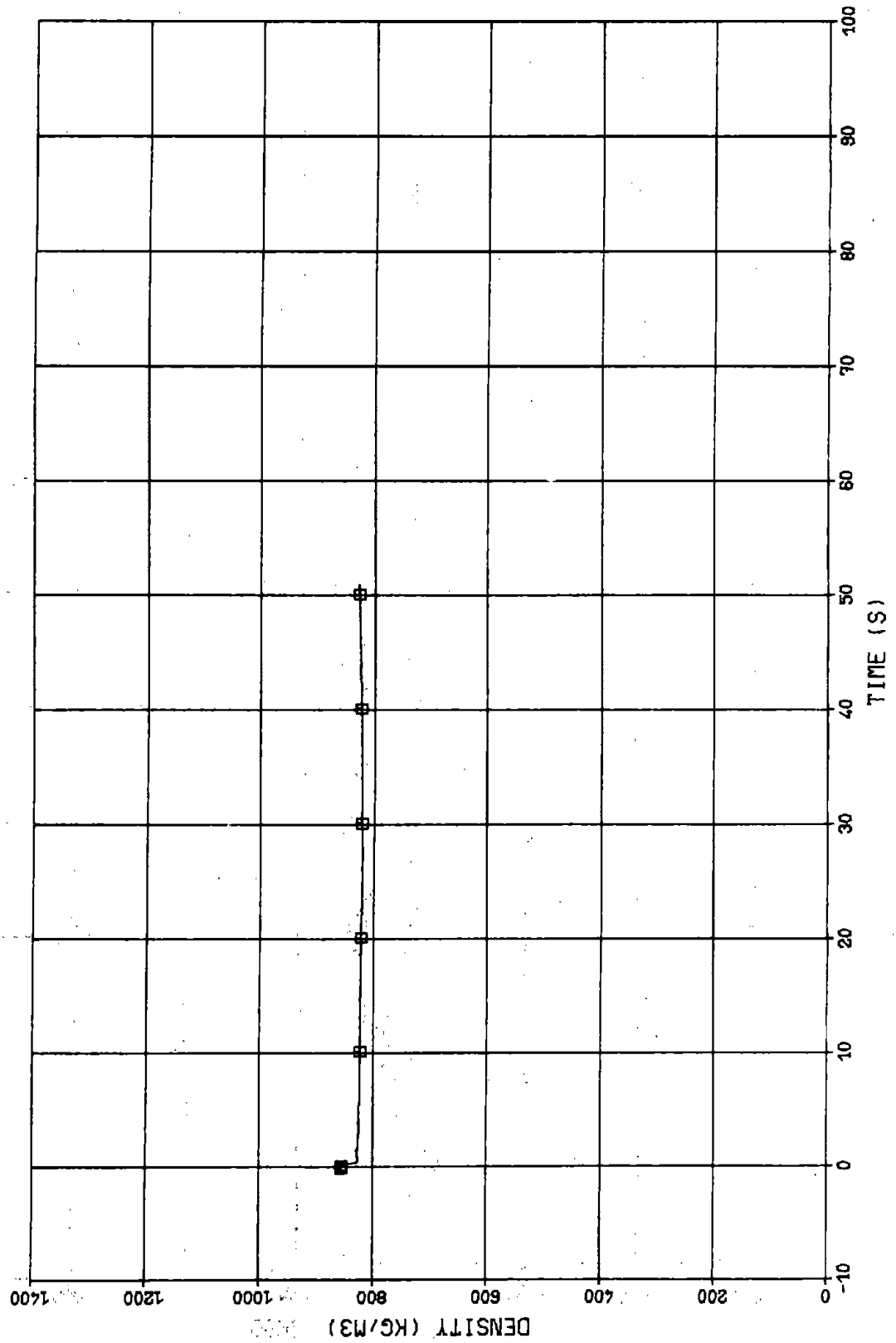
Plot C:20 - Test 20

TEST 20
INLET DENSITY

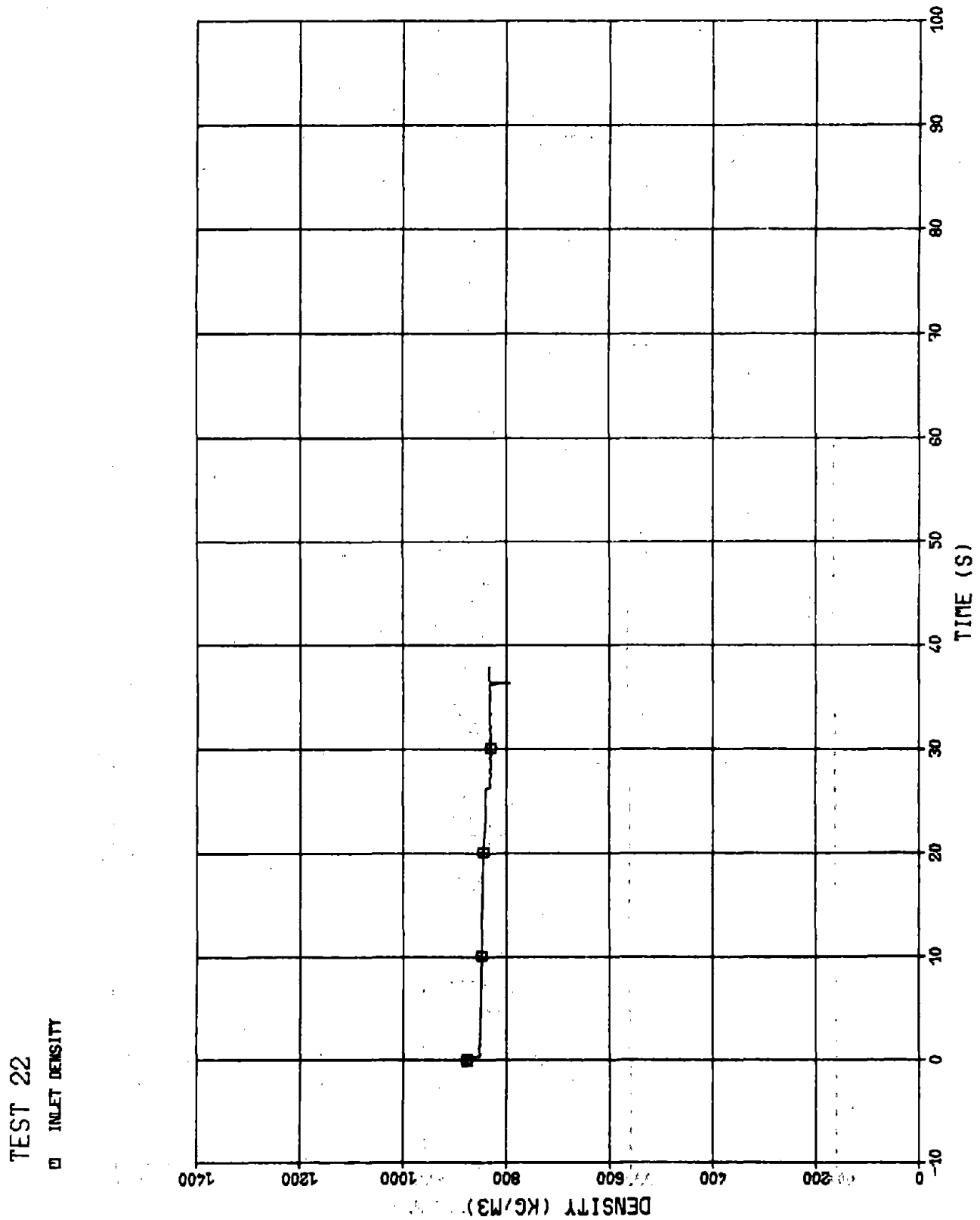


Plot C:21 - Test 21

TEST 21
INLET DENSITY

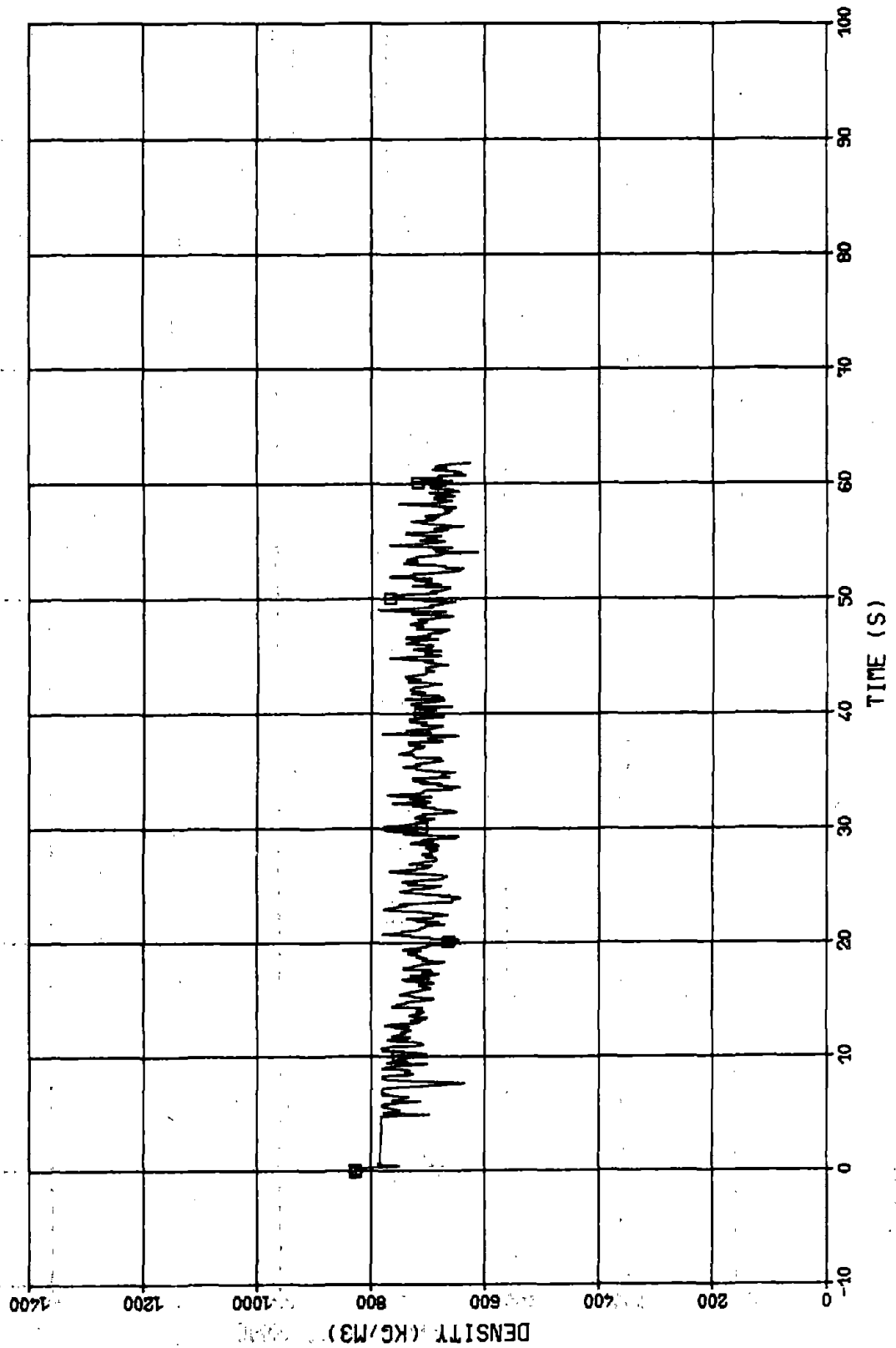


Plot C:22 - Test 22

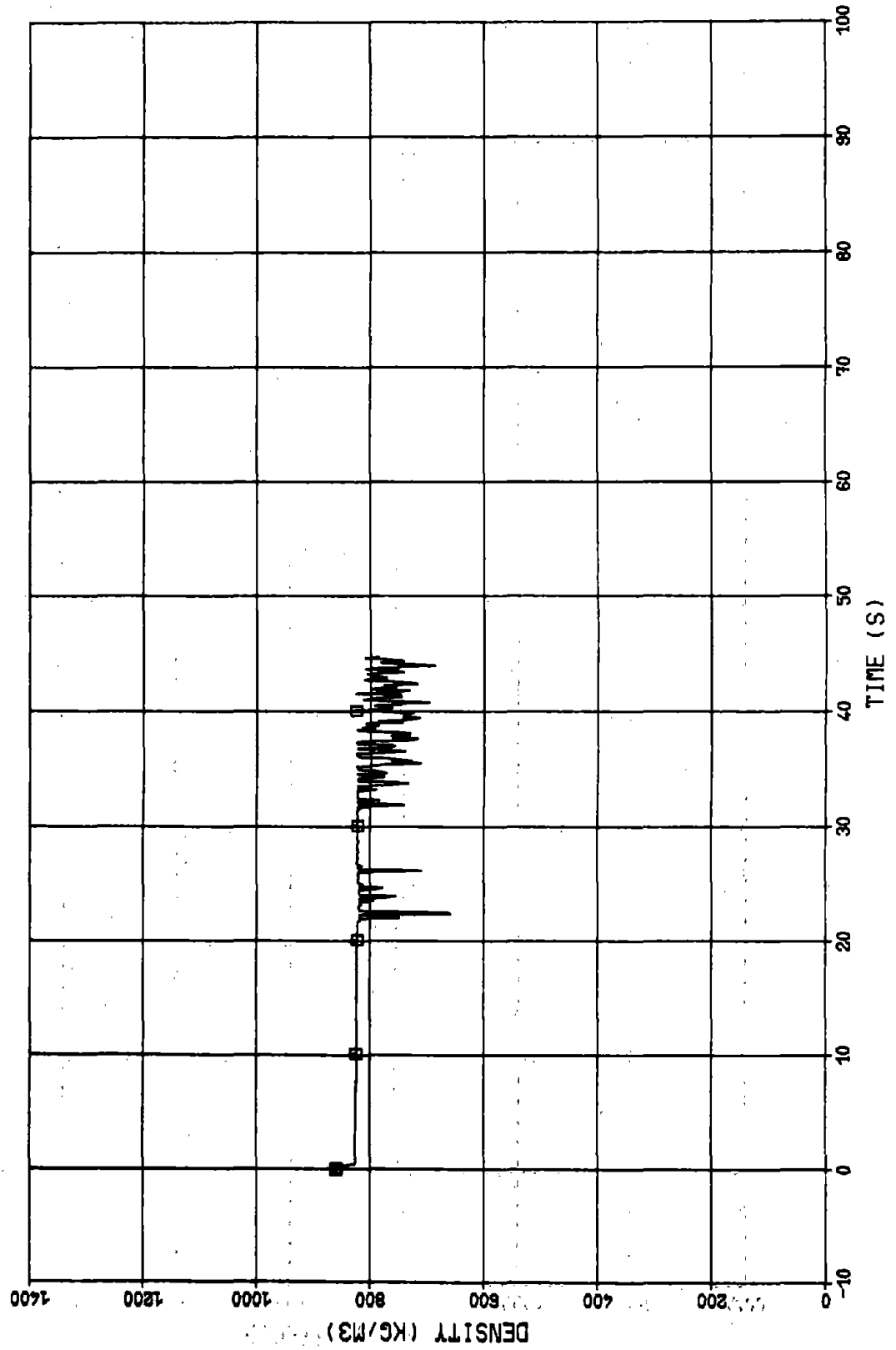


Plot C:23 - Test 23

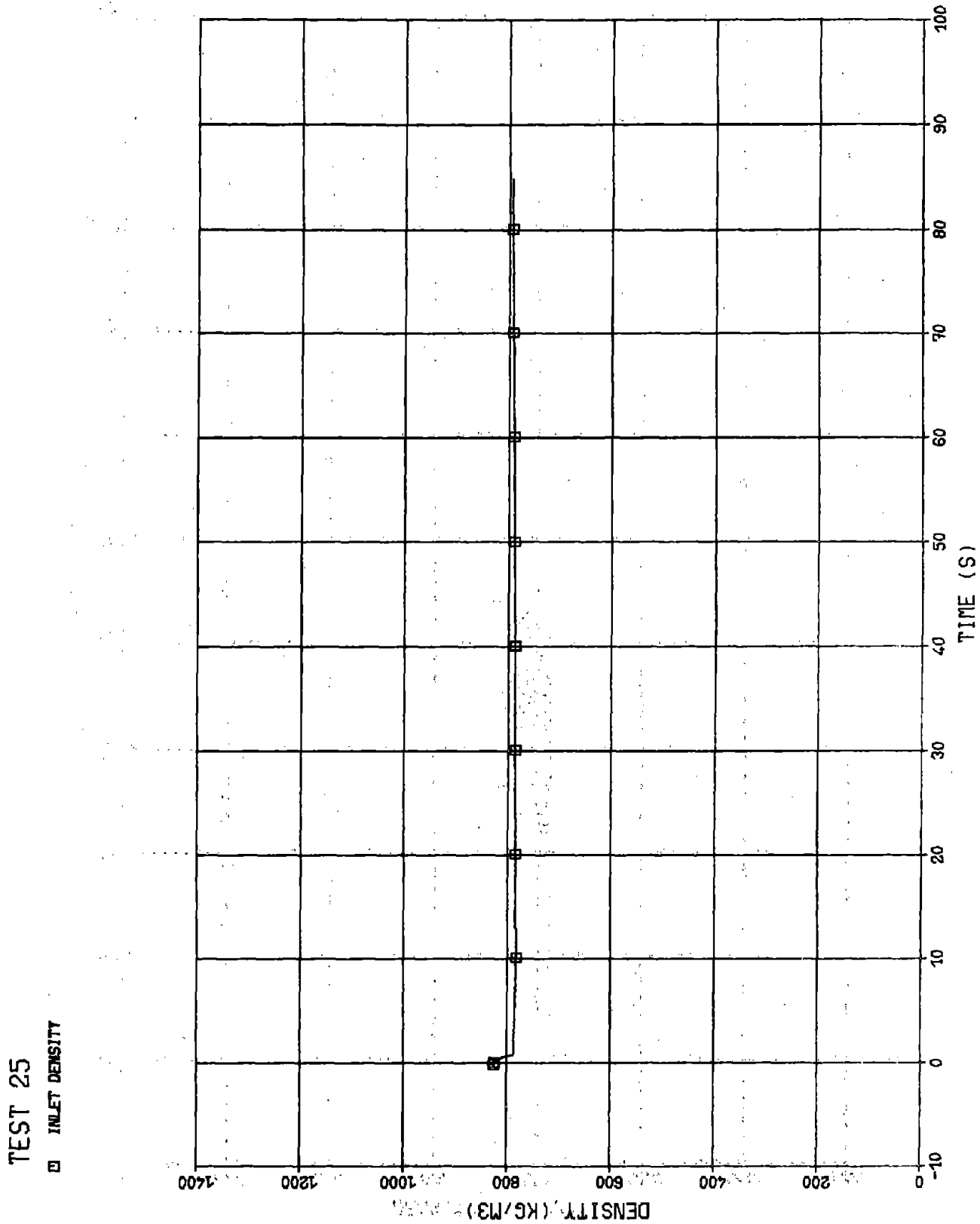
TEST 23
□ INLET DENSITY



Plot C:24 - Test 24

TEST 24
B INLET DENSITY

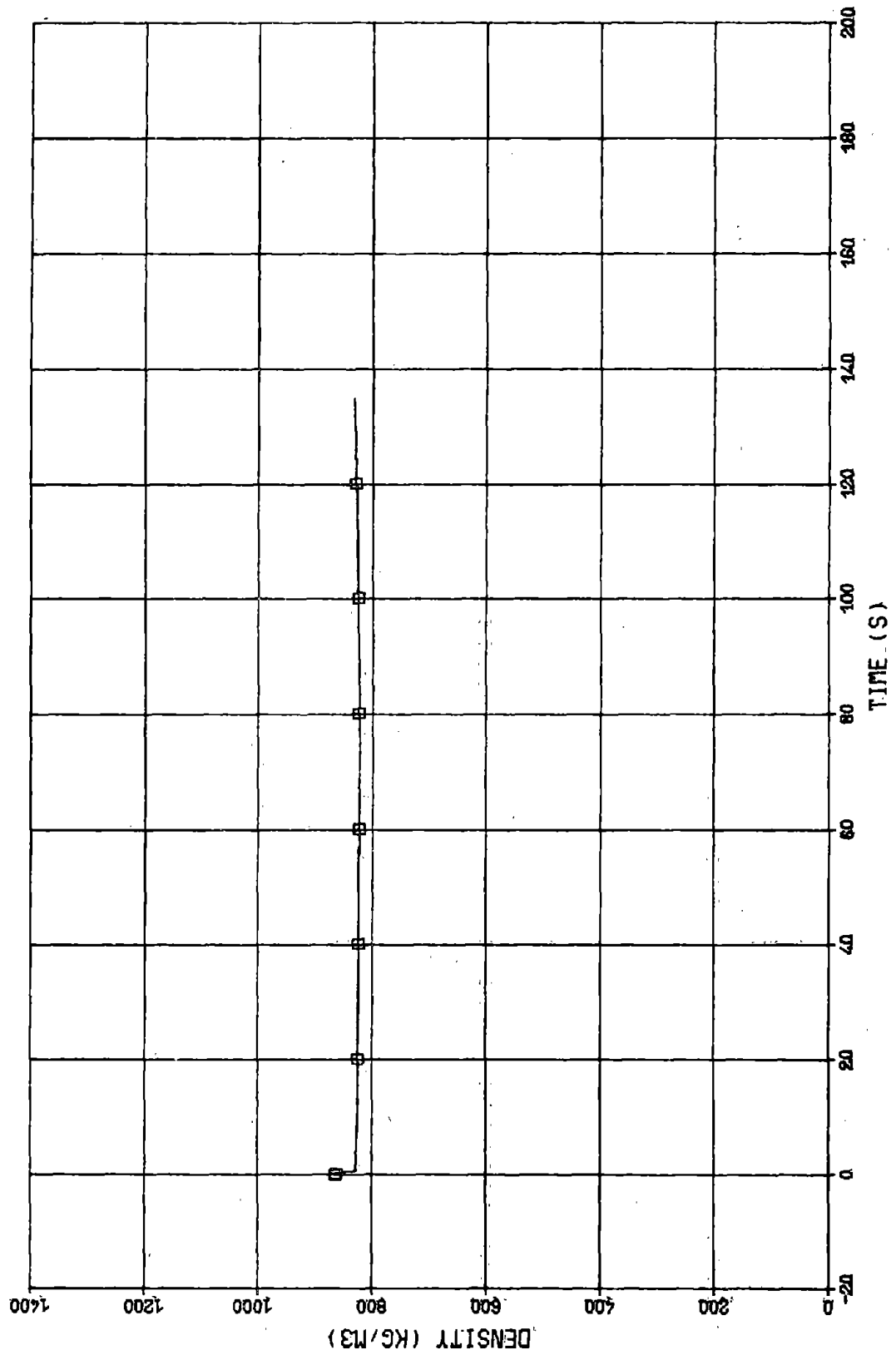
Plot C:25 - Test 25



Plot C:26 - Test 26

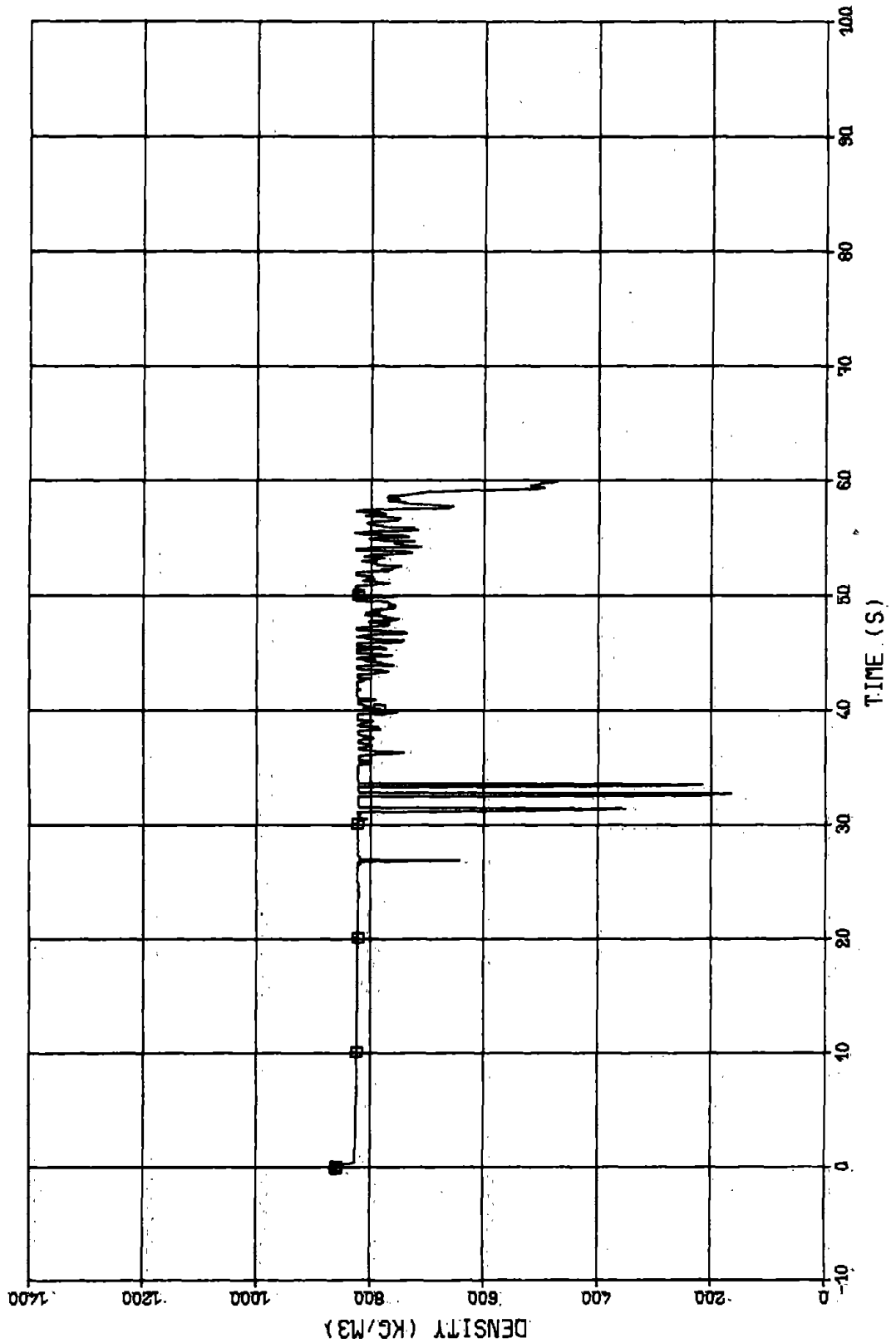
TEST 26

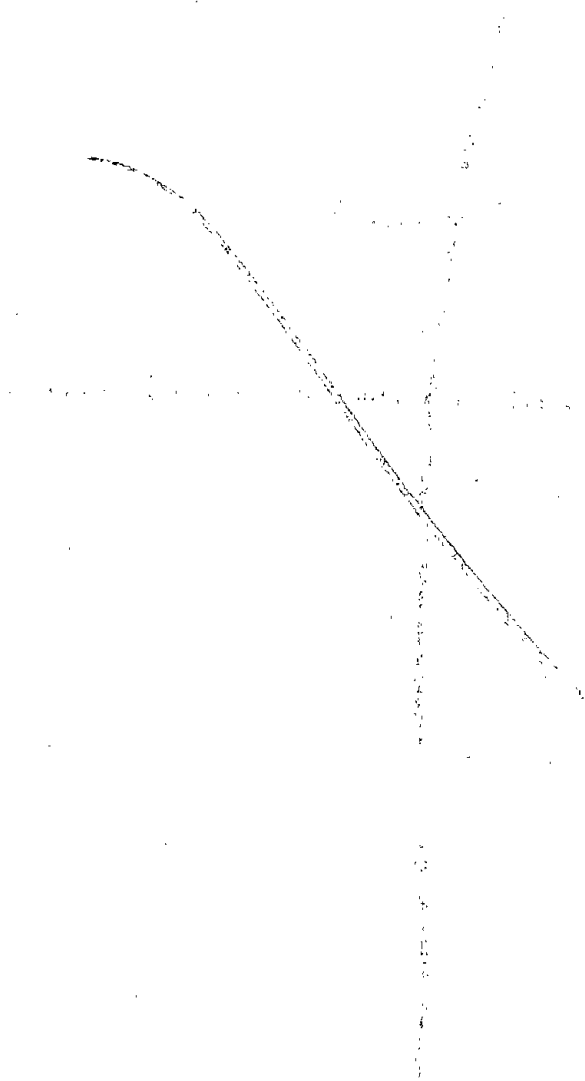
□ INLET DENSITY



Plot C:27 - Test 27

TEST 27
□ INLET DENSITY





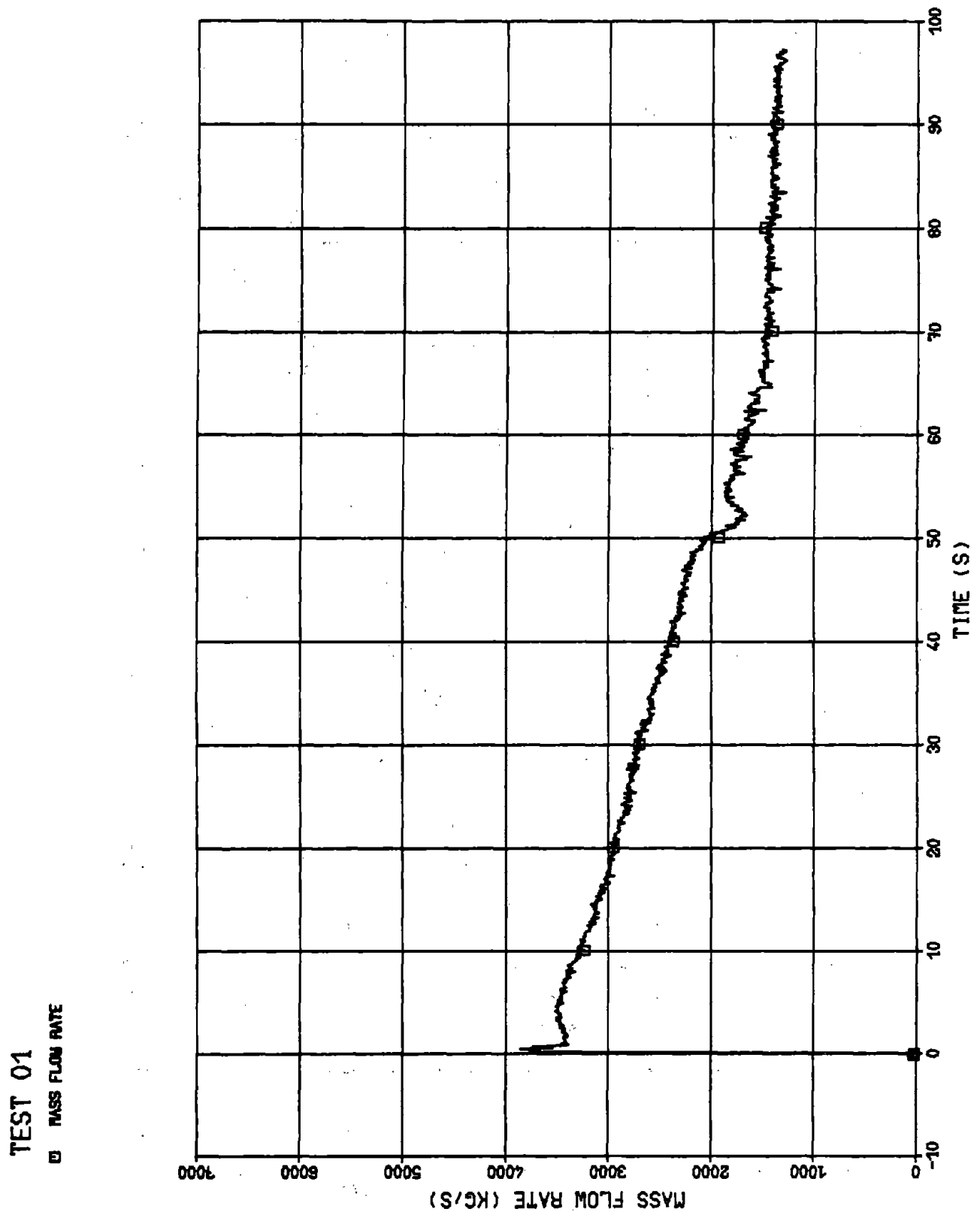
APPENDIX D

Nozzle mass flow rate: Tests 1 through 27

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Plot D:1 - Test 1

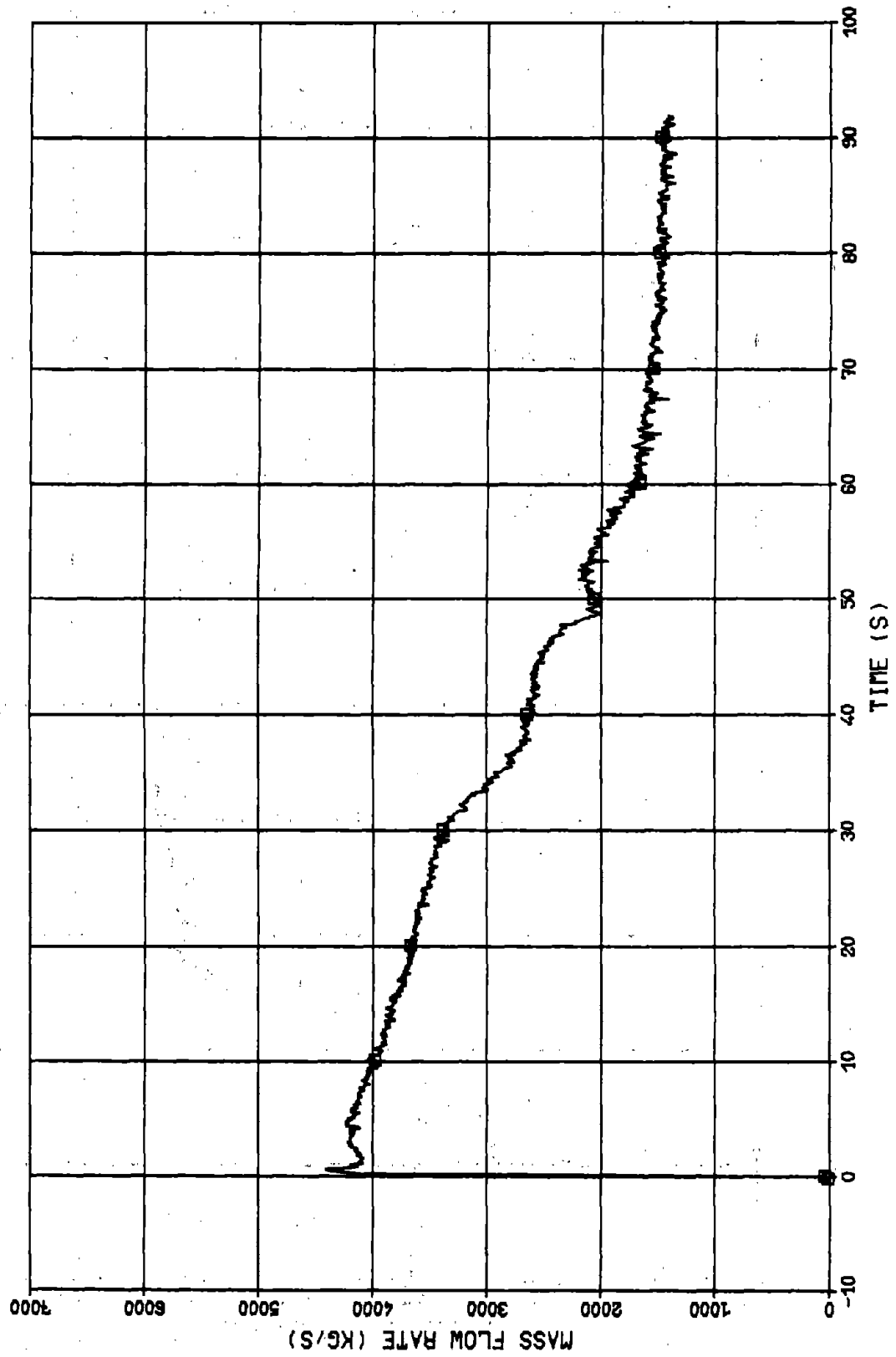


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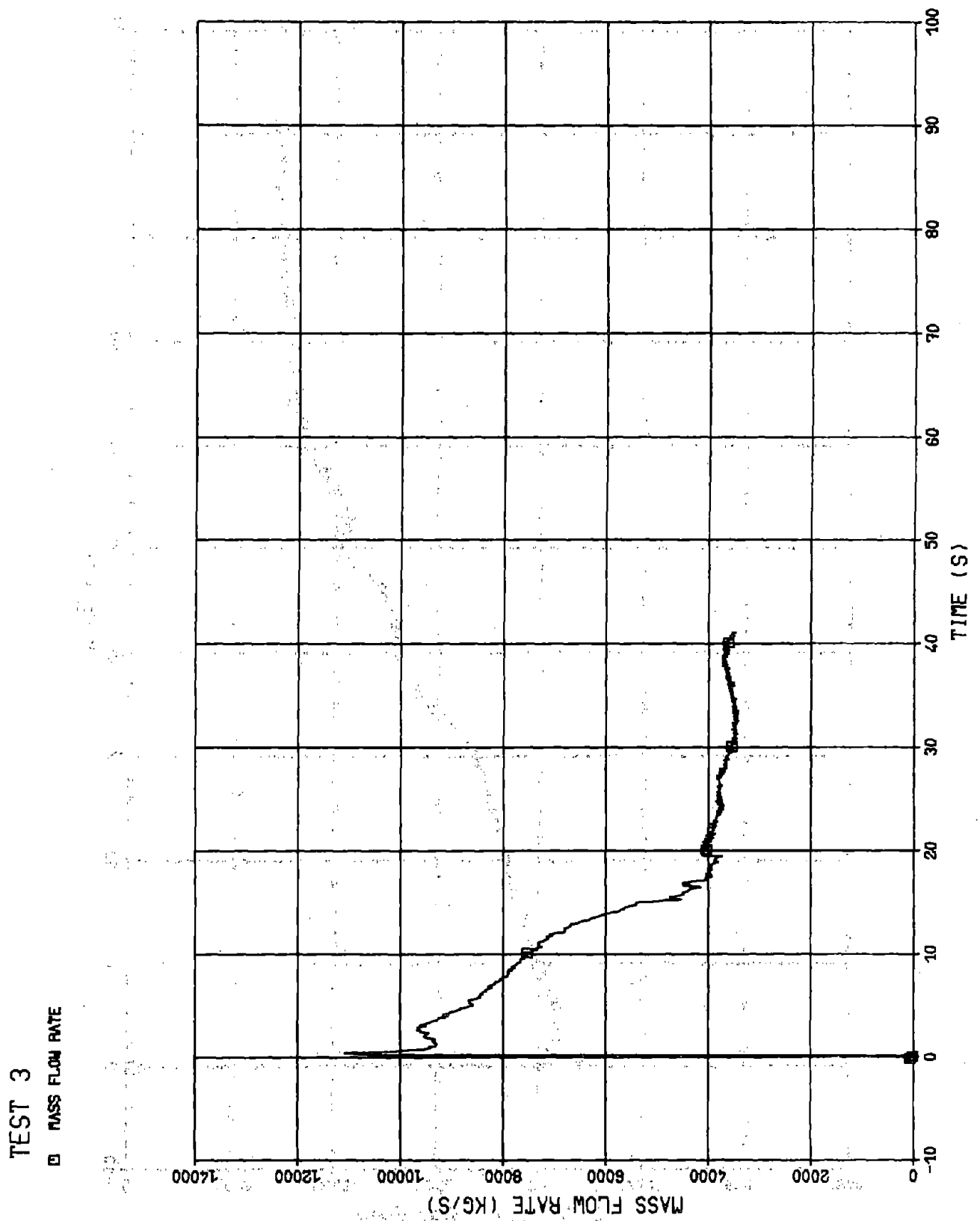
Plot D:2 - Test 2

TEST 2

□ MASS FLOW RATE

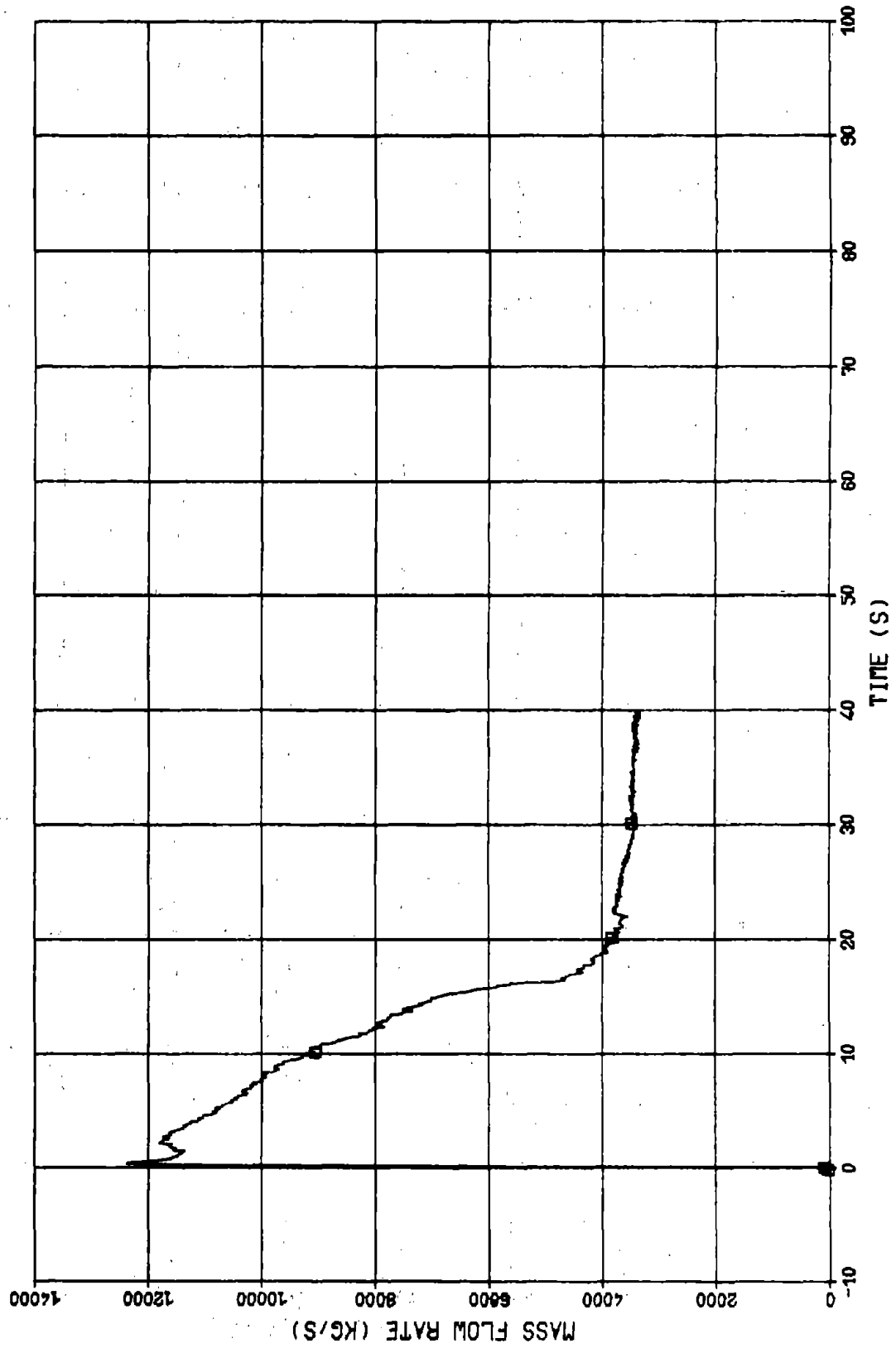


Plot D:3 - Test 3

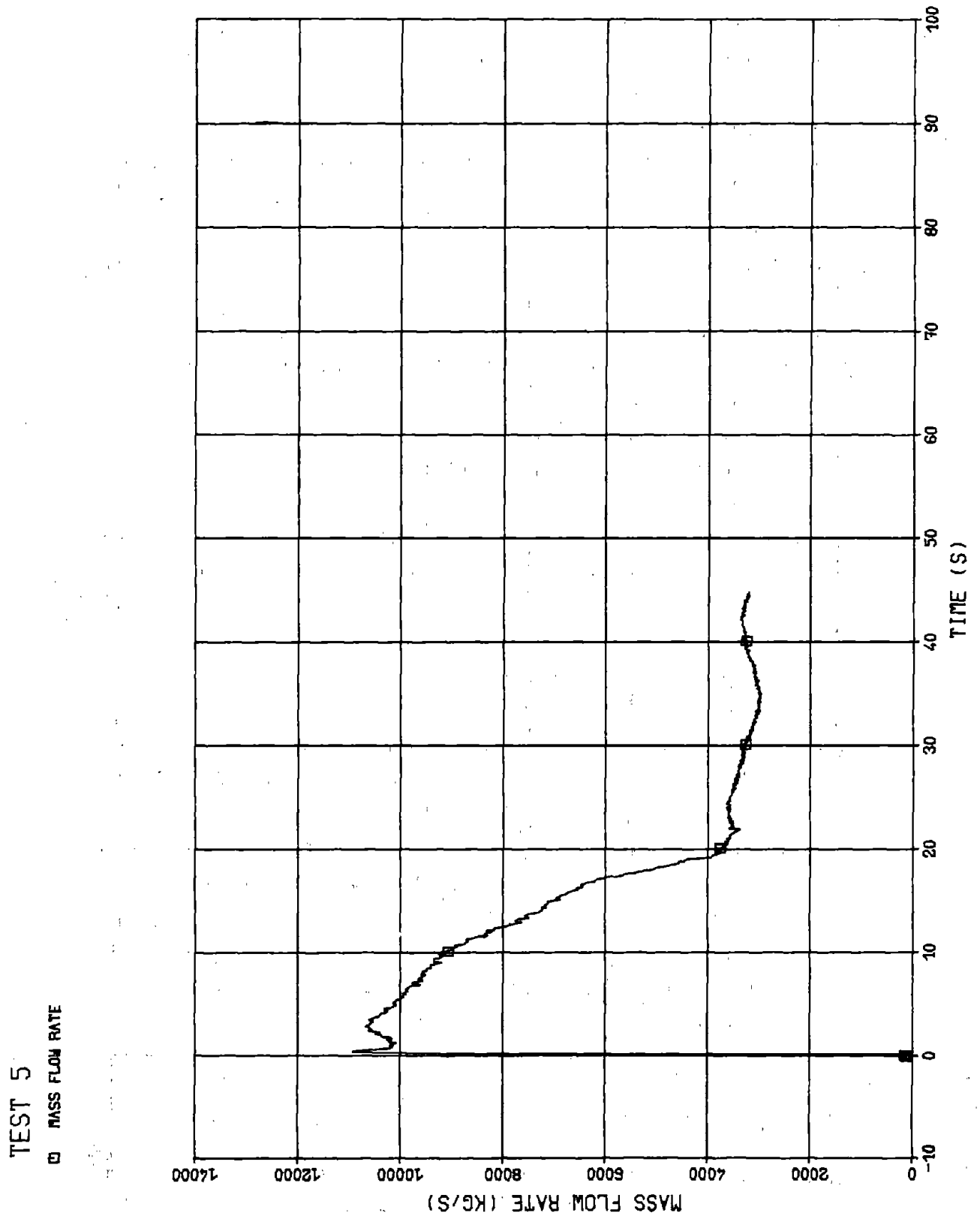


Plot D:4 - Test 4

TEST 4
□ MASS FLOW RATE



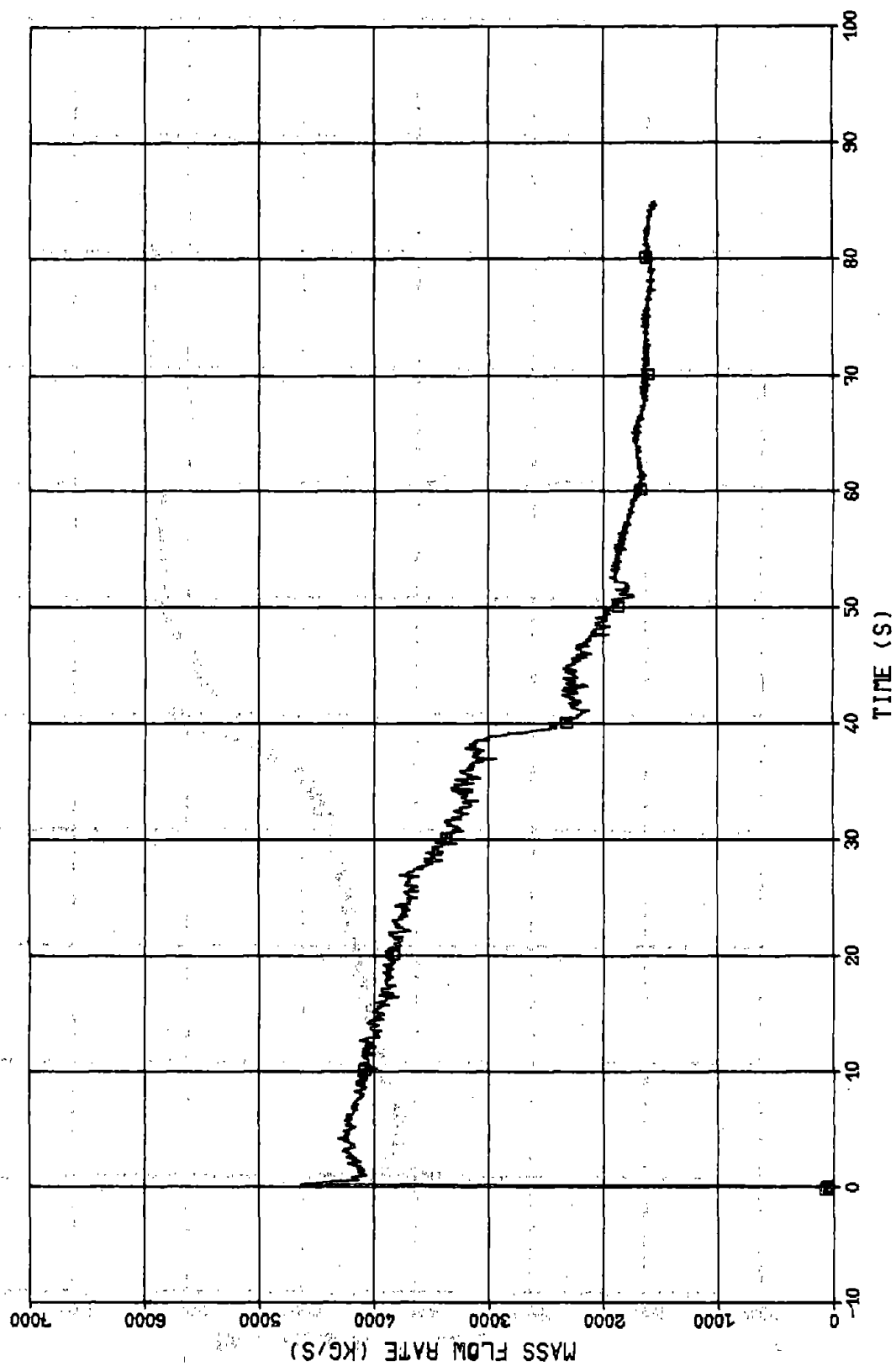
Plot D:5 - Test 5



Plot D:6 - Test 6

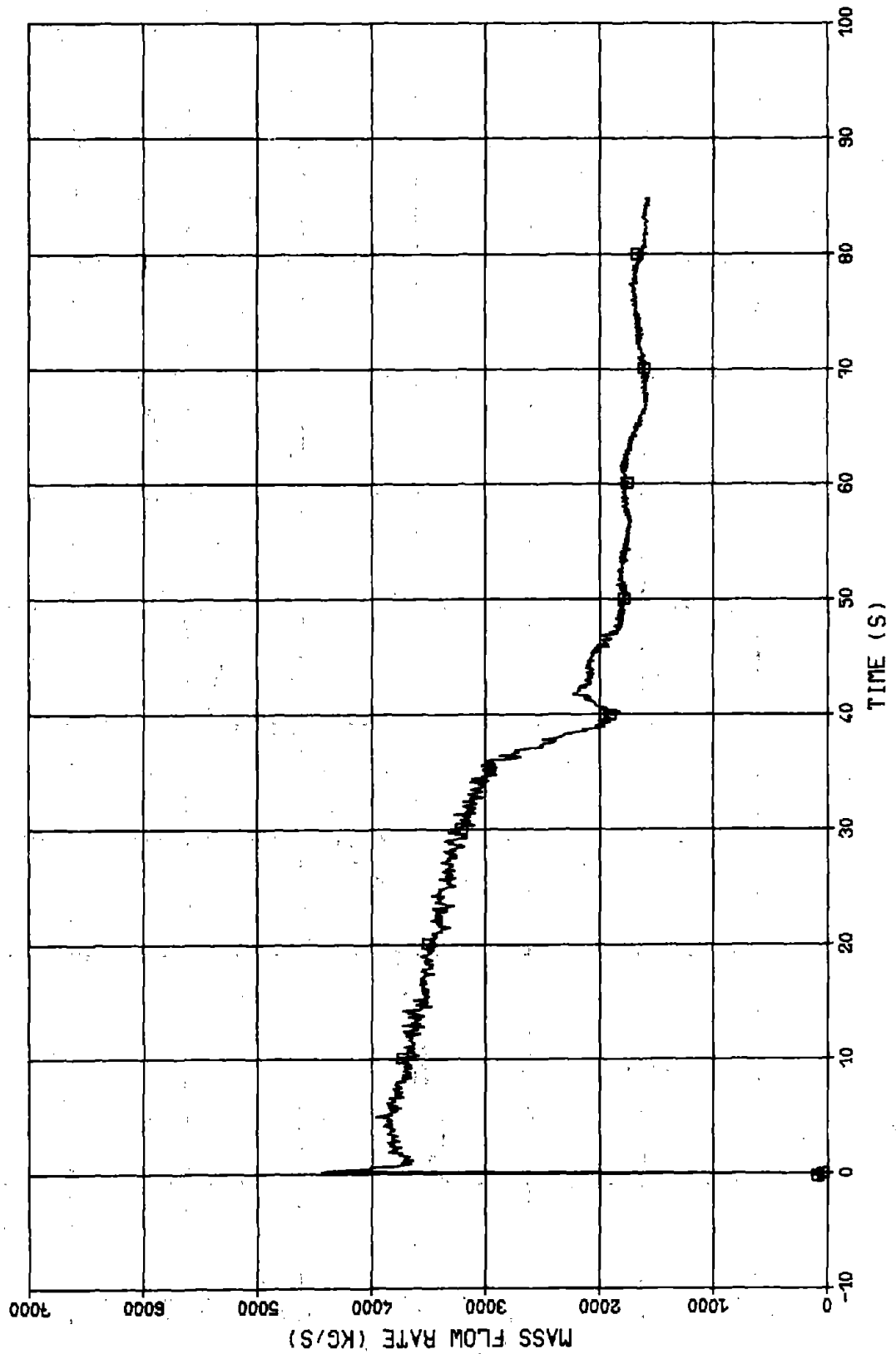
TEST 6

□ MASS FLOW RATE

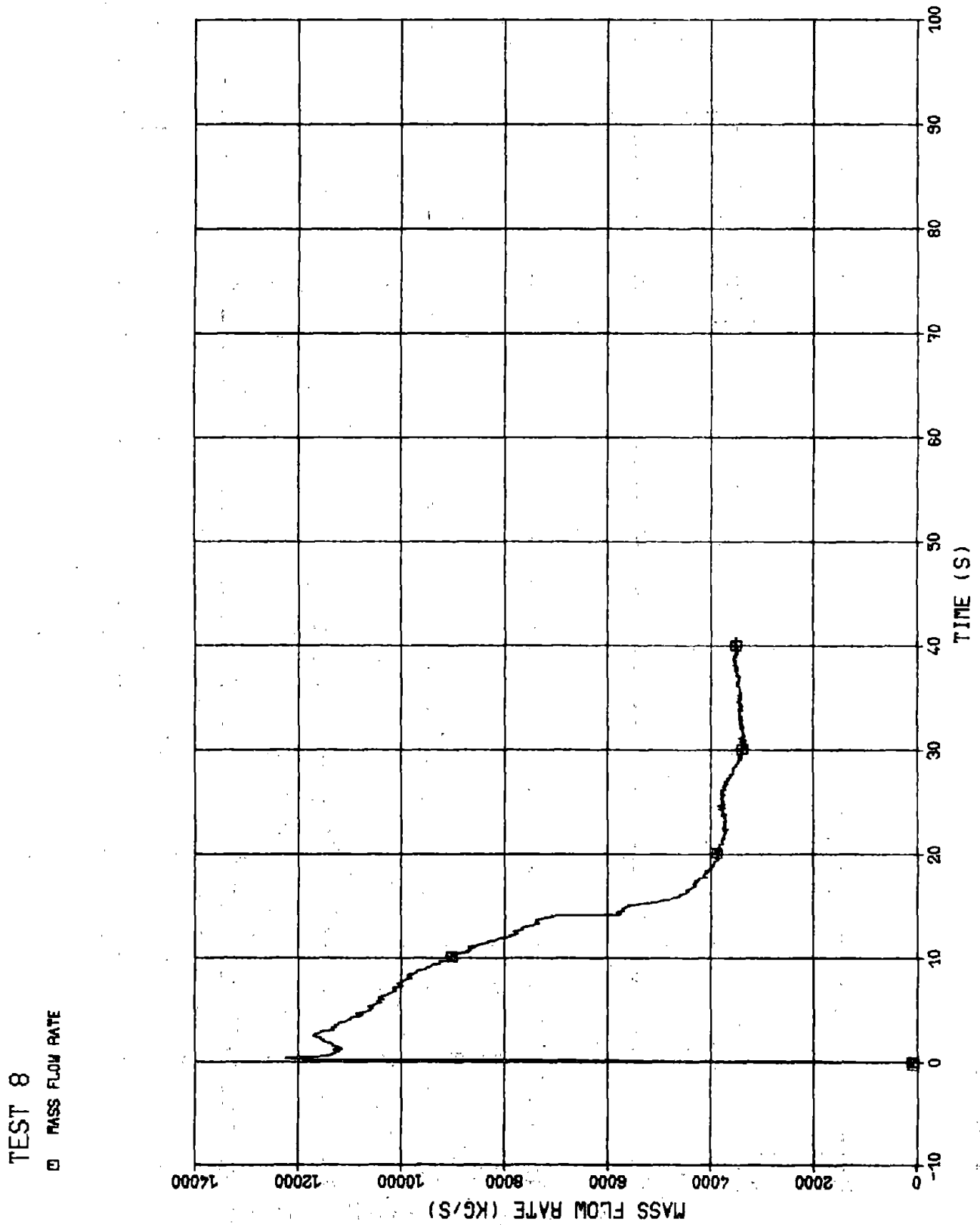


Plot D:7 - Test 7

TEST 7
□ MASS FLOW RATE



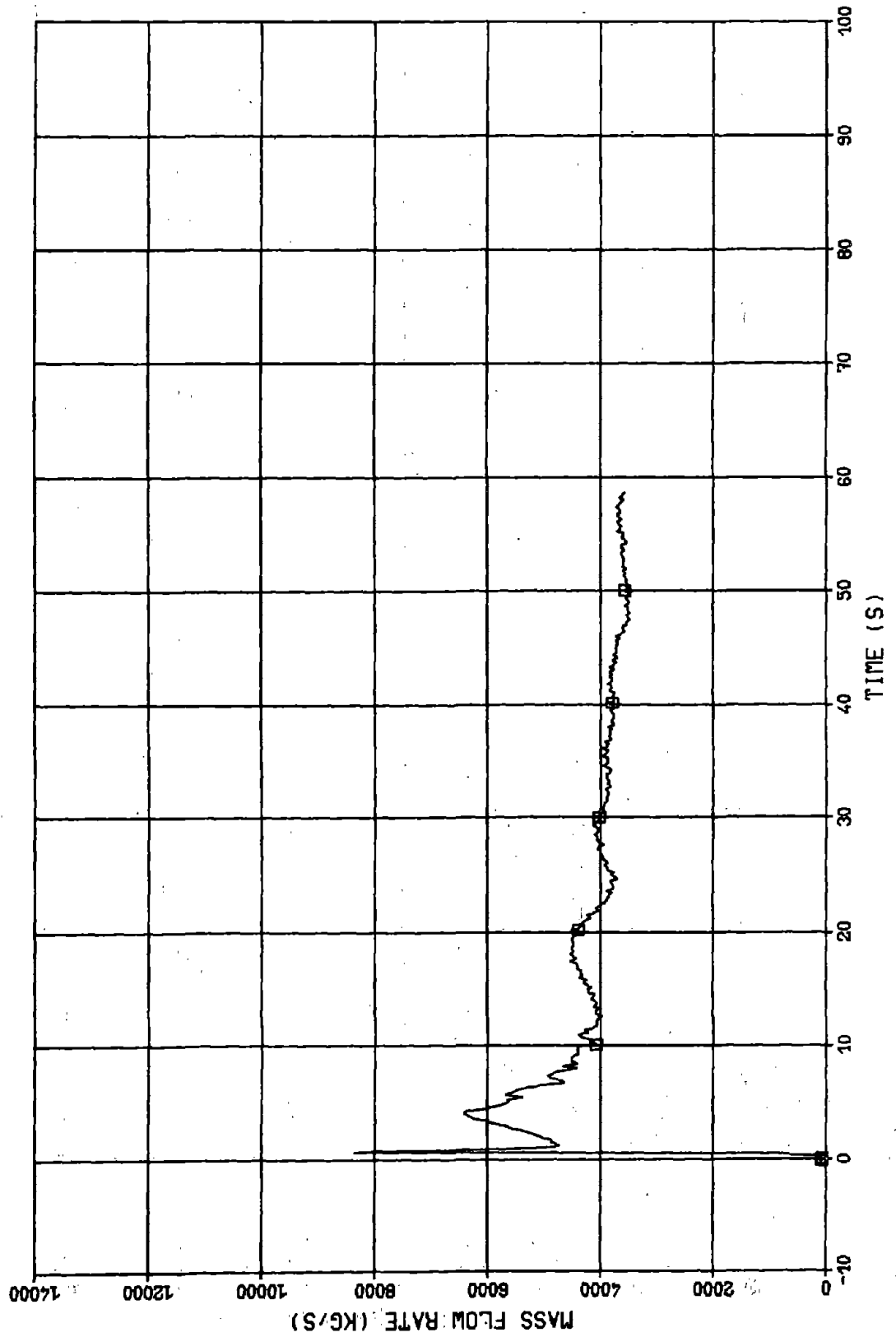
Plot D:8 - Test 8



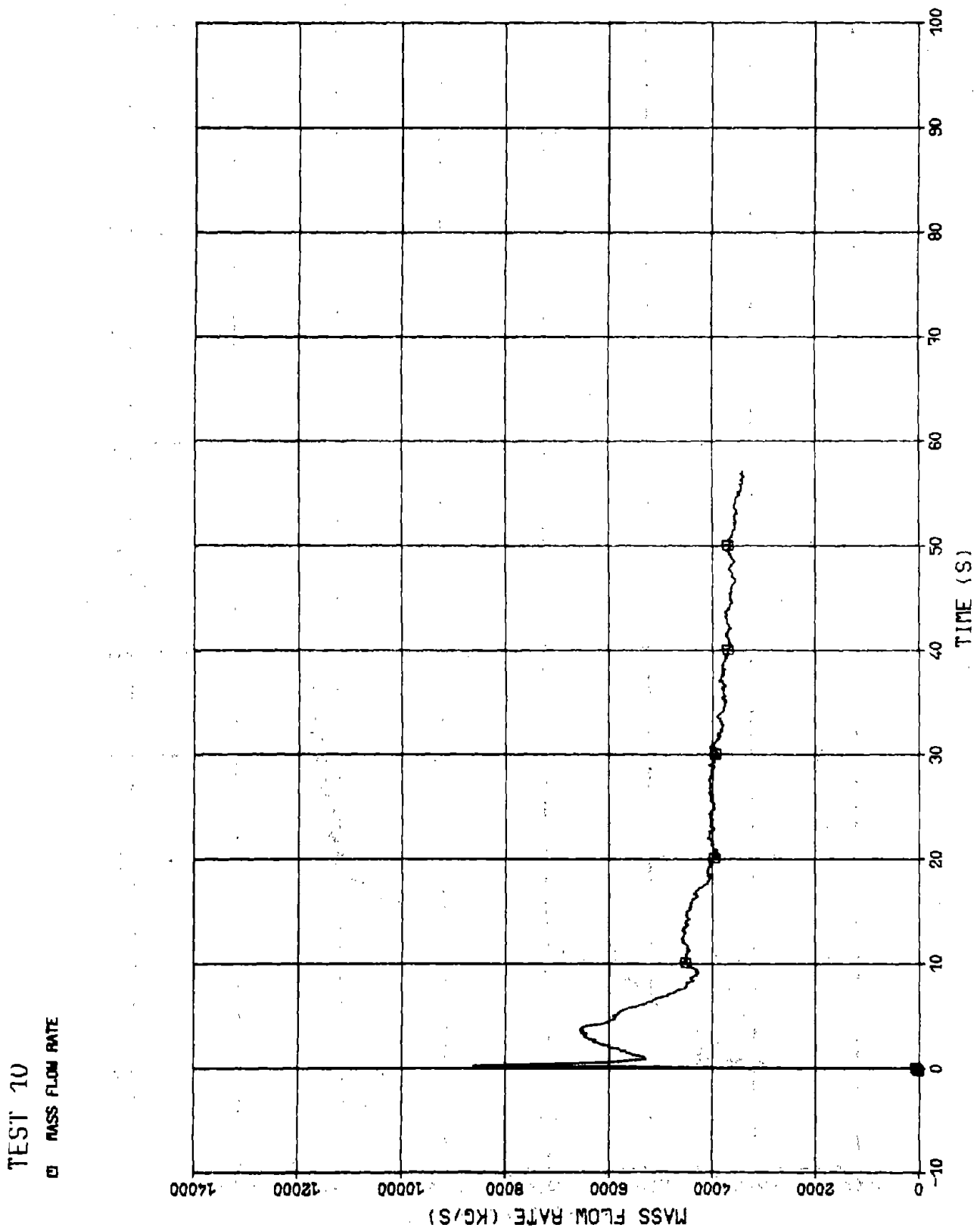
Plot D:9 - Test 9

TEST 9

□ MASS FLOW RATE

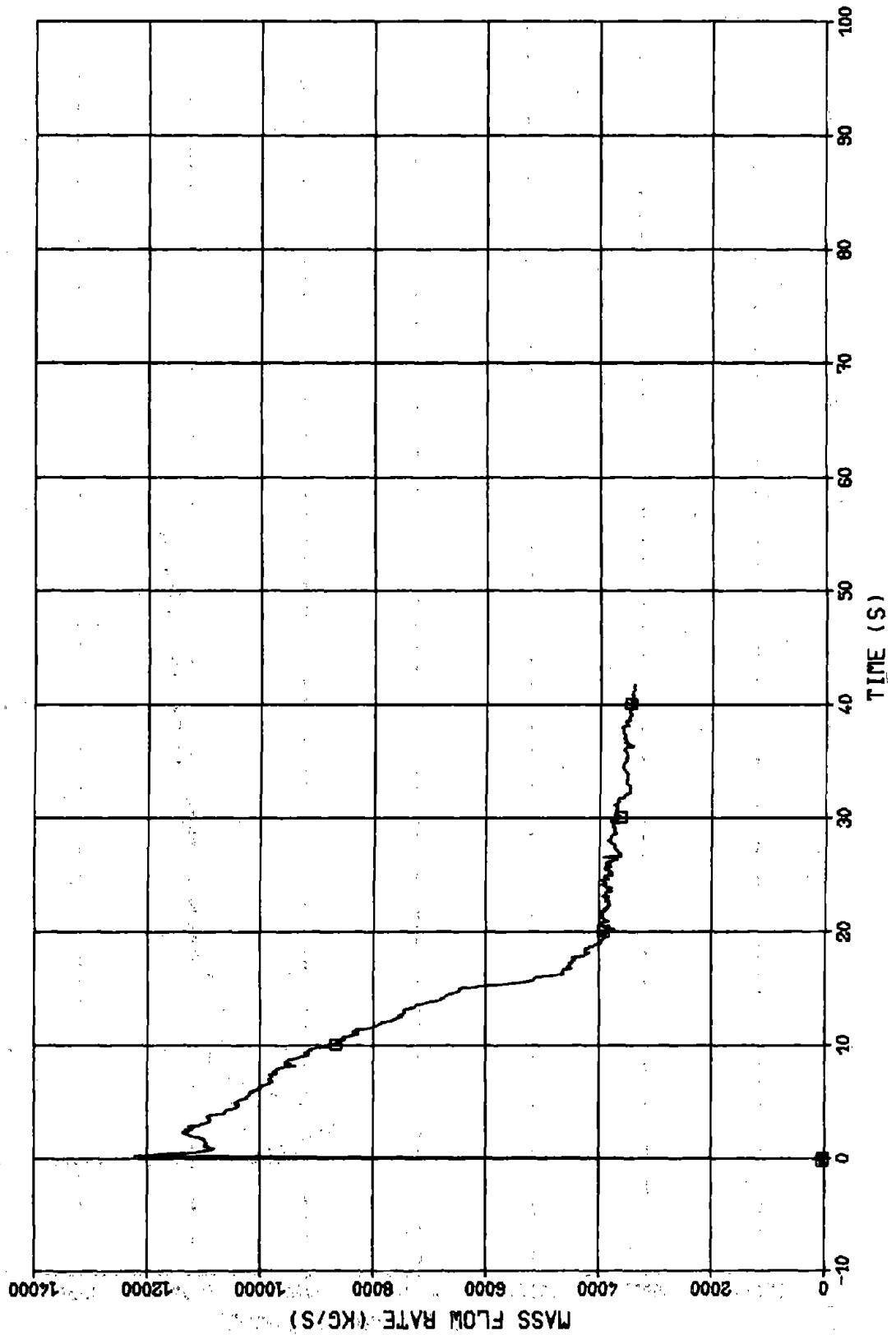


Plot D:10 - Test 10



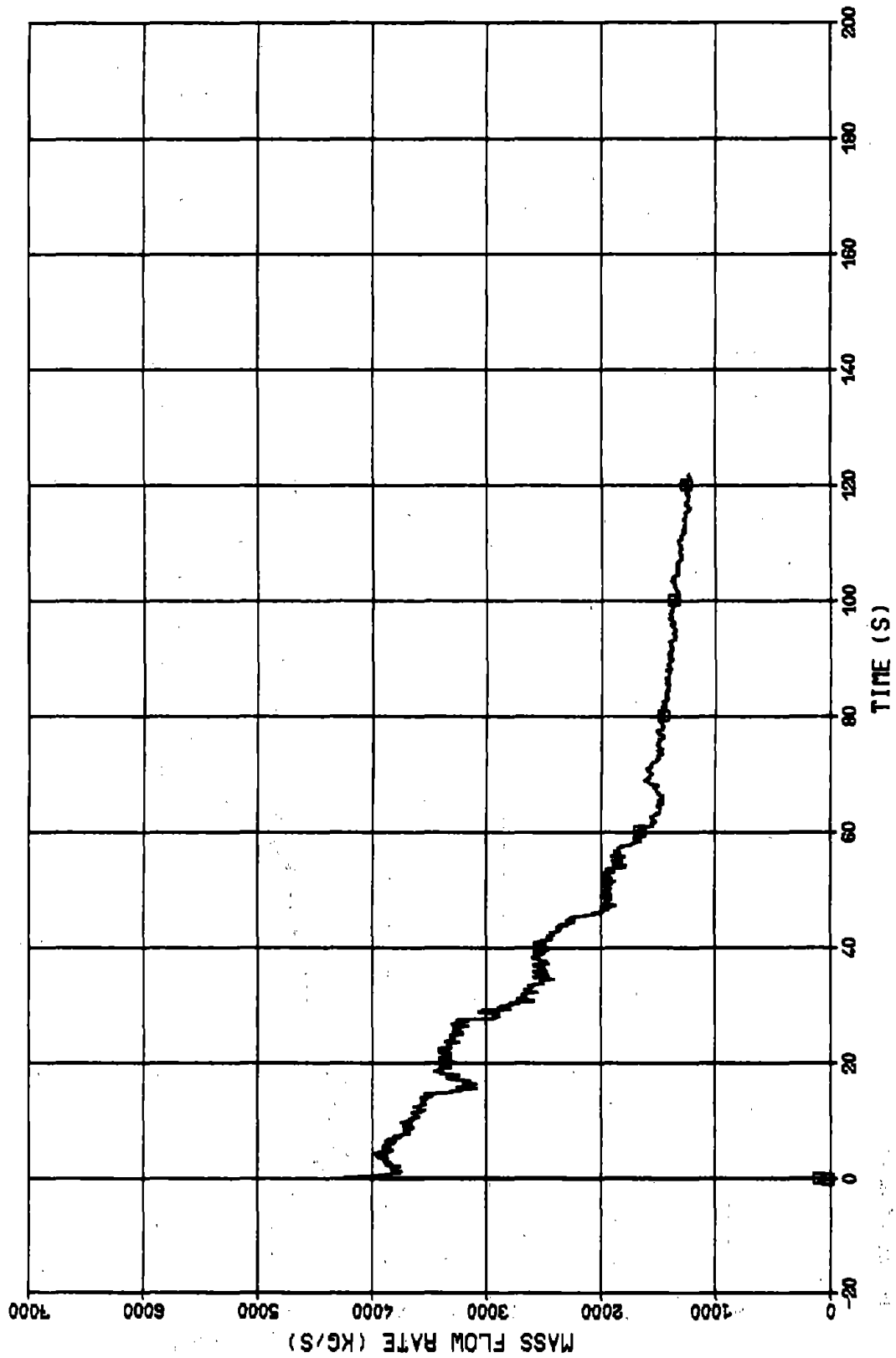
Plot D:11- Test 11

TEST 11
□ MASS FLOW RATE

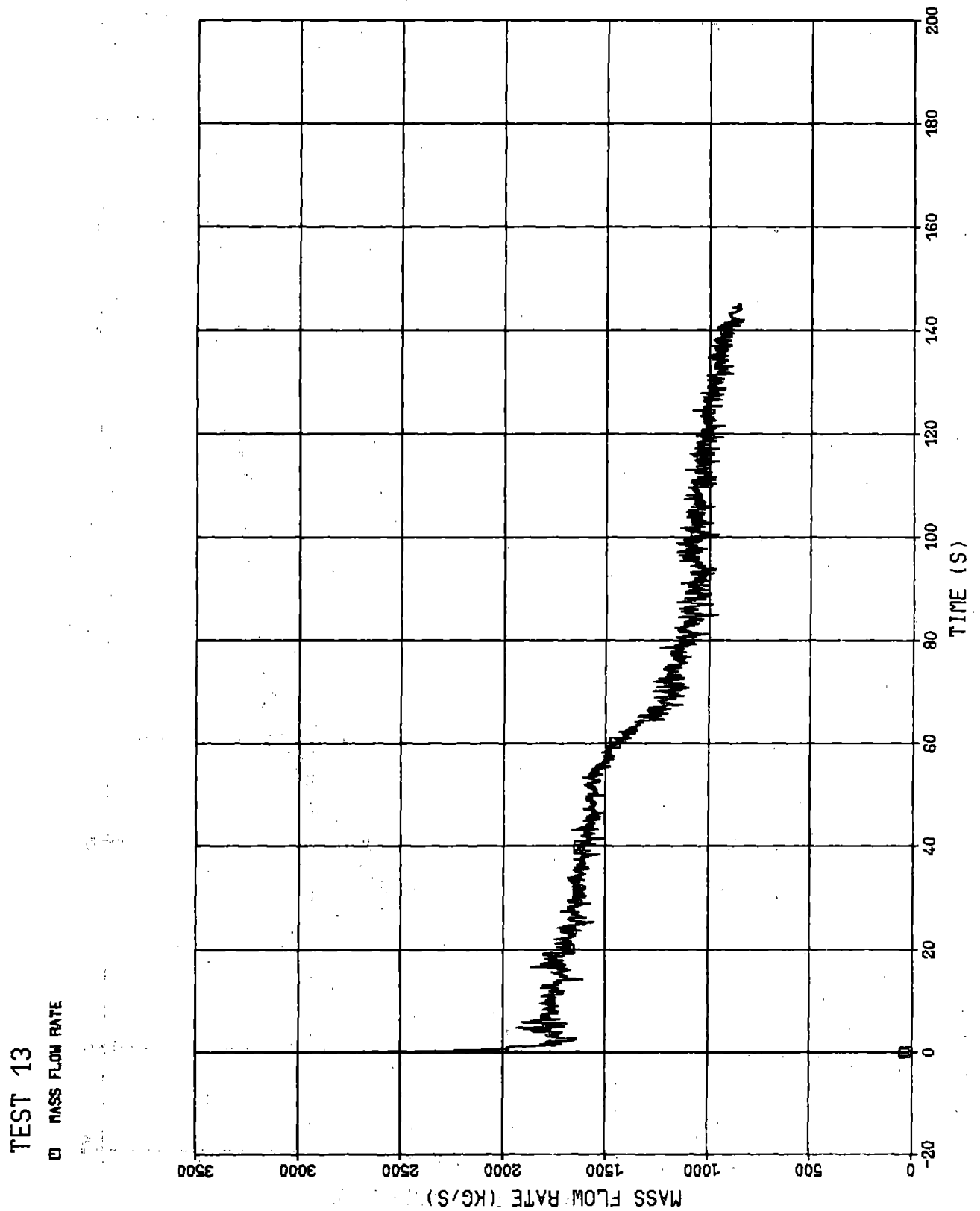


Plot D:12 - Test 12

TEST 12
□ MASS FLOW RATE



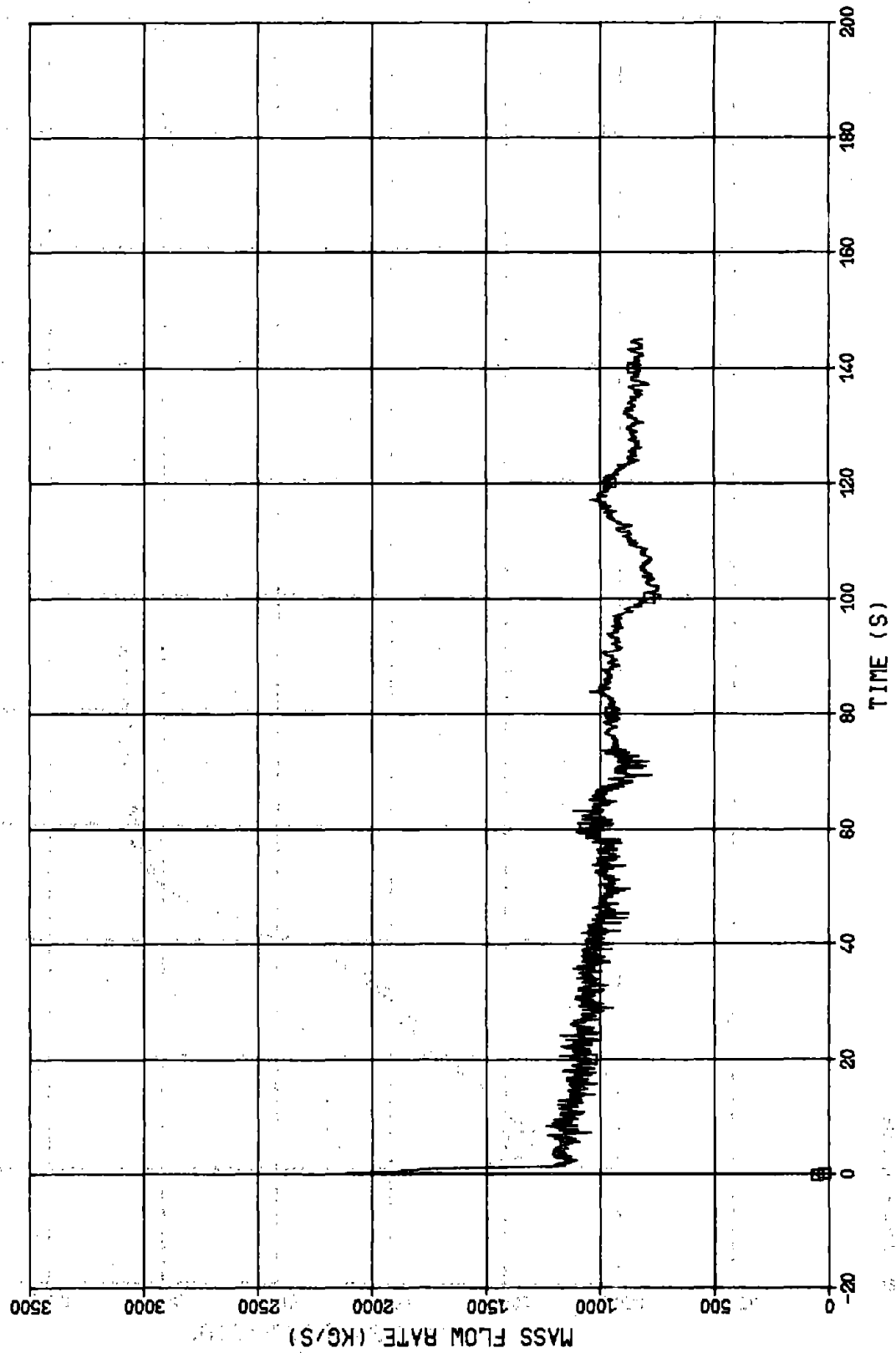
Plot D:13 - Test 13



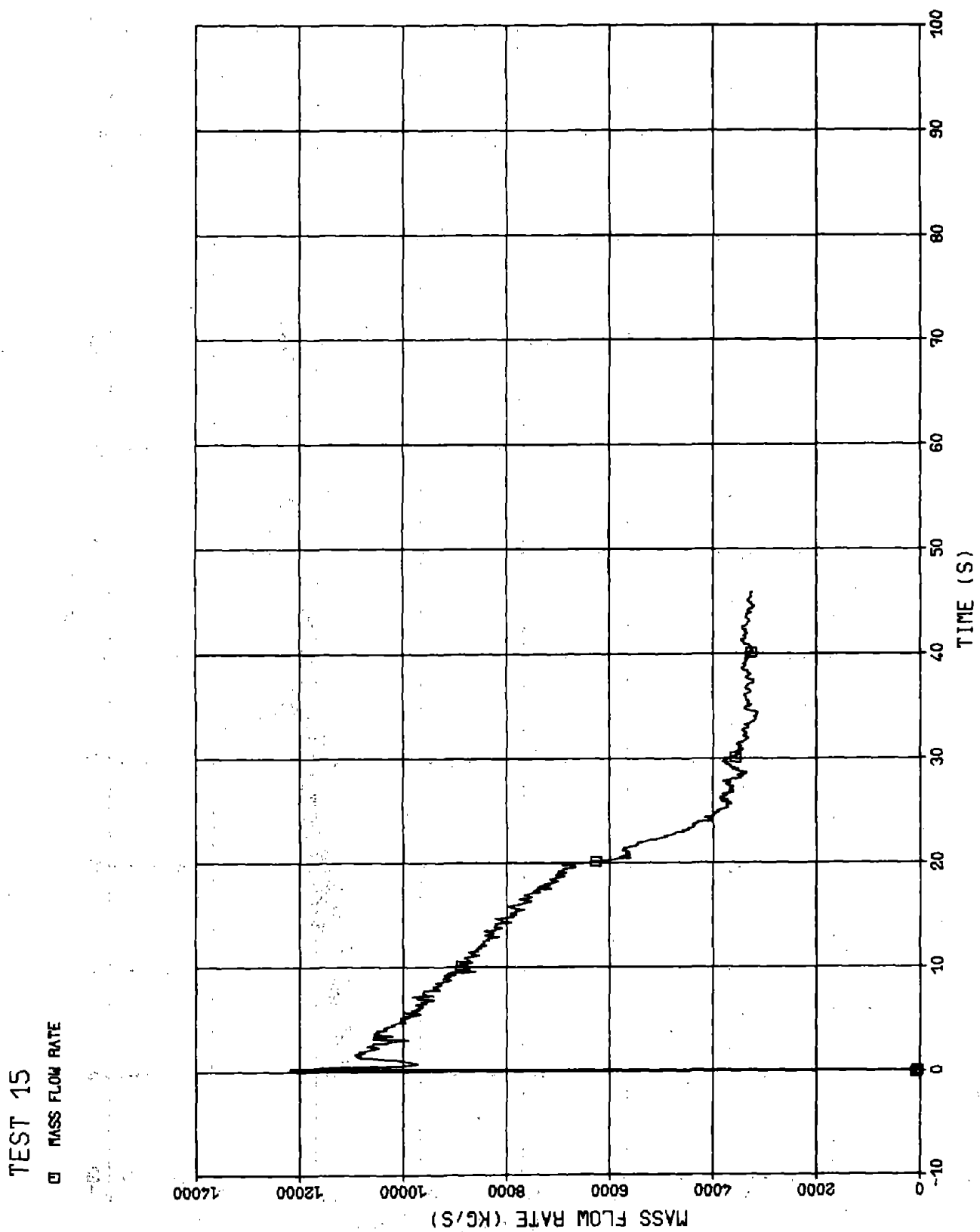
Plot D:14 - Test 14

TEST 14

□ MASS FLOW RATE

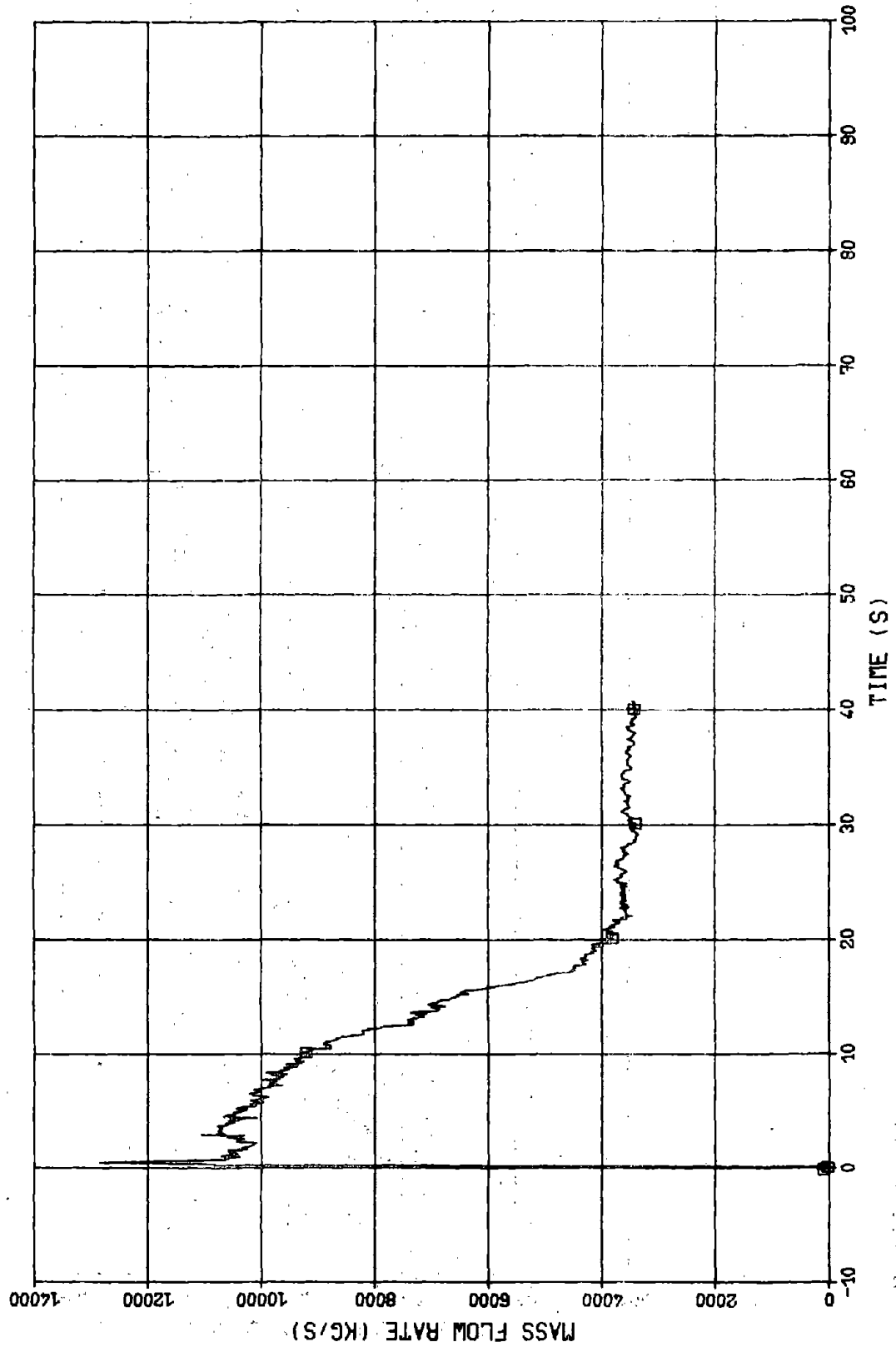


Plot D:15 - Test 15



Plot D:16 - Test 16

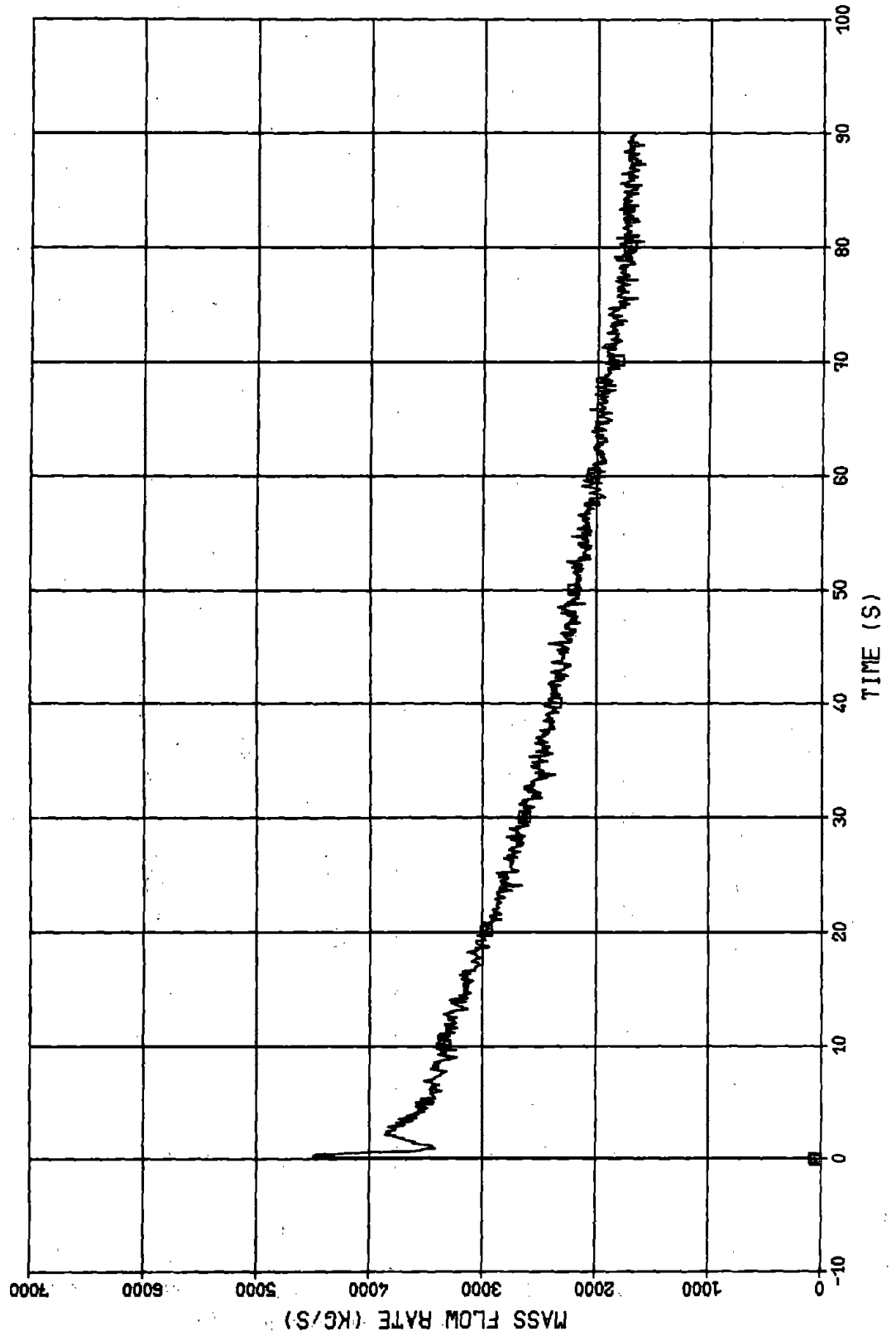
TEST 16
□ MASS FLOW RATE



Plot D:17 - Test 17

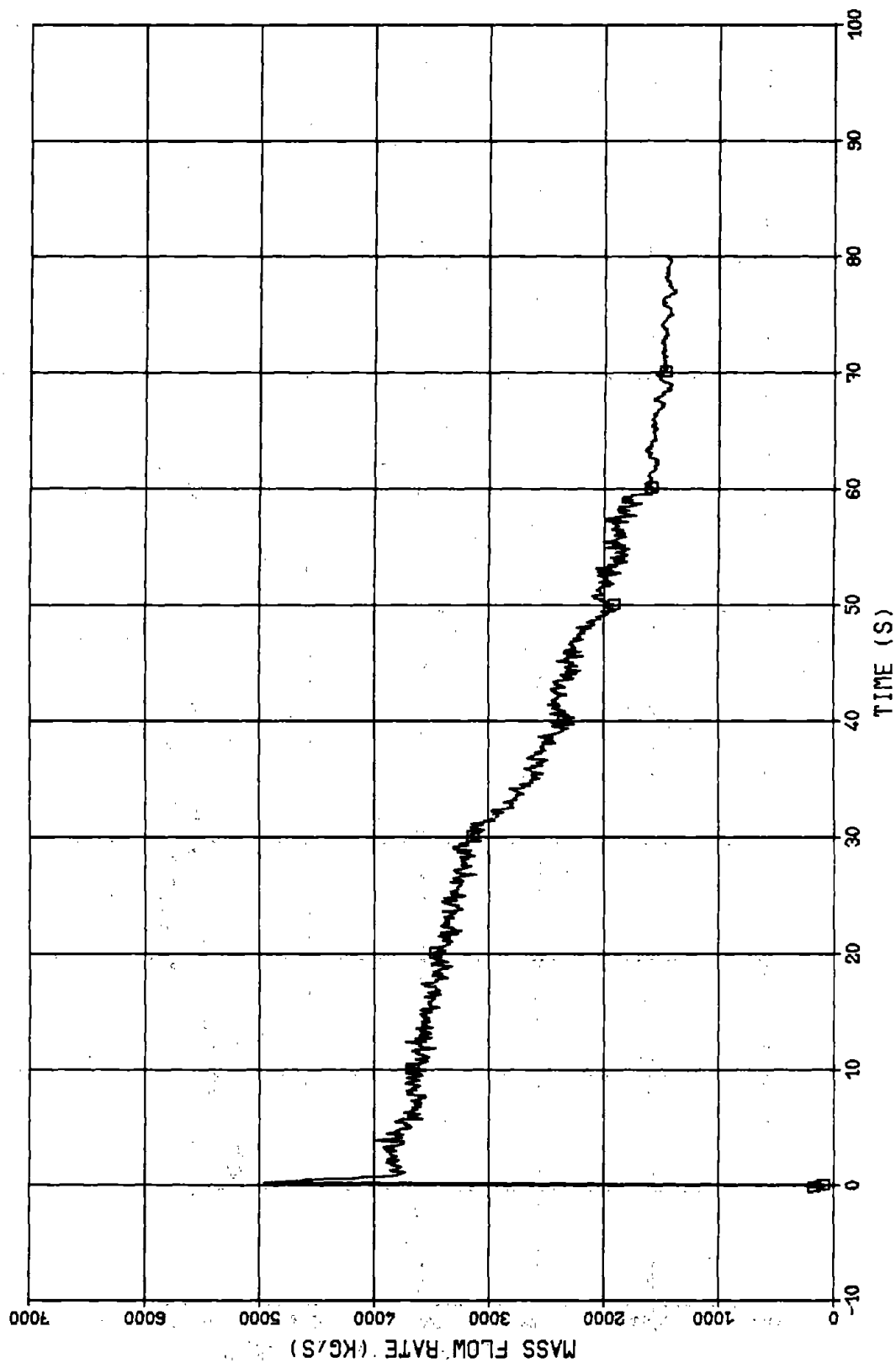
TEST 17

□ MASS FLOW RATE



Plot D:18 - Test 18

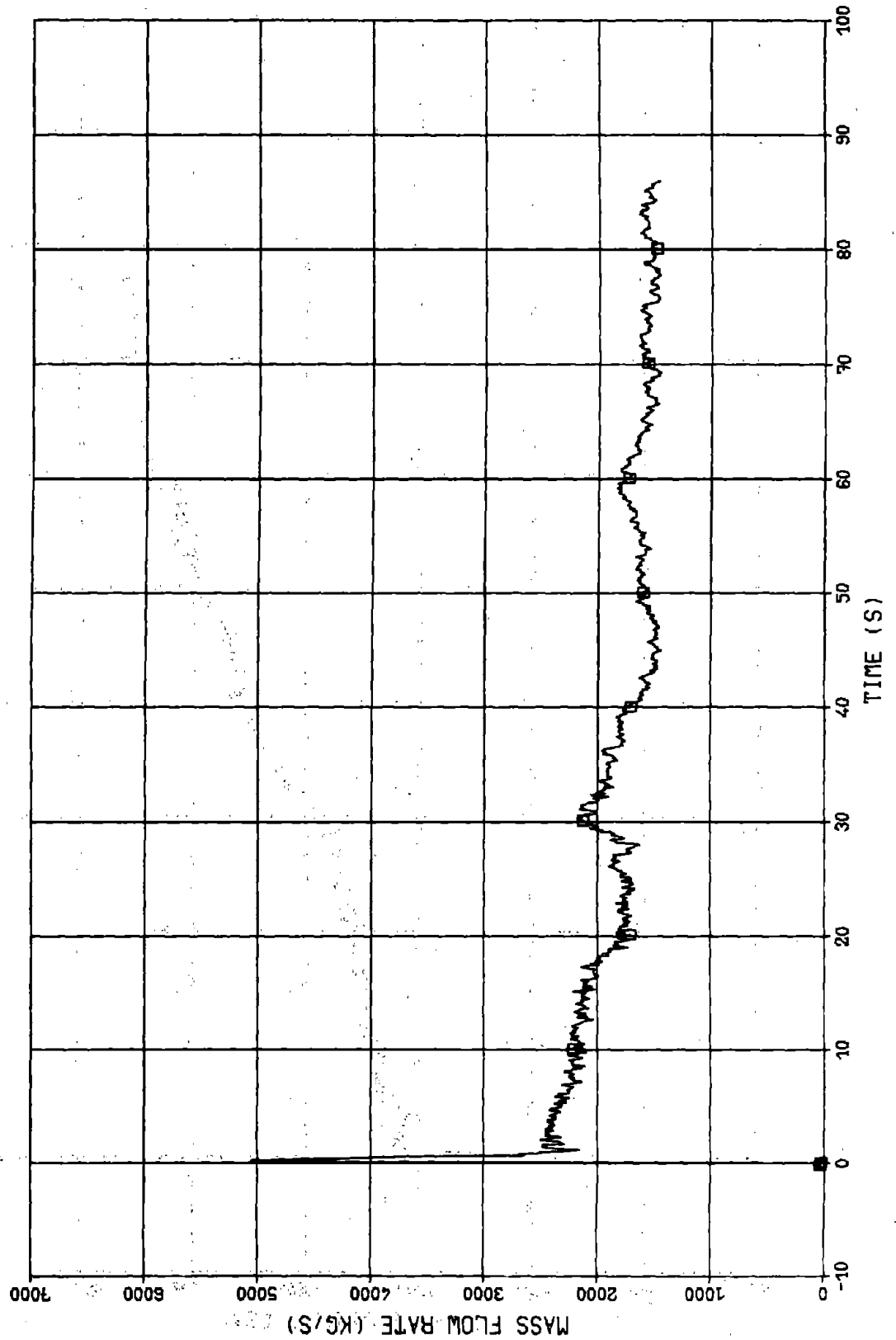
TEST 18
□ MASS FLOW RATE



Plot D:19 - Test 19

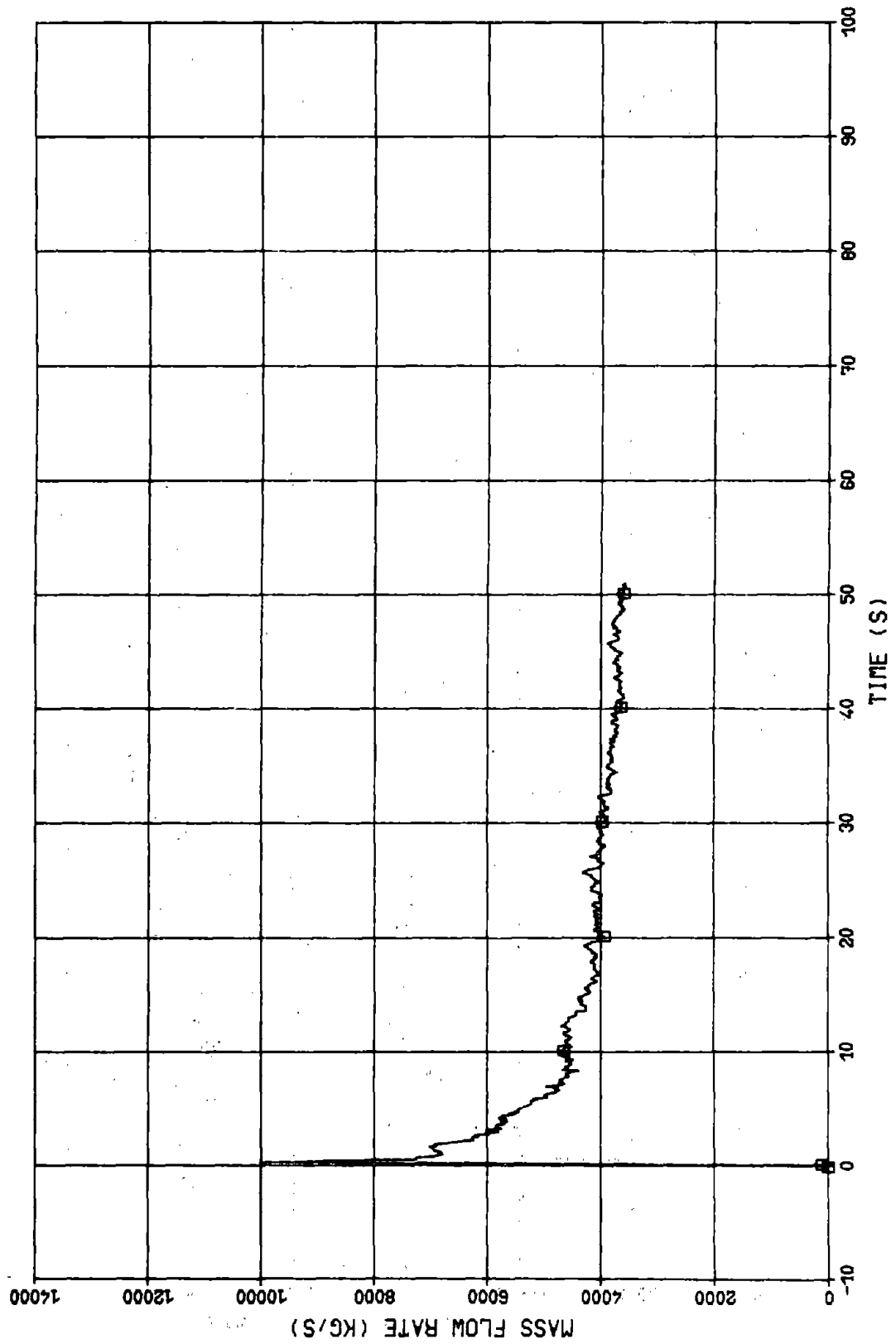
TEST 19

□ MASS FLOW RATE



Plot D:20 - Test 20

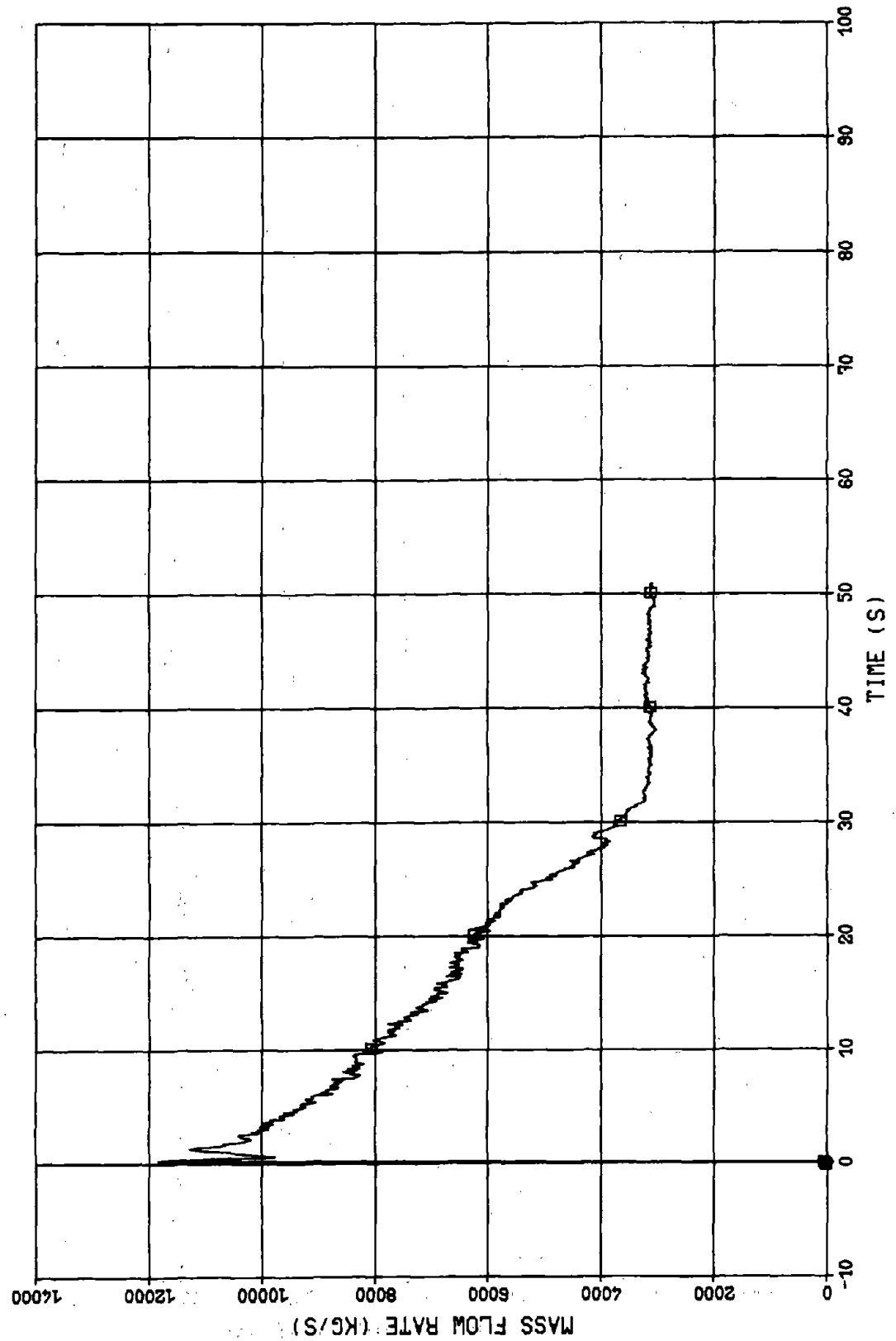
TEST 20
□ MASS FLOW RATE



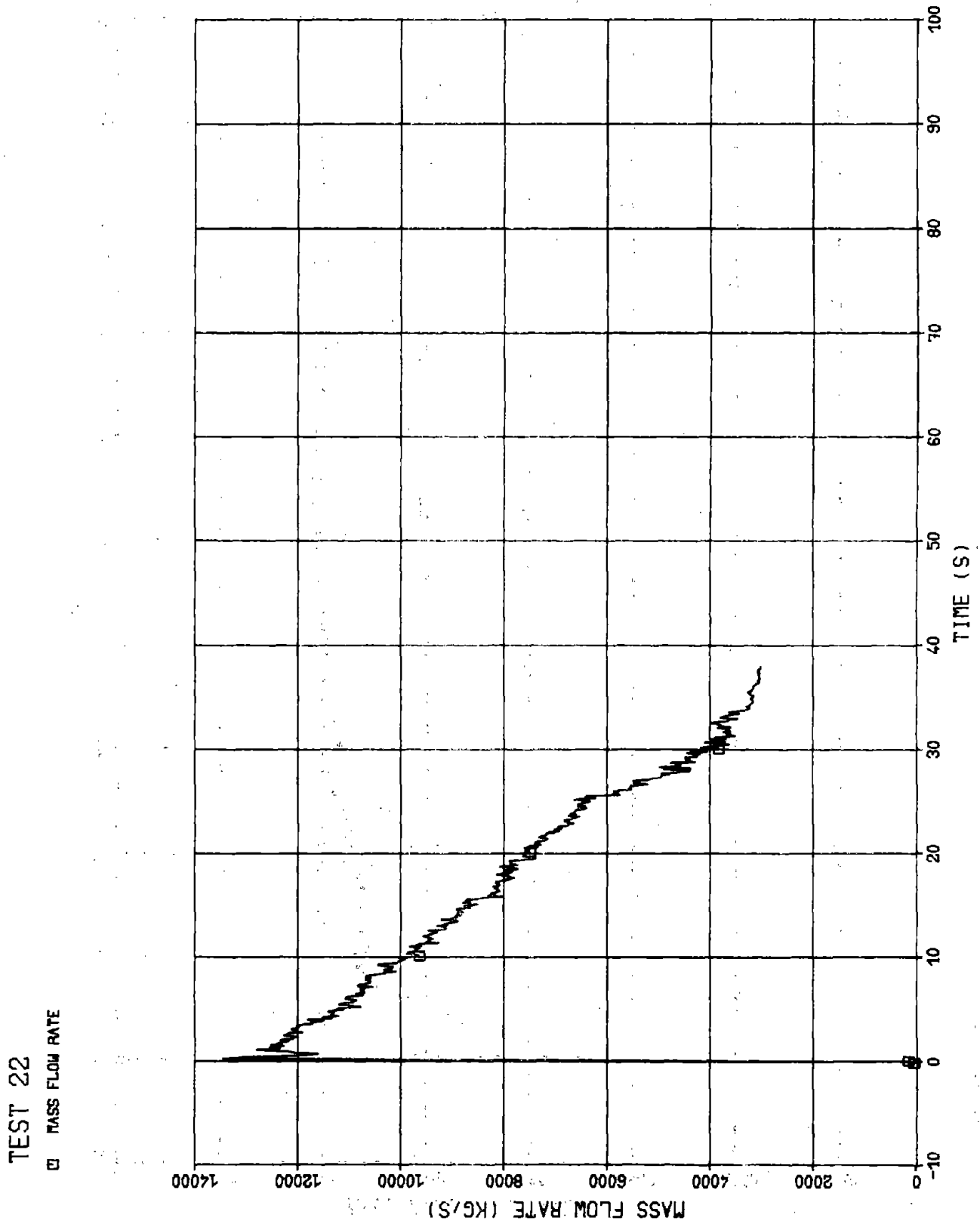
Plot D:21 - Test 21

TEST 21

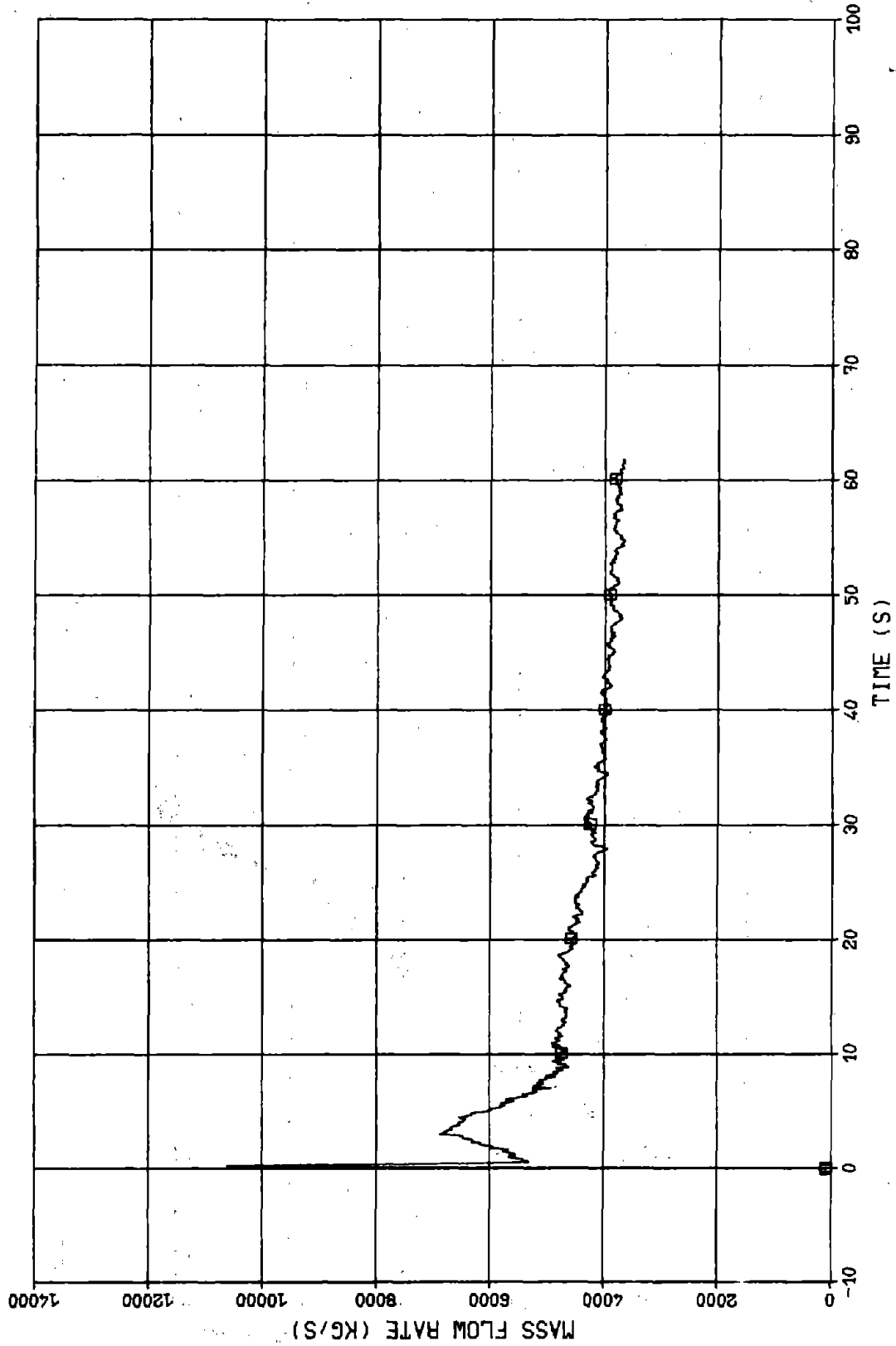
□ MASS FLOW RATE



Plot D:22 - Test 22

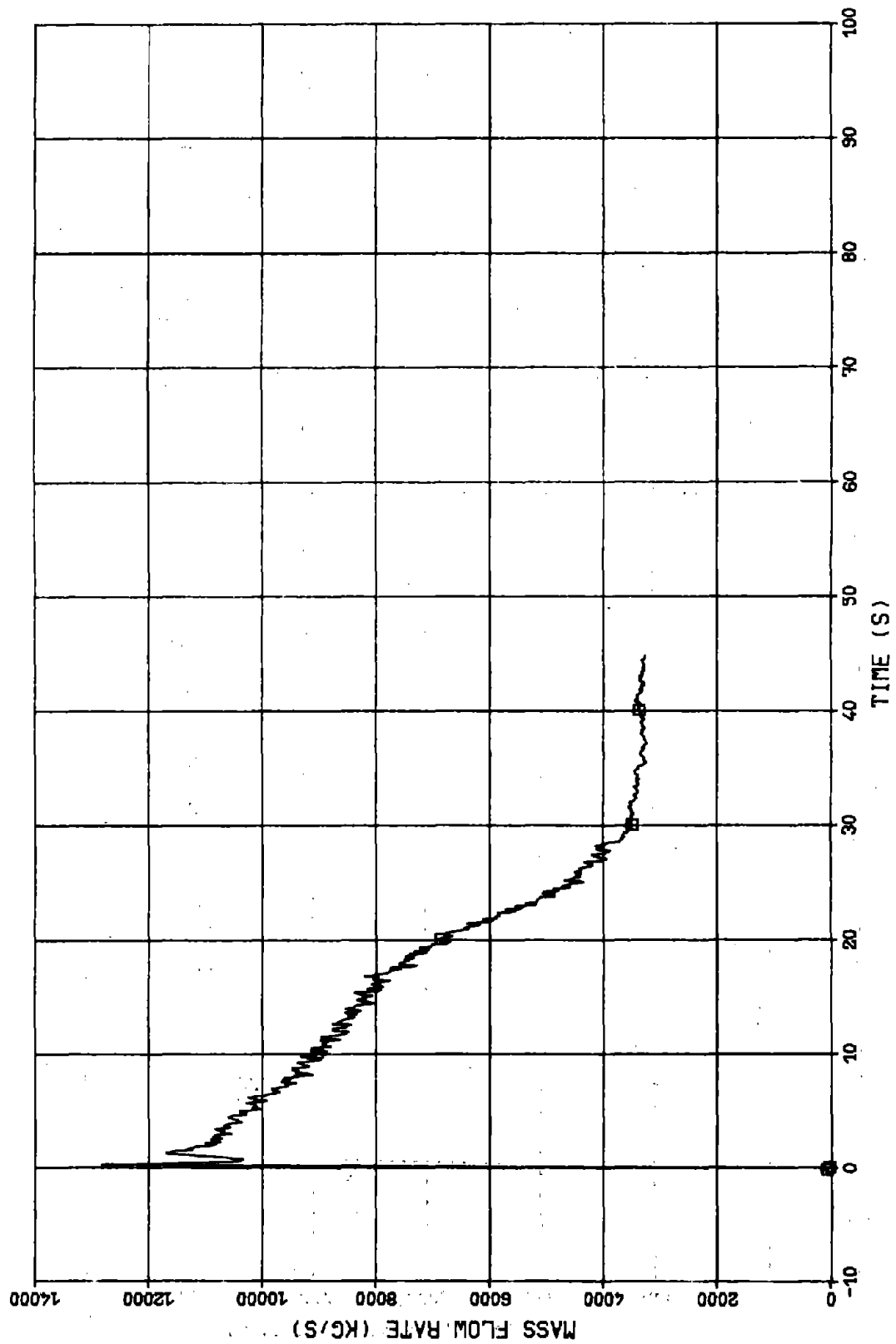


Plot D:23 - Test 23



Plot D:24 - Test 24

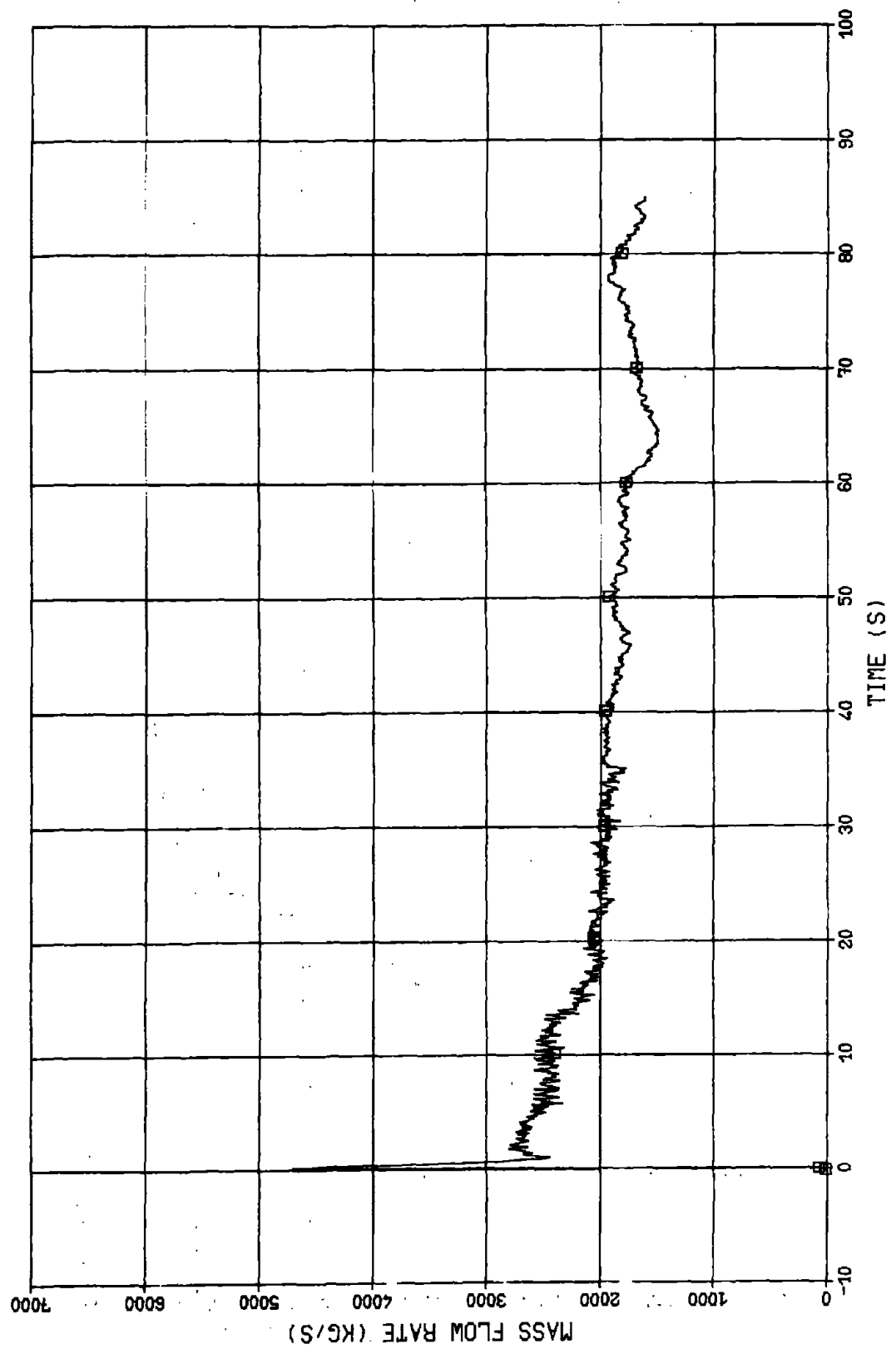
TEST 24
□ MASS FLOW RATE



Plot D:25 - Test 25

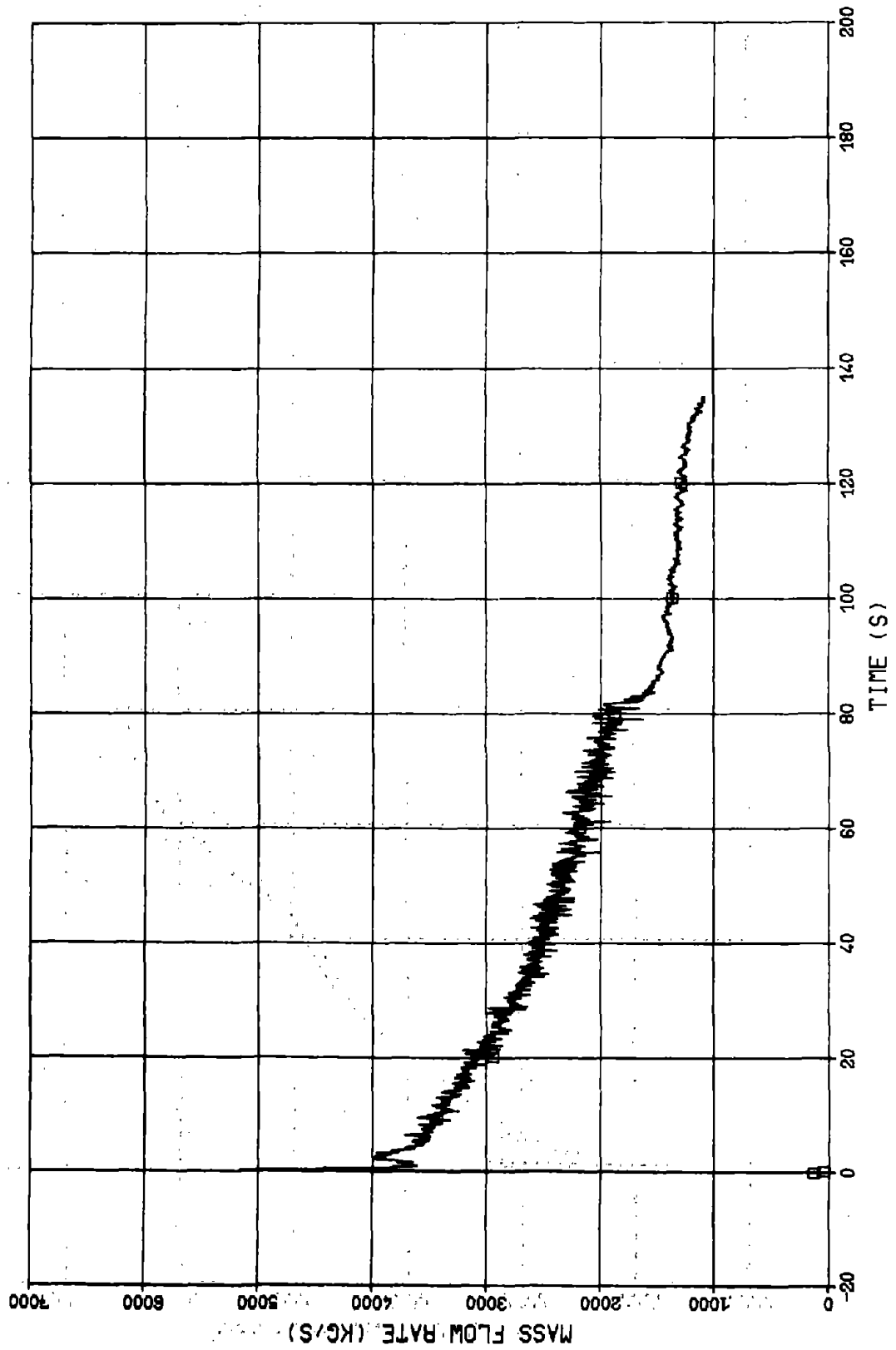
TEST 25

□ MASS FLOW RATE



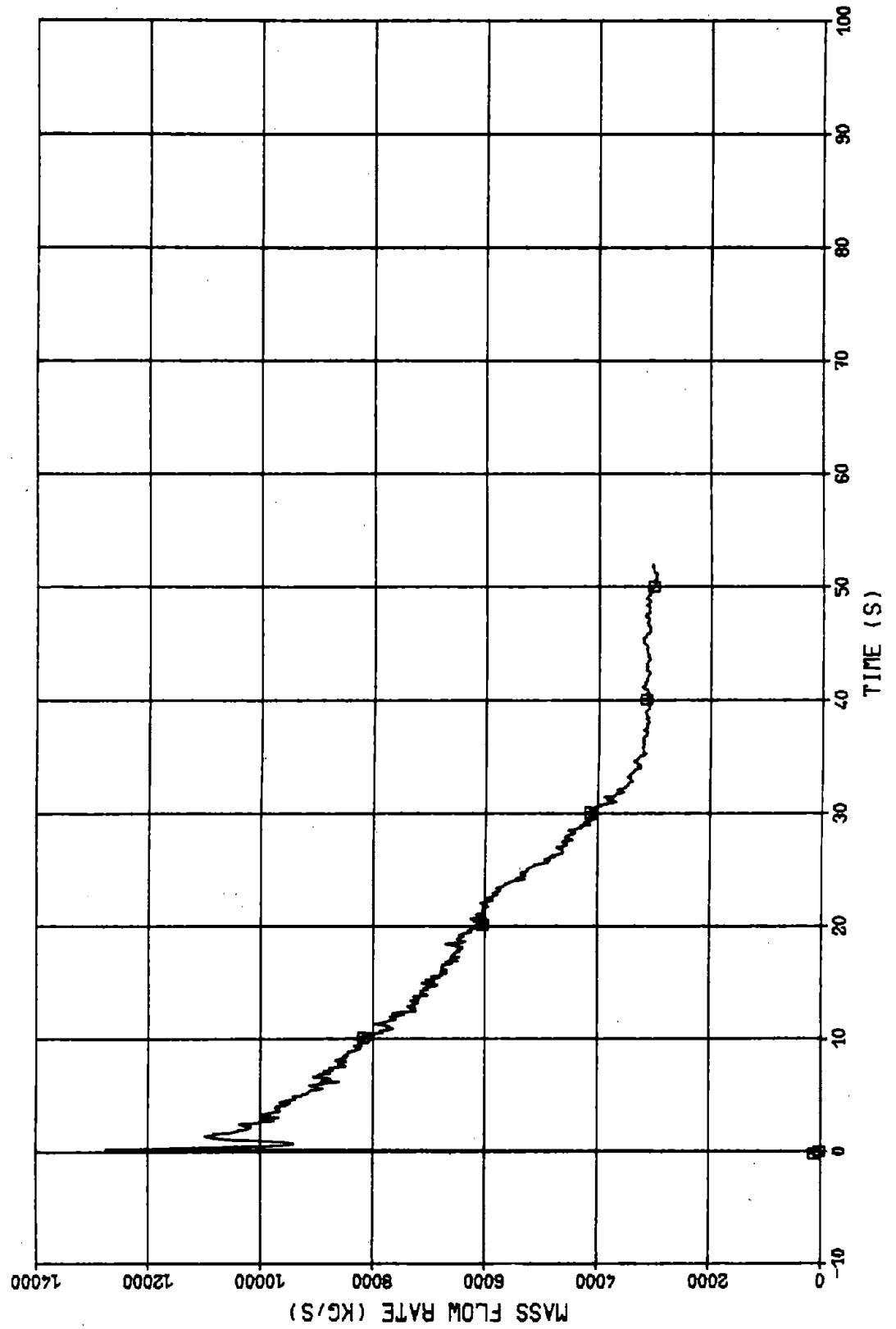
Plot D:26 - Test 26

TEST 26
□ MASS FLOW RATE



Plot D:27 - Test 27

TEST 27
□ MASS FLOW RATE



APPENDIX E

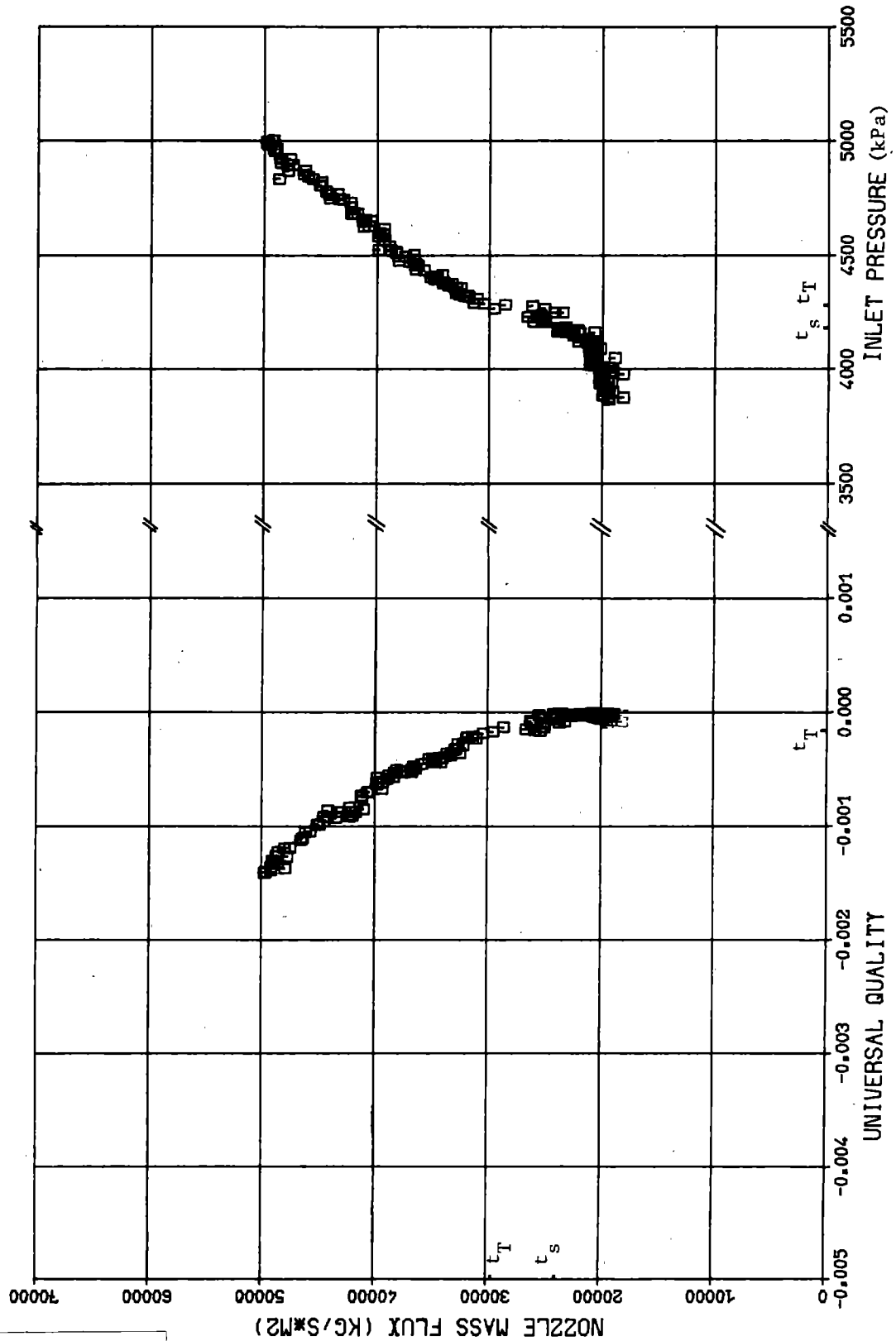
Nozzle mass flux and inlet stagnation pressure as a
function of the universal stagnation quality:
Tests 1 through 27

Plot E:1 - Test 1

NOTE: 1. Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY
AND INLET STAGNATION PRESSURE

T, TEST 1



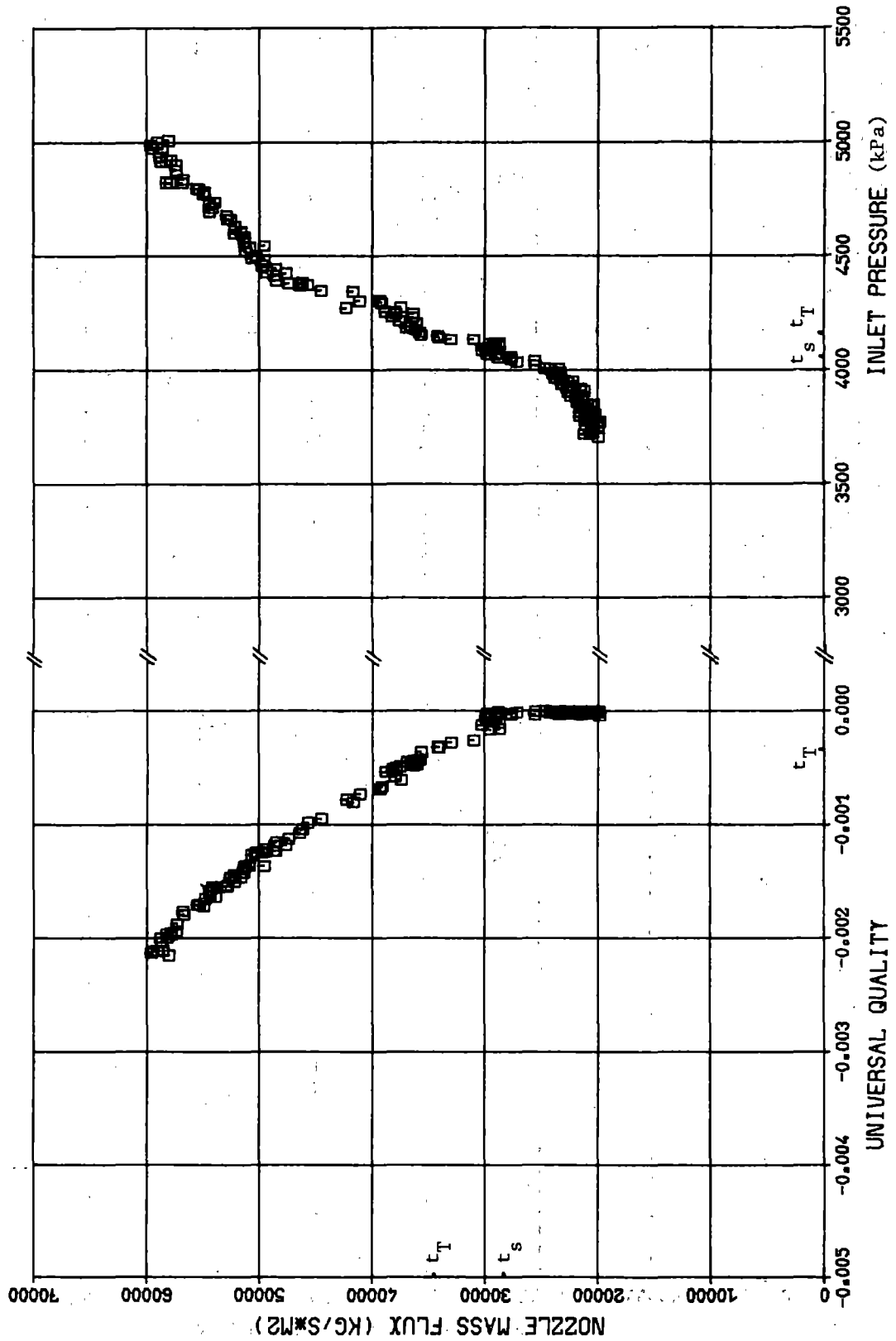
Preceding page blank

Plot E:2 - Test 2

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY
AND INLET STAGNATION PRESSURE

□ TEST 2

NOTE: 1. Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

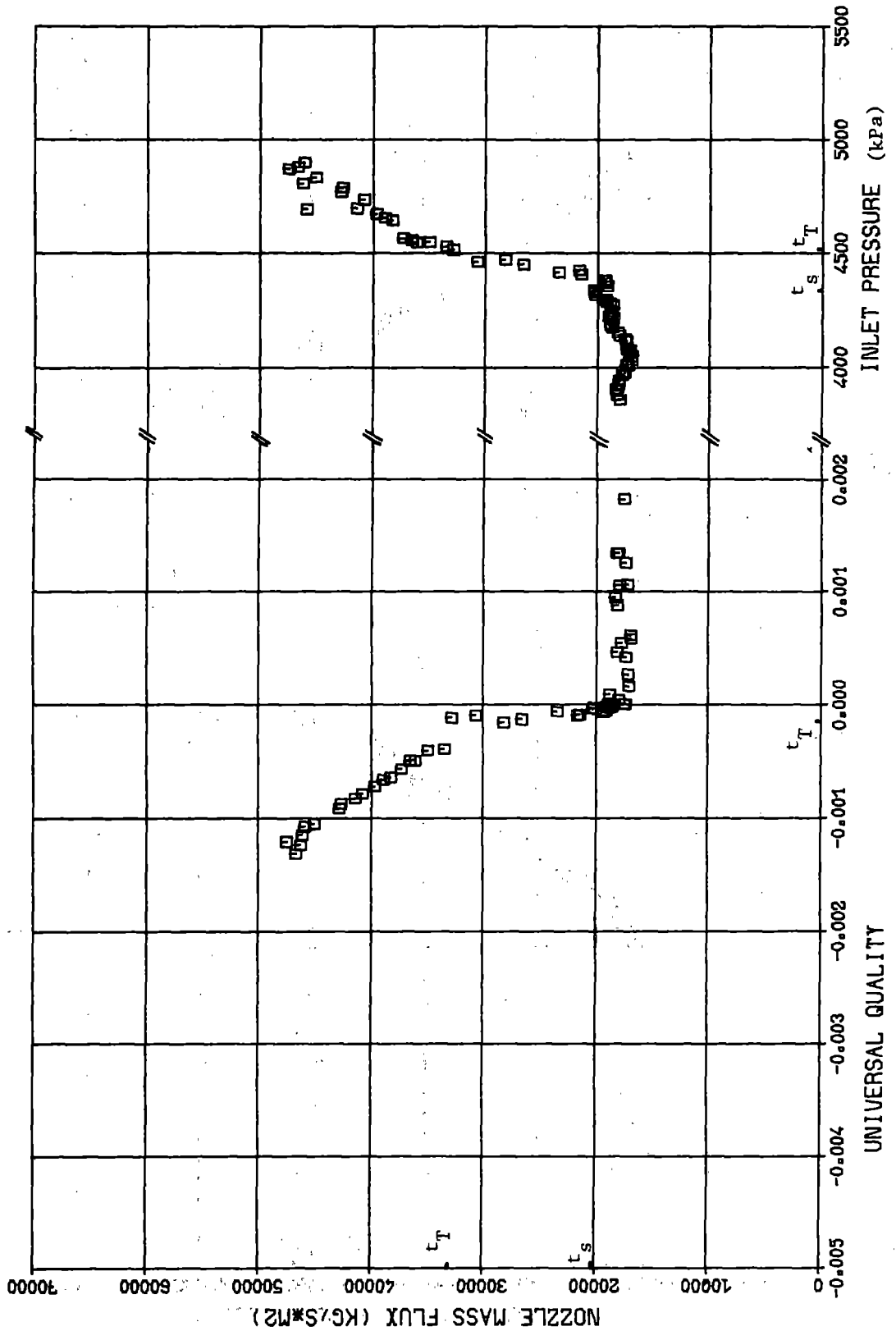


Plot E:3 - Test 3

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY
AND INLET STAGNATION PRESSURE

□ TEST 3

NOTE: Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_S (see Table 8:1) indicate the start of the transition and saturated data respectively.

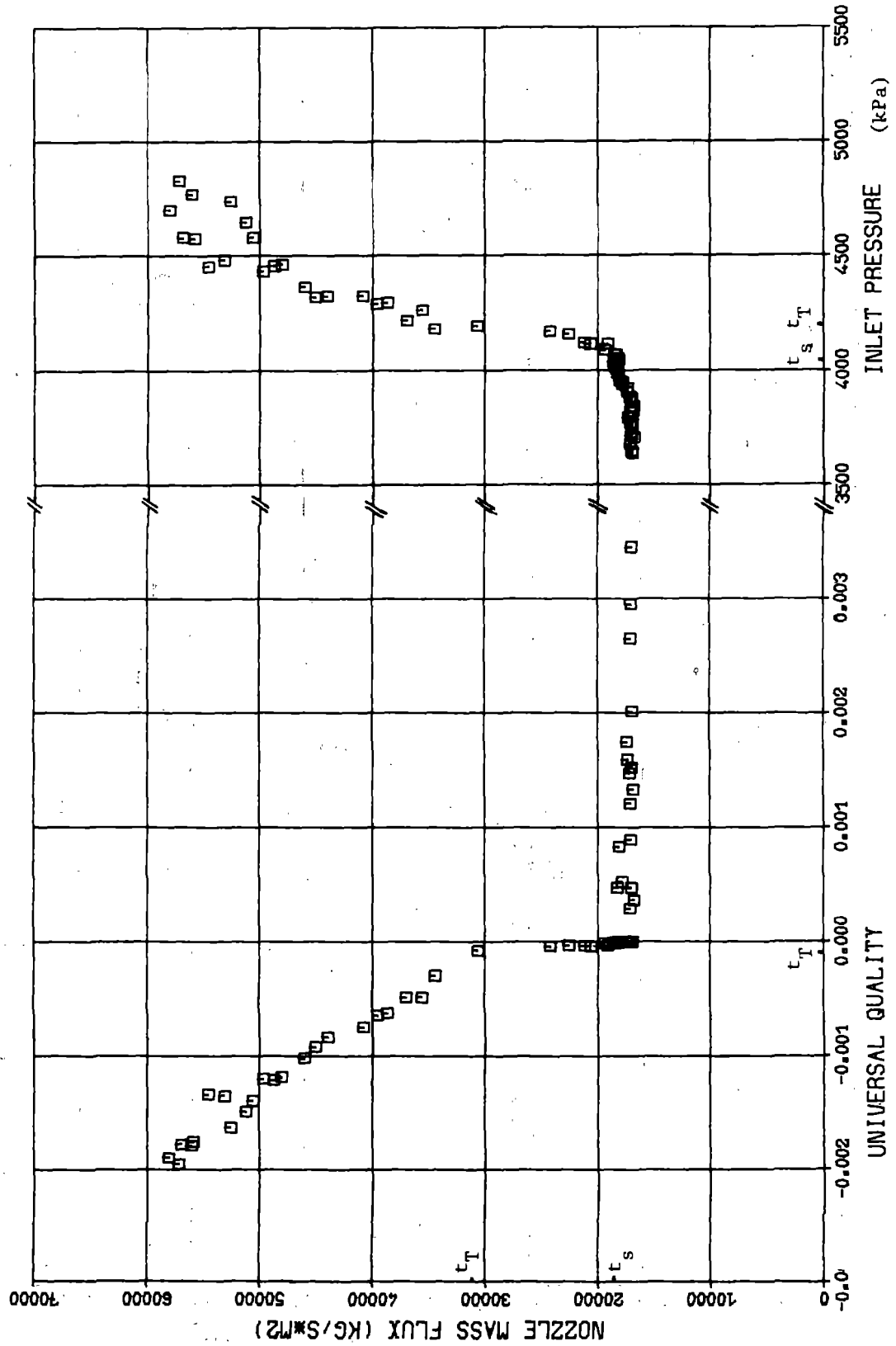


Plot E:4 - Test 4

NOTE: Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY
AND INLET STAGNATION PRESSURE

□ TEST 4

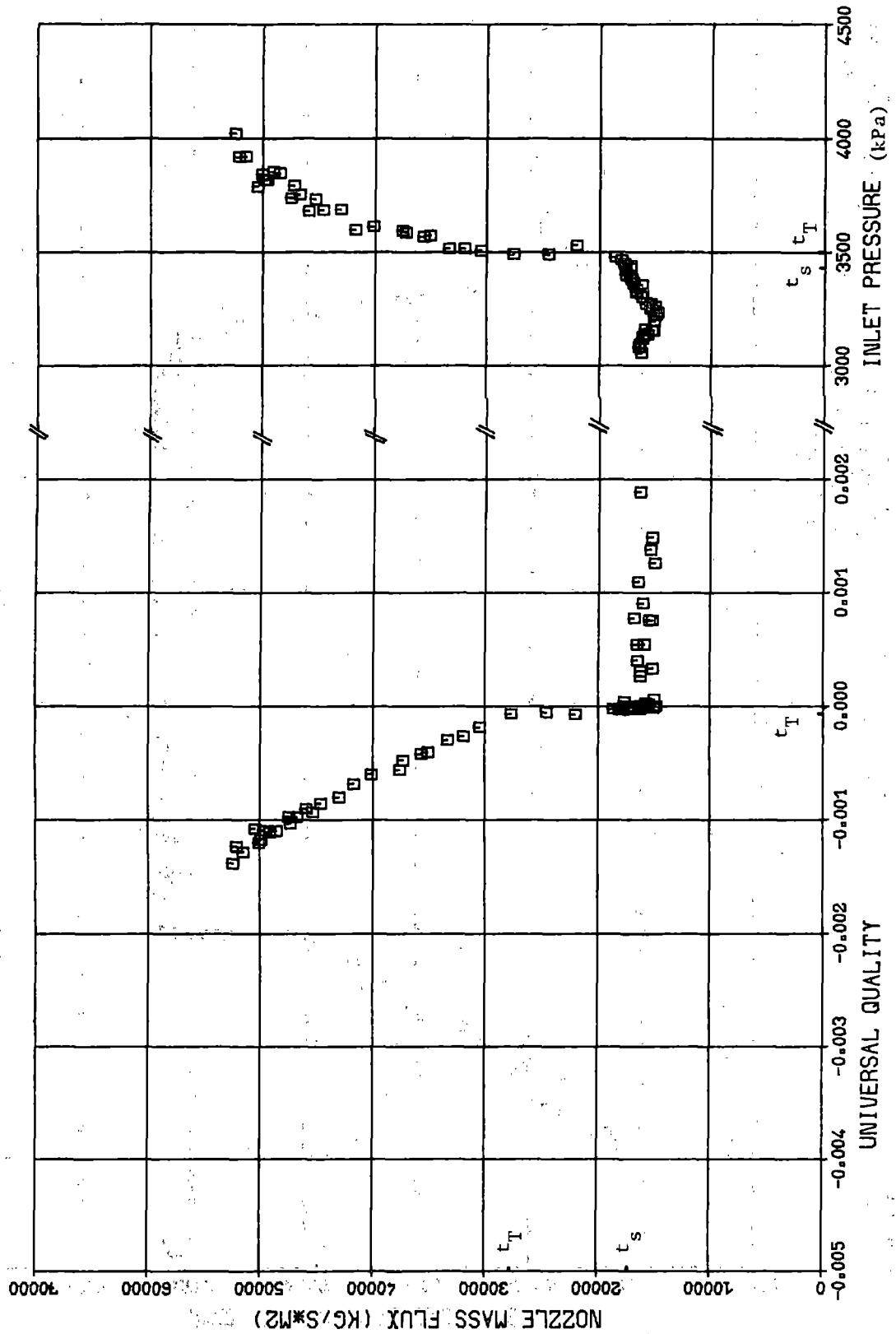


Plot E:5 - Test 5

NOTE: Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY
AND INLET STAGNATION PRESSURE

□ TEST 5

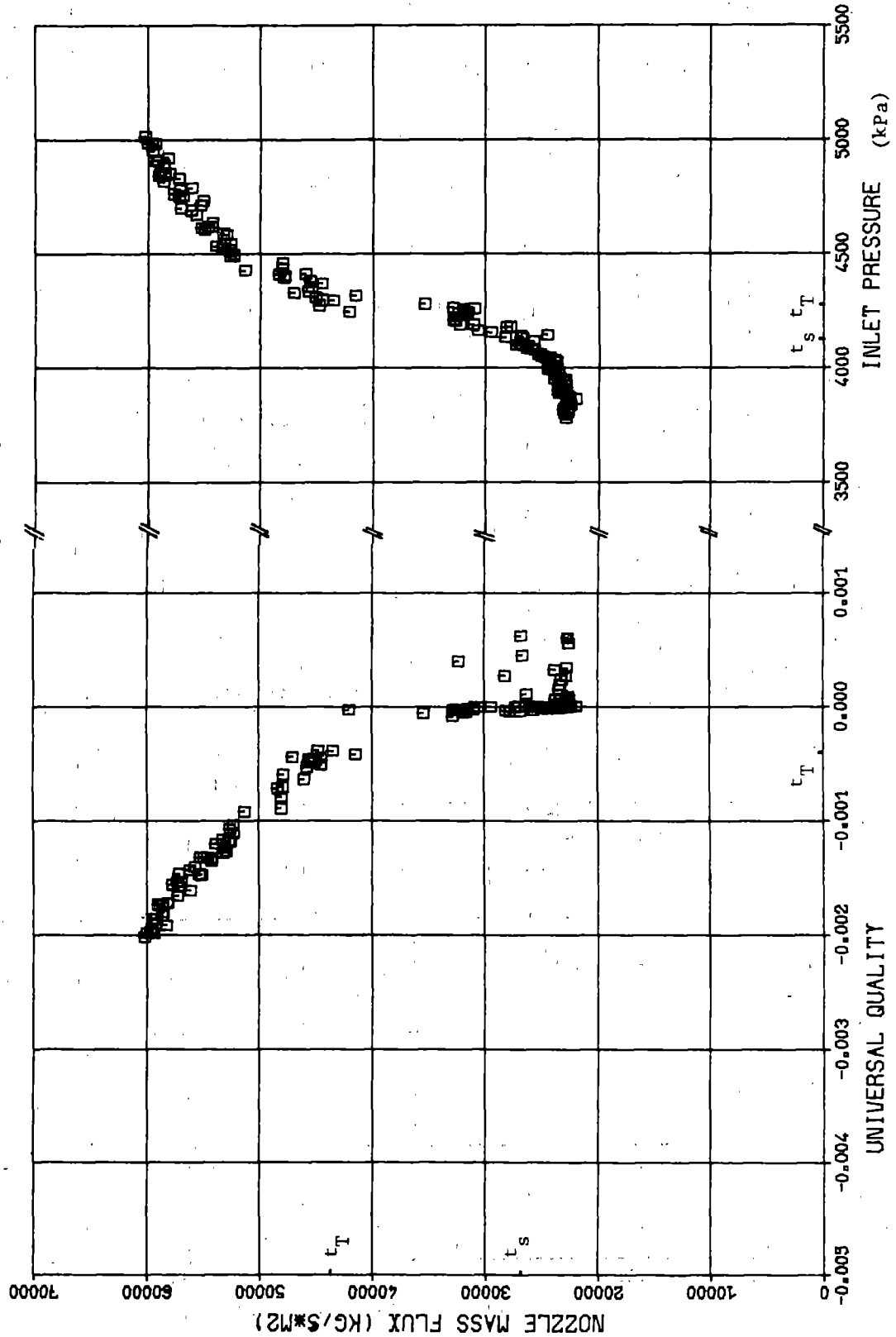


Plot E:6 - Test 6

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY
AND INLET STAGNATION PRESSURE

□ TEST 6

NOTE: Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

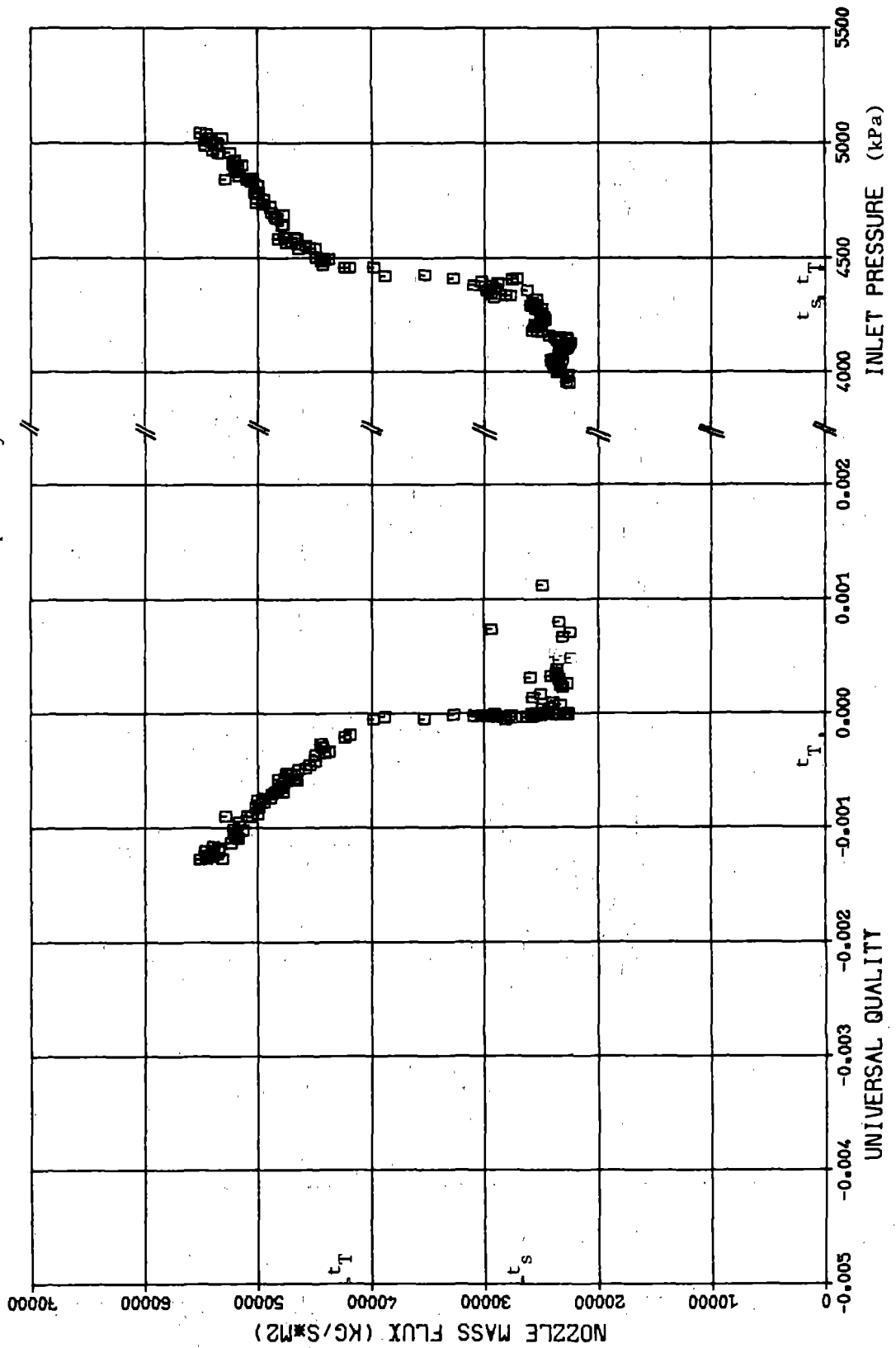


Plot E:7 - Test 7

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY
AND INLET STAGNATION PRESSURE

□ TEST 7

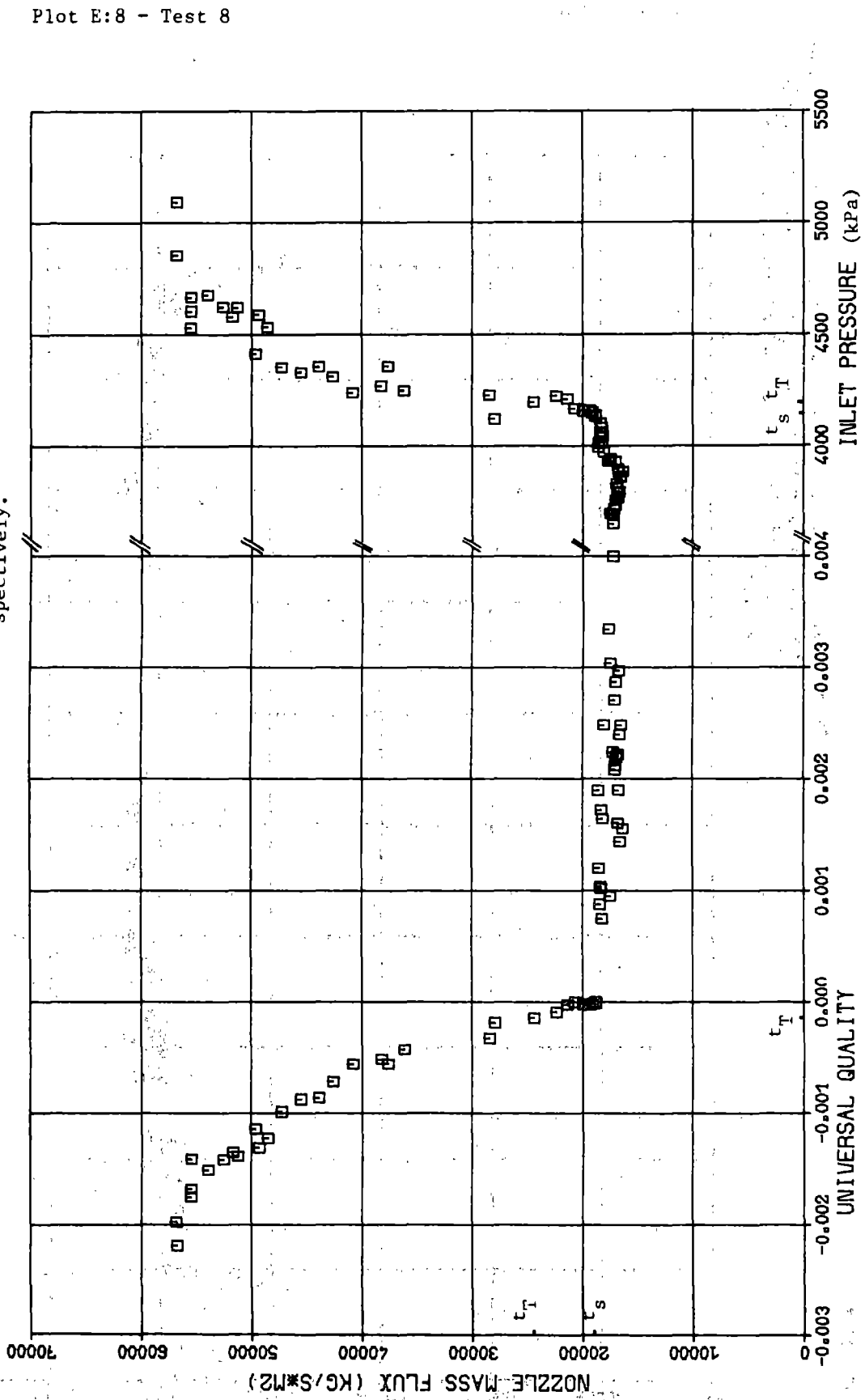
NOTE: Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.



NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY AND INLET STAGNATION PRESSURE

□ TEST 8

NOTE: Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

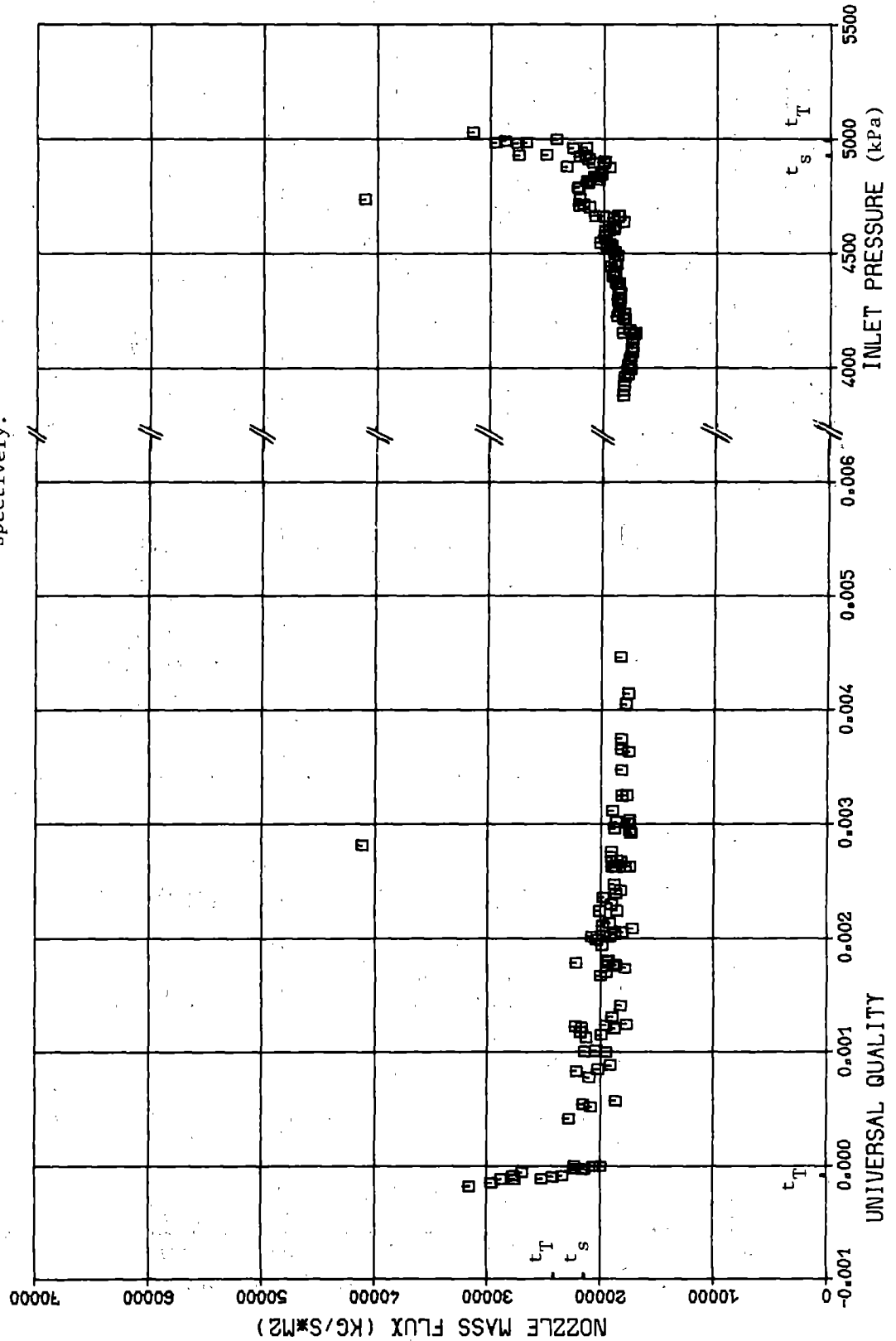


Plot E:9 - Test 9

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY
AND INLET STAGNATION PRESSURE

□ TEST 9

NOTE: Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

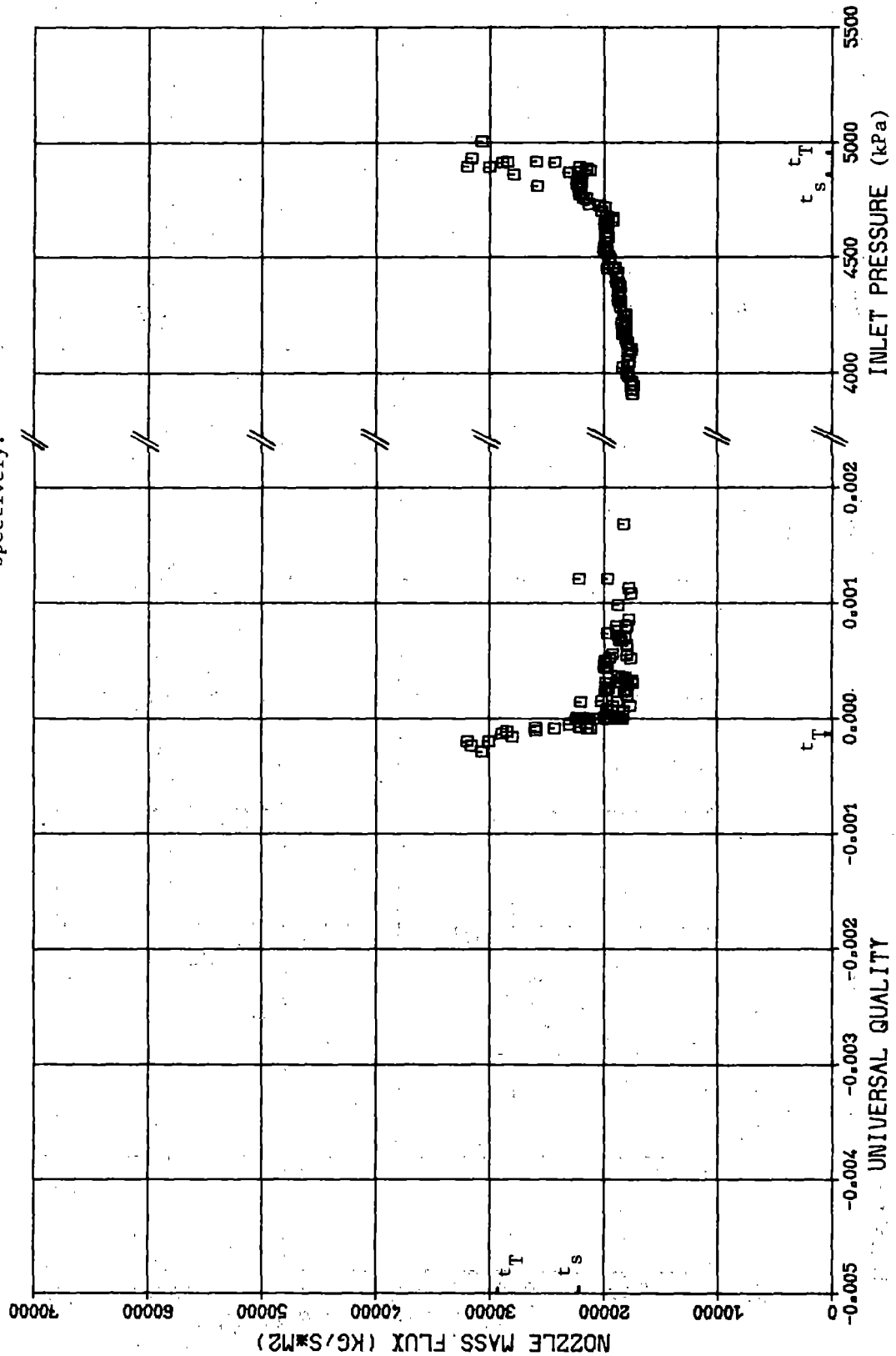


Plot E:10 - Test 10

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY
AND INLET STAGNATION PRESSURE

□ TEST 10

NOTE: Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

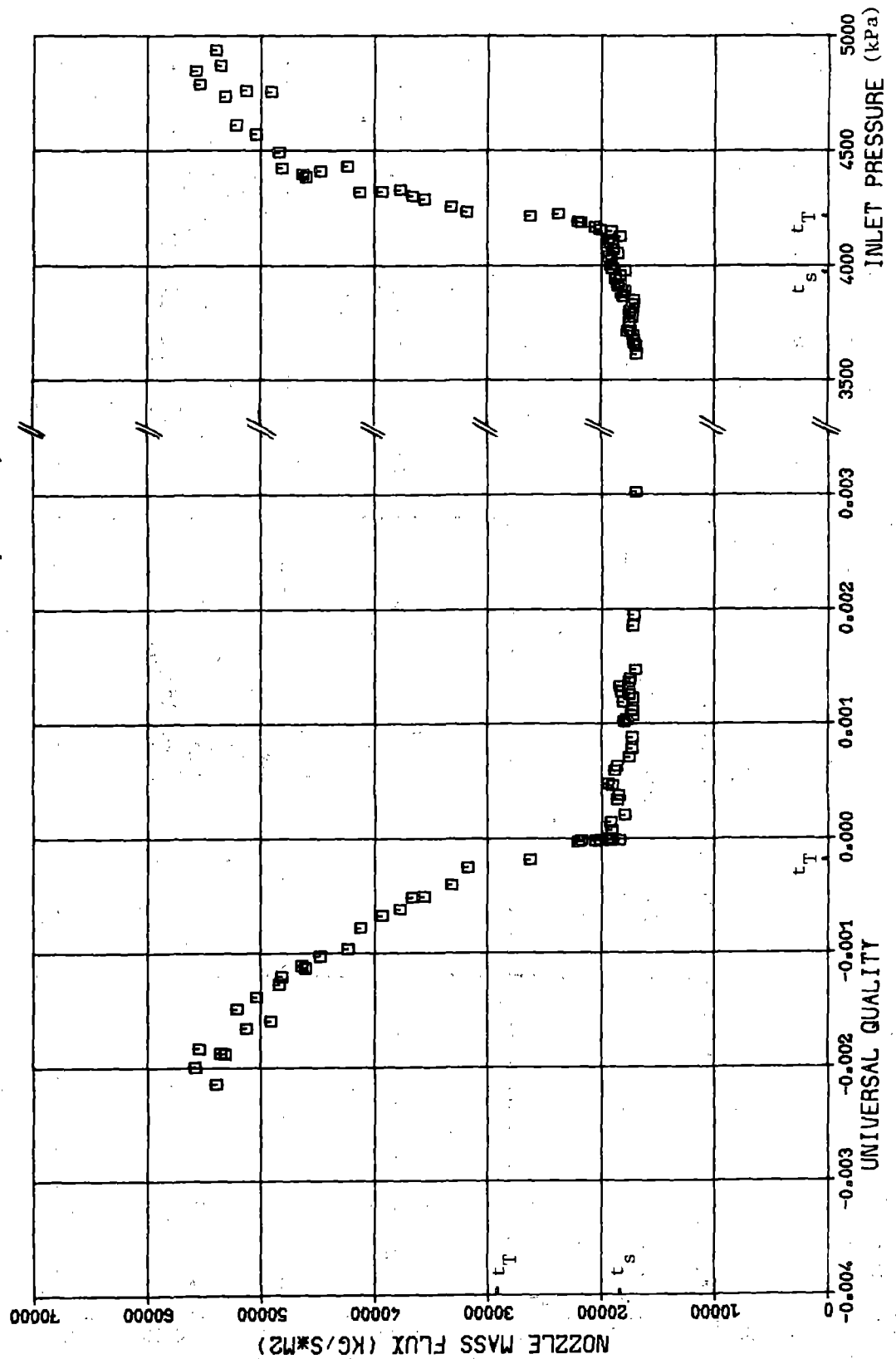


Plot E:11 - Test 11

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY AND INLET STAGNATION PRESSURE

□ TEST 11

NOTE: Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

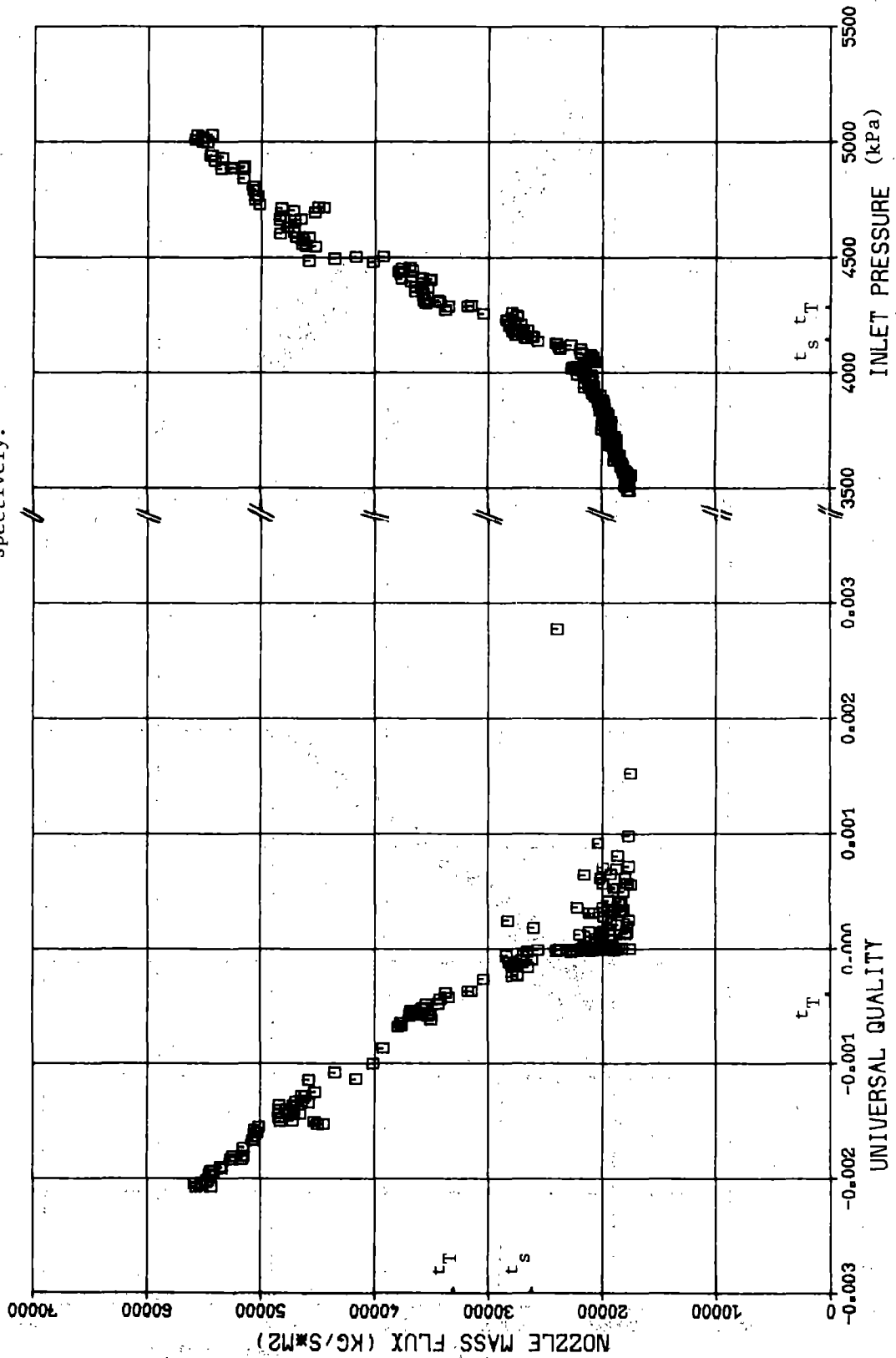


Plot E:12 - Test 12

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY
AND INLET STAGNATION PRESSURE

□ TEST 12

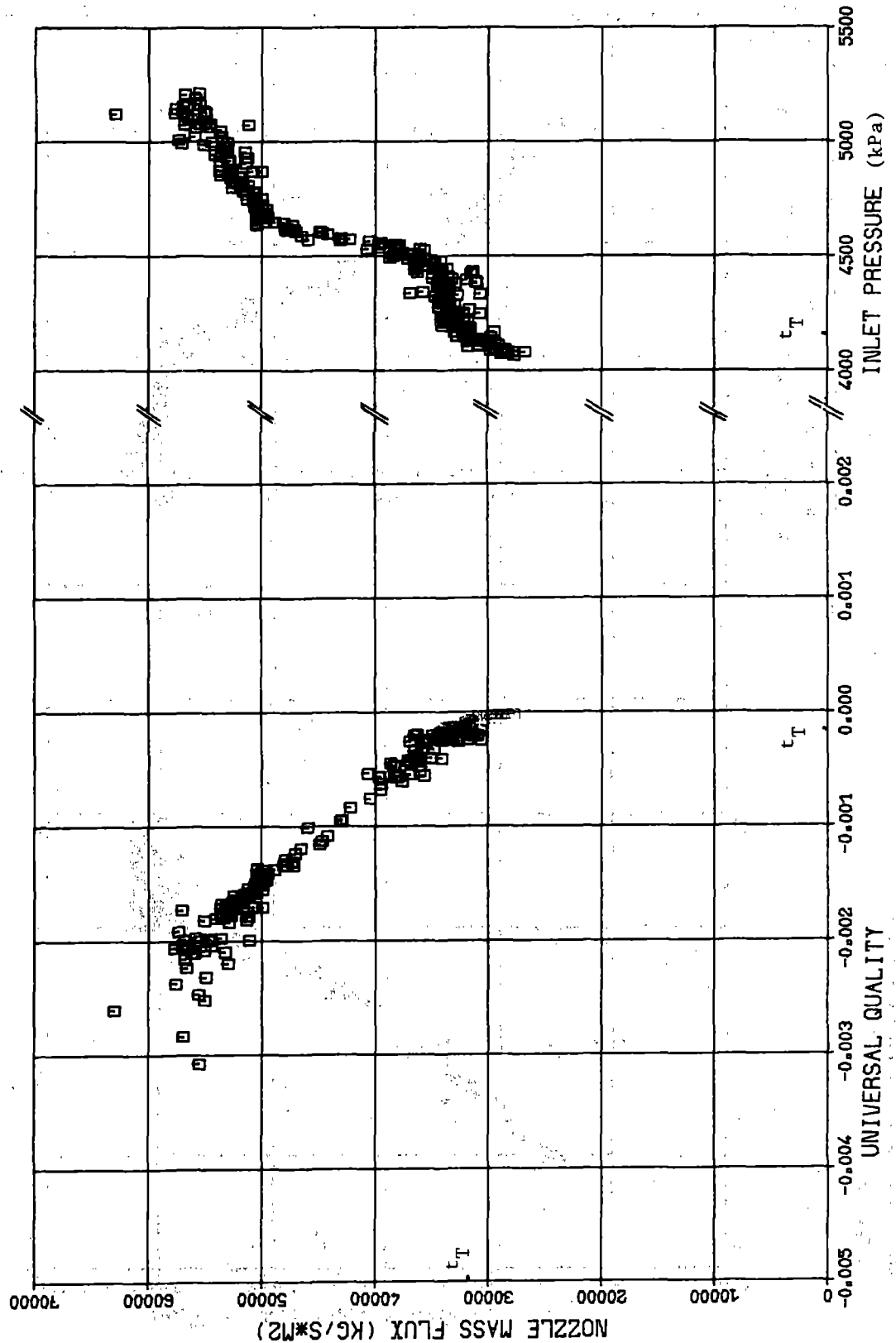
NOTE: Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.



Plot E:13 - Test 13

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY AND INLET STAGNATION PRESSURE

NOTE: 1. Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

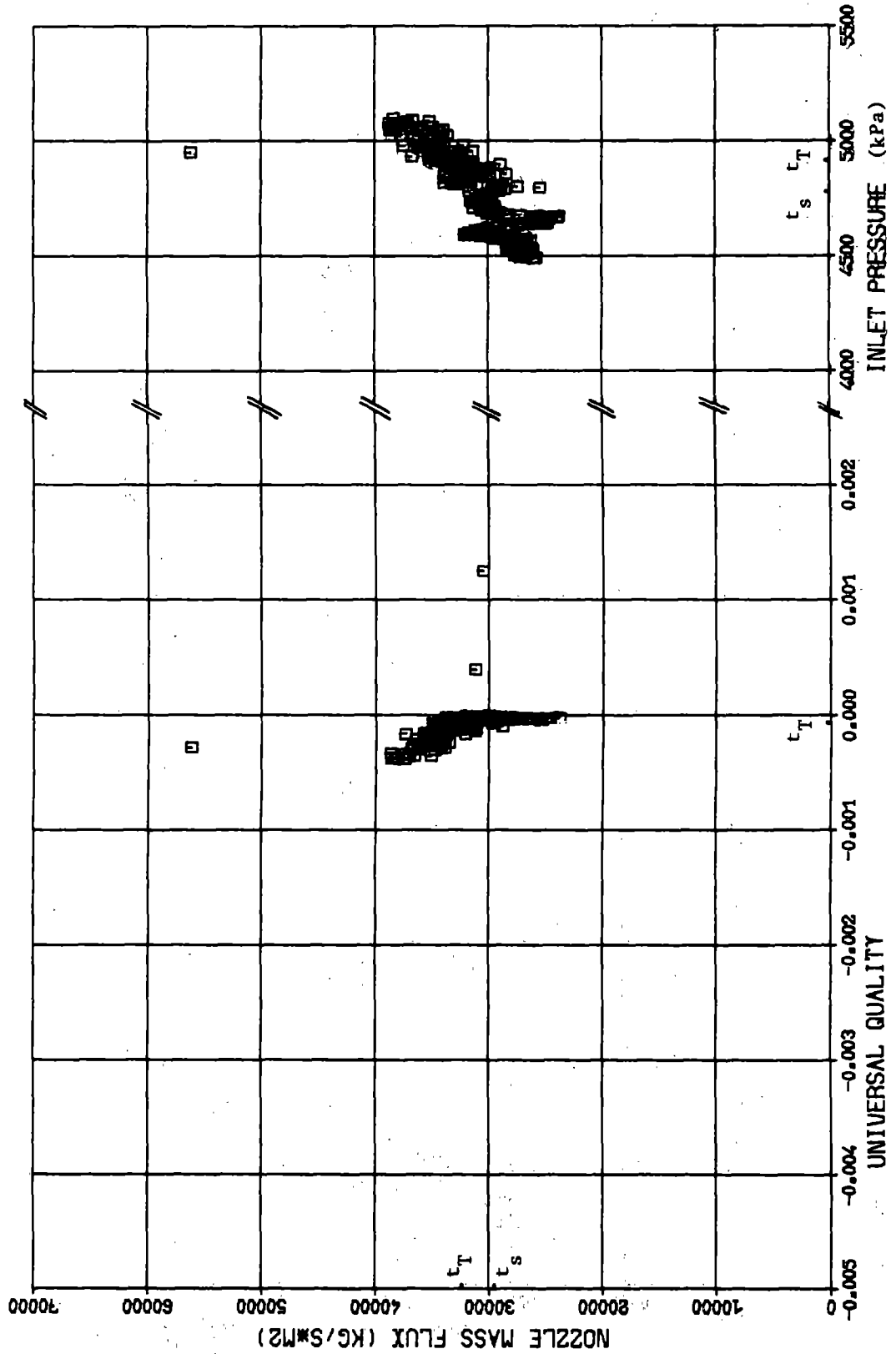


Plot E:14 - Test 14

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY AND INLET STAGNATION PRESSURE

TEST 14

NOTE: 1. Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

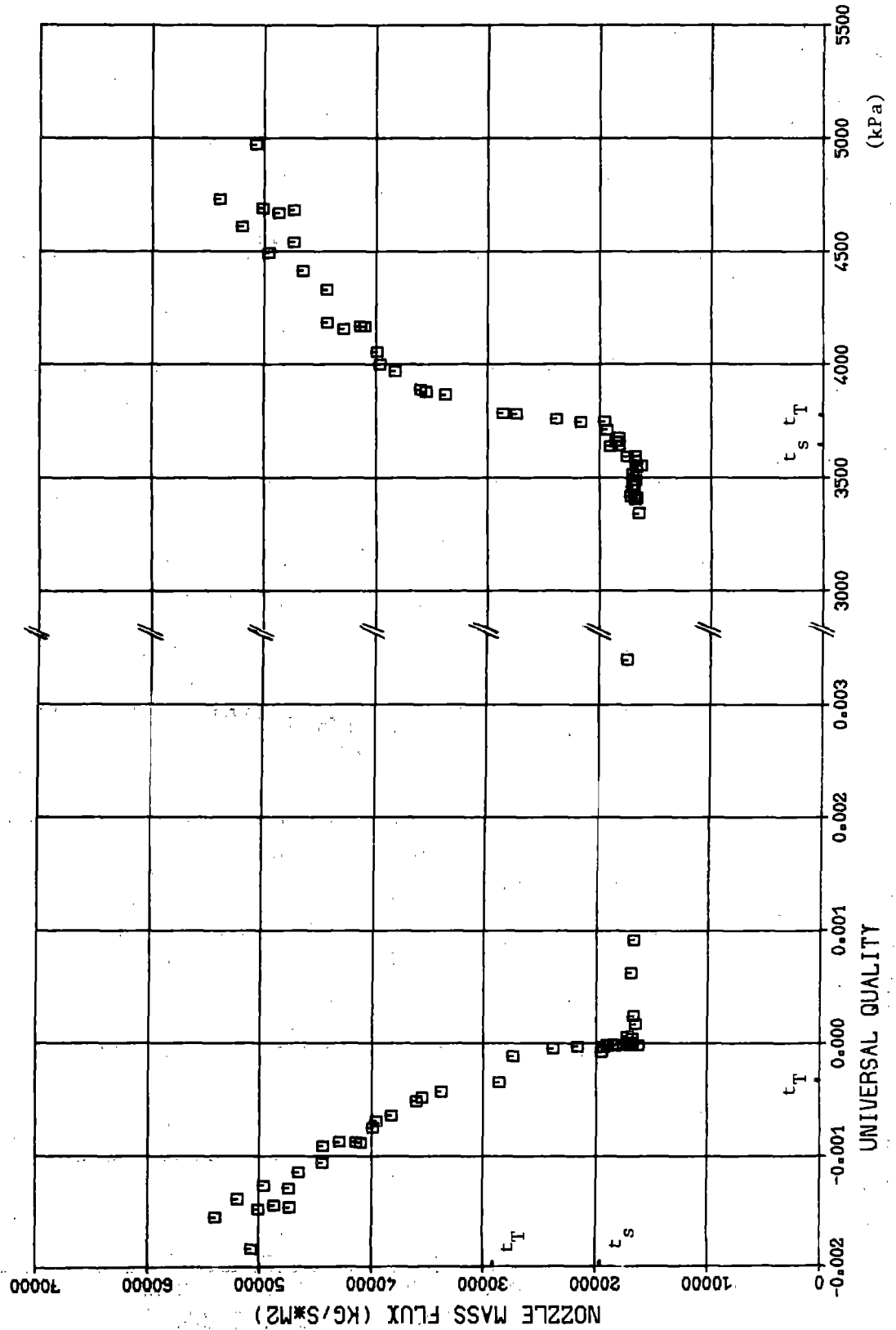


Plot E:15 - Test 15

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY AND INLET STAGNATION PRESSURE

□ TEST 15

NOTE: Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

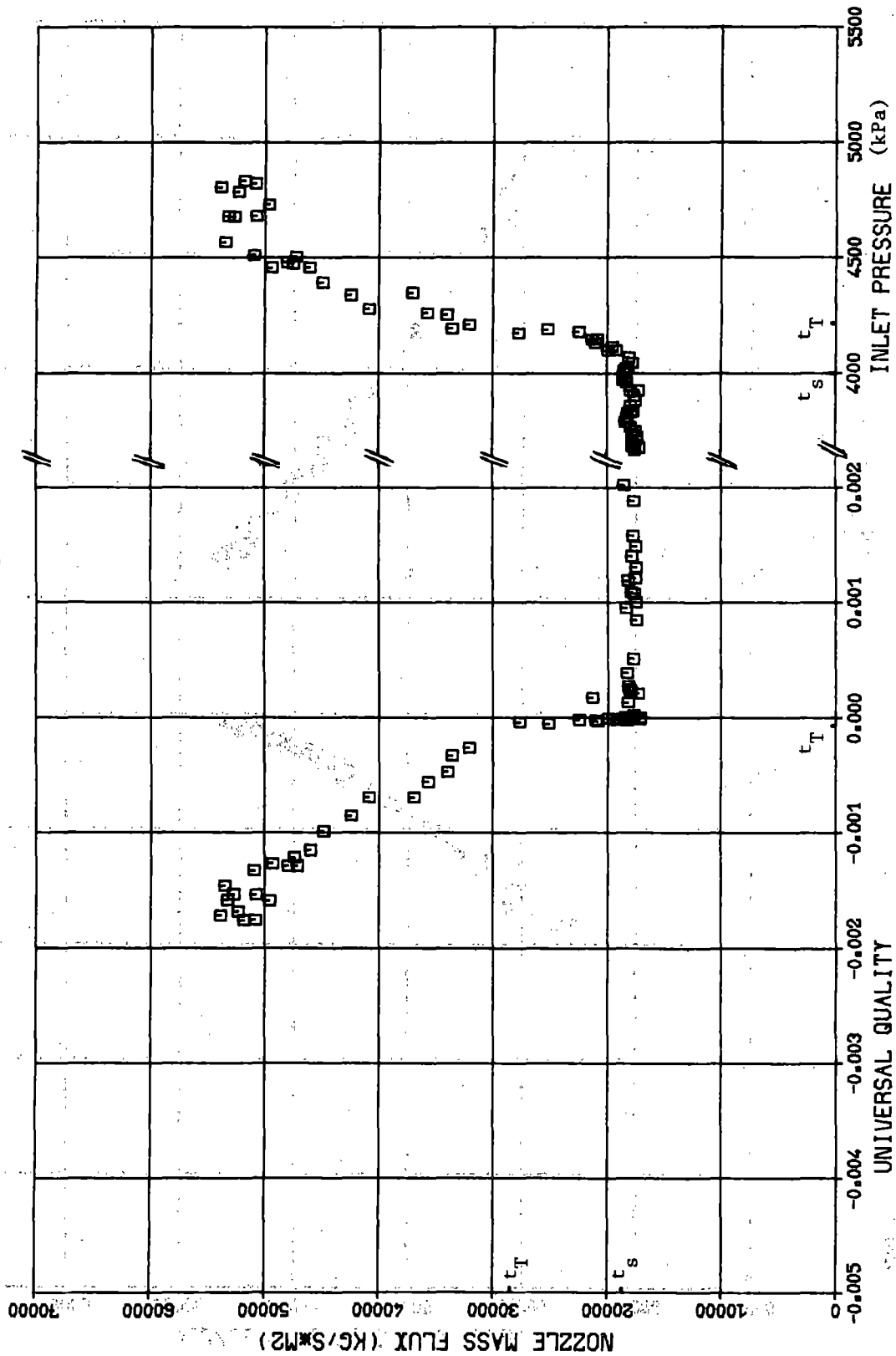


Plot E:16 - Test 16

NOTE: Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY
AND INLET STAGNATION PRESSURE

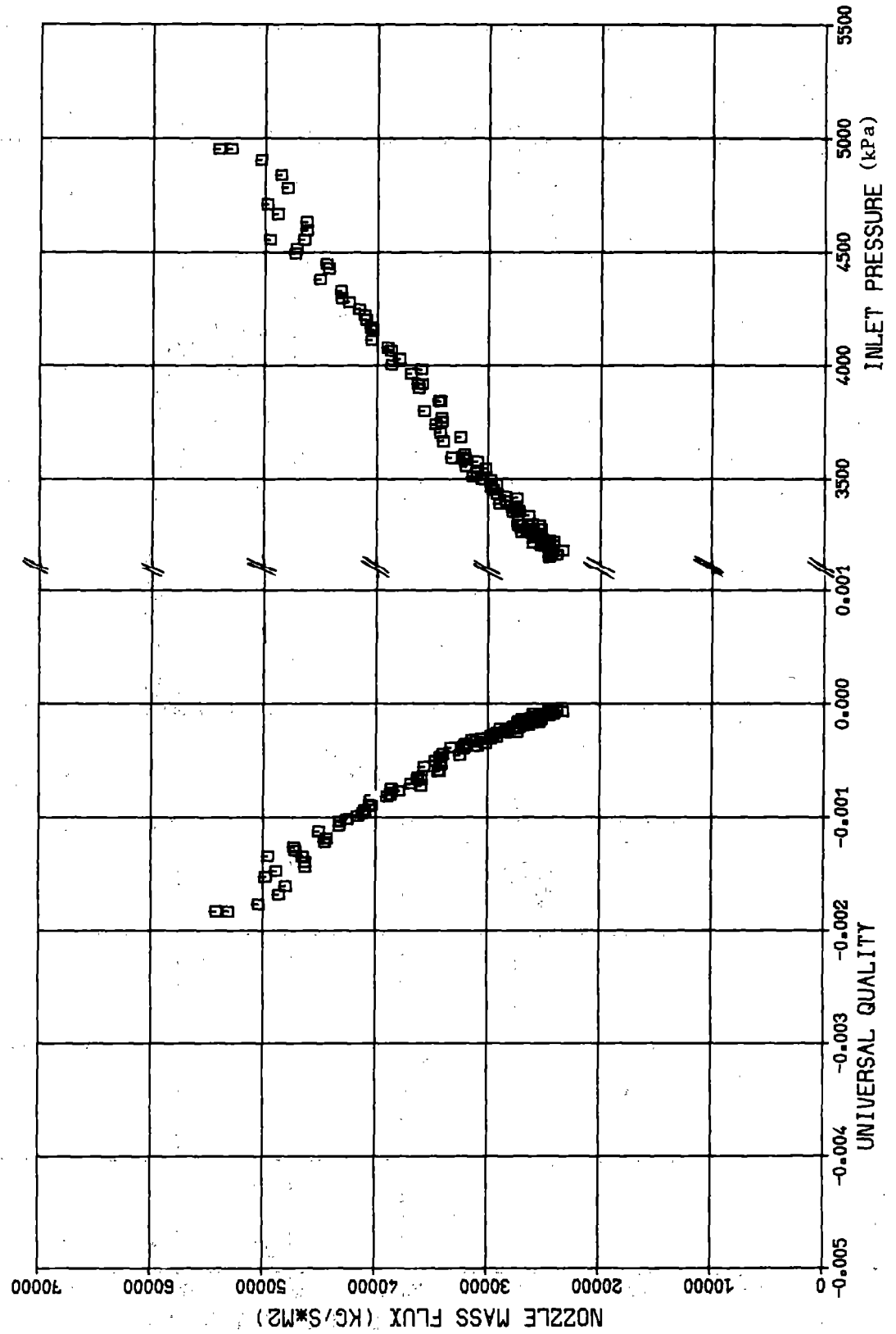
□ TEST 16



Plot E:17 - Test 17

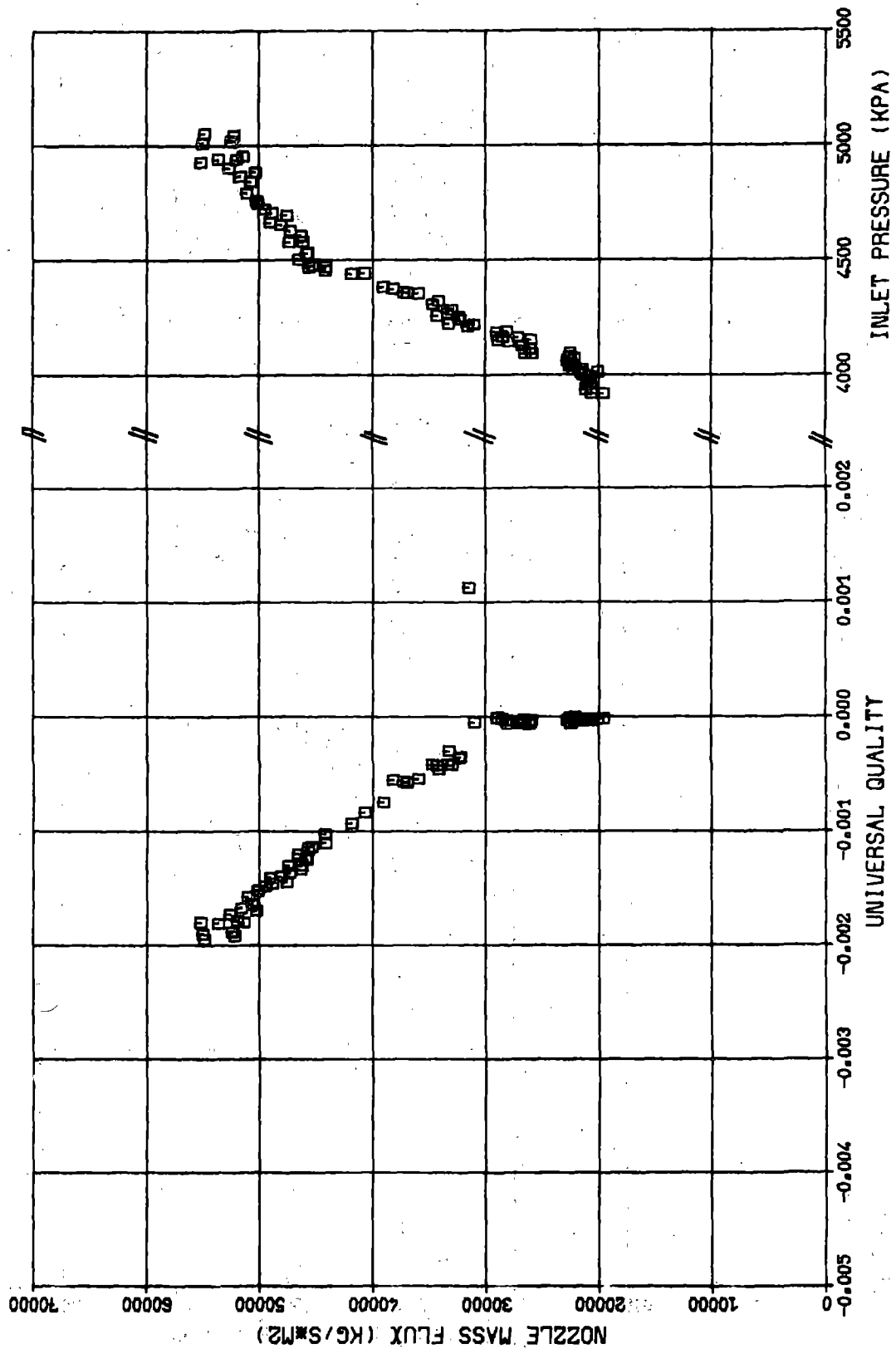
NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY
AND INLET STAGNATION PRESSURE

□ TEST 17



Plot E:18 - Test 18

Note: Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_S (see Table 8:1) indicate the start of the transition and saturated data respectively.



NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY
AND INLET STAGNATION PRESSURE

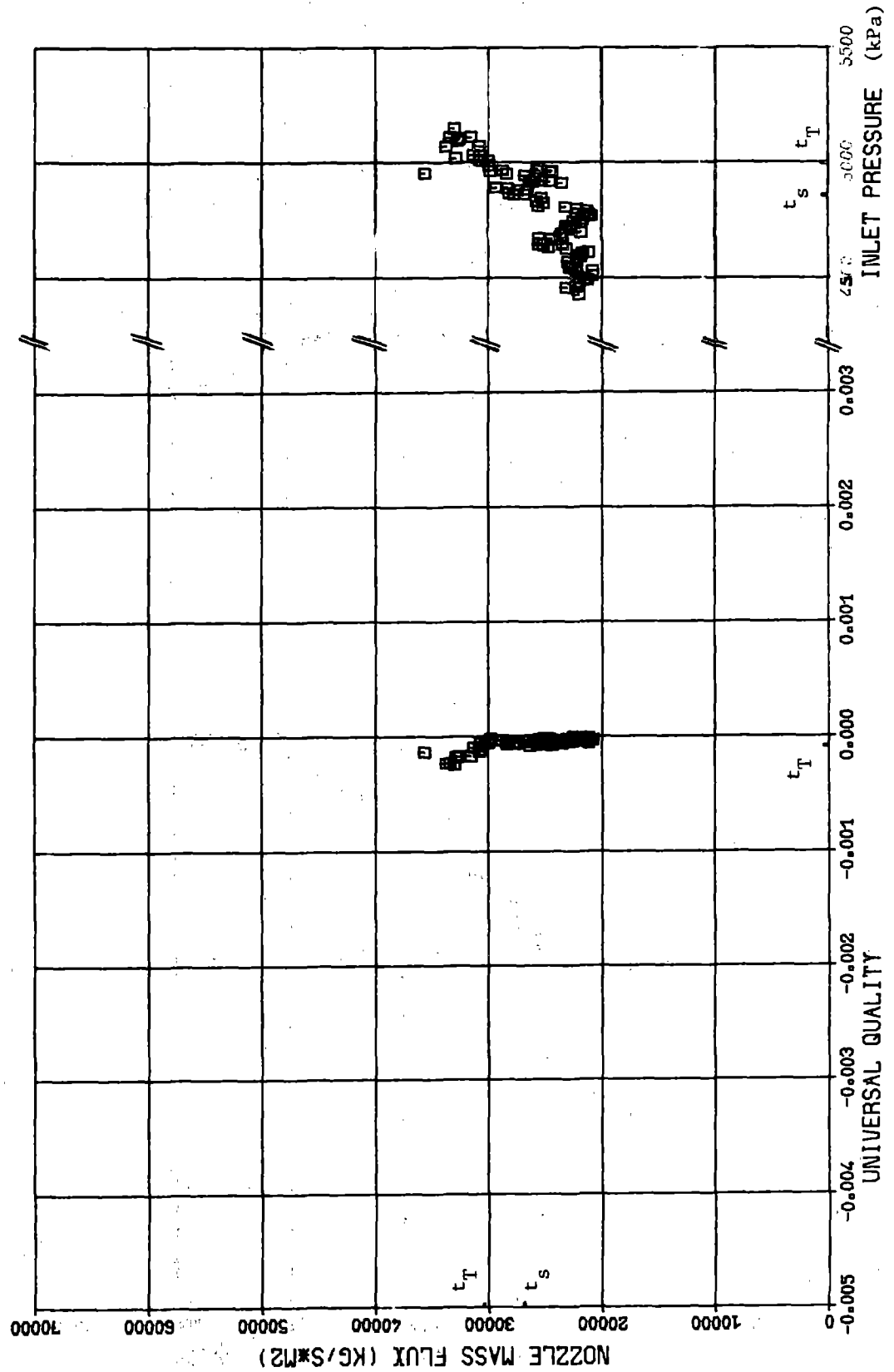
□ TEST 18

Plot E:19 - Test 19

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY
AND INLET STAGNATION PRESSURE

IN TEST 19

NOTE: 1. Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

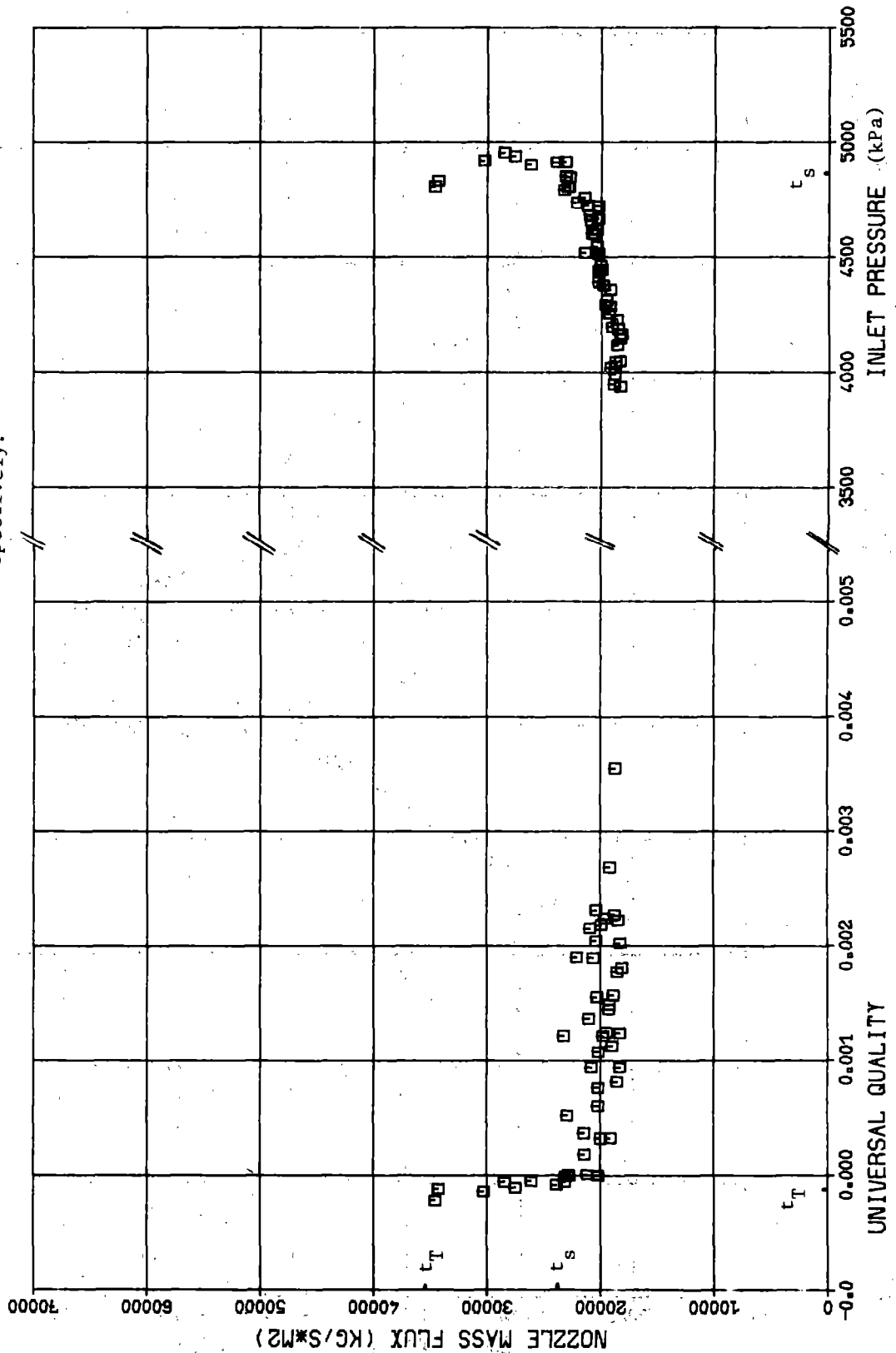


Plot E:20 - Test 20

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY
AND INLET STAGNATION PRESSURE

□ TEST 20

NOTE: Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

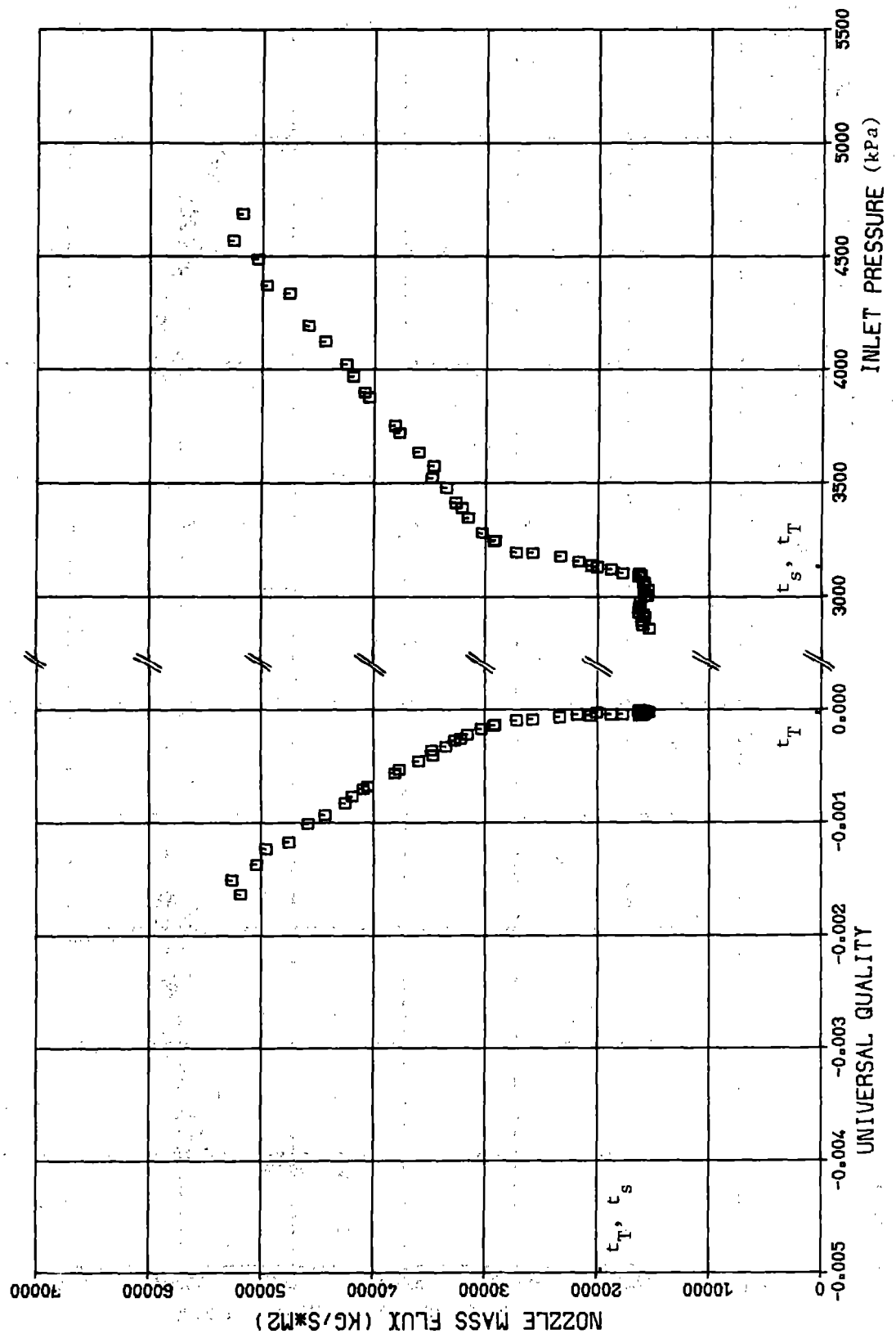


Plot E:21 - Test 21

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY AND INLET STAGNATION PRESSURE

NOTE: 1. Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

□ TEST 21

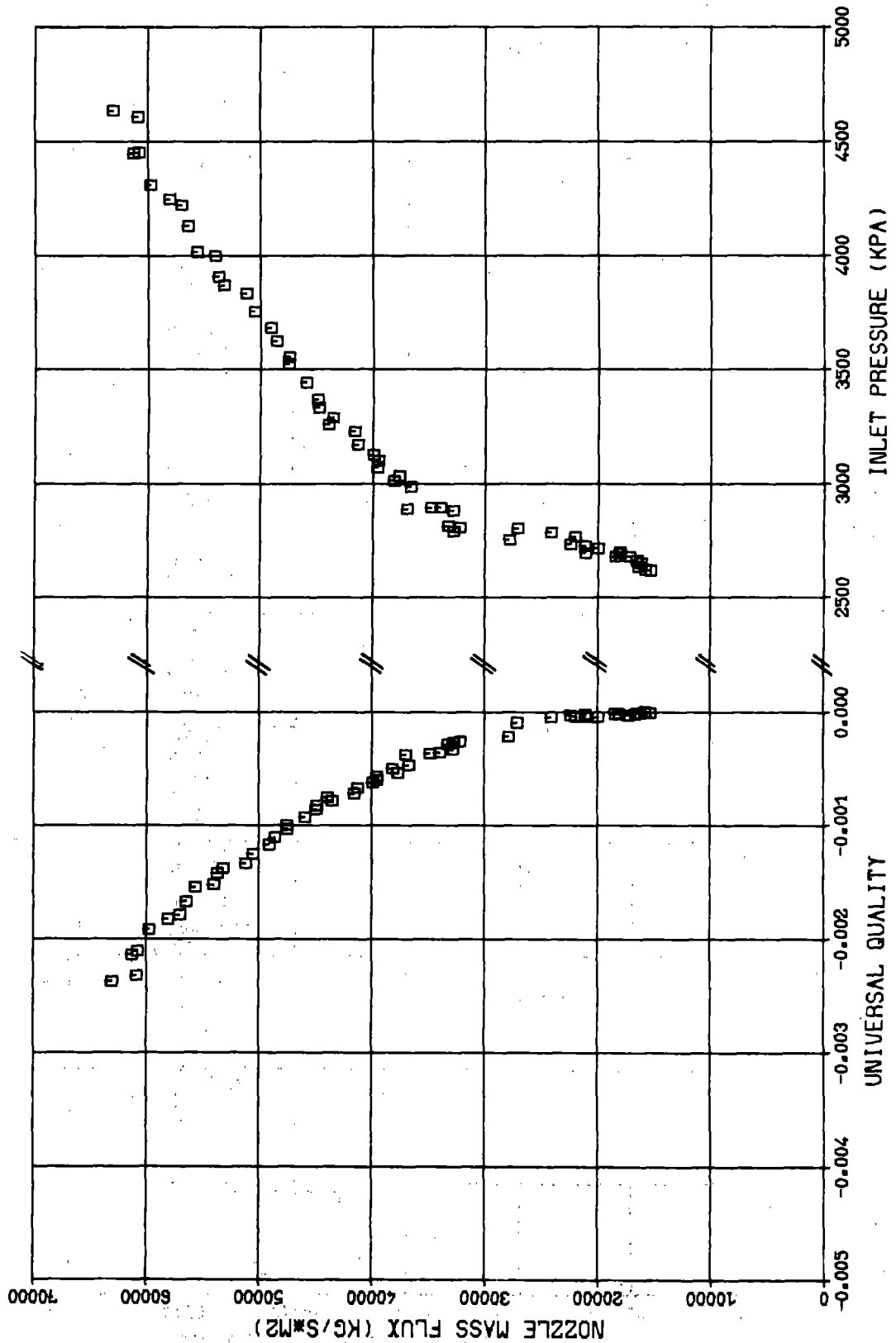


Plot E:22 - Test 22

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY
AND INLET STAGNATION PRESSURE

□ TEST 22

Note: Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_S (see Table 8:1) indicate the start of the transition and saturated data re-spectively.

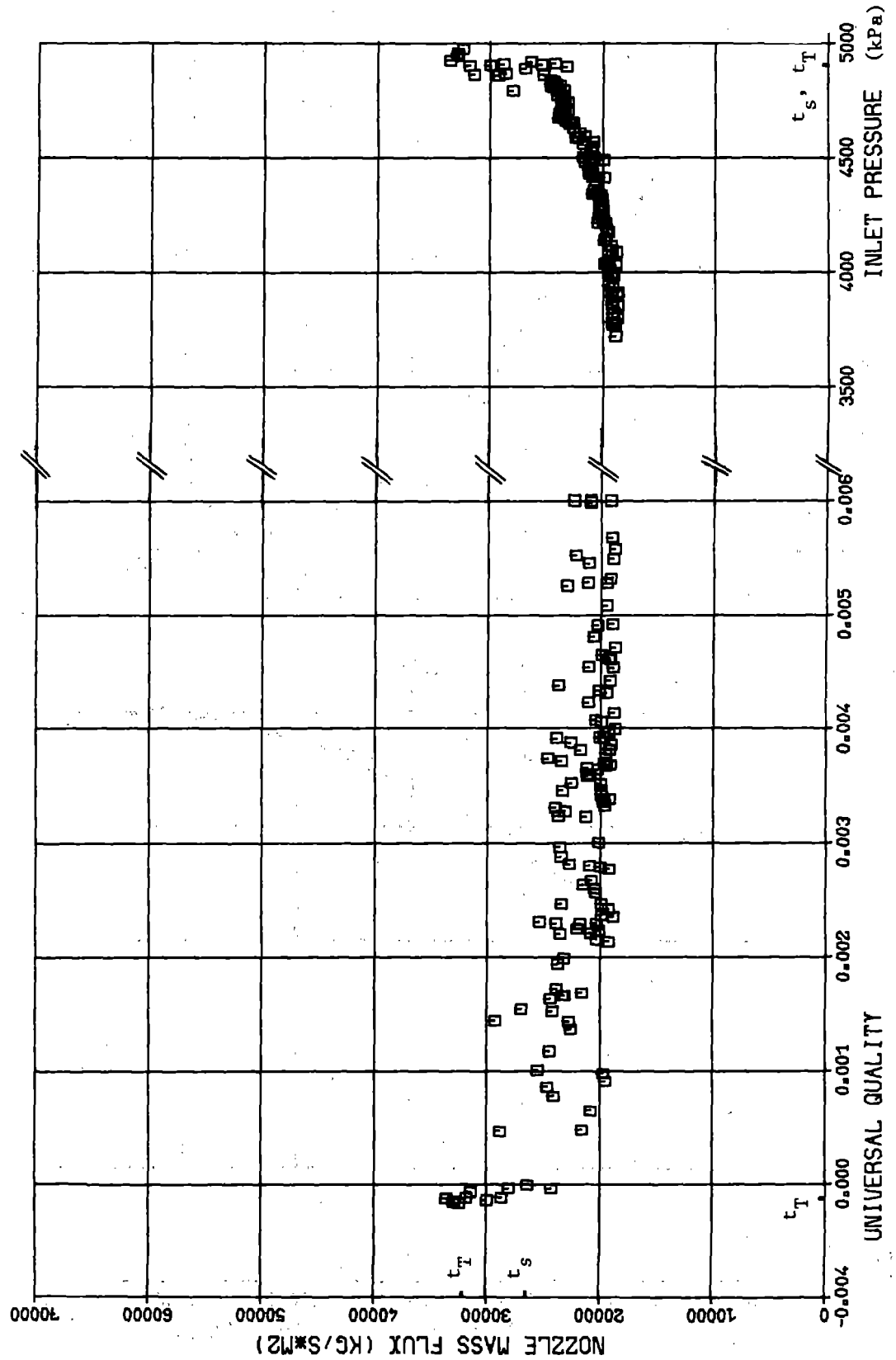


Plot E:23 - Test 23

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY
AND INLET STAGNATION PRESSURE

□ TEST 23

NOTE: Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

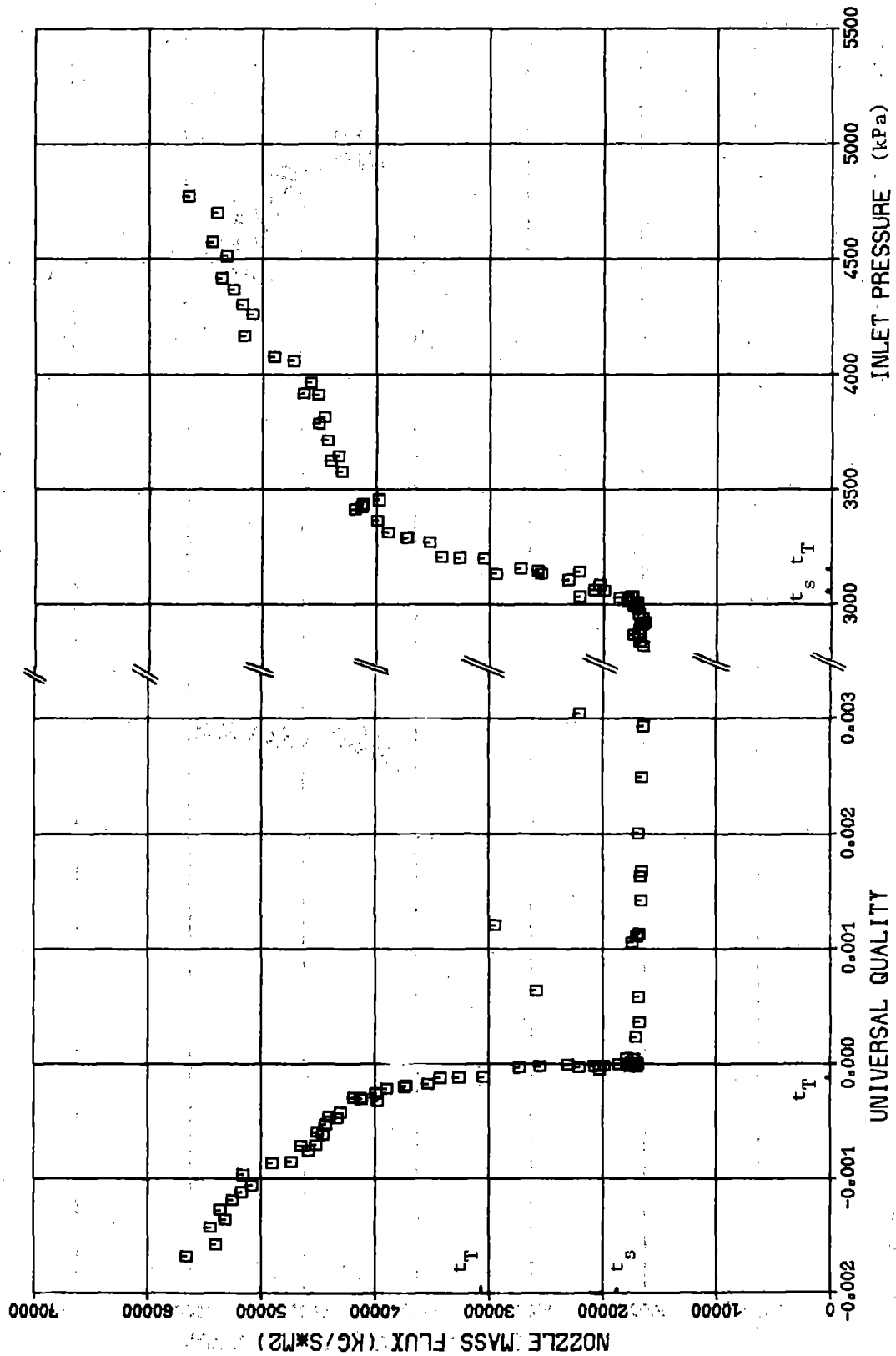


Plot E:24 - Test 24

NOTE: Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY AND INLET STAGNATION PRESSURE

□ TEST 24

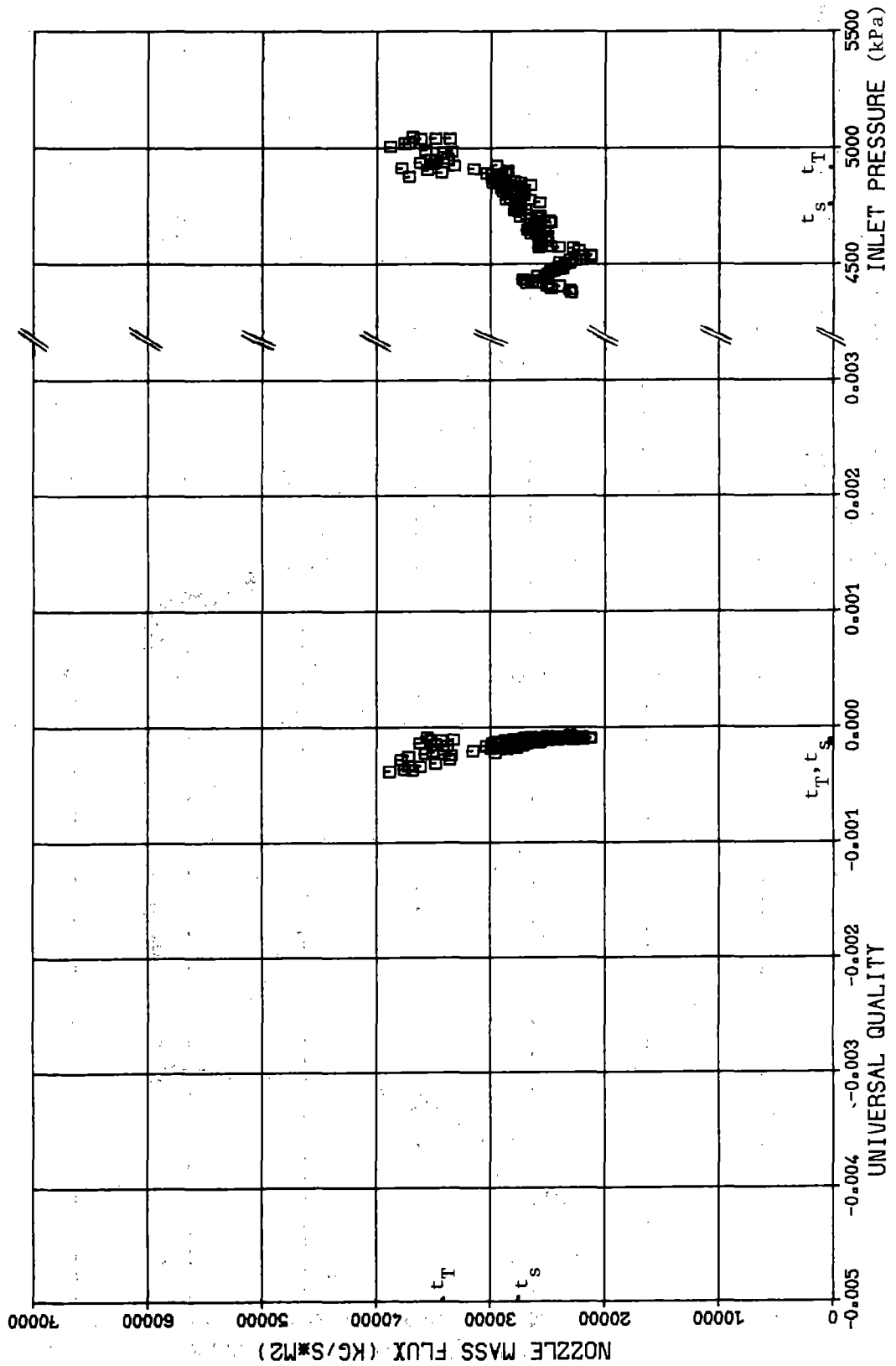


Plot E:25 - Test 25

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY
AND INLET STAGNATION PRESSURE

NOTE: 1. Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

Q TEST 25

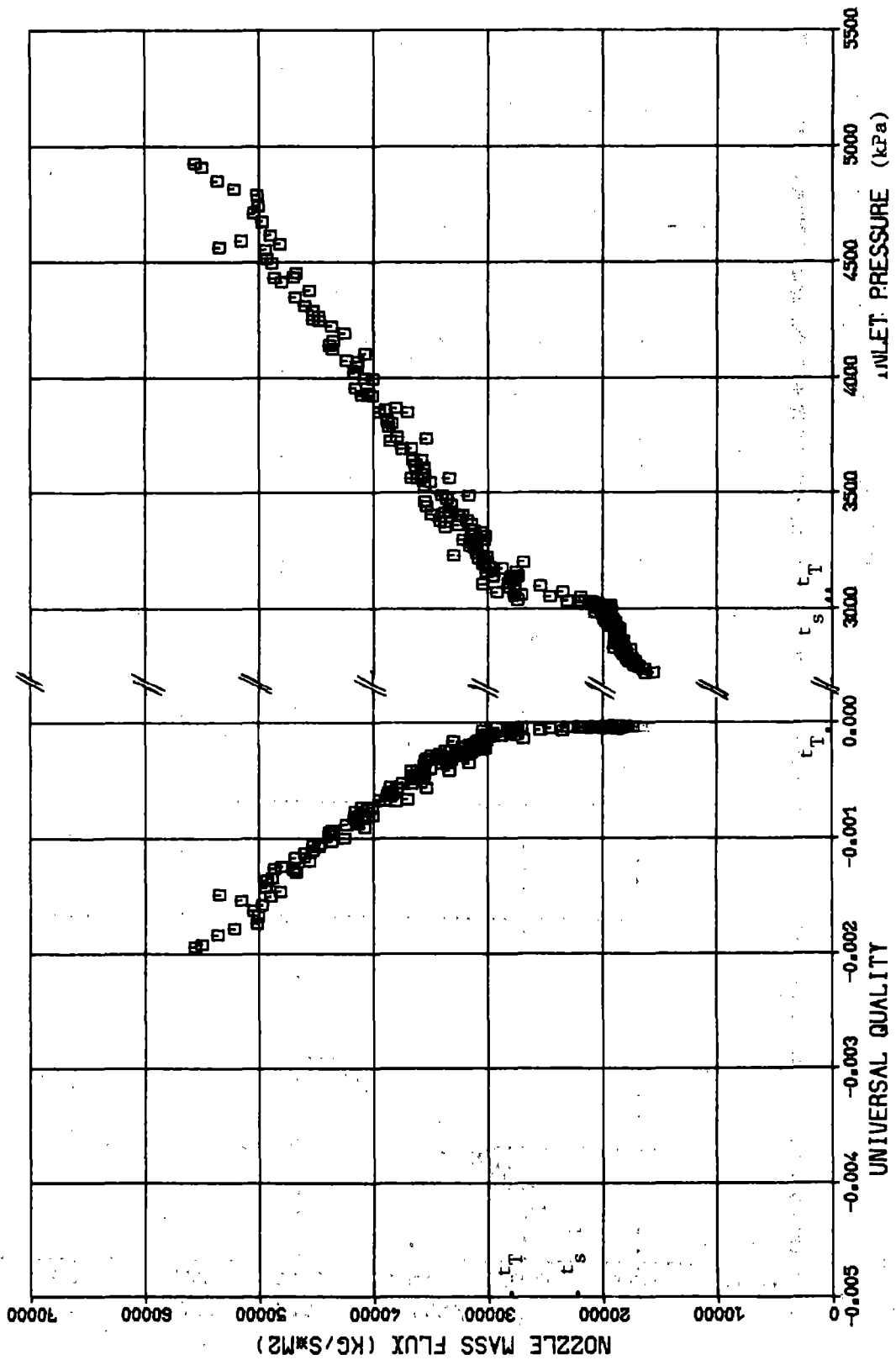


Plot E:26 - Test 26

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY
AND INLET STAGNATION PRESSURE

TEST 26

NOTE: 1. Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.



Plot E:27 - Test 27

NOZZLE MASS FLUX VS UNIVERSAL STAGNATION QUALITY AND INLET STAGNATION PRESSURE

□ TEST 27

NOTE: Quantities of the nozzle mass flux, quality and inlet pressure marked by t_T and t_s (see Table 8:1) indicate the start of the transition and saturated data respectively.

